

151-0591-00L

Control Systems I

Lecture Notes

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Preface

Control systems are ubiquitous in modern technology, from thermostats to complex aerospace guidance systems. This course introduces the mathematical foundations and design principles that enable engineers to model, analyze and synthesize systems.

These lecture notes have been created with two aims in mind. Firstly, to serve as another source of information that covers the [151-0591-00L](#) course, where great care was given to provide a clear narrative through the course material. The aim here is to help contextualize and ease the understanding of the course content. Secondly, a significant portion of the course content is visual and interactive in nature. Hence, reading from a (static) script may not be the best form to communicate certain ideas. In said cases, we have provided adjoining interactive diagrams and plots, which are mentioned and referenced within the script accordingly.

Although great effort was made to cover the course content, some course topics may be missing. Hence, it is encouraged to refer to this document as a supplement to the official lectures and problem sets, going to the respective section when in doubt, seeking clarification, or looking for a different perspective.

Lastly, this document is provided without guarantee of correctness – it is possible that errors remain. For this reason, we are very grateful for and open to feedback, which will help us improve the quality of these lecture notes.

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Chapter 1

Modeling

1.1 Introduction

Many of the systems we interact with every day — from cruise control on cars to insulin pumps that regulate blood sugar — rely on mathematical descriptions of the underlying physical system. In this chapter, we explore how to build such a model for a physical system. This model not only helps us understand and predict how the system behaves, but also serves as the foundation for designing controllers that ensure the system achieves its goals safely and efficiently.

In creating a model of a physical system, we want to mathematically represent the behaviour of the given system. More precisely, we want to write down equations that express the outputs of a system, as a function of the inputs. These systems, that map some input to an output, are referred to as *dynamical systems*, and are described by an input-output model.

Definition 1.1 (Input-Output model) *An input-output model is a map, Σ , that maps an input signal $u : t \mapsto u(t)$ to an output signal $y : t \mapsto y(t)$.*

Usually this input-output relationship is described by one or more differential equations, typically ordinary differential equations (ODEs) or partial differential equations (PDEs) — *Note that in this course we will only consider systems described by ODEs.*

The input-output relation can often be illustrated with the help of *block diagrams*. These diagrams provide an intuitive visual description of how the inputs are transformed into outputs, giving a good overview over complex interconnections and relationships of various subsystems, inputs, and outputs. The most simple case is some system Σ with a single input u and a single output y , as shown in Figure 1.1. Here the block Σ represents a dynamical system that receives a single input u and produces a single output y .

The system in Figure 1.1 is referred to as a *single input single output (SISO) system* — as it has a single input and a single output. Note that in general dynamical system may have multiple inputs and multiple outputs (MIMO), but these systems will not be covered in this

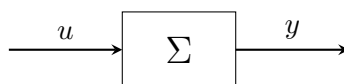


Figure 1.1: Block diagram showing input-output relation of system Σ

course and are instead considered in [Control Systems II](#).

Throughout the rest of this chapter, in [Section 1.2](#), we begin by exploring how to derive a mathematical model of a physical system, and how to obtain Σ . Here, we outline a general procedure for doing so, which you will study in greater depth in courses such as Thermodynamics I and Mechanics III. Often it is useful to work with a particular representation of the model, such as the state-space form — as introduced in [Section 1.3](#). Furthermore, many of the tools introduced later in the course require systems of a specific type (e.g., linear and time-invariant). We therefore discuss how to classify systems in [Section 1.4](#), and introduce the system class that will serve as our main focus in [Section 1.5](#). Not all systems naturally belong to this class, however, so in [Section 1.6](#) we consider how to convert more general systems into a suitable form. Finally, because real-world systems are often composed of multiple interacting subsystems, it can be complex to derive analytical formulations for the entire system. Hence, in [Section 1.7](#) we expand on how to represent and analyse such a system by independently deriving the model for each subsystem and then combining them using block diagrams.

1.2 System Modeling

In the following [Sections 1.2](#) and [1.3](#), we will outline a general procedure for deriving a mathematical model of a physical system and writing it in a general form. The procedure is as follows:

1. Identify the system boundaries: inputs, outputs, state and parameters of the system.
2. Write down the governing differential equations of the system.
3. Transform the system into the state-space form.

1.2.1 Identifying system boundaries

This first step in setting up a model is to identify the system boundaries, the inputs, outputs, state and parameters of the model. By asking how the world affects our system we can identify the inputs $u(t)$ and by looking at what we can observe about the system we obtain the outputs $y(t)$. Another important quantity is the *state* $x(t)$ of the system, which tells us how the system evolves internally over time. Together these three quantities will completely describe the behaviour of the dynamical system at hand. Finally, to fully describe a system, other system specific parameters have to be defined. These do not change over time and are fundamental to the system.

Inputs: The inputs to a system define how the outside world affects the system itself. This includes both, factors from the general environment and that of the user (who may have the goal

of controlling the system for a specific task). For example while controlling the temperature of the water coming out of a faucet, it is affected by both the user controlling the knob and the heat dissipation in the plumbing. In general, inputs can be classified as:

- *Endogenous*: Inputs that can be controlled. These are in general the control inputs (for example the faucet knob or the gas throttle in a car).
- *Exogenous*: Inputs that the user has no control over. These are what we will refer to as disturbances (for example the scale building up inside the pipes resulting in non-uniform water flow).

Outputs: Are the measurements or observations we make of the system. These can be separated into:

- Measured Outputs: What we can measure (from sensors), e.g. the speed of a car.
- Performance Outputs: not directly measurable, but we want to control, e.g. avg. Fuel consumption.

State: Encompasses everything that tells us how the system behaves internally over time. The state of a system, summarizes the effects of all past inputs. We will sometimes refer to this as the memory of the system, and is often represented as a vector (examples will be given later on).

Parameters: These are attributes/features that are specific to the problem, and will not change over time. For example, this could be the size of the faucet — but varying across different faucets, the values of these parameters may vary.

Example

As a first example consider a pendulum — we will use this as a running example throughout the course. Pendulums appear frequently in real-life applications, clocks, metronomes, and even amusement park rides. For the sake of the example, let's say that we have a metronome.

A schematic of the pendulum is shown in Figure 1.2. The pendulum consists of a point mass m attached to the tip of a massless rod length l . The point mass is free to rotate around a hinge point. In addition to gravity g , friction at the hinge also influences the systems motion (let c denote the friction coefficient). Through a time-varying external force $F(t)$ the pendulum/metronome is able to keep a constant rhythm. This force always acts horizontally on the point mass. There is also a sensor measuring the angle of the pendulum θ .

Let's identify the system boundaries: the inputs, outputs, state and parameters of the system.

The input $u(t)$ to the system is the external push force $F(t)$, and it is an endogenous input as this is something that we can manipulate (at least in our example). The output $y(t)$

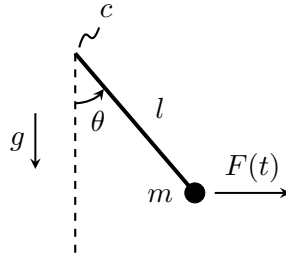


Figure 1.2: Schematic of a pendulum

is what we can observe about the system, in this case θ , which is measured by a sensor. The state of the system, $x(t)$, is not unique and depends on the way we model the system (more on that later), the thing to keep in mind is that the state has to capture the complete behaviour of the system. For the pendulum the state could be given as

$$x(t) = [\theta(t), \dot{\theta}(t)]^\top,$$

where θ is the angle θ and $\dot{\theta}$ is the angular velocity of the pendulum, or it could be the 2d-position of the mass,

$$x(t) = [l \cos \theta(t), l \sin \theta(t)]^\top.$$

The choice depends on the application; you will be able to practice this in *Problem Set 2*. The system parameters are all other quantities such as the mass m , length l , friction coefficient c , and gravity g — These do not vary over time and are treated as constants.

1.2.2 Deriving the governing equations

After identifying the inputs and outputs of our system we are ready to write down the governing equations that will describe the systems' behaviour. Firstly, a system will often comprise a so-called storage quantity (e.g., energy, temperature, mass, momentum, etc.). In the case of a heating system of a house, this storage quantity could be the temperature of the house or a specific room. Note that it is sometimes possible to consider multiple storage quantities for a given scenario, nevertheless sometimes it is easier to pick one option over another. Deciding which storage quantity is appropriate for a given context will mostly come through practice and experience. Secondly, the storage quantity will typically vary, induced by the flow of some other quantities. In the heating system example, the rate of change of heat transfer is equal to the rate of change of heat inflows (e.g., radiator) subtracted by the rate of change of heat outflows (e.g., losses). Thus, we can formulate a differential equation as

$$\frac{d}{dt} \text{storage} = \sum \text{inflows} - \sum \text{outflows}.$$

Usually the governing physical law, such as Newtons 2nd Law, mass conservation, or the conservation of energy, are applied directly as it is clear from the context which law to use. Nevertheless, most physical laws can be interpreted or rewritten as a variation of this storage approach.

In any case, in the scope of this course, we will end up with one (or more) ODEs describing the system at hand. The simplest version of which being a first order ODE of the form

$$\dot{x}(t) = ax(t). \quad (1.1)$$

We can see that for this first order ODE a change in x (given by its derivative $\dot{x}$) is a function of x itself. For higher order differential equations the relation is expanded to higher order derivatives, but the principle remains the same. We can conclude that for systems described by ODEs, the change of the system is related to its current state. Note that other variables can also impact the change of a system, e.g. the input. But in the most simple way we can say that

$$\text{How the system changes} = f(\text{current state}).$$

Example - Mass spring system

Consider a mass spring system, where a mass m is attached to a spring with spring constant k . Additionally, an external force $u(t)$ is applied — For clarity, we will start using $u(t)$ to denote the input to the system. A sensor gives us information about the speed of the mass. The schematic is shown in Figure 1.3.

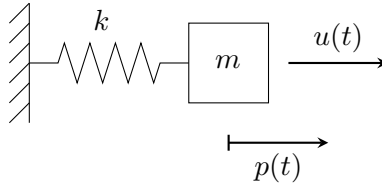


Figure 1.3: Disturbed mass spring system

The position of the mass, and consequently the displacement of the spring, is given by $p(t)$. In order to derive the governing differential equations of this system we can use Newton's 2nd Law and the formula for the restoring force in the spring (Hooke's Law) given by

$$F_{\text{net}}(t) = ma(t), \quad (1.2)$$

$$F_{\text{spring}}(t) = -kp(t), \quad (1.3)$$

respectively. Considering all forces acting on the mass, the input $u(t)$ and Equation (1.3), substituting the relevant quantities into Equation (1.2) gives

$$u(t) - kp(t) = m\ddot{p}(t),$$

which can be rewritten to,

$$\ddot{p}(t) = -\frac{k}{m}p(t) + \frac{1}{m}u(t). \quad (1.4)$$

Equation (1.4) describes the dynamic behaviour of this system (this is equivalent to finding the equation of motion) and is the governing ODE of the physical system in Figure 1.3.

Since k and m are constants, we can observe that the rate of change $\ddot{p}(t)$ is proportional to $p(t)$, which is in line with the behaviour of ODEs introduced above.

Lastly, in the prompt above, we stated that the system has a sensor that measures the velocity of the mass, i.e. $\dot{p}(t)$. Therefore, the output is then given by

$$y(t) = \dot{p}(t). \quad (1.5)$$

Example - Pendulum

As an example let's revisit the pendulum introduced in Figure 1.2 and derive the governing ODE. Here, the storage quantity is angular momentum, with Newtons Second Law for rotations being the relevant physical law. This is given by

$$I\ddot{\theta} = M_B, \quad (1.6)$$

where I is the moment of inertia of the point mass about the axis of rotation, M_B is the sum of all torques acting on the hinge point, and $\ddot{\theta}$ the angular acceleration.

To understand what torques are being applied to the point mass, consider the free body diagram in Figure 1.4.

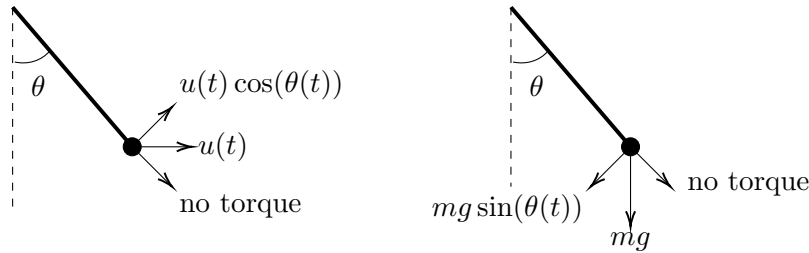


Figure 1.4: Decomposition of forces acting on the system that will result in torque around hinge point. Left is the decomposition of the input force, and right of the gravitational force.

Additionally, the torque induced by the friction is proportional to the angular velocity and scaled by the friction coefficient c . Thus, the total torque acting on the hinge point is given by

$$M_B = \underbrace{\text{gravitational torque}}_{-lmg \sin(\theta(t))} + \underbrace{\text{friction torque}}_{-c\dot{\theta}(t)} + \underbrace{\text{input torque}}_{u(t)l \cos(\theta(t))}.$$

Additionally, the moment of inertia for this setup is $I = ml^2$. Putting everything together results in the following differential equation:

$$ml^2\ddot{\theta}(t) = -lmg \sin(\theta(t)) - c\dot{\theta}(t) + u(t)l \cos(\theta(t)) \quad (1.7)$$

$$\ddot{\theta}(t) = -\frac{g}{l} \sin(\theta(t)) - \frac{c}{ml^2} \dot{\theta}(t) + \frac{1}{ml} u(t) \cos(\theta(t)). \quad (1.8)$$

We commonly rearrange Equation (1.7) to obtain the form in Equation (1.8), the reason for this will become clear in Section 1.3.

Lastly, we define the output to be the sensor measurement, i.e. the angle θ . Therefore, the output is then given by

$$y(t) = \theta. \quad (1.9)$$

It is important to note that any dynamical model is a simplification of reality; it can never capture every detail of the physical system. The required level of detail depends on the application, but every model remains an approximation. Thus, it should be kept in mind that

“All models are wrong, but some are useful.”
— George E. P. Box

1.3 State Space Representation

Many physical systems are described by second- or higher-order ODEs, e.g. the pendulum seen in Equation (1.8). While these higher order ODEs might be directly solvable, it is often not practicable to do so. Instead, a general approach is to represent an n^{th} order ODE by transforming it into a system of n first order ODEs.

1.3.1 Transforming higher order ODE's into first order

To transform any second-, or higher-order, ODE of the general form

$$x^{(n)}(t) = f(x(t), \dot{x}(t), \ddot{x}(t), \dots, x^{(n-1)}(t)),$$

to a system of first order ODEs, we make use of the substitution

$$\begin{aligned} x_1(t) &= x(t) \\ x_2(t) &= \dot{x}(t) \\ x_3(t) &= \ddot{x}(t) \\ &\vdots \\ x_n(t) &= x^{(n-1)}(t). \end{aligned} \quad (1.10)$$

By substituting the new variables x_i for the derivatives of the original equation we obtain first order system.

Example - Mass spring system

Recall the second order ODE from the previously introduced mass spring system, given by Equation (1.4):

$$\ddot{p}(t) = -\frac{k}{m}p(t) + \frac{1}{m}u(t).$$

We recognize that it can be re-written as a system of first order ODEs. With the substitution

$$\begin{aligned} x_1(t) &= p(t) \\ x_2(t) &= \dot{p}(t), \end{aligned}$$

we obtain a system of first order differential equations:

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\frac{k}{m}x_1(t) + \frac{1}{m}u(t). \end{aligned} \tag{1.11}$$

Additionally, we know the output to be given by the speed $\dot{p}(t)$ of the mass. This can also be expressed with the new variables, and so we can write the output to be

$$y(t) = x_2(t).$$

Example - Pendulum

Similarly, we can convert the ODE for the pendulum, Equation (1.8), into a system of first order ODEs. Recall that Equation (1.8) is given by:

$$\ddot{\theta}(t) = -\frac{g}{l} \sin(\theta(t)) - \frac{c}{ml^2} \dot{\theta}(t) + \frac{1}{ml} u(t) \cos(\theta(t)).$$

This second order equation can be transformed into a first order system with the substitution

$$\begin{aligned} x_1(t) &= \theta(t), \\ x_2(t) &= \dot{\theta}(t). \end{aligned}$$

By applying the substitution, we arrive at the first order system

$$\begin{aligned} \dot{x}_1(t) &= x_2(t), \\ \dot{x}_2(t) &= -\frac{g}{l} \sin(x_1(t)) - \frac{c}{ml^2} x_2(t) + u(t) \frac{1}{ml} \cos(x_1(t)). \end{aligned} \tag{1.12}$$

Recall that the output was defined by the sensor that measures the angle θ , which, through the substitution, is given by x_1 , i.e.

$$y(t) = x_1(t).$$

1.3.2 General State Space Formulation

If we take a closer look at the systems of first order ODEs in Equations (1.11) and (1.12), we can see that the variables $x_i(t)$ play a fundamental role in describing the system. The collection of all these variables is called the state (as briefly mentioned in the introduction), and the dimension n of the state ($x(t) \in \mathbb{R}^n$) is equal to the dimension/order of the system. Importantly, the state $x(t)$ cannot be manipulated or accessed directly, it can only be controlled through the input and observed through the output. It tells us how the system evolves internally over time. One way to think about it is as a memory, containing a summary of how the system behaved in the past. Given the state $x(t)$ and the current input $u(t)$, we can uniquely predict any future behaviour. It turns out that any dynamical system can be described in this way, and expressed in the form

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t)).\end{aligned}\tag{1.13}$$

This is called the state space form, since it describes how the state $x(t)$ changes over time. The function f gives information on how the state vector itself changes, and the function h describes how the output y behaves. The output is defined through the sensor configuration and can be any combination of the states and inputs, depending on what we want to observe in our system. Importantly, to obtain a representation of the output $y(t)$ we first require an explicit solution of the state $x(t)$ given by the solution of the first order system of ODEs. The choice of a systems state is not unique and can vary depending on how we model the system. Every different choice of states is called a realization (there are infinitely many realizations). If we choose a realization with the smallest possible state vector, it is called minimal realization. We can also change the state/realization by performing a coordinate transformation, e.g. as seen in Linear Algebra (more on that later).

1.4 System Classification

With the modeling approach introduced above we are able to model a large variety of systems. However, many of the resulting models are complicated to deal with — in some cases there may not even exist analytical solutions. To avoid such system and since the tools taught in this course will often require systems to be of a specific type, it is important to classify them. This classification will apply to the input output relationship we obtain from the model by solving the system of first order ODEs and the consequent solution for the output $y(t)$. In particular, we will distinguish systems according to whether they are:

- Linear or Nonlinear,
- Causal or Noncausal,
- Static (memoryless) or Dynamic,

- Time invariant or Time variant.

1.4.1 Linearity

For an input-output system Σ to be linear two conditions have to be fulfilled:

1. Additivity: $\Sigma(u_1 + u_2) = \Sigma u_1 + \Sigma u_2$
2. Homogeneity: $\Sigma(\alpha u) = \alpha \Sigma u$, $\alpha \in \mathbb{R}$

We can summarize both conditions to:

$$\Sigma(\alpha u_1 + \beta u_2) = \alpha \Sigma u_1 + \beta \Sigma u_2, \quad \alpha, \beta \in \mathbb{R}. \quad (1.14)$$

The concept of linearity also implies the idea of superposition — which is central in the analysis of linear systems. It can be summarized as follows.

For a linear system the superposition principle implies that we can:

- Break down complicated input signals into simpler components

$$u = \alpha u_1 + \beta u_2$$

- Compute the output for each simple input separately

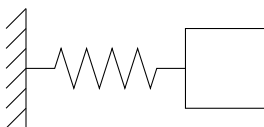
$$y_1 = \Sigma u_1, \quad y_2 = \Sigma u_2$$

- Sum all the simple outputs together to obtain the output to the complex input

$$y = \alpha y_1 + \beta y_2$$

Linear Example

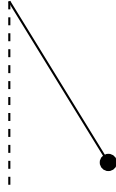
The mass spring system from before is a prime example for a linear system. You can check all the conditions listed above and convince yourself. For a system like this the superposition principle will be fulfilled.



$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\frac{k}{m}x_1(t) + \frac{1}{m}u(t) \\ y(t) &= x_2(t) \end{aligned}$$

Nonlinear Example

The swing model is a nonlinear system. You can check the conditions for linearity and convince yourself. Usually sine and cosine terms are a good indicator for nonlinearity.



$$\dot{x}_1(t) = x_2(t)$$

$$x_2(t) = -\frac{g}{l} \sin(x_1(t)) - \frac{c}{ml^2} x_2(t) + u(t) \frac{1}{ml} \cos(x_1(t))$$

$$y(t) = x_1(t)$$

1.4.2 Causality

A system is said to be causal, if future inputs do not affect the present or past outputs. All practically realizable systems are causal — Otherwise, you could predict the future. Consider a thermostat in your home. It responds to current and past temperature readings, turning the heating on once the temperature drops below a certain threshold. The thermostat's actions are based solely on present and past inputs, never on future inputs. While true non-causal systems don't exist in real-time physical world, we can find examples in data processing. For example a video editing software applying a blur effect to an entire video clip is effectively non-causal. When blurring a frame at time t , it uses information from frames before and after t . This process “knows” about future frames, making it non-causal from the perspective of real-time video processing.

Examples

Causal systems include:

- $y(t) = u(t)$
- $y(t) = u(t - \tau), \forall \tau > 0$
- $y(t) = \cos(3t + 1)u(t - 1)$
- $y(t) = \int_{-\infty}^t u(\tau) d\tau$

Non-Causal systems include:

- $y(t) = u(t - a), \forall a < 0$
- $y(t) = u(t + 1) + u(t) + u(t - 1)$
- $y(t) = u(bt), \forall b > 1$
- $y(t) = \int_{-\infty}^{t+1} u(\tau) d\tau$

1.4.3 Memoryless vs. Dynamic

An input-output system is memoryless or static if the output $y(t)$ at some specific time $t = t^*$ is only a function of the input of that exact same time $u(t^*)$. In other words, the present output only depends on the present input and not on past or future inputs. Systems described by ODEs are always dynamic, whereas static systems are usually described by algebraic equations. You can think of system from Mechanics I as static and those from Mechanics III as dynamic. Another example of a static system is a regular kitchen scale. It will always show the apparent weight of the object placed upon it, regardless of what object was placed on it an hour, two days, or a month ago. An example of a dynamic system is the longitudinal speed of a car. A specific

depression of the gas pedal, e.g. 50% of the maximum, does not correspond to one specific speed, because the car's speed depends on both how much the gas pedal is currently pressed and how it has been pressed before. Other factors such as inertia and friction also contribute, meaning the speed evolves dynamically based on the system's history. In the context of state space models we saw that the state is an internal memory of the system, keeping track of its past behaviour. For a system with a zero dimensional state this memory does not exist, and thus the system must be memoryless.

Examples

Memoryless (or static) systems include:

- $y(t) = 3u(t)$
- $y(t) = t^2 u(t)$
- $y(t) = 2^{-(t+1)} u(t)$
- $y(t) = \sqrt{\sin(u^2(t))}$

Dynamic systems include:

- $y(t) = \int_{-\infty}^t u(\tau) d\tau$
- $y(t) = \dot{u}(t)$
- $y(t) = u(t^2)$
- $y(t) = u(t - a), \forall a \neq 0$

1.4.4 Time Invariant vs. Time Variant

A time invariant system will always have the same output to a certain input, independent of when the input is applied. Formally this means that we can shift the input in time and the output will also be shifted by the same amount. An example for a time invariant system is an ideal light bulb. It doesn't matter if you turn it on at night or during the day, it will always produce the same amount of light. On the other hand a rocket is a time-varying system. During liftoff at sea level a rocket is very heavy since all the tanks are filled and gravity has a very strong effect, thus a lot of thrust is needed to propel the rocket upwards. Once the rocket expels most of its fuel and gravity gets weaker less thrust is needed to propel the rocket by the same amount. So the same input (thrust) generates different outputs (distance moved) depending on when the input is applied. Figure 1.5 shows this in a graphical way.

Examples

Time varying systems include:

- $y(t) = \cos(t)u(t)$
- $y(t) = u(3t + 1)$
- $y(t) = \sqrt{t - \cos^2(t)} u(t^2)$
- $y(t) = \int_{-\infty}^{t-2} e^{-\tau} u(\tau + 2) d\tau$

Time invariant systems include:

- $y(t) = u(t - 1)u(t + 2)$
- $y(t) = 1 + u(t - 1) + 2u(t)$
- $y(t) = 2 \int_{-\infty}^t u^2(\tau) d\tau$
- $y(t) = \cos(u(t)) \int_{-\infty}^{t-2} u(\tau) d\tau$

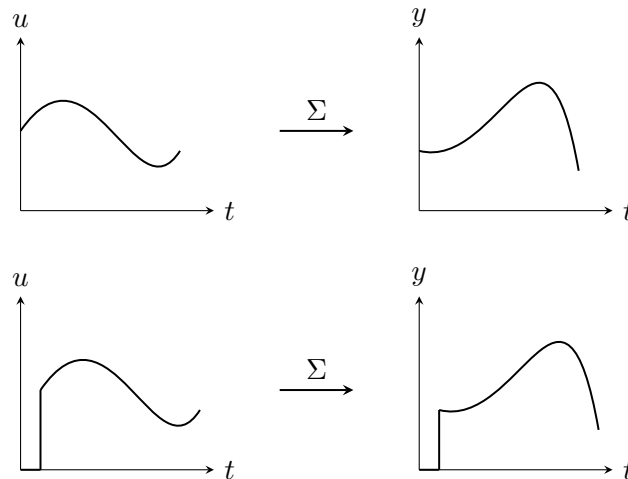


Figure 1.5: Visual representation of a time invariant system, where a delayed input yields the same output as the regular input

1.5 Linear Time Invariant Systems

Now we know how to classify a system, we can start to look at what types of systems we are interested in. For the majority of the course, we will consider systems that are:

- Single Input, Single Output (SISO)
- Linear
- Time invariant
- Causal

Systems that are Linear and Time-Invariant are referred to as LTI systems. This is a very restrictive class of systems and most real systems are in fact not LTI. Nevertheless, real systems can often be approximated very well by LTI systems. Furthermore, most of the control systems in use today are even designed by the principles we will learn through the LTI systems in this course.

One very nice characteristic of LTI systems is that we can always write their state space model in the form

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t),\end{aligned}\tag{1.15}$$

where A , B , C , and D are constant matrices. As discussed in Section 1.3 the dimension/order of a system is given by the dimension of the state, often denoted as n . Based on that we can always know the dimensions of A , B , C , and D .

For a LTI SISO system with $x \in \mathbb{R}^n$, the dimensions of the matrices A, B, C , and D are given by:

$$A \in \mathbb{R}^{n \times n}, B \in \mathbb{R}^{n \times 1}, C \in \mathbb{R}^{1 \times n}, D \in \mathbb{R} \quad (1.16)$$

Note that the above is only true for SISO systems. For multiple inputs, $u(t) \in \mathbb{R}^m$, $B \in \mathbb{R}^{n \times m}$, and for multiple outputs, $y(t) \in \mathbb{R}^p$, $C \in \mathbb{R}^{p \times n}$.

Example - Mass spring system

Recall the state space representation of the mass spring system given by Equation (1.11),

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\frac{k}{m}x_1(t) + \frac{1}{m}u(t). \end{aligned}$$

Writing the state as a vector: $x(t) = [x_1(t) \ x_2(t)]^\top$ leads us to the LTI state space form

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t) + 0 \cdot u(t). \end{aligned}$$

By comparison with Equation (1.15) we can directly identify the matrices A, B, C , and D to have the values

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}, C = \begin{bmatrix} 0 & 1 \end{bmatrix}, D = 0.$$

Example - Pendulum

Let's look at the other example from before, the pendulum, and try to find the A, B, C , and D matrices for this system. Recall that the state space representation is given by Equation (1.12):

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\frac{g}{l} \sin(x_1(t)) - \frac{c}{ml^2} x_2(t) + u(t) \frac{1}{ml} \cos(x_1(t)) \\ y(t) &= x_1(t). \end{aligned}$$

But how do we write this in the LTI state space form? As we have seen in Section 1.4.1, this is not a linear system. Thus, we can not write it in the LTI form. Hence, in the following section we will learn to approximate it as an LTI system and find an LTI model that represents our original system well enough.

1.6 Linearization

Unfortunately most real life system are not linear and thus not representable in the LTI form. Even a “simple” system as the pendulum is nonlinear. Luckily we can approximate finite-dimensional, time-invariant, causal nonlinear system as LTI systems through *linearization*. With this approximation we can extend the use of the tools designed for LTI systems to nonlinear systems.

The first part of linearization consists of finding an equilibrium point, we can later use to linearize the system around. An equilibrium point is some combination of state and input where the system is static — i.e., does not change.

Definition 1.2 (Equilibrium Point) *A state-input pair (x_e, u_e) is called an equilibrium point of the system*

$$\dot{x}(t) = f(x(t), u(t)), \quad (1.17)$$

if, when the input and state are held constant $x(t) = x_e, u(t) = u_e$, the pair satisfies the system dynamics. Equivalently,

$$f(x_e, u_e) = 0. \quad (1.18)$$

Example - Pendulum

Let's try to find an equilibrium point for the pendulum example from before. Recall the state space form from Equation (1.12) given by

$$\begin{aligned} \dot{x}_1(t) &= x_2(t), \\ \dot{x}_2(t) &= -\frac{g}{l} \sin(x_1(t)) - \frac{c}{ml^2} x_2(t) + u(t) \frac{1}{ml} \cos(x_1(t)), \\ y(t) &= x_1(t), \end{aligned}$$

where $f(x(t), u(t))$ is the vector-valued function given by

$$f_1(x(t), u(t)) = x_2(t), \quad (1.19)$$

$$f_2(x(t), u(t)) = -\frac{g}{l} \sin(x_1(t)) - \frac{c}{ml^2} x_2(t) + u(t) \frac{1}{ml} \cos(x_1(t)). \quad (1.20)$$

By inspection, we can see that $f_1(t) = x_2(t) = 0$. Hence, $x_{2,e} = 0$. Intuitively, this makes sense, since for the system to be not changing, the mass must be at rest (not moving). We can then plug in $x_2(t) = 0$ into $f_2(x(t), u(t))$,

$$f_2(x(t), u(t)) = -\frac{g}{l} \sin(x_1(t)) + u(t) \frac{1}{ml} \cos(x_1(t)) = 0.$$

Here we see that we have one equation, $f_2(\cdot, \cdot) = 0$, but two unknowns, $x_1(t)$ and $u(t)$. Instead of finding all equilibrium points, for the sake of the example, let us consider those

where $u(t) = u_e = 0$. Where would you expect these to be intuitively? Mathematically we can evaluate the above expression to be

$$f_2(x(t), u(t)) = -\frac{g}{l} \sin(x_1(t)) - \frac{c}{ml^2} \overset{0}{x_2(t)} + \frac{1}{ml} \overset{0}{u(t)} \cos(x_1(t)) = 0.$$

This leaves us with

$$-\frac{g}{l} \cdot \sin(x_1(t)) = 0.$$

Resulting in the two equilibrium points

$$\begin{aligned} 1 : \quad x_e &= [0, 0]^\top, \quad u_e = 0 \\ 2 : \quad x_e &= [\pi, 0]^\top, \quad u_e = 0. \end{aligned}$$

This aligns with our intuition, since the first equilibrium point is equivalent to the pendulum hanging straight down, and the second one is pointing vertically up. If we don't perturb the system, i.e. $u(t) = 0$, it should not change and stay in those configurations for all times.

1.6.1 Jacobian Linearization Procedure

After finding the relevant equilibrium points we are now ready to linearize around them. To that end we use the Jacobian linearization procedure. Where we do a Taylor series expansion around (x_e, u_e) of the nonlinear system's dynamics. Here we only use first order terms and neglect all higher order terms. Additionally, since $f(x_e, u_e) = 0$ the 0-order terms are all equal to zero. The linearized LTI SISO system matrices are then given by

$$\begin{aligned} A &= \left. \frac{\partial f(x, u)}{\partial x} \right|_{(x_e, u_e)} = \left[\begin{array}{ccc} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{array} \right]_{(x_e, u_e)} & A \in \mathbb{R}^{n \times n} \\ B &= \left. \frac{\partial f(x, u)}{\partial u} \right|_{(x_e, u_e)} = \left[\begin{array}{c} \frac{\partial f_1}{\partial u} \\ \vdots \\ \frac{\partial f_n}{\partial u} \end{array} \right]_{(x_e, u_e)} & B \in \mathbb{R}^{n \times 1} \\ C &= \left. \frac{\partial h(x, u)}{\partial x} \right|_{(x_e, u_e)} = \left[\begin{array}{ccc} \frac{\partial h}{\partial x_1} & \dots & \frac{\partial h}{\partial x_n} \end{array} \right]_{(x_e, u_e)} & C \in \mathbb{R}^{1 \times n} \\ D &= \left. \frac{\partial h(x, u)}{\partial u} \right|_{(x_e, u_e)} = \left[\frac{\partial h}{\partial u} \right]_{(x_e, u_e)} & D \in \mathbb{R}^{1 \times 1}. \end{aligned} \tag{1.21}$$

Graphically, and in 2 dimensions, you can think of this approach as fitting a line onto a curve at one specific (equilibrium) point. Figure 1.6 graphically visualizes this approach. The figure also shows that as we leave the vicinity of the equilibrium point, the linear approximation

will get worse. So our approximation is only valid if we remain close to the equilibrium point.

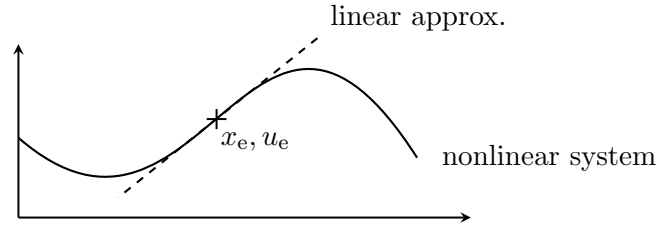


Figure 1.6: Graphical representation of the linearization process

Example

Let's apply this procedure to the pendulum example. We already have the equilibrium points, so we just have to plug everything into Equation (1.21). The functions f_1, f_2 and h are given by

$$\begin{aligned} f_1 &= x_2(t) \\ f_2 &= -\frac{g}{l} \sin(x_1(t)) - \frac{c}{ml^2} x_2(t) + u(t) \frac{1}{ml} \cos(x_1(t)) \\ h &= x_1(t). \end{aligned}$$

In this 2-dimensional case the Matrix A is given by

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}.$$

If we evaluate the partial derivatives at the equilibrium point $x_{e,1} = (0, 0)$, $u_e = 0$ the matrix A of the linearized system results in

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{c}{ml^2} \end{bmatrix}.$$

The same can be done for B, C, and D to get a complete LTI state space model. In our case they are given by

$$B = \begin{bmatrix} 0 \\ \frac{1}{ml} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

Important: this approximation is only valid in a small neighborhood around the equilibrium point $x_{e,1} = (0, 0)$. This means that if we linearize around the other equilibrium point we will get a different linearized state space model. Doing the same procedure as above with the other equilibrium point $x_{e,2} = (\pi, 0)$, $u_e = 0$ results in a new set of matrices given

by

$$A = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{c}{ml^2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{ml} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

Depending on at what angle we want to operate the system at we have to choose different linearized models to represent the system.

1.7 Block Diagrams

In the sections above we described how to construct a model of a system that takes some input and, considering the current state of the system, produces some output. We can use these models to predict how a system will react given some input. Consequently, by choosing the input carefully, we can make the system behave in a desired way. In an abstract way we can visualize this relationship by representing the model of our physical system, also called plant, as a block containing the ODEs describing the system and the corresponding inputs and outputs as arrows entering/leaving the system respectively as illustrated in Figure 1.7.

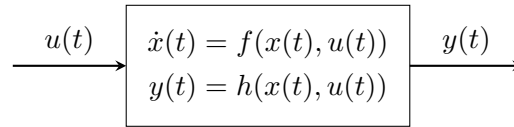


Figure 1.7: Block diagram showing the state space form.

Usually it is not convenient to write out the entire system of ODEs inside small blocks, so we simplify the notation and represent the whole model as an operator Σ . Importantly Σ and the state space representation map the input to the output in the exact same way. With this in mind we can now simplify the block diagram from Figure 1.7 and express the relationship between the input and output with the help of the operator Σ as shown in Figure 1.8.



Figure 1.8: Input-output relationship of system Σ

Both block diagrams in Figures 1.7 and 1.8 represent the same thing and are just different ways to represent a system. Now imagine that instead of having just one system in isolation we consider a second system. As a simple example we can take the pendulum from before and place it next to a bell. As we excite the pendulum it starts swinging and at some point it will hit the bell producing a sound. Effectively we can conclude that if the pendulum reaches some angle a sound will be produced. So the output of the first system (pendulum) directly influences the second system (bell). This can be interpreted as a connection in series, and we can represent it with the help of block diagrams. Figure 1.9 shows such an interconnection of two systems Σ_1 and Σ_2 in series. To obtain a similar input-output relationship, to the one in Equation (1.22),

we introduce an intermediate signal a and write out the relations for both Σ_1 and Σ_2 separately. By substituting a we can then obtain a direct relation between u and y — neglecting time for simplicity of exposition.

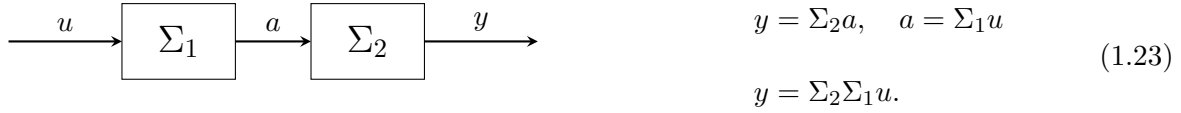


Figure 1.9: Two systems in series

Similar to the series connection we can also connect systems in parallel. The corresponding block diagram is shown in Figure 1.10. To obtain an input-output relationship we sum both outputs resulting from Σ_1 and Σ_2 .



Figure 1.10: Two systems in parallel

We can also combine the parallel and series connections to analyse more intricate diagrams. Figure 1.11 shows a mixture of series and parallel interconnections. To find the mathematical relation between u and y we proceed in the same way as above. The systems in series are multiplied and the ones in parallel are added.

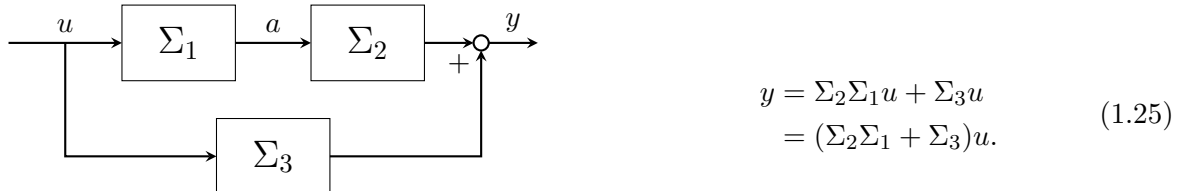
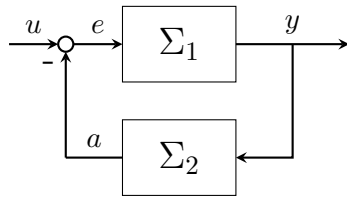


Figure 1.11: Series and Parallel interconnection

A very important configuration, that is fundamental for control systems, is the feedback architecture. Here we feed the output back to the input. The exact implications will be discussed in detail in following chapters, but for now let's analyse its effects on the block diagrams. A possible form of feedback is shown in Figure 1.12. Notice that the parallel interconnection has a negative sign. This has to be taken into account when writing down the input-output relation. As before we can define intermediate signals e.g. a and e . By substituting and rearranging we can obtain the input-output relationship given by Equation (1.26).

Note that the systems Σ_i can be matrices or scalars. Since matrices do not commute, we have to be careful with the order of multiplication. That's why, in general, you always have to respect the order of the systems. This also implies that, when dealing with matrices, we can't do division, but have to multiply with the inverse instead. Only if explicitly stated, that all



$$y = \Sigma_1 e, \quad a = \Sigma_2 y, \quad e = u - a,$$

therefore

$$e = u - \Sigma_2 y,$$

and

$$y = \Sigma_1(u - \Sigma_2 y).$$

Rearranging

$$\begin{aligned} y + \Sigma_1 \Sigma_2 y &= \Sigma_1 u, \\ (1 + \Sigma_1 \Sigma_2) y &= \Sigma_1 u, \end{aligned}$$

which finally yields

$$y = (1 + \Sigma_1 \Sigma_2)^{-1} \Sigma_1 u. \quad (1.26)$$

Figure 1.12: Negative Feedback

systems are scalars, we can do division and change the order of multiplication. E.g. the feedback interconnection described by Equation (1.26) can be rewritten in the scalar case to

$$y = \frac{\Sigma_1}{1 + \Sigma_1 \Sigma_2} u.$$

Chapter 2

Time Response

2.1 Introduction

In the previous chapter, we learned how to describe real-world systems with ODEs and express them in state-space form. We also saw how to classify these systems, and that from now on we will focus exclusively on LTI systems. However, while the state-space form provides an analytic description of a systems' behaviour, it does not yield an explicit input–output relation. Therefore, the next step is to solve the LTI system and obtain an explicit expression for the system output $y(t)$ — That is, determine the mapping Σ in the block diagram.

Consider, the general LTI state-space model

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t).\end{aligned}$$

The analytical expression of $y(t)$ is referred to as the *time response* of the system, as it tells us how the system responds to a specific initial condition and input over time. How to obtain this expression is the main topic of this chapter. Crucially, the presented approach can be applied to any LTI system.

The approach relies on obtaining an expression for the state $x(t)$ and then substituting it into the output equation to obtain $y(t)$. To do so, we can take advantage of the fact that the system is linear and therefore the principle of superposition applies — i.e., we decompose the response into two parts: the initial condition response and the forced response. This is analogous to solving an ODE by decomposing it into a homogenous and particular solution.

- Initial condition response:

$$\begin{cases} x_{\text{IC}}(0) = x_0 \\ u_{\text{IC}}(t) = 0, \quad t \geq t_0, \end{cases} \rightarrow x_{\text{IC}}(t), y_{\text{IC}}(t).$$

This is the systems response driven by a non-zero initial condition and zero input. For

example, in a pendulum, this corresponds to releasing it from a specific starting point and letting it move freely.

- Forced response:

$$\begin{cases} x_F(0) = 0, \\ u_F(t) = u(t), \quad t \geq t_0, \end{cases} \rightarrow x_F(t), y_F(t).$$

This is the response due solely to the external input and zero initial condition. In the pendulum example, this corresponds to applying a torque to a pendulum whose bob is at rest.

To obtain the response to any initial condition and any input, we use the concept of superposition and solve for the initial condition $y_{IC}(t)$, and forced response $y_F(t)$ separately. Then the complete response is given by the sum of the two cases

$$y(t) = y_{IC}(t) + y_F(t).$$

This enables us to analyze the effects of initial conditions and inputs independently, while also framing the problem in a known way.

In Section 2.2 we look at obtaining the complete response of a first order system, and then apply the same methodology to higher order systems in Section 2.3. Next we study the time responses to common inputs in Section 2.4, because they are standard test signals used to probe system behaviour and provide us with an intuition for the concept of *stability*. Finally, we establish the connection between the long-term behaviour of the time response and fundamental system properties, leading to the introduction of *stability* in Section 2.5.

2.2 First Order Systems

A first-order system is one whose dynamics are governed by a first-order ODE of the form $\dot{x}(t) = f(x(t), u(t))$, in which the rate of change of the state depends on the state and the input. No higher-order derivatives, such as $\ddot{x}(t)$, appear. In most cases, the order of the system can be identified by looking at the dimension of the state. For the first-order case this would mean that $x \in \mathbb{R}$, whilst for a second order system, $x \in \mathbb{R}^2$.

In general, provided the state-space representation is minimal (i.e. it uses the smallest possible state vector that fully describes the system's input-output behaviour), the order coincides with the dimension of the state. The precise definition of a minimal realization will be introduced in the next chapter — for now, you can think of it as the representation of a system using the smallest possible A, B, C, D matrices that still capture its behaviour completely.

Considering a SISO LTI first order system, the general state-space form is

$$\begin{aligned} \dot{x}(t) &= ax(t) + bu(t), \\ y(t) &= cx(t) + du(t). \end{aligned} \tag{2.1}$$

For a first-order SISO system, both the state and the input are scalars. Consequently, the state-space matrices A, B, C, D reduce to scalar quantities. For clarity of exposition, we will therefore denote them by lowercase letters a, b, c, d .

Example - Mass Spring/Damper Systems

Consider again the mass spring system, as shown in Figure 1.3 and its equation of motion given by

$$\ddot{p}(t) = -\frac{k}{m}p(t) + \frac{1}{m}u(t).$$

We can recognize that the second derivative $\ddot{p}(t)$ is a function of $p(t)$, to write this system in a state-space representation we introduced a second state, the velocity $\dot{p}(t)$. In this case we have a two-dimensional state vector and hence a second order system. Considering a different system, where we replace the spring with a damper (also referred to as a dashpot), and where we are interested solely in the velocity of the system, as shown in Figure 2.1, the equation of motion of the system changes.

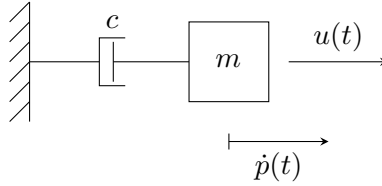


Figure 2.1: Disturbed mass damper system

The resulting force from the damper is equal to the velocity scaled by the coefficient c , i.e.

$$F_{\text{Damper}}(t) = -c\dot{p}(t).$$

Analogous to the mass spring system — using Newton's 2nd law — we can then obtain the equation of motion by considering all forces acting on the system resulting in

$$\ddot{p}(t) = -\frac{c}{m}\dot{p}(t) + \frac{1}{m}u(t).$$

We can see that $\ddot{p}(t)$ is only a function of $\dot{p}(t)$ which is in line with the general first order form introduced in Equation (2.1). In this case we could also express this ODE as a function of the velocity as

$$\dot{v}(t) = -\frac{c}{m}v(t) + \frac{1}{m}u(t),$$

since the dynamics are not dependent on the position anymore. We can further conclude that in this case the state of this system is one-dimensional and only comprised of the velocity.

2.2.1 Initial Condition Response

We will now start solving the first-order LTI form by first considering the initial condition response. To obtain a general expression for the initial condition response set the input to zero, $u(t) = 0$, and the initial condition to some arbitrary value $x(0) = x_0$. This allows us to simplify the LTI state space form in Equation (2.1) to

$$\begin{aligned}\dot{x}(t) &= ax(t), \\ y(t) &= cx(t).\end{aligned}$$

First, we solve for $x(t)$ by solving the homogenous, first-order ODE resulting in

$$x_{\text{IC}}(t) = e^{at}x_0. \quad (2.2)$$

Consequently, the output is then given by

$$y_{\text{IC}}(t) = ce^{at}x_0. \quad (2.3)$$

2.2.2 Forced Response

Next, let us consider the forced response where the initial condition is $x(0) = 0$, and the input $u(t) \neq 0$. The resulting LTI state space form is then given by

$$\begin{aligned}\dot{x}(t) &= ax(t) + bu(t), \\ y(t) &= cx(t) + du(t).\end{aligned}$$

Just like before we first need to solve the ODE to get an expression for $x(t)$. In this case, since the ODE is not homogenous anymore, it is not as straight forward to solve as the initial condition response. One approach is to first multiply both sides of the ODE by e^{-at} and integrate over $[0, t]$

$$\int_0^t e^{-a\tau} \dot{x}(\tau) \, d\tau = \int_0^t (e^{-a\tau} ax(\tau) + e^{-a\tau} bu(\tau)) \, d\tau.$$

Then we integrate by parts on the left

$$e^{-a\tau}x(\tau)\Big|_0^t + \int_0^t e^{-a\tau}ax(\tau) \, d\tau = \int_0^t (e^{-a\tau}ax(\tau) + e^{-a\tau}bu(\tau)) \, d\tau$$

to arrive at

$$e^{-at}x(t) - e^{-a0}x(0) = \int_0^t e^{-a\tau}bu(\tau) \, d\tau,$$

and since $x(0) = 0$ we finally get

$$x_{\text{F}}(t) = \int_0^t e^{a(t-\tau)}bu(\tau) \, d\tau.$$

From the solution of $x_F(t)$ we can then get the forced response

$$y_F(t) = c \int_0^t e^{a(t-\tau)} b u(\tau) d\tau + d u(t).$$

2.2.3 Complete Response

Finally, we can now sum up both solutions, i.e. $x(t) = x_{IC}(t) + x_F(t)$ and $y(t) = y_{IC}(t) + y_F(t)$. Which yields the complete time response of a first-order LTI system to some arbitrary initial condition and input

$$\begin{aligned} x(t) &= e^{at} x_0 + \int_0^t e^{a(t-\tau)} b u(\tau) d\tau, \\ y(t) &= c e^{at} x_0 + c \int_0^t e^{a(t-\tau)} b u(\tau) d\tau + d u(t). \end{aligned} \tag{2.4}$$

Remember that for now a, b, c , and d are scalars.

2.3 Higher Order Systems

The response for higher order systems is obtained analogously to the first-order case — except that in general A, B, C , and D are matrices and thus $x(t) \in \mathbb{R}^n$ unlike in the first-order case. Fortunately, the solution looks very similar and is given by

$$\begin{aligned} x(t) &= e^{At} x_0 + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau, \\ y(t) &= C e^{At} x_0 + C \int_0^t e^{A(t-\tau)} B u(\tau) d\tau + D u(t). \end{aligned} \tag{2.5}$$

Keep in mind that matrices do not commute, and hence whenever A, B, C , and D are matrices you have to maintain the order of multiplication. Furthermore, note that sometimes the matrix exponentials in Equation (2.5) are expressed as the state transition function $\phi(t) := e^{At}$ and similarly $\phi(t - \tau) := e^{A(t-\tau)}$.

2.3.1 Similarity Transformations

When analyzing the general response of an LTI system, given by Equation (2.5), we see that some terms contain the matrix exponential e^{At} . The evaluation of this exponential is more complicated than in the scalar case. One approach is to use the Taylor-Series to define the matrix exponential as

$$e^{At} = \sum_{n=0}^{\infty} \frac{1}{n!} (At)^n = I + At + \frac{1}{2!} (At)^2 + \cdots + \frac{1}{n!} (At)^n.$$

However, obtaining an exact solution generally requires computing infinitely many terms of the series expansion. There are, nevertheless, two important cases where the matrix exponential admits a closed-form expression: when the matrix is diagonalizable, and when it is in Jordan canonical form.

- Diagonal matrices: when the matrix $A \in \mathbb{R}^{n \times n}$ is diagonal, i.e. $A = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, the matrix exponential is given by

$$\exp \left(\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} t \right) = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix}. \quad (2.6)$$

- Jordan form: when the matrix $A \in \mathbb{R}^{2 \times 2}$ contains repeated eigenvalues, i.e. is in Jordan form, the matrix exponential is given by

$$\exp \left(\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix} t \right) = \begin{bmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix}. \quad (2.7)$$

Note that Jordan forms exist for matrices of any size, and every square matrix can be transformed into a corresponding Jordan form.

However, in general, the matrix A , when derived from the system's dynamics, is not diagonal or in Jordan form. Therefore, we need to find a way to transform the matrix A into a diagonal or Jordan form.

To do so, we use a coordinate transformation T (also referred to as the similarity transform) such that $x = T\tilde{x}$. In the context of LTI systems, a similarity transformation means transforming the coordinates of the state vector, i.e. choosing a different realization of the system. Let's apply the transformation $x = T\tilde{x}$ to the general LTI system

$$\begin{aligned} \left\{ \begin{array}{l} \dot{x}(t) = Ax(t) + Bu(t), \\ y(t) = Cx(t) + Du(t) \end{array} \right\} &\longrightarrow \left\{ \begin{array}{l} T\dot{\tilde{x}}(t) = AT\tilde{x}(t) + Bu(t), \\ y(t) = CT\tilde{x}(t) + Du(t) \end{array} \right\} \\ &\longrightarrow \left\{ \begin{array}{l} \dot{\tilde{x}}(t) = (T^{-1}AT)\tilde{x}(t) + (T^{-1}B)u(t), \\ y(t) = CT\tilde{x}(t) + Du(t) \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \dot{\tilde{x}}(t) = \tilde{A}\tilde{x}(t) + \tilde{B}u(t), \\ y(t) = \tilde{C}\tilde{x}(t) + Du(t) \end{array} \right\}. \end{aligned}$$

One important property of the similarity transformation is that it does not change the system's response. This can be seen by observing that the output $y(t)$ remains unchanged by the transformation¹. Through the transformation we simply use a different realization of the system, i.e. a different state vector and system matrices $\tilde{A}$, $\tilde{B}$, and $\tilde{C}$, but the response of the system remains unchanged.

¹ $\tilde{C} = CT$ and $\tilde{x}(t) = T^{-1}x(t)$, hence $\tilde{C}\tilde{x}(t) = Cx(t)$.

We can use both the original (A, B, C, D) and the transformed $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D})$ realizations to compute the time response of the system.^a

^aSometimes $\tilde{D}$ is used instead of D to represent the feedthrough term in the transformed realization, i.e., $\tilde{D} = D$.

One particular case, known from Linear Algebra II, is where the matrix $A \in \mathbb{R}^{n \times n}$ is diagonalizable. This means that A has n linearly independent eigenvectors, or equivalently, each eigenvalue λ_i has the same algebraic- and geometric multiplicities (*Halbeinfach*).

In this case we can always find some invertible transformation matrix T such that

$$\tilde{A} = T^{-1}AT = \text{diag}(\lambda_1, \dots, \lambda_n), \quad (2.8)$$

where $\tilde{A}$ is a diagonal matrix with the eigenvalues of A on the diagonal and T a transformation matrix having the eigenvectors $v_1, \dots, v_n$ of A as columns.

By using Equation (2.6) the matrix exponential for diagonalizable matrices can thus be simplified to

$$e^A = Te^{\tilde{A}}T^{-1} = T \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})T^{-1}. \quad (2.9)$$

Similarly, there are transformations that bring matrices into the Jordan form. These are especially useful when a given matrix is not diagonalizable. However, computing the transformation for the Jordan form is more complicated and will not be covered in this course. Both of these transformations can then be used to evaluate the time response given by Equation (2.5).

Example - Pendulum

Let's consider the pendulum system from before and compute its initial condition response. We have already derived an equation of motion and linearized it around both equilibrium points. For the equilibrium point $x_{e,2} = (\pi, 0)$, $u_e = 0$ we obtained

$$A = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{c}{ml^2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{ml} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

Assuming the following numerical values for the internal parameters: $g = 10$, $l = 5$, $m = 1$, and $c = 25$; the LTI state space form is given by

$$\begin{aligned} \dot{x}(t) &= \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}}_A x(t) + \underbrace{\begin{bmatrix} 0 \\ 0.2 \end{bmatrix}}_B u(t), \\ y(t) &= \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_C x(t). \end{aligned}$$

Now we can solve for the initial condition response, i.e. $x(0) = x_0$ and $u(t) = 0$. From Equation (2.5) we know the general solution of the initial condition response to be

$$\begin{aligned} x(t) &= e^{At} x_0 \\ y(t) &= C e^{At} x_0. \end{aligned} \quad (2.10)$$

In order to evaluate the matrix exponential in the expression above we will diagonalize the matrix A to then use Equation (2.9). To that end we need the eigenvalues and eigenvectors of A , which, in this case, turn out to be

$$\lambda_1 = 1, \quad \lambda_2 = -2, \quad v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix},$$

and since v_1 and v_2 are linearly independent we can also confirm that this matrix is indeed diagonalizable. There are now two ways to compute the output. One is to substitute the simplified expression for the matrix exponential into Equation (2.9), whereas the other consist of performing the similarity transform on the whole system. Both result in the same output but one can potentially be easier to evaluate than the other. Below we will present both approaches.

Solving by Substitution From Equation (2.9) we know that the matrix exponential is given by

$$e^{At} = T \operatorname{diag}(e^{\lambda_1 t}, e^{\lambda_2 t}) T^{-1} = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & e^{-2t} \end{bmatrix} \frac{1}{3} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix},$$

where T is a transformation matrix containing the eigenvectors as columns. Finally, let $x_0 = [x_{0,1}, x_{0,2}]^\top$ be the initial condition, then the solution to our initial condition problem is

$$\begin{aligned} x(t) &= \underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}}_T \underbrace{\begin{bmatrix} e^t & 0 \\ 0 & e^{-2t} \end{bmatrix}}_{e^{\tilde{A}t}} \underbrace{\frac{1}{3} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}}_{T^{-1}} x_0, \\ x(t) &= \begin{bmatrix} e^t & -e^{-2t} \\ e^t & 2e^{-2t} \end{bmatrix} \begin{bmatrix} \tilde{x}_{0,1} \\ \tilde{x}_{0,2} \end{bmatrix}, \end{aligned}$$

where $\tilde{x}_{0,i}$ are the components of the transformed initial condition, i.e. $\tilde{x}_0 = T^{-1}x_0$. The output is then given by

$$\begin{aligned} y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} e^t & -e^{-2t} \\ e^t & 2e^{-2t} \end{bmatrix} \begin{bmatrix} \tilde{x}_{0,1} \\ \tilde{x}_{0,2} \end{bmatrix} \\ &= \tilde{x}_{0,1} e^t - \tilde{x}_{0,2} e^{-2t}. \end{aligned} \quad (2.11)$$

Solving with Similarity Transform Using the general similarity transform from above we can see that the solution to our initial condition response can also be expressed by

$$\begin{aligned}\tilde{x}(t) &= e^{\tilde{A}t} \tilde{x}_0, \\ y(t) &= \tilde{C} e^{\tilde{A}t} \tilde{x}_0.\end{aligned}$$

With $\tilde{x}_0 = T^{-1}x_0$ being the transformed initial condition and $\tilde{A}, \tilde{C}$ the transformed system matrices. More specifically $\tilde{A}$ is the diagonal matrix containing the eigenvalues on the diagonal and

$$\tilde{C} = CT = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \end{bmatrix}.$$

Putting everything together yields

$$\begin{aligned}y(t) &= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} \tilde{x}_{0,1} \\ \tilde{x}_{0,2} \end{bmatrix} \\ &= \tilde{x}_{0,1}e^t - \tilde{x}_{0,2}e^{-2t}.\end{aligned}$$

For an LTI system with A diagonalizable, the general solution to the initial condition problem is given by

$$x(t) = \tilde{x}_{0,1}v_1e^{\lambda_1 t} + \tilde{x}_{0,2}v_2e^{\lambda_2 t} + \cdots + \tilde{x}_{0,n}v_ne^{\lambda_n t}.$$

Where $\tilde{x}_{0,i}$ are the components of the transformed initial condition $\tilde{x}_0 = T^{-1}x_0$ and v_i, λ_i the eigenvectors and eigenvalues of A respectively.

2.4 Response to common inputs

In the previous section we developed a general formula, Equation (2.5), for computing the response to any input to our system. Unfortunately the integral makes it hard to develop an intuition for the behaviour of the system and the effects any input might have. To get a better feeling for how a system responds, we can apply specific inputs that help us understand how a given system might behave. Especially useful are the step (Section 2.4.1) and impulse (Section 2.4.2) responses.

2.4.1 Step Response

The step response is the complete response of a system to a step input. A step input is defined as an input that is constant for $t \geq 0$ and zero for $t < 0$, i.e., $u(t) = h(t)$ with $h(t)$ being the Heaviside function defined as

$$h(t) = \begin{cases} \alpha, & t \geq 0, \\ 0, & t < 0, \end{cases} \quad (2.12)$$

where α is the magnitude of the step. In the case that $\alpha = 1$ we call the step input a unit step input.

Recall the general expression for the time response of a first order system

$$y(t) = ce^{at}x_0 + c \int_0^t e^{a(t-\tau)}bu(\tau)d\tau + du(t).$$

Plugging in the unit step input $u(t) = h(t)$, where $\alpha = 1$, results in

$$\begin{aligned} y_{\text{step}}(t) &= ce^{at}x_0 + c \int_0^t e^{a(t-\tau)}bd\tau + d, \\ &= ce^{at}x_0 + \frac{cb}{a}(e^{at} - 1) + d. \end{aligned} \quad (2.13)$$

Since x_0 and d only apply a shift to the system's response and do not influence the behaviour of the response itself, for sake of simplicity we will assume $x_0 = 0$ and $d = 0$. We can further substitute $k = -\frac{cb}{a}$, which yields a compact expression for the step response

$$y_{\text{step}}(t) = k(1 - e^{at}). \quad (2.14)$$

Note that, in this scalar case, the value of a also corresponds to the eigenvalue of the system. Given the equation for the step response, we can now plot it as a function of time for various values of a — or better, various eigenvalues of the system. The resulting responses are shown in Figure 2.2.

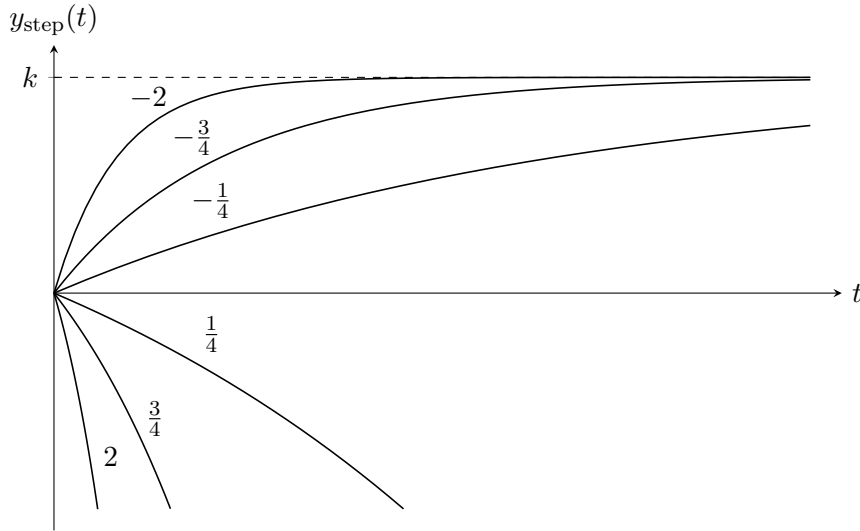


Figure 2.2: Step responses of first order systems with different eigenvalues, indicated on the respective curves.

We can observe that for negative eigenvalues, the step response asymptotically approaches the value k , whereas positive eigenvalues drive the response towards negative infinity. This is already a very insightful conclusion we can take from considering the step response as it seems that the eigenvalues of the system are responsible for the behaviour of the step response, and hence the stability of the system.

2.4.2 Impulse Response

Another important response we will consider, is the impulse response. It is the response of a system to zero initial condition $x(0) = 0$ and a unit impulse as input $u(t) = \delta(t)$. The unit impulse (Dirac delta) is defined as

$$\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0, \end{cases} \quad (2.15)$$

such that

$$\int_0^\varepsilon \delta(t) dt = 1, \quad \varepsilon > 0, \quad (2.16)$$

with the important characteristic

$$\int_0^t f(\tau) \delta(\tau) d\tau = f(0). \quad (2.17)$$

Again we can consider a first order system with the general response

$$y(t) = ce^{at}x_0 + c \int_0^t e^{a(t-\tau)} bu(\tau) d\tau + du(t),$$

and solve for the impulse response. Again we set $d = 0$ and plug in all conditions, resulting in

$$y_{\text{imp}}(t) = ce^{at} \int_0^t e^{-a\tau} b \delta(\tau) d\tau,$$

which, by using Equation (2.17), and setting $k = cb$ results in

$$y_{\text{imp}}(t) = ke^{at}.$$

As in the step response, the impulse response also has an exponential with a in the exponent. Now the response is a simple exponential, but we can still observe that for positive values of a , i.e. the systems eigenvalues, the impulse response will diverge and for negative values it will converge to zero, as indicated in Figure 2.3. In the following section we will formalize this observation, and relate the long-term response of a system to arbitrary inputs to the eigenvalues of the system.

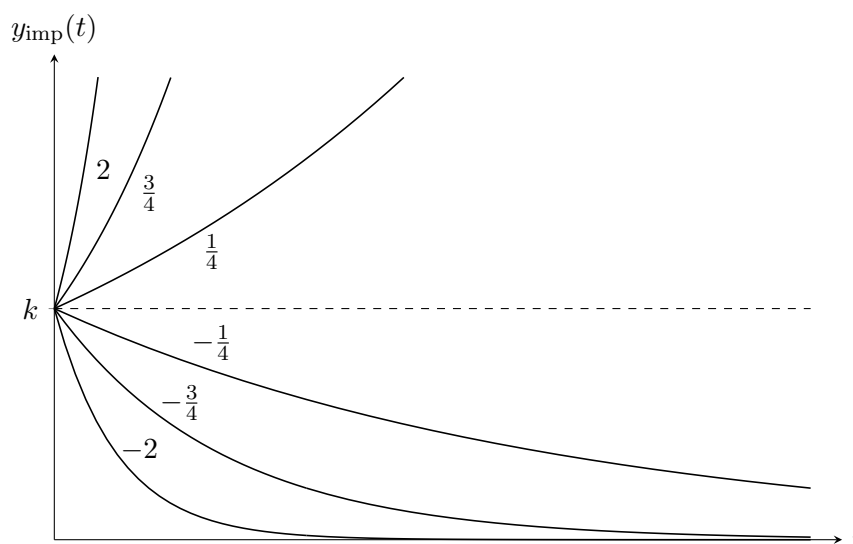


Figure 2.3: Impulse responses of first order systems with different eigenvalues. The respective values are indicated on the curves.

To gain a better understanding of the influence of the eigenvalues on response to step, impulse or arbitrary inputs, we will plot the response for the pendulum linearized around the downright equilibrium and vary the eigenvalues.

Recall that the pendulum linearized around the downright equilibrium is given by:

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{c}{ml^2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{ml} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

The eigenvalues are given by:

$$\lambda_{1,2} = -\frac{1}{2} \frac{c}{ml^2} \pm \sqrt{\left(\frac{c}{ml^2}\right)^2 - 4\frac{g}{l}}.$$

As we wish to investigate the influence of eigenvalues on the time response, let's set $g = 10, l = 1, m = 1$, and vary the damping coefficient c .

Using the tool ([MyBinder hyperlink](#)), plot the step response of the system for the two eigenvalues for different values of c .

Hint: Select step input, and place the following into the non-linear box:

$$[x[1], -10*x[0] - c*x[1] + u]$$

Set the Time T to be 50, and click **Show Non-Linear**. Then vary the values of c in the range $0 \leq c \leq 100$. You should notice that as c increases, the response takes longer to reach its asymptotic value. However, the response is always converging towards the same value. Now try varying the damping coefficient such that $c > 0$. You should notice that the response is now diverging. Feel free to test that this behaviour is consistent across various inputs, initial conditions, and systems.

2.5 Stability

One important property of a system is whether the system's response to an input converges to a finite value, remains bounded, or diverges to infinity. For example, we would not like a cruise control system to result in the vehicle's velocity diverging to infinity. In this section we will formalize this notion of stability and discuss the conditions under which a system is stable, i.e. it does not diverge to infinity. In Section 2.5.1 we will generalize the observations made for the step and impulse responses to the general case for any LTI system subject to an arbitrary input. Then we will introduce the various notions of stability in Sections 2.5.2 to 2.5.4. Finally, although in this course we focus on LTI systems, most systems are not LTI. Consequently, we also discuss the conditions under which the introduced stability notions are applicable to non-linear systems in Section 2.5.5.

2.5.1 Initial Condition Response

We now extend the observations made for the step and impulse responses of first order systems in Section 2.4 to the general case for any LTI system. As a first step, we examine the initial condition response, which describes the system's natural evolution in the absence of inputs, i.e., for $u(t) = 0$ and a prescribed initial condition $x(0) = x_0$.

Without loss of generality, let's consider an LTI system with state vector $x(t) \in \mathbb{R}^2$ and the real, diagonal matrix $A = \text{diag}(\lambda_1, \lambda_2)$. Recall from Equation (2.5) that the initial condition response is given by

$$x(t) = e^{At}x_0, \quad \text{with } A = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad \lambda_i \in \mathbb{R},$$

$$y(t) = Ce^{At}x_0,$$

Let's investigate the behaviour of the output more closely. Since A is diagonal we can use Equation (2.6) to simplify the matrix exponential and write out all other terms to obtain

$$y(t) = Ce^{At}x_0 = \begin{bmatrix} c_1 & c_2 \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} \begin{bmatrix} x_{0,1} \\ x_{0,2} \end{bmatrix}, \quad (2.18)$$

$$y(t) = c_1 e^{\lambda_1 t} x_{0,1} + c_2 e^{\lambda_2 t} x_{0,2},$$

where c_i are the entries of the matrix C . We see that for this diagonal, real matrix the initial condition response is a linear combination of exponential functions $e^{\lambda_i t}$. Importantly the growth/decay of the exponential functions is given by the eigenvalues of A — just like in the first order step and impulse responses. Note that this also extends to higher order systems, where each eigenvalue just adds another exponential to the response.

Let us now also look the case where A has complex eigenvalues. More specifically, consider

the case where A is a complex, diagonal 2×2 matrix². Similarly to before the initial condition response is obtained by solving

$$x(t) = e^{At}x_0, \quad \text{with } A = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad \lambda_i \in \mathbb{C},$$

$$y(t) = Ce^{At}x_0.$$

Note that complex eigenvalues always appear as complex-conjugate pairs, i.e. $\lambda_i = \sigma \pm j\omega$ where σ is the real- and ω the imaginary part. Again we can write out all terms, simplify for A being diagonal, and use Euler's formula ($e^{j\varphi} = \cos(\varphi) + j\sin(\varphi)$) to obtain

$$\begin{aligned} y(t) &= c_1 e^{\sigma t} e^{j\omega t} x_{0,1} + c_2 e^{\sigma t} e^{-j\omega t} x_{0,2}, \\ &= e^{\sigma t} (c_1 e^{j\omega t} x_{0,1} + c_2 e^{-j\omega t} x_{0,2}), \\ &= e^{\sigma t} (\alpha_1 \sin(\omega t) + \alpha_2 \cos(\omega t)), \end{aligned}$$

which finally yields

$$y(t) = \alpha e^{\sigma t} \sin(\omega t + \varphi). \quad (2.19)$$

The time response is now an exponentially growing/decaying sinusoidal function of the form $e^{\sigma t} \sin(\omega t + \varphi)$ — examples of which are shown in Figure 2.4. Here we can also see that the

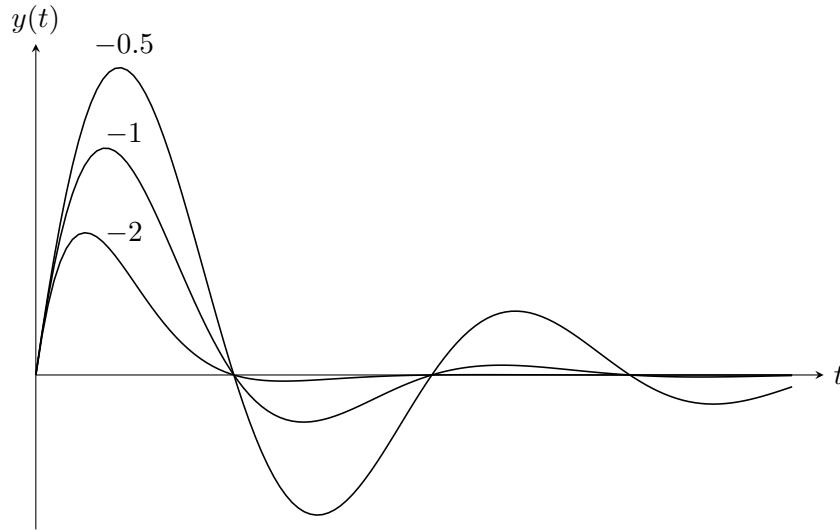


Figure 2.4: Initial condition responses of second order systems with complex eigenvalues. The imaginary part ω is kept constant while the real part σ varies. The corresponding values of σ are indicated on the curves.

overall response is dominated by the exponential term $e^{\sigma t}$. The growth rate of this exponential is, just as before, given by the eigenvalues of A — more precisely, it is given by its real part σ (the imaginary part ω gives us information about the frequency of the sinusoid). From this we can conclude that both the real and complex case, the growth of the output is dictated by the real part of the eigenvalues of A . For higher order systems this also holds true, we just have to

²Note that the diagonalized matrix with complex eigenvalues has the following form: $A = \begin{bmatrix} \sigma & \omega \\ -\omega & \sigma \end{bmatrix}$.

consider more eigenvalues, each adding another (complex) exponential.

As in the step and impulse responses we observe that, if the eigenvalues have a positive real part, the output will grow exponentially over time. We will refer to these systems as *unstable*. Conversely, if the real part of the eigenvalues is negative, the output will asymptotically converge to some finite value or remain constant, and thus the system is *stable*. Formally we distinguish between three types of stability: *Lyapunov*, *asymptotic*, and *bounded-input, bounded-output (BIBO)* stability.

2.5.2 Lyapunov Stability

A system is Lyapunov stable if, for any bounded initial condition, and zero input, the state remains bounded, i.e.

$$\|x_0\| < \epsilon, \text{ and } u = 0 \implies \|x(t)\| < \delta \quad \forall t \geq 0.$$

Figure 2.5 depicts this behaviour, where a state trajectory is depicted in a 2-dimensional state space. We observe that the trajectory, starting from a bounded initial point, always has a magnitude smaller than some finite value δ .

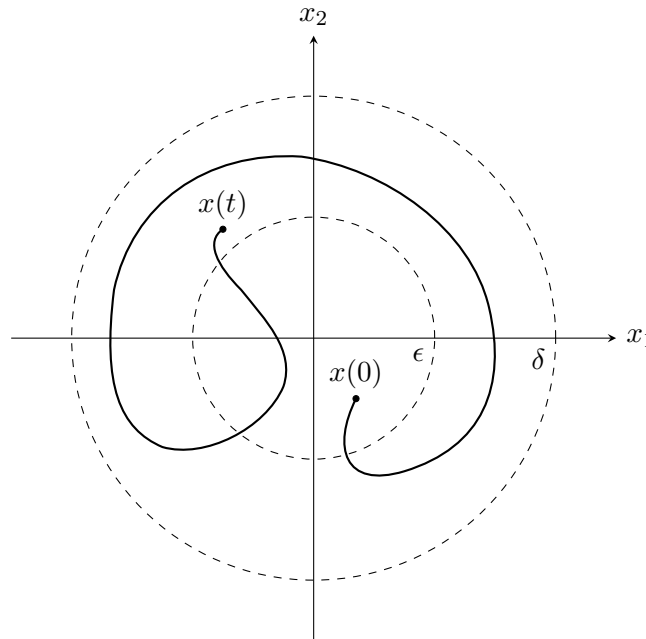


Figure 2.5: Illustration of Lyapunov stability in the 2-dimensional state space.

Lyapunov Stability for Diagonalizable Systems

By considering the cases described by Equations (2.18) and (2.19) we can see that for a bounded output all eigenvalues of A have to have a negative or zero real part. The condition for Lyapunov stability is thus stated as

Definition 2.1 *An LTI system with diagonalizable A matrix is Lyapunov stable if and only if*

$$\operatorname{Re}(\lambda_i) \leq 0, \quad \forall i$$

Lyapunov Stability for Non-Diagonalizable Systems

Above we have shown that for systems with a diagonalizable A matrix, the output will be given by some linear combination of exponential functions and/or exponentially growing/decaying sinusoids, i.e. terms of the form

$$e^{\lambda_i t}, \quad e^{\sigma_i t} \sin(\omega_i t + \varphi_i).$$

Unfortunately not every matrix is diagonalizable. For such systems the output is additionally composed of time scaled exponential functions. They take the form

$$t^{m_i} e^{\lambda_i t}, \quad t^{m_i} e^{\sigma_i t} \sin(\omega_i t + \varphi_i),$$

where m_i is the algebraic multiplicity of the associated eigenvalue λ_i . This means that we have to be a bit more careful to assess the Lyapunov stability of such systems.

Definition 2.2 *An LTI system with non-diagonalizable A matrix is Lyapunov stable if and only if*

$$\operatorname{Re}(\lambda_i) \leq 0, \quad \forall i$$

and there are no repeated eigenvalues with 0 real part.

2.5.3 Asymptotic Stability

A system is asymptotically stable if, for any bounded initial condition, and zero input, the state converges to zero, i.e.

$$\|x_0\| < \epsilon, \text{ and } u = 0 \Rightarrow \lim_{t \rightarrow \infty} \|x(t)\| = 0.$$

This condition is graphically depicted in Figure 2.6, where the state trajectory, starting from a bounded initial condition converges to zero.

As we know from above each eigenvalue has a corresponding exponential in the time response. Consequently, for the response to converge to zero, all eigenvalues of A have to have a strictly negative real part.

Definition 2.3 *An LTI system A, B, C, D is asymptotically stable if and only if*

$$\operatorname{Re}(\lambda_i) < 0, \quad \forall i$$

From this definition we can also conclude that every asymptotically stable system is also

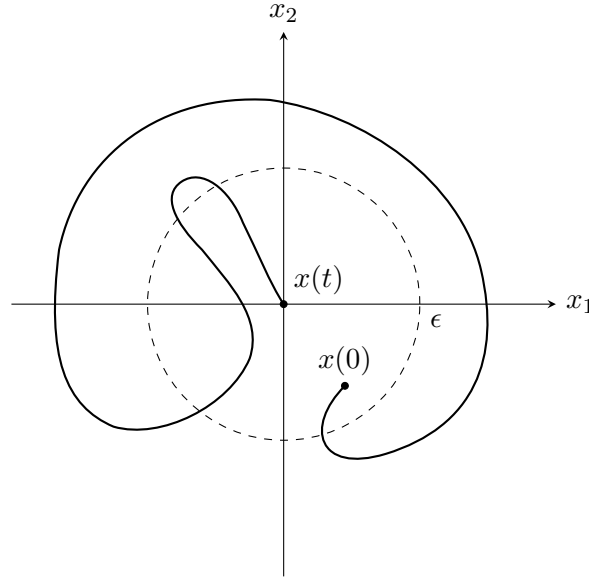


Figure 2.6: Illustration of asymptotic stability in the 2-dimensional state space.

Lyapunov stable — but not the other way around. Furthermore, it is important to note that asymptotic stability does not imply monotonicity of the output.

Example - Pendulum

Let's again consider the pendulum system from before and analyze its stability. First, let's consider the system resulting from the linearization around $x_{e,1} = (0, 0)$, $u_e = 0$. Assuming the following values for the internal parameters: $g = 10$, $l = 5$, $m = 1$, and $c = 25$; the LTI state space form is then given by

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0.2 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} x(t),\end{aligned}$$

which has the eigenvalues

$$\lambda_1 = \frac{1}{2}(-1 + j\sqrt{7}), \quad \lambda_2 = \frac{1}{2}(-1 - j\sqrt{7}).$$

In this case all eigenvalues have a negative real part, i.e. $\text{Re}(\lambda_i) < 0$, and the system is therefore asymptotically stable.

We can also check the stability of the system resulting from the linearization around the other equilibrium point $x_{e,2} = (\pi, 0)$, $u_e = 0$. Assuming the same model parameters as

before, the corresponding LTI state space system is given by

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0.2 \end{bmatrix} u(t), \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} x(t),\end{aligned}$$

which has the eigenvalues

$$\lambda_1 = 1, \quad \lambda_2 = -2.$$

We can see that the system is unstable since $\text{Re}(\lambda_1) = 1 > 0$, therefore the output grows exponentially over time. This is also visible by inspecting the resulting output obtained in Equation (2.11)

$$y(t) = \tilde{x}_{0,1}e^t - \tilde{x}_{0,2}e^{-2t}.$$

The first exponential term will grow indefinitely over time making the system unstable.

Feel free to verify this behaviour by using the tool ([MyBinder hyperlink](#)). Add in the corresponding A, B, C, D matrices. Independently of the selected initial condition and input, in the stable case, the response will converge to a finite value. In the unstable case, the response will diverge to infinity.

2.5.4 Bounded-Input, Bounded Output Stability

A system is BIBO-stable if, for any bounded input, the output remains bounded, i.e.

$$\|u(t)\| < \epsilon \quad \forall t \geq 0, \text{ and } x_0 = 0 \Rightarrow \|y(t)\| < \delta \quad \forall t \geq 0.$$

This type of stability does not have a concrete interpretation through the eigenvalues and is harder to identify. Nevertheless, in the case that the LTI system is a minimum realization, i.e., the state vector is the smallest possible vector that can represent the system, then if the system is asymptotically stable it is also BIBO stable.

2.5.5 Stability of Non-Linear Systems

It is important to keep in mind that we are often working with linearized approximations of real-world systems. Conclusions drawn from a linearized LTI model cannot always be directly applied to the original nonlinear system. As an example take the pendulum example from above. Here the stability properties of the linearized model vary depending on the point we linearized the system about. So it is important to remember that any linearized LTI model is only valid in a small neighborhood around the equilibrium point about which the system was linearized. As a consequence, we must be careful when drawing conclusions about the stability of the real-world system based solely on its linearized model. However, the Hartman-Grobman theorem provides a way to draw conclusions about the stability of the nonlinear system based

on the stability of the linearized system around hyperbolic equilibrium points.

Theorem 2.4 (Hartman-Grobman) *If the linearization of a nonlinear system around an equilibrium point is (asymptotically) stable, then the nonlinear system is also locally (asymptotically) stable at that equilibrium point.*

Note: The Hartman-Grobman Theorem only applies to hyperbolic equilibrium points. An equilibrium point is hyperbolic if all eigenvalues of the linearization have non-zero real parts (i.e., $\text{Re}(\lambda_i) \neq 0, \forall i$). Thus, no claims about Lyapunov stability (i.e. where $\text{Re}(\lambda_i) \leq 0$) can be made.

Chapter 3

Transfer Functions

3.1 Introduction

In the previous sections we looked at the time domain solution of the general LTI model and through the concept of stability, started to develop some understanding for the behavior of a system based on its A matrix. More precisely we looked at the natural dynamics of the system, i.e. the initial condition response when $u(t) = 0$ and whose dynamics resembled $x(t) = e^{At}x_0$. But what about the case where $u(t) \neq 0$? Consider the complete output

$$y(t) = Ce^{At}x_0 + C \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau + Du(t).$$

Here, we see that, since the input $u(t)$ is contained inside the convolution integral, the effect of the input is much harder to interpret. Consequently, in this chapter, we will introduce an approach that will allow us to more easily analyze and interpret the effects of inputs.

To achieve this, we will change the domain in which we describe and analyze the system, its input, and its output. Instead of working only in the time domain, we introduce the s-domain, where signals are represented as functions of a complex variable s . In this representation, the system will be described by its *transfer function*. This makes it possible to analyze the response to input signals by treating them as combinations of simpler building blocks.

For this whole chapter it is important to keep in mind that we are working with linear systems. This means that we can exploit the concept of superposition, to rewrite any arbitrary input $u(t)$ as a linear combination of “simpler bits” $u_i(t)$, i.e.

$$u(t) = \alpha_1 u_1(t) + \alpha_2 u_2(t) + \cdots + \alpha_m u_m(t),$$

then apply each input $u_i(t)$ to the system separately and sum all resulting outputs

$$y(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t) + \cdots + \alpha_m y_m(t). \quad (3.1)$$

To help us understand the s -domain, we will briefly introduce the Fourier and Laplace transforms in Section 3.1.1, that will provide a foundation for the concepts introduced in the following sections. In Sections 3.2 to 3.5 we introduce the transfer functions for LTI systems, how to compute them and the various forms that we will use later on in the course. Then in Section 3.6 we introduce the poles and zeros of the transfer function, and discuss how they influence the behaviour of the system. Finally, since we are interested in the time domain behaviour of the system, in Section 3.7 we introduce two approaches to determine the value of the time domain output by using transfer functions.

3.1.1 Fourier and Laplace Transform

This section on the Fourier and Laplace transforms is not specifically exam relevant; nonetheless it is useful in the context of Control Systems. Since these transforms are covered in much more detail in other courses (Analysis III) the following is not a complete and rigorous description but rather a short overview highlighting the most important characteristics for our specific applications.

As we have seen in Chapter 1, real world systems are usually described by differential equations. The solutions to these differential equations are typically some combination of sinusoids and exponentials, i.e. functions with derivatives that are scaled versions of themselves. So when dealing with differential equations — or specifically in our case, LTI systems — we constantly encounter sinusoidal and exponential functions. Consequently, in the following we introduce a way to think about and represent these types of functions, making them easier to work with.

Frequency Domain

Let's begin by considering a general sinusoid in the time domain, which is fully described by

$$f(t) = M \sin(\omega t + \varphi), \quad (3.2)$$

where M is the magnitude (amplitude), ω the frequency, and φ the phase shift — for the sake of exposition, we will consider functions with zero phase-shift, i.e. $\varphi = 0$. By keeping track of these three parameters we can represent any general sinusoid without loss of information. As an example, let's consider the time domain function

$$f_1(t) = 2 \sin(4t), \quad (3.3)$$

as shown in Figure 3.1 on the left. We will now represent this function by only keeping track of its amplitude and frequency. Concretely we introduce the function $M(\omega)$ that tells us how big the amplitude of a sine function at each frequency is. In our case it is zero everywhere except at $\omega = 4$ where we have a sine with amplitude 2. The resulting function $M(\omega)$ is shown in Figure 3.1 on the right.

Both graphs in Figure 3.1 now contain the same information about the function f_1 , they are just expressed in different domains. On the left in the time domain, and on the right in the “frequency” domain. The value of $M(\omega)$ in the frequency domain tells us how big the magnitude



Figure 3.1: Time- and “frequency” domain representation of the function f_1 .

of a sine with frequency ω is. If we now want to reconstruct the time domain function from its frequency representation we need to go through all the frequencies and multiply the value of $M(\omega)$ with a sine of the same frequency. This is written mathematically as

$$f(t) = \sum_{\omega=0}^{\infty} M(\omega) \sin(\omega t). \quad (3.4)$$

In this simple case of f_1 this sum results in

$$f(t) = \sum_{\omega=0}^{\infty} M(\omega) \sin(\omega t) = M(4) \sin(4t) = 2 \sin(4t).$$

For one sine function it might not be apparent why this is useful, but consider now some other function f_2 and its time domain representation on the left side of Figure 3.2. As before we also plot the frequency domain representation, given by $M(\omega)$, on the right. By comparing the two



Figure 3.2: Time- and “frequency” domain representation of function f_2 .

plots we can see that it is hard to extract meaningful information from the time domain plot, whereas in the frequency domain we can clearly see that f_2 is composed of three sines, one at each non-zero value of $M(\omega)$. By the same logic as above, we can use Equation (3.4) to retrieve the time domain function from $M(\omega)$ resulting in

$$f_2(t) = M(2) \sin(2t) + M(4) \sin(4t) + M(6) \sin(6t).$$

Note that in general we also would like to include information about the phase φ of the sinusoid in its frequency domain representation. To do this we can introduce another function $\varphi(\omega)$ that, similar to $M(\omega)$, keeps track of the phase shift as a function of frequency. It can be interpreted analogously to the magnitude function, and we can also extend Equation (3.4) to include the

phase shift resulting in

$$f(t) = \sum_{\omega=0}^{\infty} M(\omega) \sin(\omega t + \varphi(\omega)). \quad (3.5)$$

As the number of sine functions in f grows, the frequency representation becomes more powerful and easier to work with, when compared to the time domain. Instead of considering how every t gets mapped on to the time-dependent function $f(t)$, we can summarize the entire function using $M(\omega)$ and $\varphi(\omega)$. Consequently, we arrive at the following statement.

Any function $f(t)$ that is a summation of sine waves can be uniquely characterized by the functions $M(\omega)$ and $\varphi(\omega)$.

Note that even though we refer to ω as a frequency, it is strictly speaking an angular velocity; but since the frequency f and angular velocity only differ by the constant 2π , i.e. $\omega = 2\pi f$, we will use the terms interchangeably.

Fourier Transform

A key question is, how do we map from one domain to the other given any kind of arbitrary function? This is where the Fourier transform, denoted with $\mathcal{F}$, comes in. It transforms some function from the time domain $f(t)$ to the frequency domain $F(\omega)$. It is defined as

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt, \quad (3.6)$$

and its inverse, that transforms back from the frequency domain to the time domain:

$$f(t) = \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega. \quad (3.7)$$

We see that the transform takes some function of time $f(t)$ and produces its counterpart as a function of frequency $F(\omega)$. Now, if we think about the term $e^{j\omega t}$ as being “half” of a sinusoid (since: $2 \cos(\omega t) = e^{j\omega t} + e^{-j\omega t}$), then the inverse Fourier transform from Equation (3.7) is similar to the result we derived in Equations (3.4) and (3.5). It essentially tells us that we can write any function, as an infinite sum (integral) of weighted $e^{j\omega t}$ -terms, weighted by the corresponding $F(\omega)$. By using the integral instead of a sum we can consider ω to be continuous and not just integer valued. Overall this means that we can express a huge class of functions f as a linear combination of sinusoids and the weights of this linear combination are given by $F(\omega)$.

Note that in general the function $F(\omega)$ is complex valued, which makes it easier to account for phase shifts since they can be expressed by a multiplication with a complex exponential. This can be interpreted as combining the functions $M(\omega)$ and $\varphi(\omega)$ into a single complex valued function, or mathematically

$$F(\omega) = |M(\omega)| e^{j\varphi(\omega)}.$$

For an intuitive visual explanation of the Fourier Transform, check out this helpful video by 3Blue1Brown — [But what is the Fourier Transform? A visual introduction.](#)

Laplace Transform

It is important to note that not all functions can be expressed as an infinite sum of sinusoids. A prominent exception are exponential functions, which are especially important for us since they are present in the solutions of the LTI state-space form. To tackle this problem we will make use of the Laplace Transform $\mathcal{L}$, which is conceptually similar to the Fourier transform, but instead of decomposing functions in terms of $e^{j\omega t}$, we will do so in terms of $e^{(\sigma+j\omega)t}$. Due to the introduction of σ this enables us to consider exponential functions. For convenience, we will use a new variable that keeps track of σ and ω at the same time,

$$s = \sigma + j\omega. \quad (3.8)$$

That is $s \in \mathbb{C}$ is a complex variable. The Laplace transform $\mathcal{L}$ is then given by:

$$F(s) = \int_0^\infty f(t)e^{-st}dt, \quad (3.9)$$

and its inverse $\mathcal{L}^{-1}$ as

$$f(t) = \frac{1}{2\pi j} \lim_{\omega \rightarrow \infty} \int_{\sigma-j\omega}^{\sigma+j\omega} F(s)e^{st}ds. \quad (3.10)$$

Similarly to before, the inverse Laplace transform $\mathcal{L}^{-1}$ tells us that we can write any function, as an infinite sum/integral of weighted e^{st} -terms, weighted by the corresponding value of $F(s)$. This generalizes the Fourier transform, enabling us to represent a broader spectrum of functions. This transform will be the fundamental to all future methods we will develop in frequency domain.

We can write (almost) any function $f(t)$ as a combination of complex exponentials e^{st} .

Now remember what the initial goal was. We aimed at finding some fundamental inputs u_i , which when linearly combined, could represent any other, arbitrary input u . This is exactly what we found with e^{st} .

s -Domain

In the introduction to the frequency domain we saw that the function $M(\omega)$ contains information about the amplitude of sinusoids. So by looking at the peaks of $M(\omega)$ in the frequency domain we are able to identify which frequencies are contained in a given signal and reconstruct the function. We can now ask, what does the corresponding function $F(s)$, in the s -domain, tell us about the contents of the function? Recall that in the complex variable $s = \sigma + j\omega$ the exponential terms are represented by the real part σ and the sinusoidal terms are represented

by the imaginary component ω . Following the same logic as in the frequency domain we will conclude that peaks in the function $F(s)$ correspond to components of exponentials and/or sinusoids being present in a certain signal. Let's look at some examples.

Consider Figure 3.3 where, on the left the function $f(t) = e^{-t}$ is plotted in the time-domain and on the right the corresponding Laplace transform

$$F(s) = \frac{1}{s + 1}$$

is shown in the s -domain. For clarity, we use crosses to indicate only the peaks of $F(s)$ instead of plotting the whole function in three dimensions. The peaks arise at the places where the denominator of $F(s)$ goes to zero and thus the magnitude goes to infinity. In our case if $s \rightarrow -1$,

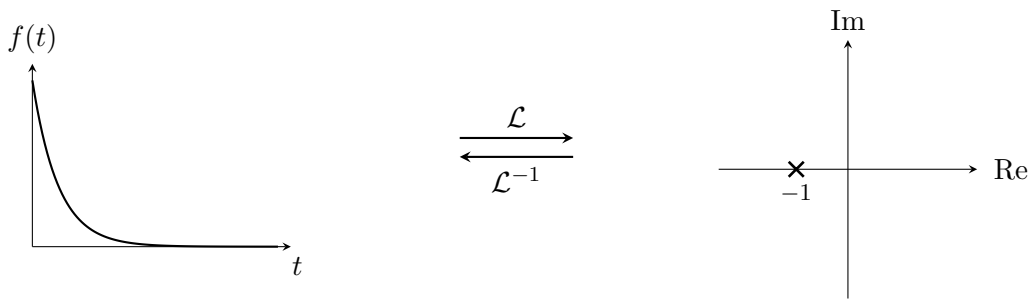


Figure 3.3: Time and s -domain representation of function f

then $|F(s)| \rightarrow \infty$. Thus, $F(s)$ has one peak at $s = -1$, i.e. $\sigma = -1$, $\omega = 0$. This directly translates to the time domain function that is composed of one exponential with exponent $\sigma = -1$.

Consider another function $g(t) = \sin(3t)$. As before, shown in Figure 3.4, we plot the time domain representation on the left and mark the peaks of the Laplace transform

$$G(s) = \frac{3}{s^2 + 9} = \frac{3}{(s + 3j)(s - 3j)},$$

on the right. Notice that peaks with imaginary components always appear as complex-conjugate pairs. We can make a similar observation as before. The s -domain representation now has two

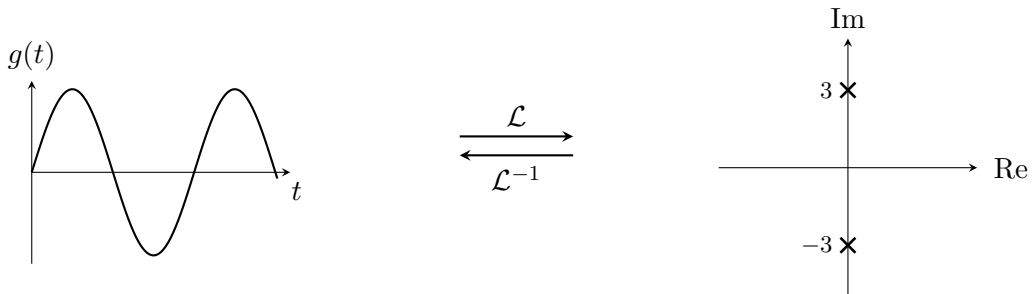


Figure 3.4: Time and s -domain representation of function g

peaks at $s = \pm 3j$, i.e. $\sigma = 0$, $\omega = \pm 3$. This directly translates to the time domain function that is composed of a sinusoid with frequency $\omega = 3$.

Lastly we consider the function $h(t) = e^{-t} \sin(3t)$ and its Laplace transform

$$H(s) = \frac{3}{(s+3)^2 + 9}$$

as shown in Figure 3.5. The s -domain representation in this case has two peaks at $s = -1 \pm 3j$,

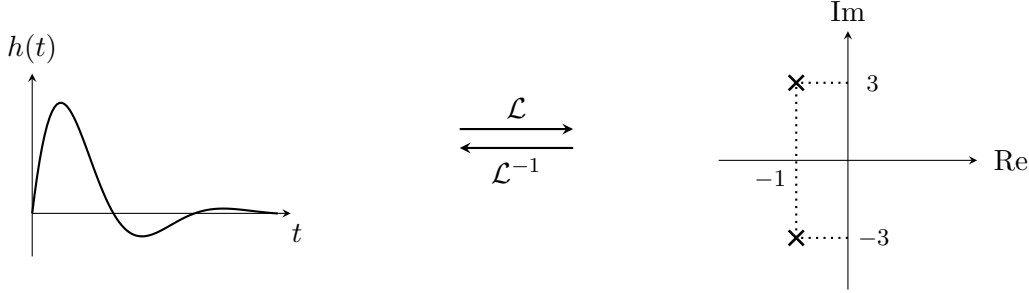


Figure 3.5: Time and s -domain representation of function h

i.e. $\sigma = -1$, $\omega = \pm 3$. This directly translates to the time domain function that is composed of an exponential with exponent $\sigma = -1$ and a sinusoid with frequency $\omega = 3$.

In summary, you can think of the s -domain as an extension to the frequency domain described before. In fact if you only consider the imaginary axis $s = j\omega$ in the s -domain, by setting $\sigma = 0$, you end up with the frequency domain again.

If you want to strengthen your intuition about the Fourier Transform, the Laplace Transform, or the s -domain, there are many great visual explanations on YouTube. Channels like 3Blue1Brown or Brian Douglas offer clear and engaging introductions to these topics.

3.2 Transfer Functions of LTI Systems

3.2.1 Solving the Convolution Integral

By leveraging the fact that any input $u(t)$ can be expressed as a linear combination of e^{st} we can now compute the response $y_i(t)$ to the input $u_i(t) = e^{st}$ and, due to superposition, the response to any other input, will be a linear combination of the outputs $y_i(t)$. As a result we only have to solve the convolution integral once. So let's compute the response of a general system to e^{st} . Recall the general output of an LTI system to be

$$y(t) = Ce^{At}x_0 + C \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau + Du(t).$$

We substitute $u(t) = e^{st}$ and get

$$y(t) = Ce^{At}x_0 + C \int_0^t e^{A(t-\tau)} Be^{s\tau} d\tau + De^{st},$$

which we rearrange to

$$y(t) = Ce^{At}x_0 + Ce^{At} \int_0^t e^{(sI-A)\tau} B d\tau + De^{st}.$$

Assuming $(sI - A)$ is invertible, we solve the integral

$$y(t) = Ce^{At}x_0 + Ce^{At} \left((sI - A)^{-1} e^{(sI-A)t} B \right) \Big|_0^t + De^{st}$$

rearranging further

$$y(t) = \underbrace{Ce^{At}x_0}_{\text{Initial condition response}} + \underbrace{Ce^{At} (sI - A)^{-1} \left(e^{(sI-A)t} - I \right) B + De^{st}}_{\text{Forced response}},$$

and finally

$$y(t) = \underbrace{Ce^{At} [x_0 - (sI - A)^{-1} B]}_{\text{Transient response}} + \underbrace{[C(sI - A)^{-1} B + D] e^{st}}_{\text{Steady state response}}.$$

The first part represents the transient response of the system. The transient usually dominates the system behaviour at the beginning, but, if the system is asymptotically stable, it eventually goes to zero and the steady state response will dominate.

The complex function $G(s)$ is known as the transfer function, and it describes how a stable system G , with zero initial condition, transforms an input e^{st} into the output $G(s)e^{st}$.

$$G(s) = C(sI - A)^{-1} B + D. \quad (3.11)$$

Furthermore, in the case that the system is asymptotically stable, the steady state response $y_{ss}(t)$ can be written as

$$y_{ss}(t) = G(s)e^{st} \quad (3.12)$$

3.2.2 Deriving the Transfer Function with the Laplace Transform

Another way to arrive at the transfer function in Equation (3.11) is by solving the general LTI state-space form using the Laplace transform. We can make use of the differentiation property

$$\mathcal{L} \left\{ \frac{d^n x(t)}{dt^n} \right\} = s^n X(s) - \sum_{i=1}^n s^{n-i} \left. \frac{d^{(i-1)} x(t)}{dt^{(i-1)}} \right|_{t=0},$$

which tells us that a time derivative in t is transformed to a multiplication in s , turning differential equations in t to algebraic equations in s . Let's apply this to the LTI state-space form

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t), \end{aligned}$$

given some initial condition $x(0) = x_0$. We begin by transforming both sides of the system of ODEs and use the differentiation property resulting in

$$\begin{aligned}\mathcal{L}\{\dot{x}(t)\} &= \mathcal{L}\{Ax(t)\} + \mathcal{L}\{Bu(t)\} \\ sX(s) - x_0 &= AX(s) + BU(s),\end{aligned}$$

which can be rearranged to

$$X(s) = (sI - A)^{-1}x_0 + (sI - A)^{-1}BU(s).$$

We can now plug the expression for $X(s)$ into the output equation $Y(s) = CX(s) + DU(s)$ to obtain

$$\begin{aligned}Y(s) &= C \left[(sI - A)^{-1}x_0 + (sI - A)^{-1}BU(s) \right] + DU(s) \\ &= C(sI - A)^{-1}x_0 + \underbrace{[C(sI - A)^{-1}B + D]}_{G(s)} U(s).\end{aligned}$$

Just as before, we can decompose the output into an initial condition response and one induced by the input. Note that here the output is expressed in the s -domain, whereas above it was expressed in the time-domain. With the help of the Laplace transform and its inverse we convert between the two domains.

Example - Pendulum

Consider the pendulum system introduced before. In the previous chapters we saw that when linearizing around the equilibrium point $x_{e,1} = (0, 0)$, $u_e = 0$ and assuming the following values for the parameters: $g = 10$, $l = 5$, $m = 1$, and $c = 25$, the LTI state-space matrices are given by

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0.2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0.$$

By using Equation (3.12), we can determine the transfer function $G(s)$ of the system.

$$\begin{aligned}G(s) &= C(sI - A)^{-1}B + D = \begin{bmatrix} 1 & 0 \end{bmatrix} \left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 \\ 0.2 \end{bmatrix} + 0 \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s & -1 \\ 2 & s+1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0.2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s^2 + s + 2} \begin{bmatrix} s+1 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 0.2 \end{bmatrix} \\ &= \frac{1}{s^2 + s + 2} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0.2 \\ 0.2s \end{bmatrix}\end{aligned}$$

$$G(s) = \frac{0.2}{s^2 + s + 2}. \quad (3.13)$$

This example illustrates how to derive the transfer function $G(s)$ from the state-space matrices (A, B, C, D) , linking the time-domain and frequency-domain representations.

To gain some intuition about the time-response associated with a transfer function, using the time-response tool ([MyBinder hyperlink](#)), try plotting the step response from the above transfer function. Does it match the result from the previous chapter? In the chapters that follow, when you see a new transfer function, plot the step response of that transfer function. This will help build intuition about the behaviour of the system by solely looking at the transfer function.

3.3 Input-Output Relation in the s -Domain

So far, we have learnt how to obtain a transfer function from the LTI state-space form and how to compute the response to the fundamental input e^{st} . However, this is still not very useful since the input is usually more complex than this. The next step is thus to expand the existing concept to more complex inputs. Crucially we have to keep in mind that we selected the fundamental input e^{st} since any other input can be written as a linear combination of it. Mathematically this can be expressed through the linear combination

$$u(t) = \sum_i U_i e^{s_i t},$$

where U_i are the coefficients in the linear combination. As a concrete example let $u(t) = \cos(\omega t)$, which we can be rewritten with Euler's formula as

$$u(t) = \cos(\omega t) = \frac{1}{2}e^{j\omega t} + \frac{1}{2}e^{-j\omega t}.$$

In this case $U_{1,2} = \frac{1}{2}$ and $s_{1,2} = \pm j\omega$. This is similar to the case introduced in Equation (3.12) since we can now use superposition and the fact that $y_{ss}(t) = G(s)e^{st}$ to calculate the output to each e^{st} -term in the input. In a general case this results in

$$y_{ss}(t) = \sum_i U_i G(s_i) e^{s_i t}.$$

In our case we plug in the specific values of $s_{1,2} = \pm j\omega$ into $G(s)$, and multiply with the corresponding complex exponentials to obtain

$$y_{ss}(t) = \frac{1}{2}G(j\omega)e^{j\omega t} + \frac{1}{2}G(-j\omega)e^{-j\omega t}. \quad (3.14)$$

Lastly, we also want to get rid of the complex part in the time response. For this we use the fact that $G(\pm j\omega)$ is a complex number which can be rewritten in polar form with a magnitude

M and a phase φ . Specifically these quantities can be expressed as

$$M = |G(j\omega)| \quad \text{and} \quad \varphi = \angle(G(j\omega)),$$

resulting in the complex number $Me^{j\varphi}$. By substituting this into Equation (3.14) we get

$$y_{ss}(t) = \frac{1}{2} \underbrace{Me^{j\varphi}}_{G(j\omega)} e^{j\omega t} + \frac{1}{2} \underbrace{Me^{-j\varphi}}_{G(-j\omega)} e^{-j\omega t},$$

which can be rewritten as

$$y_{ss}(t) = M \cos(\omega t + \varphi). \quad (3.15)$$

Remarkably we see that for a sinusoidal input, the steady-state output is also a sinusoid with the same frequency ω , the magnitude M given by the magnitude of the transfer function, and a phase shift equal to the phase φ of the transfer function. Also note that, if the matrices A, B, C, D only have real entries, the transfer function has conjugate symmetry, i.e. $G(s^*) = G(s)^*$, where $*$ denotes complex conjugation.

In summary, we can state that

$$u(t) = \sum_i U_i e^{s_i t} \quad \Rightarrow \quad y_{ss}(t) = \sum_i G(s_i) U_i e^{s_i t},$$

which can be further generalized by remaining in the s -domain and using the inverse Laplace Transform to retrieve the transient response, resulting in

$$u(t) = \frac{1}{2\pi j} \lim_{\omega \rightarrow \infty} \int_{\sigma - j\omega}^{\sigma + j\omega} U(s) e^{st} ds \quad \Rightarrow \quad y(t) = \underbrace{\frac{1}{2\pi j} \lim_{\omega \rightarrow \infty} \int_{\sigma - j\omega}^{\sigma + j\omega} G(s) U(s) e^{st} ds}_{\mathcal{L}^{-1}\{G(s)U(s)\}},$$

where $U(s)$ is the Laplace transform of the input $u(t)$. Let's take a closer look at the resulting output

$$y(t) = \mathcal{L}^{-1}\{G(s)U(s)\}.$$

By applying $\mathcal{L}$ to both sides we get

$$Y(s) = G(s)U(s) \quad (3.16)$$

which means that the output in the Laplace domain $Y(s)$ can be written as the product of the transfer function and the input.

Therefore, we now have a way to compute the output to some general input by multiplication in the s -domain instead of solving the convolution integral in the time-domain. To facilitate analysis in the s -domain, Table 3.1 shows frequently used transformations.

Note that we deal with causal systems, and consequently it is not uncommon to see input signals multiplied with the Heaviside step function $h(t)$. This is a way to ensure that the input is zero for $t < 0$. However, since the Laplace transform integrates from 0 to ∞ , this has no

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot \sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$h(t) \cdot \cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$

Table 3.1: Laplace Transform Table

mathematical impact, and is purely notational. Intuitively, the Heaviside step function $h(t)$ can be thought of as “turning on” the input at $t = 0$.

Example - Pendulum with Cosine Input

Let’s look again at the example of the pendulum linearized around the stable equilibrium point. Recall the stable transfer function given by

$$G(s) = \frac{0.2}{s^2 + s + 2}.$$

If we now want to compute the systems’ response to a cosine, e.g. $u(t) = 5 \cos(2t)$, we can use Table 3.1 to obtain the Laplace transform of the input. In this case we get

$$U(s) = 5 \frac{s}{s^2 + 4}.$$

Next we know from Equation (3.16) that the output will be given by

$$\begin{aligned} Y(s) &= G(s)U(s) = \frac{0.2}{s^2 + s + 2} \frac{5s}{s^2 + 4} \\ &= \frac{s}{s^4 + s^3 + 6s^2 + 4s + 8}. \end{aligned}$$

With partial fraction decomposition this can be rewritten to

$$Y(s) = 0.25 \frac{2-s}{s^2+4} + 0.25 \frac{s-1}{s^2+s+2},$$

and further broken down to

$$Y(s) = 0.25 \frac{2}{s^2+4} - 0.25 \frac{s}{s^2+4} - 0.25 \frac{1}{s^2+s+2} + 0.25 \frac{s}{s^2+s+2}.$$

Finally, to get the time response we use the inverse Laplace transform $\mathcal{L}^{-1}\{Y(s)\}$, consid-

ering each term individually we obtain

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = 0.25 \sin(2t) - 0.25 \cos(2t) - \frac{\sqrt{7}}{14} e^{-\frac{t}{2}} \sin\left(\frac{\sqrt{7}t}{2}\right) - \frac{\sqrt{7}}{28} e^{-\frac{t}{2}} \sin\left(\frac{\sqrt{7}t}{2}\right) + 0.25 e^{-\frac{t}{2}} \cos\left(\frac{\sqrt{7}t}{2}\right).$$

We see that some terms contain an exponential with a negative exponent. For large t these tend to zero leaving us with the steady-state response

$$\begin{aligned} y_{ss}(t) &= 0.25 (\sin(2t) - \cos(2t)) \\ &= \frac{\sqrt{2}}{4} \cos\left(2t + \frac{5\pi}{4}\right). \end{aligned}$$

This way of computing the solution is the most general form but can be simplified in certain cases. For the specific case where in input is a pure sine or cosine we can use what we learned in Equation (3.15) and compute the output by evaluating the magnitude and phase of the transfer function. Recall that the steady state output can then be obtained through

$$y(t) = M \cos(\omega t + \varphi),$$

where

$$M = |G(j\omega)| \quad \text{and} \quad \varphi = \angle(G(j\omega)).$$

In our case $\omega = 2$, therefore the magnitude is

$$M = \left| \frac{0.2}{(2j)^2 + 2j + 2} \right| = \frac{|0.2|}{|-2 + 2j|} = \frac{0.2}{\sqrt{8}},$$

and the phase

$$\begin{aligned} \varphi &= \angle\left(\frac{0.2}{(2j)^2 + 2j + 2}\right) = \angle(0.2) - \angle(-2 + 2j) \\ &= 0 - \left[\arctan\left(\frac{2}{-2}\right) + \pi\right] = -\frac{3\pi}{4} \text{ or } \frac{5\pi}{4}. \end{aligned}$$

The steady state output is thus given by

$$y_{ss}(t) = 5 \frac{0.2}{\sqrt{8}} \cos\left(2t + \frac{5\pi}{4}\right) = \frac{\sqrt{2}}{4} \cos\left(2t + \frac{5\pi}{4}\right).$$

Note that in this example the amplitude of the input $u(t)$ is not 1. Therefore, the steady state response is given by $y_{ss}(t) = AM \cos(\omega t + \varphi)$, where A is the amplitude of the input. In our case, since $u(t) = 5 \cos(2t)$, the steady state response is $y_{ss}(t) = 5M \cos(\omega t + \varphi)$.

In order to evaluate the magnitude and phase of complex transfer functions the following identities can be useful. Let $A, B, C \in \mathbb{C}$ and $a, b, c, d \in \mathbb{R}$. The usage of these identities is illustrate in Lecture 5 and introduced as the graphical approach.

Magnitude

$$\left| A \cdot \frac{B}{C} \right| = |A| \cdot \frac{|B|}{|C|}, \quad \left| \frac{(a + jb)^x}{(c + jd)^y} \right| = \frac{(\sqrt{a^2 + b^2})^x}{(\sqrt{c^2 + d^2})^y}$$

Phase

$$\angle(a + jb) = \begin{cases} \arctan(b/a), & a > 0 \\ \arctan(b/a) + \pi, & a < 0, b \geq 0 \\ \arctan(b/a) - \pi, & a < 0, b < 0 \end{cases}$$

$$\angle\left(A \cdot \frac{B}{C}\right) = \angle(A) + \angle(B) - \angle(C)$$

$$\angle\left(\frac{(a + jb)^x}{(c + jd)^y}\right) = x\angle(a + jb) - y\angle(c + jd)$$

3.4 Equivalent Forms of Transfer Functions

In general, we can express any transfer functions in the form of a rational function, i.e. the ratio of two polynomials (the nominator $N(s)$ and denominator $D(s)$), plus the “direct feed-through” term (which is usually zero in our case).

$$G(s) = \frac{N(s)}{D(s)} = \frac{c_{n-1}s^{n-1} + c_{n-2}s^{n-2} + \dots + c_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d. \quad (3.17)$$

For causal, finite-dimensional LTI SISO systems the transfer function will consist of a strictly proper rational function, where the degree of the denominator is greater than the degree of the numerator, and the feed-through term d (often 0).

It is important to note that, the form of the transfer function is not unique. In fact, depending on what we want to do with the transfer function, we want to rewrite it in a different form that is most applicable for the task at hand. For example, when performing the inverse Laplace transform to obtain the time domain response, it is useful to rewrite the transfer function in partial fraction expansion form. The remaining forms introduced below will be used later on in the course.

Finally, note that for ease of exposition, without loss of generality in the remainder of the section we assume $d = 0$. For non-zero d the form is similar to Equation (3.17).

3.4.1 Partial Fraction Expansion

One way to express the transfer function is with the help of the partial fraction expansion. In this case it can be rewritten as

$$G(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \cdots + \frac{r_n}{s - p_n} + d, \quad (3.18)$$

where r_i are called the residues and p_i the *poles* (more on them in Section 3.6). It is useful to compute transient responses, and to assess how much different modes contribute to the response of a system. To convert to the partial fraction expansion, we can either use the standard approach to partial fraction decomposition, or we can use the cover-up method to compute the residues r_i . For non-repeating poles the residues r_i can be obtained through

$$r_i = \lim_{s \rightarrow p_i} (s - p_i) G(s). \quad (3.19)$$

While for repeating poles of order m the formula becomes more complicated and is given by

$$r_i = \frac{1}{(m-1)!} \lim_{s \rightarrow p_i} \frac{d^{m-1}}{ds^{m-1}} ((s - p_i)^m G(s)). \quad (3.20)$$

3.4.2 Root-Locus Form

We can also factorize the nominator and denominator. One way to do so is by writing it as

$$G(s) = \frac{k_{rl}}{s^q} \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_{n-q})}, \quad (3.21)$$

where p_i are *poles*, z_i *zeros*, k_{rl} the *root-locus gain*, and q the number of *poles* at the origin (i.e., $s = 0$). This form will be useful to construct the root-locus plot (introduced in Week 7), hence its name.

3.4.3 Bode Form

Another way to factorize the nominator and denominator is given by the Bode form

$$G(s) = \frac{k_{\text{Bode}}}{s^q} \frac{\left(\frac{s}{-z_1} + 1\right) \left(\frac{s}{-z_2} + 1\right) \cdots \left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right) \left(\frac{s}{-p_2} + 1\right) \cdots \left(\frac{s}{-p_{n-q}} + 1\right)}. \quad (3.22)$$

As before p_i are the *poles*, z_i *zeros*, k_{Bode} is the *Bode gain*, and q is the number of *poles* at the origin. This form will be useful to construct the Bode plot (introduced in Week 9).

3.5 Transfer Functions and the LTI State-Space Form

Just as the LTI state-space form, the transfer function also describes the input-output behaviour of a system. For a given system both the transfer function and LTI form contain the same information, just expressed in a different domain. Below we exploit the structure of the LTI form to facilitate the conversion between the two.

3.5.1 From State-Space to Transfer Function

Above we learned how to convert from the LTI model given by the matrices A, B, C , and D , to the transfer function by the relation

$$G(s) = C(sI - A)^{-1}B + D. \quad (3.23)$$

This can be simplified for the case where A is diagonal. By comparing coefficients the LTI form can be transformed to a transfer function as

$$A = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}, B = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}, C = [c_1 \quad \cdots \quad c_n], D = d,$$

$$\Downarrow$$

$$G(s) = \frac{c_1 b_1}{s - \lambda_1} + \frac{c_2 b_2}{s - \lambda_2} + \cdots + \frac{c_n b_n}{s - \lambda_n} + d, \quad (3.24)$$

Example - Pendulum

We have already seen the transformation of the pendulum system, linearized around $x_{e,1} = (0, 0)$, $u_e = 0$, from the state-space

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0.2 \end{bmatrix}, C = [1 \quad 0], D = 0$$

to the transfer function

$$G(s) = \frac{0.2}{s^2 + s + 2}$$

using Equation (3.23). Now, let's calculate a transfer function by diagonalizing the system. The eigenvalues of A are obtained from the characteristic polynomial

$$\det(sI - A) = s(s + 1) + 2 = s^2 + s + 2 = 0,$$

which gives

$$\lambda_{1,2} = \frac{-1 \pm j\sqrt{7}}{2}.$$

The corresponding eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ -\frac{1-j\sqrt{7}}{2} \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -\frac{1+j\sqrt{7}}{2} \end{bmatrix}.$$

Next, we can diagonalize the system by finding the transformation matrix T , whose columns are the eigenvectors of A . The transformation matrix T and its inverse T^{-1} are respectively given by

$$T = \begin{bmatrix} 1 & 1 \\ -\frac{1-j\sqrt{7}}{2} & -\frac{1+j\sqrt{7}}{2} \end{bmatrix}, \quad T^{-1} = \frac{j}{\sqrt{7}} \begin{bmatrix} \frac{1+j\sqrt{7}}{2} & 1 \\ -\frac{1-j\sqrt{7}}{2} & -1 \end{bmatrix}.$$

By diagonalizing A we obtain

$$\tilde{A} = T^{-1}AT = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -\frac{1+j\sqrt{7}}{2} & 0 \\ 0 & -\frac{1-j\sqrt{7}}{2} \end{bmatrix}.$$

To preserve the input-output behaviour, the other matrices must also be transformed accordingly:

$$\tilde{B} = T^{-1}B, \quad \tilde{C} = CT, \quad \tilde{D} = D = 0.$$

For $\tilde{B}$ we get

$$\tilde{B} = T^{-1}B = \frac{j}{\sqrt{7}} \begin{bmatrix} \frac{1+j\sqrt{7}}{2} & 1 \\ -\frac{1-j\sqrt{7}}{2} & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0.2 \end{bmatrix} = \frac{j}{\sqrt{7}} \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} = \begin{bmatrix} \frac{0.2j}{\sqrt{7}} \\ -\frac{0.2j}{\sqrt{7}} \end{bmatrix}.$$

Similarly,

$$\tilde{C} = CT = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -\frac{1-j\sqrt{7}}{2} & -\frac{1+j\sqrt{7}}{2} \end{bmatrix} = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

Using the transformation from Equation (3.24), which in this case reduces to

$$G(s) = \frac{c_1 b_1}{s - \lambda_1} + \frac{c_2 b_2}{s - \lambda_2} + \tilde{D},$$

we obtain

$$G(s) = \frac{1 \cdot \frac{0.2j}{\sqrt{7}}}{s - \lambda_1} + \frac{1 \cdot \left(-\frac{0.2j}{\sqrt{7}}\right)}{s - \lambda_2} = \frac{0.2j}{\sqrt{7}} \left(\frac{1}{s - \lambda_1} - \frac{1}{s - \lambda_2} \right).$$

We can now simplify this expression to recover the real rational form of the transfer function. Combining the two terms over a common denominator gives

$$G(s) = \frac{0.2j}{\sqrt{7}} \frac{(s - \lambda_2) - (s - \lambda_1)}{(s - \lambda_1)(s - \lambda_2)} = \frac{0.2j}{\sqrt{7}} \frac{\lambda_1 - \lambda_2}{(s - \lambda_1)(s - \lambda_2)}.$$

Since

$$\lambda_1 - \lambda_2 = \frac{-1 + j\sqrt{7}}{2} - \frac{-1 - j\sqrt{7}}{2} = j\sqrt{7},$$

we obtain

$$G(s) = \frac{0.2j}{\sqrt{7}} \frac{j\sqrt{7}}{(s - \lambda_1)(s - \lambda_2)} = \frac{0.2}{(s - \lambda_1)(s - \lambda_2)}.$$

Finally, recalling that

$$(s - \lambda_1)(s - \lambda_2) = s^2 + s + 2,$$

we arrive at the original form of the transfer function:

$$G(s) = \frac{0.2}{s^2 + s + 2}.$$

This confirms that diagonalizing the system matrix A does not change the transfer function, it only separates the system dynamics into their modal components, each associated with one eigenvalue of A .

3.5.2 From Transfer Function to State-Space

As introduced above, a transfer function $G(s)$ is generally some rational function as shown in Equation (3.17). Given such a transfer function, there are many state-space realizations such that Equation (3.23) holds. We are particularly interested in finding a minimal realization, i.e. some combination of A, B, C and D such that the order/dimension of the LTI system – size of state vector $x(t)$ – is equal to the degree of the denominator n . One such minimal realization can be obtained by inspecting the components of $G(s)$ and combining them in the following way

$$G(s) = \frac{c_{n-1}s^{n-1} + c_{n-2}s^{n-2} + \cdots + c_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d,$$

$$\Downarrow$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \ddots & & 1 \\ -a_0 & -a_1 & \cdots & & & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}, \quad C = [c_0 \quad c_1 \quad \cdots \quad c_{n-1}], \quad D = d,$$

which is also called the controllable canonical form.

If a transfer function is in its partial fraction expansion form, as seen in Section 3.4, we can directly obtain a diagonal realization of our system.

$$G(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \cdots + \frac{r_n}{s - p_n} + d$$

$\Downarrow$

$$A = \begin{bmatrix} p_1 & & \\ & \ddots & \\ & & p_n \end{bmatrix}, \quad B = \begin{bmatrix} \sqrt{r_1} \\ \vdots \\ \sqrt{r_n} \end{bmatrix}, \quad C = [\sqrt{r_1} \quad \cdots \quad \sqrt{r_n}], \quad D = d.$$

Keep in mind that the mapping from a transfer function to an LTI system is not unique and many realizations for the same transfer functions can be found. We will always aim to work with the minimal realization of a system, i.e. with the smallest possible state-space matrices A, B, C , and D .

Example - Pendulum

Consider once more the transfer function of the pendulum system linearized around its stable equilibrium point. Recall the stable transfer function given by

$$G(s) = \frac{0.2}{s^2 + s + 2}.$$

To obtain the corresponding state-space representation, we can make use of the controllable canonical form introduced above. Comparing the transfer function with the general form

$$G(s) = \frac{c_{n-1}s^{n-1} + c_{n-2}s^{n-2} + \cdots + c_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} + d,$$

we can identify the coefficients as

$$a_0 = 2, \quad a_1 = 1, \quad c_0 = 0.2, \quad c_1 = 0 \quad \text{and} \quad d = 0.$$

Furthermore, entering all the values in the controllable canonical form, we obtain the following state-space realization:

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = [0.2 \quad 0], \quad D = 0.$$

Note that this state-space realization is obtained directly from the transfer function and therefore differs from the physically linearized model of the pendulum. Also, the realization is not unique — several sets of A, B, C, D can yield the same transfer function. Nevertheless, both representations describe the same input–output behaviour.

3.6 Poles and Zeros

Let's take a closer look at the transfer functions we derived above and see what insight we can gain from them. We will take particular interest in the roots of the nominator $N(s)$ and denominator $D(s)$. The roots of the nominator are called zeros z_i , and the roots of the denominator are called poles p_i . In the following sections we will discuss what kind of impact

they have on the systems' behaviour.

3.6.1 Poles

To understand the effects of poles, it helps to recognize a fundamental connection between the transfer function and the corresponding LTI system. For that, recall that the general form of $G(s)$, for some LTI system, is given as¹

$$G(s) = C(sI - A)^{-1}B + D = \frac{N(s)}{D(s)},$$

also consider that the inverse of a matrix can be determined with

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}.$$

Using this definition, we can rewrite the general form of the transfer function to be

$$\begin{aligned} G(s) &= \frac{1}{\det(sI - A)} \underbrace{C \text{adj}(sI - A) B + D}_{N(s)} \\ &= \frac{N(s)}{\det(sI - A)}. \end{aligned}$$

Notice that the denominator of our transfer function is given as $D(s) = \det(sI - A)$, which is also the characteristic polynomial of the matrix A . This implies that the roots of the denominator p_i are equal to the roots of the characteristic polynomial of A . As we know from Linear Algebra the roots of the characteristic polynomial are also the eigenvalues λ_i of the matrix. Thus, we can relate the eigenvalues of A and the poles of the transfer function.

The denominator of a transfer function is equal to the *characteristic polynomial* of A , and the roots of the denominator, also called *poles*, are equal to the eigenvalues of A .

One major insight we can gain from this, is that the poles also contain information about stability. As an example consider a transfer function where all poles p_i have a negative real part. This also means that all eigenvalues have a negative real part, i.e. $\text{Re}(\lambda_i) < 0$, from which we can, per definition, conclude that the system is asymptotically stable.

In order to further understand how the poles/eigenvalues affect our system we will consider the impulse response introduced in Section 2.4.2. Recall that the impulse response is the response of a system to zero initial condition $x(0) = 0$ and a unit impulse as input $u(t) = \delta(t)$. Using transfer functions we can now consider the impulse response in the s -domain. From Table 3.1 we learn that the Laplace transform of the unit impulse is $U(s) = 1$. In this case the output becomes

$$Y_{\text{imp}}(s) = G(s)U(s) = G(s).$$

¹Note that D is the feed-through term and $D(s)$ is the denominator of the rational part of the transfer function.

We observe that the inverse Laplace transform of the transfer function $\mathcal{L}^{-1}\{G(s)\}$ now corresponds to the impulse response $y_{\text{imp}}(t)$, i.e.

$$y_{\text{imp}}(t) = \mathcal{L}^{-1}\{G(s)\},$$

which fully characterizes the system's response. Considering that any transfer function can be written as a partial fraction expansion of the form

$$G(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \cdots + \frac{r_n}{s - p_n} + d,$$

and according to Table 3.1, where

$$\mathcal{L}^{-1}\left[\frac{1}{s - a}\right] = e^{at},$$

we see that each term of the transfer function generates one exponential function in the impulse response, weighted by the residual r_i . Resulting in the impulse response

$$y_{\text{imp}}(t) = r_1 e^{p_1 t} + \cdots + r_n e^{p_n t}.$$

From this we can come to an important conclusion.

In the impulse response, every real pole generates an exponential function, and each complex conjugate pole pair generates a growing/decaying sinusoid.

This behaviour is illustrated in Figure 3.6 where three examples of systems, characterized by their poles in the s -domain and the corresponding impulse responses are depicted. The examples show complex conjugate pole pairs — indicated with crosses in the s -domain — and the resulting impulse responses. On the left, the poles have a negative real part and therefore the impulse response is an exponentially decaying sinusoid that converges to zero. In the middle the poles with zero real part result in a pure sinusoid that oscillates for all times. Finally, on the right, the poles have a positive real part, i.e. the impulse response will grow exponentially and become unbounded.

To further strengthen your intuition about the location of poles with the time response, try using the tool ([MyBinder hyperlink](#)) to plot the impulse response of a system with different poles. As a suggestion, try varying the parameters, τ, ζ, K, w_n of the following systems:

$$G(s) = \frac{K}{\tau s + 1}, \quad G(s) = \frac{w_n^2}{s^2 + 2\zeta w_n s + w_n^2},$$

For any selected set of parameters, compute the poles, and verify whether the behaviour matches that described in Figure 3.6.

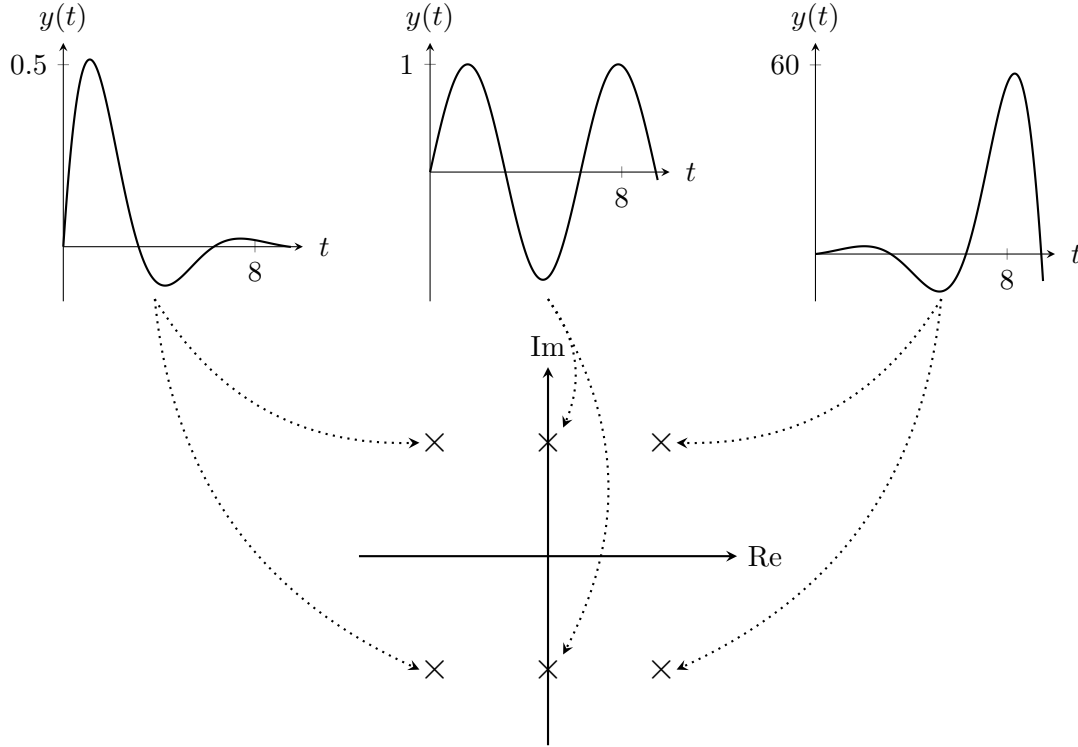


Figure 3.6: Impulse responses and the corresponding complex conjugate pole pair in the s -domain. Poles are indicated by $\times$ in the complex plane.

3.6.2 Zeros

To understand the effects of zeros, i.e. the roots of the nominator of a transfer function, we will first consider a system where the input-output relationship is described by

$$y(t) = \frac{d}{dt}u(t).$$

This type of system is also referred to as a differentiator since the output is the derivative of the input. By applying the fundamental input $u(t) = e^{st}$ to such a differentiator,

$$y(t) = \frac{d}{dt}e^{st} = se^{st},$$

we can conclude that the transfer function of such a differentiator is given by

$$G_{\text{diff}}(s) = s.$$

We can now use this fact to analyze the following example. Consider a system described by

$$G(s) = \left(\frac{s}{-z} + 1 \right) \tilde{G}(s) = \tilde{G}(s) + \frac{s}{-z} \tilde{G}(s),$$

where the transfer function $G(s)$ is obtained by adding a zero at $s = z$ to the transfer function $\tilde{G}(s)$. Now if we consider the time response of $\tilde{G}(s)$ to be given by $\tilde{y}(t)$ and using the fact that s is the transfer function of a differentiator, we can write the time domain response $y(t)$ of $G(s)$

as

$$y(t) = \tilde{y}(t) + \frac{1}{-z} \dot{\tilde{y}}(t). \quad (3.25)$$

We observe that by adding a zero to the transfer function $\tilde{G}(s)$, we effectively add a derivative term to the time domain output. This usually has an “anticipatory” effect. We can analyze this “anticipatory” effect by looking at the step response introduced in Section 2.4.1. Recall that the unit step response is the systems output given a unit step input, i.e. the Heaviside function $u(t) = h(t)$. In Figure 3.7, on the left the step response of two systems are plotted, both have the same poles as marked in the s -domain with crosses, but the dashed one has an additional zero at $s = -1$, as marked in the s -domain with a circle. This is analogous to $G(s)$ and $\tilde{G}(s)$ from above. We can see that the system with the zero reacts faster to the step input — in a way

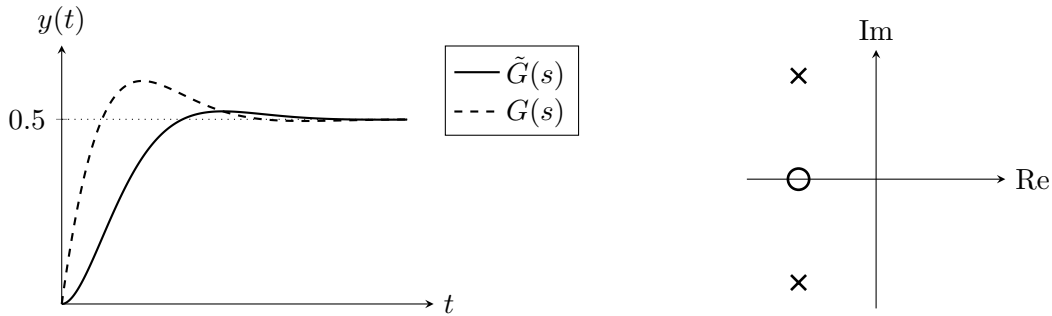


Figure 3.7: Step response of two systems on the left and the corresponding poles and zeros in the s -domain on the right. Poles are indicated with crosses and zeros with circles.

anticipating what’s ahead —, while still converging to the same steady-state value. This is also in line with Equation (3.25) where in this specific case $z = -1$. Initially, the response including the zero is larger since the derivative is large and gets added. After some time the response settles and the derivative term goes to zero and both systems approach the same steady-state value.

Zeros with a negative real part, i.e. when they are in the left half plane, are called *minimum phase zeros*.

Next, let’s consider the case where the added zero has a positive real part, i.e. it is in the right half plane. In Figure 3.8 shows the step responses of two systems both have the same poles, but the dashed one has an additional zero at $s = 1$. We observe that both still reach the same steady-state value, but initially the response moves in the wrong direction. This is also in line with Equation (3.25) since now we subtract the derivative, as z is positive.

Zeros, with a positive real part, i.e. they are in the right half plane, are called *non-minimum phase zeros*. We can interpret them as having some sort of “negative” derivative action.

Figure 3.9 illustrates the effect that the placement of zeros has on the step response for a variety of values, given the same poles.

Note that zeros don’t always have a clear interpretation like the poles have. This is just one

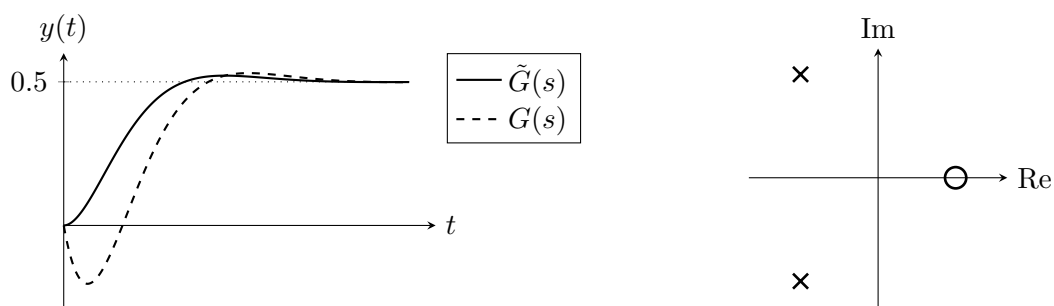


Figure 3.8: Step response of two systems on the left and the corresponding poles and zeros in the s -domain on the right. Poles are indicated with crosses and zeros with circles.

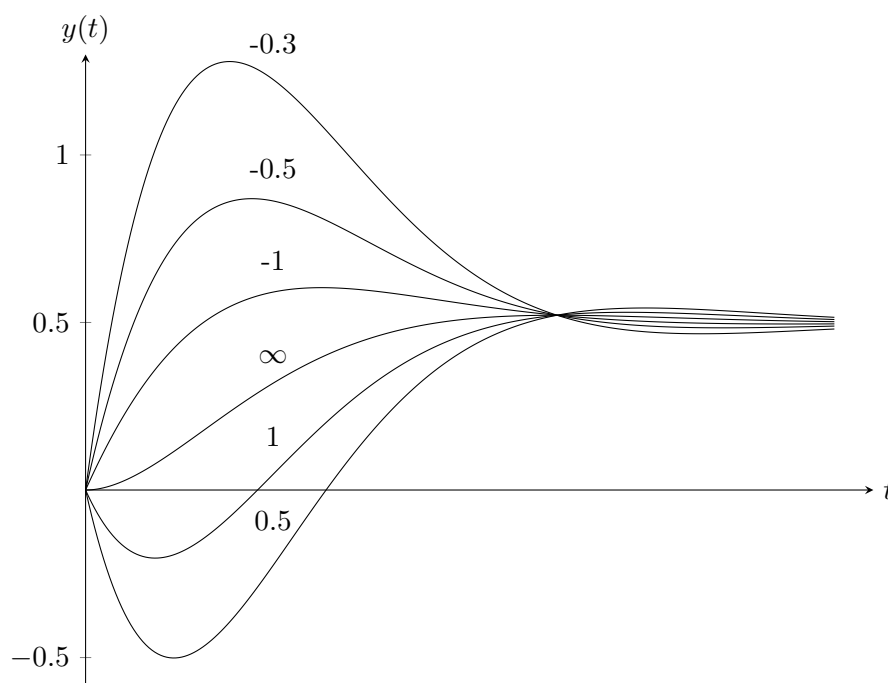


Figure 3.9: Influence of the location of the zeros z_i on the step response of a second order system. The values on the curves represent the location of the zeros on the real axis.

way of thinking about them.

To further strengthen your intuition about the location of zeros with the time response, try using the tool ([MyBinder hyperlink](#)) to plot the impulse response of a system with different zeros. As a suggestion, try varying the parameters, ζ, w_n, z_1 of the following systems:

$$G(s) = \frac{w_n^2(s + z_1)}{s^2 + 2\zeta w_n s + w_n^2},$$

For any selected set of parameters, verify whether the time-response for that zero matches the behaviour described in Figure 3.9.

3.6.3 Pole Zero Cancellation

Now let's look at the effects that poles and zeros have on each other, more specifically we want to understand what happens when a pole and a zero are very close to each other. To analyze this, consider the transfer function $G(s)$ that has three poles in the left half plane and a minimum phase zero very close to one of its poles on the real axis as illustrated in Figure 3.10. We will investigate the influence that varying the parameter ε has on the system.

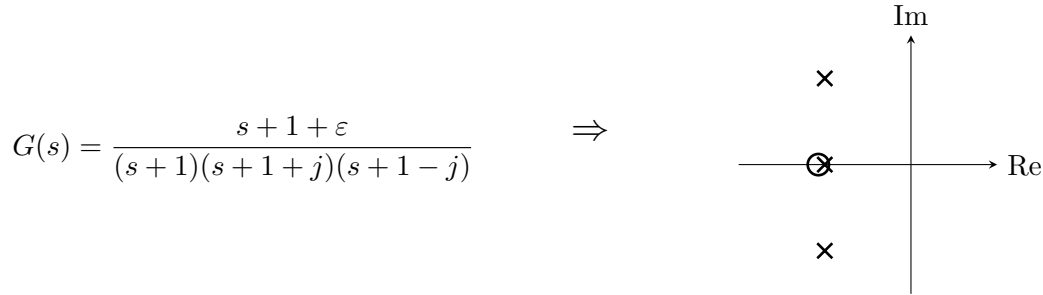


Figure 3.10: Transfer function $G(s)$ and the corresponding pole zero plot in the complex plane

We start by rewriting the transfer function $G(s)$ using partial fraction expansion. From Section 3.4 we know that, the denominators of the resulting fractions are given by the poles of the transfer function, so we only need to compute the residues. For this we can use the cover-up method to compute the residues r_i as

$$r_i = \lim_{s \rightarrow p_i} (s - p_i)G(s).$$

For our transfer function G we can then compute the residues to be

$$\begin{aligned} r_1 &= \lim_{s \rightarrow -1} (s + 1)G(-1) = \varepsilon \\ r_2 &= \lim_{s \rightarrow -1-j} (s + 1 + j)G(-1 - j) = \frac{j - \varepsilon}{2} \\ r_3 &= \lim_{s \rightarrow -1+j} (s + 1 - j)G(-1 + j) = \frac{-j - \varepsilon}{2} \end{aligned}$$

Resulting in the partial fraction expansion

$$G(s) = \frac{\varepsilon}{s + 1} + \frac{0.5(j - \varepsilon)}{s + 1 + j} - \frac{0.5(j + \varepsilon)}{s + 1 - j}.$$

In this form we can see that for $\varepsilon \rightarrow 0$, $r_1 \rightarrow 0$ and hence the effect of the pole at $s = -1$ becomes smaller. When $\varepsilon = 0$, the effect of the pole is completely cancelled out. This cancelling can also be observed in the original transfer function, since if $\varepsilon = 0$ the terms in the nominator and denominator cancel out completely. This can be of big concern when working with a system. Remember that the transfer function only describes the input output behaviour, so when we cancel poles in the transfer function we can't see them in the output anymore. But the effect of

that pole, i.e. the associated exponential, is still present in the real system. If we cancel out an unstable pole the system might “blow up” internally, but we can’t see it in the output, which could become a problem.

To better understand this internal “blow-up”, we can look at the system’s internal dynamics. When a pole-zero pair is cancelled out and the pole is unstable (i.e. located in the right half-plane), the internal state associated with that pole still evolves according to its eigenvalue p_i . Specifically, its internal state grows as $x_i(t) = e^{p_i t} x_i(0)$, leading to an exponential increase over time. Although this unstable mode is not observable at the output due to the zero cancelling its contribution, it still affects the internal energy of the system. In a physical or numerical implementation, this may result in the internal states becoming unbounded, saturating actuators damaging components or causing numerical overflow. This phenomenon is referred to as the system *blowing up internally* - the external response appears stable, but the internal dynamics diverge without bound.

3.7 Final and Initial Value Theorem

Often, we are interested in the steady state behaviour of a system. For example, we might want to know the final velocity of the metronome, is it keeping the correct tempo? However, the transfer function, being in the s -domain, makes it hard for us to interpret this behaviour. Consequently, we introduce a theorem that relates the steady state behaviour of a system in the time domain to the s -domain. This allows us to calculate the steady state value of the output by using the transfer function. For completeness, we also introduce the initial value theorem, which allows us to calculate the initial value of the output by using the transfer function.

Theorem 3.1 (Final Value Theorem) *Let $f(t)$ be a continuous function with Laplace transform $F(s)$. Then the final value theorem states that if all poles of $sF(s)$ have negative real part, then*

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s).$$

Note that the final value theorem only holds for asymptotically stable systems, i.e. all poles of $G(s)$ have real part smaller than zero or if $G(s)$ has at most one pole at the origin (Visit the optional question on the problem to see why this is the case). Similarly to the final value theorem the initial value theorem relates the behaviour of a function in the time domain at zero to the frequency domain.

Theorem 3.2 (Initial Value Theorem) *Let $f(t)$ be a continuous function with Laplace transform $F(s)$. Then the initial value theorem states that*

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s).$$

In the context of transfer functions these theorems are very helpful since the output of a system in the s -domain is given by the multiplication of the transfer function and the input. As a

consequence we can write the steady state output as

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} sG(s)U(s).$$

In the special case of the step response where the input is the Heaviside function $u(t) = h(t)$ with the Laplace transform $U(s) = 1/s$, the steady state value of the step response can be calculated as

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sG(s) \frac{1}{s} = \lim_{s \rightarrow 0} G(s).$$

Example - Final Value Theorem

Consider again the linearized pendulum with transfer function

$$G(s) = \frac{0.2}{s^2 + s + 2}$$

The poles of this system are located at $\frac{-1 \pm j\sqrt{7}}{2}$, both in the left half-plane, which means the system is stable and the Final Value Theorem is applicable.

Apply a Dirac impulse $u(t) = \delta(t)$. From Table 3.1, we know that its Laplace transform is $U(s) = 1$. The output in the s -domain is therefore given by

$$Y(s) = G(s)U(s) = G(s) = \frac{0.2}{s^2 + s + 2}.$$

Using the Final Value Theorem, we can determine the steady state value of the output. According to the theorem,

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} sG(s).$$

Substituting the expression for $G(s)$,

$$sG(s) = s \cdot \frac{0.2}{s^2 + s + 2} = \frac{0.2s}{s^2 + s + 2}$$

Then, taking the limit as $s \rightarrow 0$, we get

$$\lim_{s \rightarrow 0} \frac{0.2s}{s^2 + s + 2} = 0.$$

Hence,

$$\lim_{t \rightarrow \infty} y(t) = 0.$$

This result shows that, after an impulse, the system eventually returns to its stable equilibrium. Physically, this means that the impulse gives a brief kick, but due to damping, all energy dissipates and the pendulum returns to rest.

Chapter 4

Introduction to Feedback

4.1 Introduction

So far, we have introduced tools to model and analyze the behaviour of physical systems. We have learned how to derive mathematical representations of these systems – such as the state-space model or transfer function – from physical descriptions, and how to study system properties including stability and response to inputs.

However, our ultimate goal is to make a system follow a specified objective—for example, ensuring that a metronome oscillates at a constant rate, or that a pendulum in a theme park ride reaches and maintains a desired position. To achieve this, we need to design *controllers* (mechanisms) that influence the system’s behaviour actively. This is the essence of control.

One way of thinking about control is to extend the system architecture we have been using so far (Figure 4.1a) by adding a controller C — i.e., a dynamical system that we design — that takes some desired reference signal $r(t)$ and produces an input $u(t)$. This input is then applied to the *plant* (physical system) and results in an output $y(t)$ (Figure 4.1b). In the case of our pendulum-themed theme park ride, the reference signal $r(t)$ would be the desired angle of the pendulum bob, the input $u(t)$ would be the torque applied to the pendulum, and the output $y(t)$ would be the actual angle of the pendulum bob.

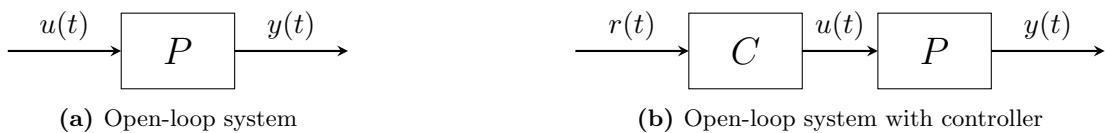


Figure 4.1: Comparison of open-loop configurations: (a) plant only and (b) plant with controller.

In this type of control architecture (Figure 4.1b), called *open-loop architecture*, the controller is not aware of the output of the plant. Although for some systems it is possible to design provably correct controllers for which the open-loop architecture is suitable, this is rarely the case. This is in part due to imprecise models, external disturbances, noisy hardware, and often

analytical assumptions not being held (E.g., most systems are not linear).

Consequently, a common approach is to close the loop by adding a feedback loop that measures the output of the plant and compares it to the reference signal (Figure 4.2). The difference now is that the controller evaluates an input as a function of the error. It can therefore adapt the input according to the size of the mismatch of output and reference. Intuitively, for our theme park ride example, when the pendulum is not at the desired angle, the controller will apply a torque to the pendulum to bring it closer to the desired angle. This is the essence of *closed-loop control* (also known as *feedback control*).

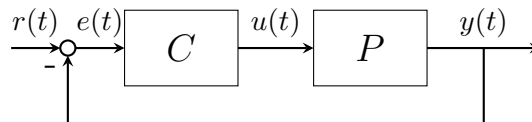


Figure 4.2: Standard feedback architecture

Closed-loop control offers several advantages over open-loop control. Firstly, it can stabilize an otherwise unstable system. Secondly, since the controller continuously adjusts the input based on the observed output, it can automatically compensate for disturbances and measurement noise. Thirdly, it provides a degree of robustness to modeling inaccuracies — even when the mathematical model of the plant is simplified or approximate —, feedback control can still achieve desirable performance by correcting deviations from the desired behaviour in real time. This property is particularly valuable for complex systems, where obtaining a perfectly accurate model is often impractical or impossible.

That said, feedback control is not a universal remedy. A poorly designed controller can introduce instability, oscillations, or excessive sensitivity to noise. Moreover, achieving good closed-loop performance requires careful analysis and tuning, taking into account real-world constraints such as actuator limits, time delays, and nonlinear effects. In short, feedback control is a powerful means of achieving stability and robustness — but it must be applied judiciously as closed-loop control does not in general guarantee that any controller will be able to achieve the control objectives.

In the chapters that follow, we will introduce methods to analyze and synthesize controllers for various types of objectives (such as stability, time to reach a desired state, or tracking a desired reference signal). In particular, the methods we introduce will *use the open-loop system to analyze the closed-loop behaviour of the system*. This is particularly powerful as it allows us to reason about the closed-loop behaviour of a system based only on the information of the open-loop system, thus simplifying our analysis.

In this chapter, we focus on proportional feedback control, the case where the controller is a scalar gain. In Section 4.2, we further solidify the concept of feedback control, and introduce how to compute various closed-loop transfer functions. Then in Section 4.3 we introduce the proportional feedback controller and how it can be used to stabilize a system. Finally, in Section 4.4, we introduce the root locus method, a tool used to analyze the closed-loop behaviour of a system based only on the information of the open-loop system.

To help build intuition over what a controller could be and why feedback control is useful, consider the following example.

Example - Closing the Loop

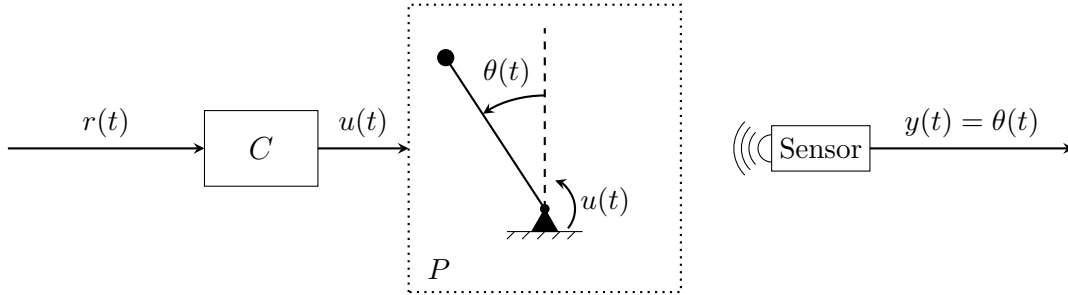


Figure 4.3: Visualization of an open-loop system for the inverted pendulum.

Suppose we want to position the bob of an inverted pendulum at an angle θ_r using the open-loop architecture in Figure 4.3. The controller is provided with a reference $r(t) = \theta_r = \epsilon^\circ$ and must generate a control input $u(t)$ that drives the pendulum. For small deviations around equilibrium, the linearized model provides a reasonable approximation of the true plant dynamics, so let's assume the pendulum starts near equilibrium, at $\theta(0) = \epsilon^\circ - \delta(0)$. To move the pendulum toward the desired position, we apply a positive input $u(t) = a > 0$. However, since friction and rod mass are neglected, the actual position soon diverges from the expected position, leaving us uncertain of the true state and the proper torque direction.

Now consider adding feedback control as in Figure 4.4. Here, the pendulum angle $\theta(t)$ is measured continuously and compared to the desired reference $r(t)$. The resulting error $e(t) = r(t) - y(t)$ is fed to a controller C that adjusts $u(t)$ accordingly. With a suitable controller, the inverted pendulum can be stabilized despite model simplifications—for example, one can always ensure that $u(t) = a > 0$ when $\theta(t) < \epsilon^\circ$ and $u(t) = -a < 0$ when $\theta(t) > \epsilon^\circ$.

One of the key ideas behind controlling the output of the plant is to add a controller C that takes the error $e(t) = r(t) - y(t)$ between the reference and the output and produces an input

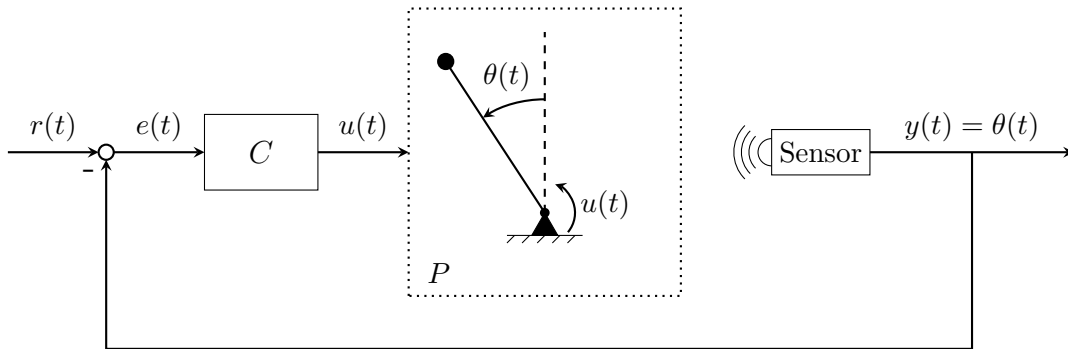


Figure 4.4: Visualization of a standard feedback loop for the inverted pendulum.

$u(t)$. This input is then applied to the plant and results in an output $y(t)$. The controller C can similarly be thought of as a dynamical system – one that we are designing – and thus can be modelled as a transfer function or state-space model. In practice, the controller will be software running on a microcontroller attached to the plant, but it can also be a physical system (see [Toilets: Behind the Flush](#)).

4.2 Transfer Functions of Closed-loop Systems

Up to this point, we derived the *transfer function* of a system as the relationship between an input signal $u(t)$ and an output signal $y(t)$. However, once we introduce feedback, several related but distinct transfer functions arise, each describing how different signals within the closed-loop system are related. Namely,

- The *open-loop transfer function* that maps from the error signal $e(t)$ to the output $y(t)$;
- The *closed-loop transfer function* that maps from the reference $r(t)$ to the output $y(t)$;
- The *sensitivity function* that maps from the reference $r(t)$ to the error $e(t)$.

The remainder of this section is dedicated to deriving expressions for these transfer functions in general form. To do so, we will begin by deriving the respective expression using the block diagram algebra we have seen in Section 1.7. Then, due to linearity, we are able to extend this relationship to the laplace domain to obtain the corresponding transfer function. The following derivations are based off the feedback architecture in Figure 4.5.

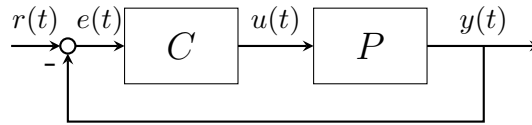


Figure 4.5: Standard feedback configuration

We begin by deriving the open-loop mapping L , that maps from an error signal $e(t)$ to an output signal $y(t)$.

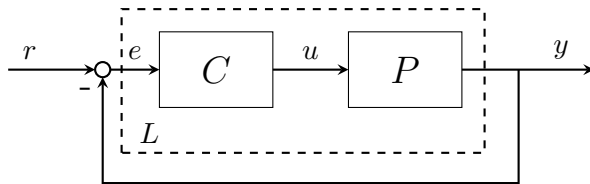


Figure 4.6: Open-loop system L

$$u = Ce, \quad (4.1)$$

$$y = Pu. \quad (4.2)$$

By substituting (4.1) into (4.2), we obtain

$$y = CPe, \implies \frac{y}{e} = L = CP. \quad (4.3)$$

As we will see Section 4.4, the open-loop mapping L is useful because it is rich in information about a systems closed-loop behaviour. However, the tools we introduce are based on the *s-domain* representation of this mapping. Consequently, we need to obtain the transfer function of L . Owing to linearity, we can apply the Laplace transform directly to Equation (4.3) and

obtain the transfer function of L .

$$Y(s) = C(s)P(s)E(s), \quad \implies \quad L(s) = \frac{Y(s)}{E(s)} = C(s)P(s). \quad (4.4)$$

Definition 4.1 (Open-Loop Transfer Function) *The open-loop transfer function $L(s)$ of a standard feedback control system (Figure 4.5) is defined as*

$$L(s) = C(s)P(s). \quad (4.5)$$

By following the same procedure, we can derive the closed-loop mapping T , that maps from a reference $r(t)$ to an output $y(t)$.

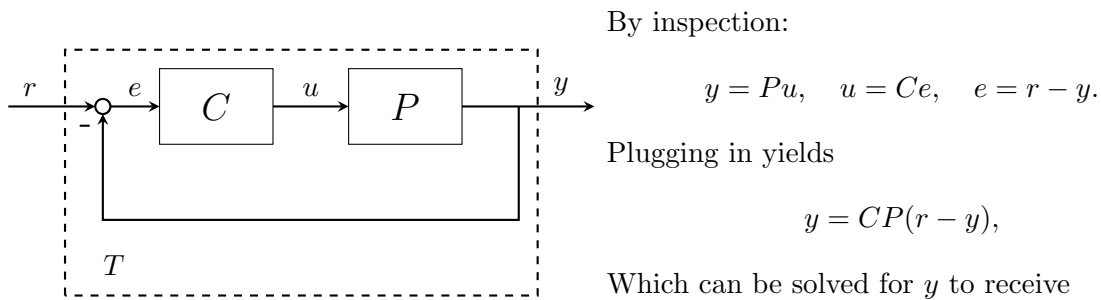


Figure 4.7: Closed-loop system T

$$y = \frac{CP}{1 + CP}r. \quad (4.6)$$

Definition 4.2 (Complementary Sensitivity) *The transfer function of a standard closed-loop system is called the complementary sensitivity function $T(s)$ and is defined as*

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)}. \quad (4.7)$$

This complementary sensitivity transfer function describes the full closed-loop behaviour from r to y of the feedback control loop. A visual representation of this closed-loop system is shown by the dashed line in Figure 4.7.

In a similar manner, the sensitivity function $S(s)$ can be obtained, describing how the error signal e responds to the reference input r .

Definition 4.3 (Sensitivity Function) *The sensitivity function $S(s)$ is defined as*

$$S(s) = \frac{1}{1 + L(s)} = \frac{1}{1 + C(s)P(s)}. \quad (4.8)$$

4.3 Introduction to Proportional Feedback Control

In a feedback system, the goal of the controller is to generate an input $u(t)$ that makes the output $y(t)$ follow the desired reference signal $r(t)$. Intuitively, when the output drifts away from the reference (i.e., the error is large), the controller should respond strongly to correct the error, and when the output matches the reference, the control effort should fade away. This reasoning leads to the idea of *proportional control*, where the control input is proportional to the instantaneous error between the desired and measured signals.

This behaviour is achieved by defining the controller C to be a constant gain $k \in \mathbb{R}$. Mathematically, proportional control is described by the relationship

$$u(t) = C e(t) = k e(t) = k (r(t) - y(t)), \quad (4.9)$$

where $k \in \mathbb{R}$ is the *proportional gain*. The proportional gain determines how strongly the controller reacts to the error: larger values of k produce stronger, faster corrections, while smaller values result in gentler but slower responses.

In section Section 4.2, we have derived a general formula that described the the closed-loop system behaviour from reference to output as Equation (4.7), the complementary sensitivity function. We will use that function to find the feedback description for the case, where the controller is a proportional gain. First, take a look at Figure 4.8, which shows the standard feedback architecture with a proportional controller.

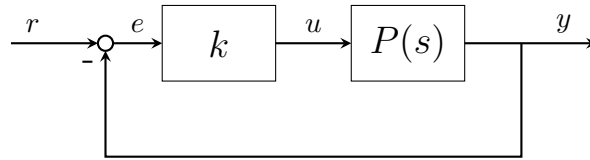


Figure 4.8: Feedback control architecture with a proportional gain.

The transfer function of the proportional controller is given by $C(s) = k$. Therefore, the open-loop transfer function according to Equation (4.5) is given by

$$L(s) = kP(s). \quad (4.10)$$

From this, the closed-loop transfer function can be found according to Equation (4.7) as

$$T(s) = \frac{kP(s)}{1 + kP(s)}. \quad (4.11)$$

The design challenge is to select a suitable value of k such that the resulting closed-loop system satisfies the desired performance and stability requirements. By knowing the general structure of the closed-loop, we can infer many properties only from knowledge of the open-loop configuration. In the remainder of this section we analyze the influence of the proportional gain on the closed-loop properties of certain systems. This will set the foundation for Section 4.4

where we will introduce a more general approach that visualizes how varying the gain influences closed-loop stability.

4.3.1 Single Pole System

Let's first consider the case of a plant with a single pole. Assume an unstable open-loop plant $P(s)$ with an unstable pole at $p = 1$ given by

$$P(s) = \frac{1}{s-1}. \quad (4.12)$$

The goal is to determine whether the plant can be stabilized by introducing proportional feedback. To do so, we compute the closed-loop transfer function using Equation (4.7) as

$$T(s) = \frac{k \frac{1}{s-1}}{1 + k \frac{1}{s-1}} = \frac{k}{s+k-1}. \quad (4.13)$$

Note that the closed-loop transfer function has a pole at $p = 1 - k$. Therefore, we can control the position of the systems pole by changing the feedback gain k . From Equation (4.13), we can conclude that it is possible to stabilize the plant by choosing a value of $k \in [1, \infty)$. Figure 4.9 illustrates how choosing various values for the feedback gain k affects $T(s)$ and therefore the location of the closed-loop poles. This can be extended to Figure 4.10, where each possible location of the pole is shown for all values of k .

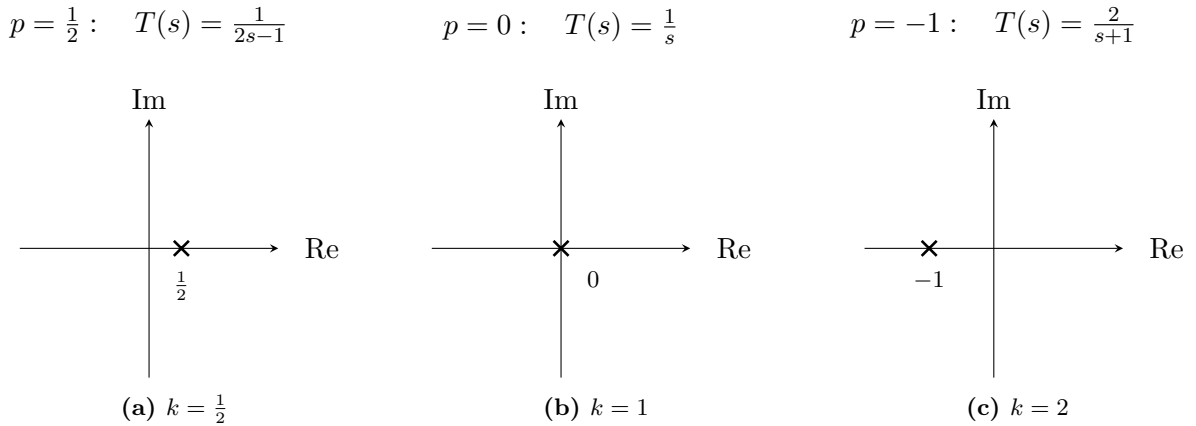


Figure 4.9: Pole location for varying feedback gains. (a) the closed-loop system remains unstable, whereas increasing the gain to the case in (b) stabilizes the plant. A further increase as shown in (c) results in an asymptotically stable closed-loop system.

The key takeaway from Figure 4.10 is that an unstable system can be stabilized by selecting an appropriate feedback gain k . Since stability requires all poles to lie in the left half-plane, by plotting the locations of the closed-loop poles as k varies, we can immediately observe that the system becomes stable once the closed-loop poles cross into the left half-plane. For the case of a single pole at $p = 1$, Equation (4.13), this occurs for all $k \in [1, \infty)$.

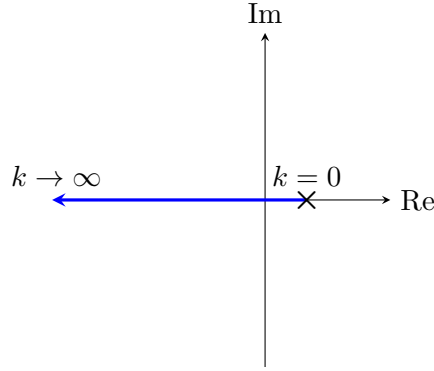


Figure 4.10: Closed-loop poles of Equation (4.13) for varying feedback gains $k \in [0, \infty)$

4.3.2 Multiple Pole Systems

However, most systems have multiple poles. To determine whether a system with multiple poles can be stabilized by a proportional controller, we can apply the same approach as in Section 4.3.1. Let's take the plant $P(s)$ given by

$$P(s) = \frac{1}{s(s+4)}, \quad (4.14)$$

which has open-loop poles at $p_1 = -4, p_2 = 0$. We now look to find an expression for the closed-loop poles. To do so, we plug $P(s)$ into the equation for $T(s)$ and obtain

$$T(s) = \frac{k}{s^2 + 4s + k},$$

from which we find the closed-loop poles as

$$p_1 = -2 + \sqrt{4-k}, \quad p_2 = -2 - \sqrt{4-k}.$$

Plotting the position of the poles of $T(s)$ for varying k produces the plot shown in Figure 4.11. The closed-loop poles start on the real axis. As k increases, the closed-loop poles move closer to each other, until, at $k = 4 \implies p_1 = p_2 = -2$, i.e., the poles meet. At this point, since poles always appear in complex-conjugate pairs, the poles then branch off to $\pm\infty$ on the imaginary axis. The point where the poles meet and then branch outward is called the *breakaway point*.

In summary, introducing proportional feedback in systems with multiple poles may result in real (stable) poles becoming complex (conjugate) poles. As we saw in Section 3.6, complex poles are associated with oscillatory behaviour. Therefore, a proportional controller can introduce oscillatory behaviour to a closed-loop system.

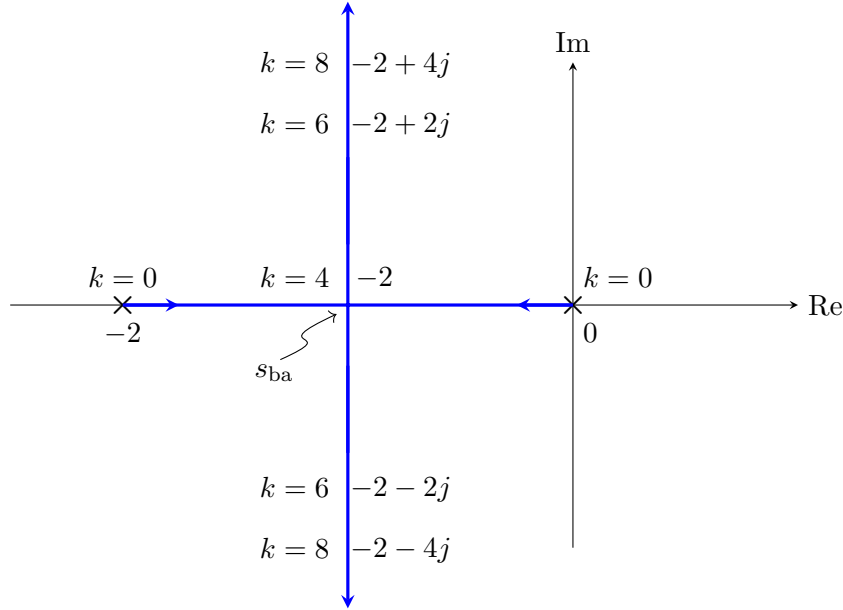


Figure 4.11: Closed-loop poles of Equation (4.14) for varying feedback gains $k \in [0, \infty)$. The breakaway point is denoted as s_{ba} .

4.3.3 Systems with Zeros

Now let's consider the case where the plant has a zero. Let the plant be given by

$$P(s) = \frac{s+1}{s-1}. \quad (4.15)$$

Similar to before, we look to find an expression for the closed-loop poles. To do so, we plug $P(s)$ into the equation for $T(s)$ and obtain:

$$T(s) = \frac{k(s+1)}{s-1+k(s+1)} = \frac{ks+k}{s(k+1)+k-1}, \quad (4.16)$$

with the closed-loop pole given by

$$p = \frac{1-k}{1+k}. \quad (4.17)$$

In this setup it is interesting to note what happens to the closed-loop pole as $k \rightarrow \infty$. By inspection,

$$\lim_{k \rightarrow \infty} \frac{1-k}{1+k} = -1. \quad (4.18)$$

Consequently, the closed-loop pole approaches $s = -1$ with increasing k . This also happens to the open-loop zero of the plant. To see why the closed-loop pole approaches the open-loop zero, consider what happens to the closed-loop transfer function as $k \rightarrow \infty$,

$$\lim_{k \rightarrow \infty} \frac{k(s+1)}{s-1+k(s+1)} \approx \frac{k(s+1)}{k(s+1)}. \quad (4.19)$$

In other words, the poles of the closed-loop transfer function approach the zero of the open-loop transfer function as $k \rightarrow \infty$. Graphically, this behaviour is illustrated in Figure 4.12.

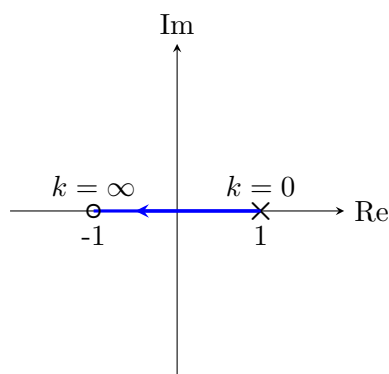


Figure 4.12: The closed-loop poles of Equation (4.16) for varying feedback gains $k \in [0, \infty)$. The closed-loop pole approaches the open-loop zero as $k \rightarrow \infty$.

In summary:

1. A scalar feedback gain does not shift the closed-loop zeros of a system.
2. Closed-loop poles approach the zeros of the plant as $k \rightarrow \infty$.

In this section we introduced the proportional controller and examined how it affects the closed-loop poles of different systems. From this, we observed that

1. Closed-loop stability depends on the location of the closed-loop poles as k varies.
2. Proportional control can introduce oscillations.
3. The closed-loop zeros are the same as the open-loop zeros.

Furthermore, as $k \rightarrow \infty$, the closed-loop poles tend to move toward the open-loop zeros¹. However, analytically analysing the closed-loop poles is cumbersome. Thus, in the next section, we introduce a graphical approach — the *root locus* method — that consists of general rules that permit us to obtain a graphical representation of the closed-loop poles directly from the open-loop transfer function.

¹In the next section we will see that when the system has more poles than zeros, only as many poles as there are zeros will approach them; the remaining poles will move toward infinity along specific asymptotic paths.

4.4 The Root Locus

In the previous section, we saw that introducing a proportional controller in the standard feedback architecture allows us to shift the poles of the closed-loop system. However,

1. Analytically determining the exact location of the closed-loop poles can be tedious, and
2. Proportional control is not always suitable — sometimes more complex controllers are required to achieve the control objectives. Thus, a quick feasibility check is desirable to determine whether the selected controller could be suitable.

Consequently, in this section, we build on the analysis from Section 4.3 by introducing a graphical method that visualizes how the closed-loop poles move as the feedback gain varies. As we will see in Section 4.4.1, this resulting plot, called the *root locus*, can be generalized to the case where the controller is more complex than a proportional gain. Using the root locus, we can then cross-reference the location of the closed-loop poles with the control objectives to determine whether the system can meet the control objectives or not.

It is important to mention that this graphical method consists of a set of rules that permit us to obtain a graphical representation of the closed-loop poles directly from the open-loop transfer function. In practice, this method allows us to quickly determine whether a feedback gain exists that places the closed-loop poles within a desirable region of the complex plane. The remainder of the section is structured as follows. In Section 4.4.1, we introduce two interpretations of how to use the root locus method. In Section 4.4.2, we derive the general rules that describe how the closed-loop poles move as the feedback gain changes. We will see that these rules are general and can be applied to the case where the controller is more complex than a proportional gain. In Section 4.4.3 we will see how to quickly determine the desired feedback gain by using the root locus plot. Finally, in Section 4.4.4, we provide a general recipe for constructing the root locus plot and an example that applied the knowledge gained from the previous subsections.

The root locus, being a graphical tool, is easier to understand through visualization and practice. To help with this process, we have created a root-locus tool that visualizes the root locus for a given open-loop transfer function. This tool can be found at [MyBinder](#) as well as on the moodle course page under *Week 7 - Root Locus*. Feel free to use it to practice the drawing the root locus.

4.4.1 Two Interpretations of the Root Locus Method

When we think about adding a feedback gain to the open-loop system, it can be thought of in two ways:

- **Interpretation 1: Proportional Control** As visible in Figure 4.13, we consider the standard feedback architecture with a proportional controller. In this case we look to identify the feedback gain k that results in a closed-loop system that satisfies the control objectives. This is done by writing the open-loop transfer function as $L(s) = kP(s)$. As

we vary k , the root locus shows how the closed-loop poles move in the complex plane for that fixed plant.

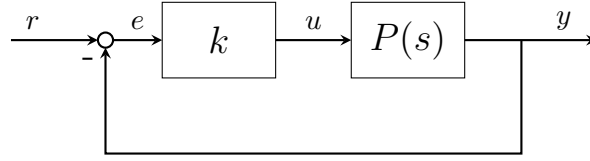


Figure 4.13: Feedback control architecture with a proportional gain.

- **Interpretation 2: General Control** As visible in Figure 4.14, we consider a more general feedback structure, where any fixed part of the controller is absorbed into $C(s)$ and only a scalar gain k is kept outside. In this case, the open-loop transfer function is written as $kL(s) = kC(s)P(s)$.

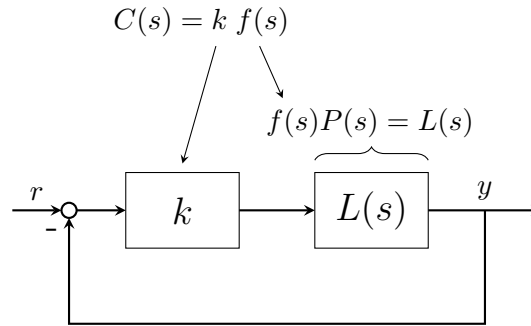


Figure 4.14: Feedback control architecture with a general controller.

Both interpretations are routinely used; it is up to you to choose which configuration best expresses your system and objectives. For the purpose of the course, we will make clear which interpretation is relevant at that point in time. However, in either case, the root locus method permits us to analyze how varying the feedback gain k influences the closed-loop poles. In the following sections, without loss of generality, we adopt the second interpretation.

4.4.2 Derivation of the Root Locus Rules

We now derive the general rules that describe how the closed-loop poles vary as a function of the proportional gain k . As previously mentioned, we will use the second interpretation of the root locus method, and hence our open-loop transfer function is written as $kL(s) = kC(s)P(s)$.

Let us start by writing the open-loop transfer function in root locus form:

$$kL(s) = k k_{rl} \frac{N(s)}{D(s)} = k k_{rl} \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)}. \quad (4.20)$$

Here, m and n denote the total number of zeros and poles of the open-loop system, respectively. The constant k_{rl} represents any fixed gain already included in $L(s)$, while k is the variable feedback gain whose variation determines the root locus. Note that for ease of exposition, and without loss of generality, we drop k_{rl} from Equation (4.20), and introduce it when required.

Using the formula for the complementary sensitivity as denoted in Equation (4.7) and bringing it into the form shown in Equation (4.20), we obtain

$$T(s) = \frac{kL(s)}{1 + kL(s)} = \frac{kN(s)}{D(s) + kN(s)}, \quad (4.21)$$

where the closed-loop poles correspond to the solution of the following characteristic equation

$$D(s) + kN(s) = 0. \quad (4.22)$$

We will use the above to derive the root locus rules.

Rule 1: Symmetry

The characteristic Equation (4.22), can be rewritten as a polynomial of the form

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0.$$

Since complex roots of polynomials with real-valued coefficients come in conjugate pairs, the root locus must always be symmetric with respect to the real axis. In other words, if $s = \sigma + j\omega$ is part of the root locus, then $s = \sigma - j\omega$ must also be part of the root locus.

Rule 2: Number of Branches

When inspecting the degree of the closed-loop characteristic polynomial $D(s) + kN(s)$, we find that its degree is equivalent to the open-loop transfer function denominator $D(s)$. From this we can conclude, that the root locus has the same number of branches² as the open-loop transfer function has poles.

Rule 3: Start and End Points of the Root Locus

To determine the start and end points of the root locus, consider the following limits:

- For $k \rightarrow 0$: The closed-loop poles approach the open-loop poles since

$$\lim_{k \rightarrow 0} D(s) + kN(s) = D(s). \quad (4.23)$$

This means that closing the loop but choosing a zero gain leads to the same overall system stability as when leaving the system in an open-loop configuration. In other words, when the feedback gain $k = 0$, the closed-loop poles are the same as the open-loop poles.

- For $k \rightarrow \infty$: Rewriting Equation (4.22) by dividing both sides by k ,

$$\frac{1}{k} D(s) + N(s) = 0. \quad (4.24)$$

²A branch can be thought of as the trajectory of a closed-loop pole as the feedback gain varies.

Taking the limit as $k \rightarrow \infty$ results in,

$$\lim_{k \rightarrow \infty} \frac{1}{k} D(s) + N(s) \approx N(s). \quad (4.25)$$

This means that the closed-loop poles approach the open-loop zeros (Recall that we saw the same behaviour in Section 4.3). If there are more poles present than zeros ($n - m > 0$), then the excess poles stray infinitely far away from their open-loop position (This is Rule 5 below).

Preliminaries to Rules 4 and 5

By analyzing the magnitude and phase of Equation (4.22), we can derive two more rules that are crucial for plotting the root locus. Firstly, we separate the characteristic equation into s -dependent and s -independent parts as follows:

$$D(s) + kN(s) = 0 \quad \implies \quad \frac{N(s)}{D(s)} = -\frac{1}{k}. \quad (4.26)$$

Applying the magnitude operation on both sides of Equation (4.26) results in

$$\left| \frac{N(s)}{D(s)} \right| = \left| -\frac{1}{k} \right| \quad \implies \quad \left| \frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)} \right| = \left| \frac{1}{k} \right|, \quad (4.27)$$

For scalars, we have that $|xy| = |x||y|$ and $\left| \frac{x}{y} \right| = \frac{|x|}{|y|}$, $x, y \in \mathbb{R}^{1 \times 1}$. Applying this to Equation (4.27) results in

$$\frac{|s - z_1| \cdot |s - z_2| \dots |s - z_m|}{|s - p_1| \cdot |s - p_2| \dots |s - p_n|} = \frac{1}{|k|}. \quad (4.28)$$

Rewriting Equation (4.28) and reintroducing the root locus gain k_{rl} results in the *Magnitude Rule*.

$$k_{rl} \prod_{i=1}^m \prod_{j=1}^n \frac{|s - z_i|}{|s - p_j|} = \frac{1}{|k|}. \quad (4.29)$$

The Magnitude Rule indicates that for any point on the root locus, the product of the magnitudes of all poles and zeros equates to the reciprocal of the absolute value of the proportional gain.

Similarly, we can obtain an *angle rule*. Recall, that when dealing with arguments:

- $\angle(x \cdot y) = \angle(x) + \angle(y)$
- $\angle\left(\frac{x}{y}\right) = \angle(x) - \angle(y)$

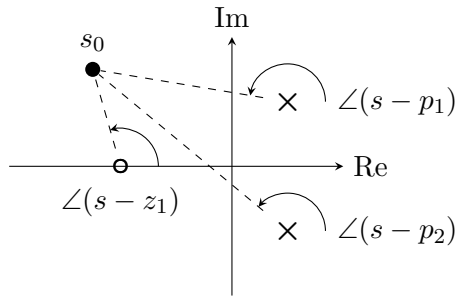
Applying these rules to Equation (4.26) results in

$$\begin{aligned}
 \angle \frac{N(s)}{D(s)} &= \angle \left(-\frac{1}{k} \right), \\
 \Rightarrow \quad & \left[\angle(s - z_1) + \angle(s - z_2) + \cdots + \angle(s - z_m) \right] \\
 & \quad - \left[\angle(s - p_1) + \angle(s - p_2) + \cdots + \angle(s - p_n) \right] \\
 &= -(\angle 1 - \angle k). \tag{4.30}
 \end{aligned}$$

Rewriting Equation (4.30) results in the *Angle Rule*.

$$\sum_{i=1}^m \angle(s - z_i) - \sum_{j=1}^n \angle(s - p_j) = \begin{cases} 180^\circ (\pm q \cdot 360^\circ, & q = 0, \dots, n - m - 1) & \text{if } k > 0, \\ 0^\circ (\pm q \cdot 360^\circ, & q = 0, \dots, n - m - 1) & \text{if } k < 0. \end{cases} \tag{4.31}$$

The Angle Rule indicates that for any point s_0 on the root locus, the sum of the angles from all zeros to s_0 minus the sum of the angles from all poles to s_0 must be equal to 180° if the feedback gain is positive, or 0° if the feedback gain is negative. The angle rule is visualized in Figure 4.15.



In this visualization of a system with two complex open-loop poles and one zero on the real axis, we can check whether any point s_0 on the complex plane is contained on the root locus by ensuring that

$$\angle(s - z_1) - \angle(s - p_1) - \angle(s - p_2) = 180^\circ.$$

Figure 4.15: Graphical interpretation of the angle rule

Rule 4: Root locus on the real axis

From the angle rule we know:

- $\angle(s - z) = 0$ if z is real and $s > z$
- $\angle(s - z) = 180^\circ$ if z is real and $s < z$
- $\angle(s - z) + \angle(s - z^*) = 0$ for complex-conjugate z, z^*
- The same holds for poles p .

Consequently, we can conclude that the root locus is on the real axis when the sum of the angles from all zeros to s_0 minus the sum of the angles from all poles to s_0 is equal to 180° if the feedback gain is positive, or 0° if the feedback gain is negative.

In other words,

- For a *positive* feedback gain k , all points on the real axis to the left of an *odd* number of poles + zeros are on the root locus.
- For a *negative* feedback gain k , all points on the real axis to the left of an *even* number of poles + zeros are on the root locus.

In Figure 4.16 an example on how to find the sections of the real axis containing points of the locus is shown. Note that the determined sections are only part of the total root locus graph. What the segments of the locus with an imaginary component look like cannot be determined using this rule — for this, a combination of the other rules must be applied.

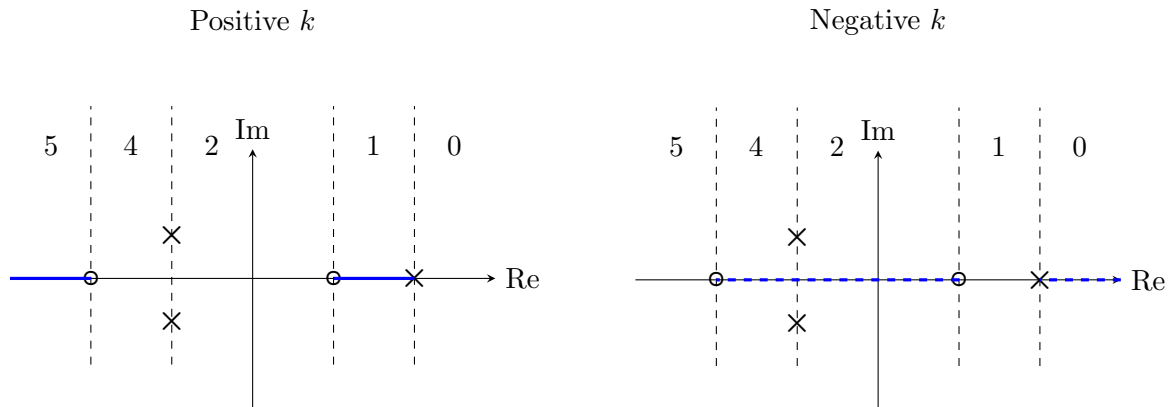


Figure 4.16: Determining sections with locus on the real axis for positive and negative feedback gain. Sections are labelled with the number of poles/zeros to the right. By convention, the negative locus is plotted as a dashed line.

However, as we saw in Section 4.3, it is not necessarily the case that the closed-loop poles always have no imaginary component – i.e., at some point, the closed-loop poles could break away from the real axis. In fact, the root locus always breaks in or out of the real axis with a 90° angle. To determine the exact location of these *break-away* or *break-in* points, we need to solve the following equation:

$$N(s)D'(s) - N'(s)D(s) = 0, \quad (4.32)$$

where the derivatives of the terms introduced in Equation (4.26) are with respect to the complex variable s .

It is important to note that the above plots do not plot the entire root locus. In fact, on the right hand side of the plot in Figure 4.16 we can observe that there is a break-in point between the two zeros. This can be inferred due to **Rule 1** that the root locus must be symmetric with respect to the real axis, and **Rule 3** that the root locus starts and ends at the open-loop poles and zeros.

Rule 5: Asymptote rule

We have previously seen that closed-loop poles approach open-loop zeros. However, as each zero always has only one pole approaching it, we must ask what happens to excess poles. As it turns out, in the case that $\#poles - \#zeros > 0$, the poles that do not approach a zero go to infinity along an asymptote. To compute the location of these asymptotes, we can use the magnitude and angle rules. Firstly, let us look at Equation (4.29) once more under the condition that there are more poles than zeros, i.e., $n > m$. For $k \rightarrow \infty$, all m zeros will be cancelled by m of the total n poles. The remaining $n - m$ poles will therefore approach infinity in some direction on the complex plane. To see that this is the case, we apply the magnitude rule,

$$\frac{|s - z_1| \cdot |s - z_2| \cdots |s - z_m|}{|s - p_1| \cdot |s - p_2| \cdots |s - p_n|} = \frac{1}{|k|} \implies \lim_{k \rightarrow \infty} \frac{1}{|s - p_{n-m}| \cdots |s - p_n|} = 0.$$

Secondly, building on the conclusion that the excess poles follow asymptotes, we must describe their location to fully understand how the closed-loop poles behave for large gains. We can find the angles of these asymptotes in the complex plane by taking another look at Equation (4.31). As k increases in size, the angles converge to one value for all the contributing poles. Since there are a total of $n - m$ poles that do not cancel zeros, we can in the limit approximate Equation (4.31) for positive k as

$$(n - m)\angle(s) = \angle - k \pm 360^\circ q,$$

which can be reformulated to

$$\angle(s) = \frac{(2q + 1)180^\circ}{n - m}, \quad (4.33)$$

where $q = 0, \dots, n - m - 1$ is the iterator for each of the asymptotes. This effect can be intuitively understood by looking at the graphical interpretation. As k increases and the root locus branch approaches the asymptote in infinity, all poles and zeros appear to shift to the origin where they all contribute with the same angle. Figure 4.17 shows this with two plots of the same system with an increasing gain. The further we ‘zoom out’ of the plot, the more the individual angles of the excess poles approach one shared value.

Finding the angles of the asymptote is not enough to fully characterize their location. Additionally, we can calculate the point where all asymptotes meet on the real axis. This point is often referred to as the ‘center of mass’ of the asymptotes and found by evaluating

$$s_{com} = \frac{\sum_{i=1}^n p_i - \sum_{j=1}^m z_j}{n - m}. \quad (4.34)$$

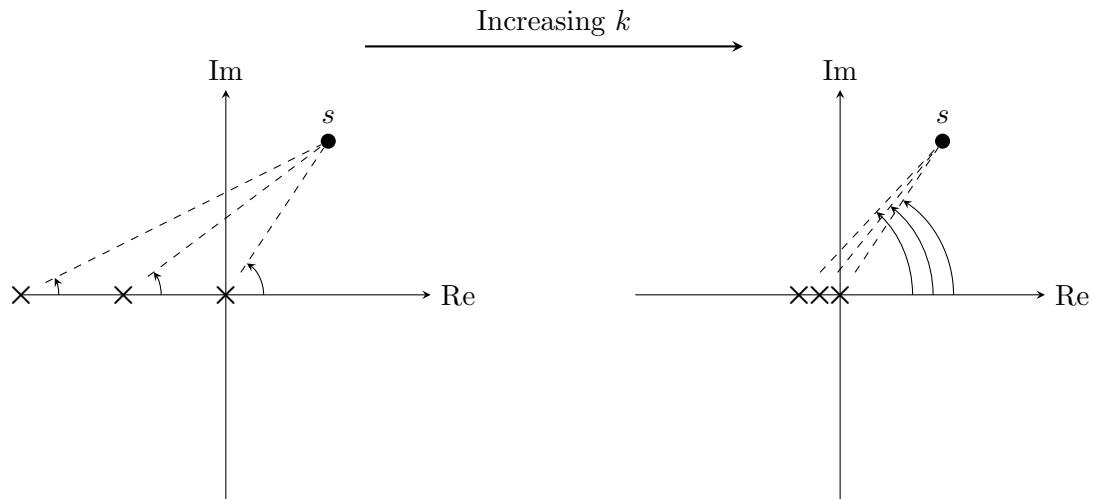


Figure 4.17: Visualization of angles shifting closer to each other when increasing feedback gain k .

To visualize these rules being applied, consider the system shown in Figure 4.18. The system has three poles located at $p \in \{-4, -2, 1\}$. Here we have $n = 3, m = 0$, which means we expect a total of $n - m = 3$ asymptotes. Evaluating Equation (4.33) for $q \in \{0, 1, 2\}$ we get $\theta_1 = 60^\circ, \theta_2 = 180^\circ, \theta_3 = 300^\circ$. With Equation (4.34) we find the center of mass as:

$$s_{com} = \frac{-4 - 3 + 1}{3} = -2,$$

which now lets us fully characterize the root locus plot as k becomes arbitrarily large.

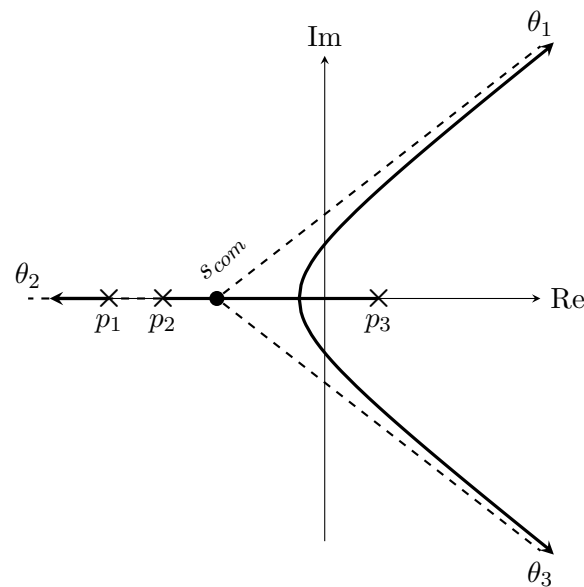


Figure 4.18: Visualization of the root locus approaching the asymptotes which meet at the center of mass.

4.4.3 Reconstructing the Feedback Gain

Once a root locus plot has been constructed it is desirable to determine the feedback gain for which the closed-loop poles are in the location of interest. This way we can, for example, find the exact feedback gain for which a system is stabilized, or find the critical gain when oscillations are introduced, or whether the closed-loop system satisfies the control objectives (introduced in future chapters).

In general, provided the open-loop poles and zeros, the feedback gain can be reconstructed by modifying Equation (4.29) as

$$k(s_0) = \frac{1}{k_{rl}} \prod_{i=1}^m \prod_{j=1}^n \frac{|s_0 - p_i|}{|s_0 - z_j|}, \quad (4.35)$$

where s_0 is the position of the point of interest on the root locus for which the value of k is desired. For a complex conjugate pair of poles, either one can be used.

Before introducing an example on how to use this method, we will first introduce the sketching rules.

4.4.4 Sketching the Root Locus

The root locus is a powerful tool for analyzing the closed-loop behaviour of a system. Although in practice, computational tools will be used to plot the root locus, it is still important to understand the underlying principles. Consequently, this section provides a step-by-step guide to sketching the root locus by hand. We note that some rules have been omitted as they are particularly difficult to apply by hand. For a complete list of rules, we refer the interested reader to [Root Locus Rules by Erik Cheever³](#).

Furthermore, although below we provide a suggested procedure to sketch the root locus, since for distinct transfer functions, the root locus will be different, you may not need to follow all the steps, or even perform them in this order.

Suppose that we have been provided with an open-loop transfer function in standard form,

$$L(s) = \frac{N(s)}{D(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_0}, \quad n \geq m, \quad n, m \in \mathbb{N}^+ \quad (4.36)$$

Steps to sketch the root locus:

1. Transform the transfer function into root locus form.
2. On the complex plane, plot the poles and zeros. By convention, denote the poles as $\times$ and the zeros as $\circ$.

³https://lpsa.swarthmore.edu/Root_Locus/RootLocusReviewRules.html

3. Using **Rule 4 - Locus on the Real Axis**, find the segments on the real axis where the root locus is present and compute any *break-in* and *break-away* points. By convention, we denote the parts of the locus associated with a positive feedback gain as solid lines, and the parts associated with a negative feedback gain as dashed lines. (Typically, in these notes we will only sketch the locus for positive feedback gains.)
4. Using **Rule 3 - Start and End Points** find the end-points of the locus. If $n - m = 0$, then all poles will end at a zero. Otherwise, use **Rule 5 - Asymptote rule** to find the center of mass of the asymptotes and the angles at which the poles stray to infinity.
5. Using a combination of **Rule 1 - Symmetry** and **Rule 2 - Number of Branches**, put the information together to sketch the remaining part of the root locus ensuring none the rules have been violated.

Example - Drawing and interpreting the root locus

Recall the pendulum linearized around the unstable equilibrium point at $(x_e = [\pi, 0]^\top, u_e = 0)$. Let's see whether we can stabilize it through proportional control, Figure 4.19. We assume $g = 10$, $l = 1$, $m = 1$, $c = 1$.

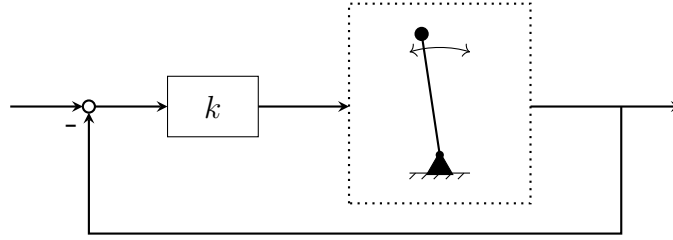


Figure 4.19: Closed feedback loop with a scalar gain k for the inverted pendulum

In order to determine whether the system can be stabilized through proportional control, we plot the root locus, and if there exist some closed-loop poles that are on the left-hand side of the s -plane, then we can conclude that the system can be stabilized.

To plot the root locus, we must first find the transfer function of the system linearized around the unstable equilibrium. From Section 1.6.1 the state-space matrices are given by

$$A = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{c}{ml^2} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ -\frac{1}{ml} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0. \quad (4.37)$$

Now we find the transfer function by plugging the results from Equation (4.37) into the

formula for transfer functions.

$$\begin{aligned}
 P(s) &= C(sI - A)^{-1}B + D, \\
 &= \begin{bmatrix} 1 & 0 \end{bmatrix} \left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{c}{ml^2} \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 \\ -\frac{1}{ml} \end{bmatrix}, \\
 &= -\frac{l}{l^2ms^2 + cs - lmg}.
 \end{aligned} \tag{4.38}$$

Since we now have the system in the desired form, we can follow the presented recipe to sketch the root locus plot.

1. By inspection of Equation (4.38), we find that there are no zeros present. The poles are all values of s , such that

$$l^2ms^2 + cs - lmg = 0.$$

The open-loop poles are given by,

$$p_{1,2} = \frac{-c \pm \sqrt{4gl^3m^2 + c^2}}{2l^2m} \implies p_1 \approx 2.7, \quad p_2 \approx -3.7.$$

$p_1 > 0$ reassures that the pendulum in upright position is unstable. In root locus form the open-loop transfer function is given by,

$$L(s) = l \frac{1}{(s - 2.7)(s + 3.7)}.$$

2. Plot the poles of the system on the s -plane.

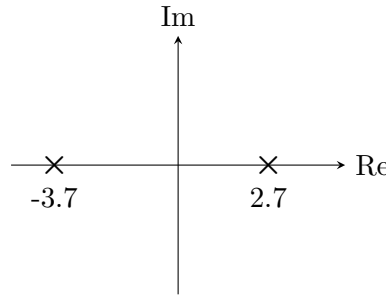


Figure 4.20: Plot of the poles of the system on the s -plane

3. Following the real axis from $+\infty$ to $-\infty$ and counting the amount of poles encountered, we find that the segment $[2.7, +\infty)$ has no poles to the left of it, $[-3.7, 2.7)$ has 1 pole to the left of it and $(-\infty, -3.7)$ has 2 poles to the left of it. We thus conclude, that there must be locus between the two poles.

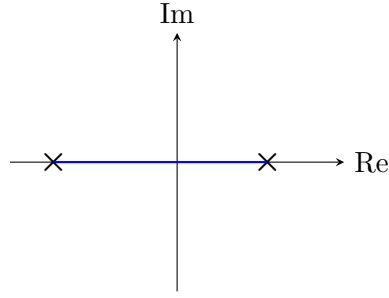


Figure 4.21: Plot of the real axis locus of the system (Rule 4)

4. Since there are no zeros present, we know that there are two asymptotes. Using Rule 5, we find that the asymptotes meet at the center of mass, which is given by

$$s_{com} = \frac{p_1 + p_2}{2 - 0} = \frac{-c + \sqrt{4gl^3m^2 + c^2} - c - \sqrt{4gl^3m^2 + c^2}}{4l^2m} = \frac{-c}{2l^2m} = \frac{-1}{2}.$$

In this case, the break-away point and the centre of mass of the asymptotes is the same. We can verify this by computing the break-away point using Equation (4.32) as

$$N(s)D'(s) - N'(s)D(s) = -l(2l^2ms + c) = 0 \implies s_{ba} = \frac{-c}{2l^2m} = -\frac{1}{2}.$$

The angles of the asymptotes are computed using Equation (4.33):

$$\begin{aligned} q = 0 : \angle(s) &= \frac{180^\circ}{2} = 90^\circ, \\ q = 1 : \angle(s) &= \frac{3 \cdot 180^\circ}{2} = 270^\circ. \end{aligned}$$

Now let's add this information to complete the root locus plot.

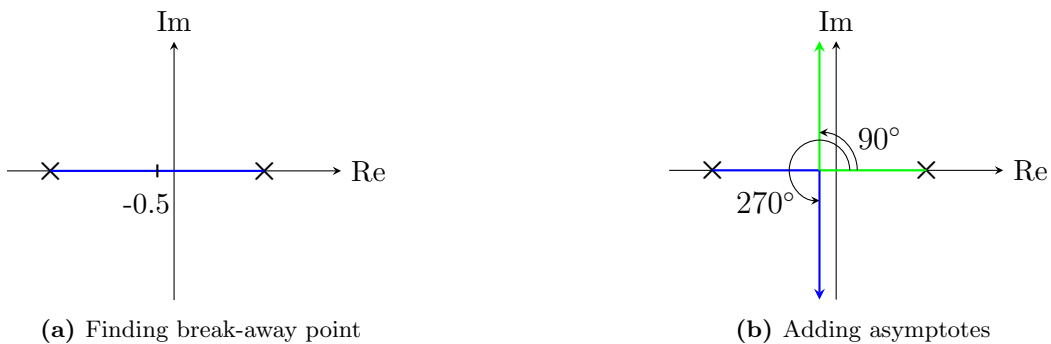


Figure 4.22: Final root locus plot for the inverted pendulum. a) visualizes the break-away points, and b) is the complete root locus plot for the positive feedback gain $k > 0$. Note that tools will often colour each closed-loop pole with a different colour. We have done the same here. Owing to the nature of conjugate poles, it does not matter whether the green line goes up or down.

By inspection of Figure 4.22b, we observe that indeed there does exist a feedback gain $k > 0$ for which the system is stabilized. Furthermore, there exists a range of gains for which the system is stabilized and no oscillations are present (all closed-loop poles have a zero imaginary part). Let's determine this range. We will do so by 2 approaches.

Approach 1: Analytical approach Recall the open-loop transfer function from Equation (4.38),

$$L(s) = \frac{l}{l^2ms^2 + cs - lmg}.$$

To compute the closed-loop poles, we first compute the closed-loop transfer function:

$$T(s) = \frac{kL(s)}{1 + kL(s)} = \frac{kl}{l^2ms^2 + cs - lmg + kl}.$$

The closed-loop poles are then given by:

$$l^2ms^2 + cs - lmg + kl = 0 \implies s = \frac{-c \pm \sqrt{c^2 + 4(l^2mg - kl)}}{2l^2m}.$$

Substituting in numerical values we obtain:

$$s = \frac{-1 \pm \sqrt{1 + 4(10 - k)}}{2}.$$

For closed-loop stability we require $\text{Re}(p) \leq 0$. Hence, we can compute the lower bound on the gain as:

$$\text{Re}(s) \leq 0 \implies \frac{-1 \pm \sqrt{1 + 4(10 - k)}}{2} \leq 0 \implies k \geq 10.$$

For no oscillation, we require that the imaginary part of the poles be zero, i.e.,

$$\text{Im}(s) = 0 \implies \sqrt{1 + 4(10 - k)} = 0 \implies k = 10.25.$$

Thus, the range of gains for which the system is stabilized and no oscillations are present is $k \in [10, 10.25]$.

Approach 2: Graphical approach Recall Equation (4.35).

$$k(s_0) = \frac{1}{k_{rl}} \prod_{i=1}^m \prod_{j=1}^n \frac{|s_0 - p_i|}{|s_0 - z_j|}.$$

For stability, we require that the real part of the poles be negative. Hence, $s_0 = 0$. Thus,

$$k_{\text{stab}} = \frac{\left| \frac{-1 + \sqrt{40+1}}{2} \right| \cdot \left| \frac{-1 - \sqrt{40+1}}{2} \right|}{1} = 10.$$

For no oscillations, we require that the imaginary part of the poles be zero. Hence, $s_0 =$

−0.5. Thus,

$$k_{\text{no-osc}} = \frac{\left| \frac{\sqrt{40+1}}{2} \right| \cdot \left| \frac{\sqrt{40+1}}{2} \right|}{1} = 10.25.$$

Therefore, the range of gains for which the system is stabilized and no oscillations are present is $k \in [10, 10.25]$.

Chapter 5

Time-Domain Specifications

5.1 Introduction

In the previous chapter, we saw that the root locus allows us to leverage information from the open loop transfer function to determine whether a feedback gain k can stabilize the closed loop system. However, stability alone is not sufficient to evaluate the performance of a controller. For example, consider a cruise control system for a car. Beyond ensuring that the system is stable, i.e., that the car does not accelerate indefinitely, we also care about *how* it reaches the target velocity; how fast, how smooth, and with how much overshoot (for instance, whether and by how much the car momentarily exceed the target velocity). Therefore, given a control objective, we need tools to quantify the controller performance, verify compliance with desired objectives, and, if necessary, adjust the controller to achieve the desired performance.

In this chapter, we begin by examining *time-domain specifications*, i.e. control objectives expressed in the time domain, how to specify them and evaluate satisfaction, such as the time required for a system to reach steady state. It should be noted that there also exist frequency-domain specifications, for example those related to noise rejection; these will be the topic of a future chapter.

In Section 5.2, we introduce the main time-domain specifications considered in this course. Since many such measures exist, we focus on a few of the most common ones: *settling time*, *time to peak*, *peak overshoot*, and *rise time*. However, note that these are not exhaustive — different systems and applications may require additional or alternative specifications. For these specifications, analytical relationships with system parameters exist for certain system types. Accordingly, we explore in Sections 5.3 and 5.4 how these relationships apply to first- and second-order systems, respectively.

As discussed in the previous chapter, one of the key advantages of the root locus is that it allows us to quickly assess stability by visualizing the closed-loop poles and their movement as the feedback gain varies. We will therefore show how time-domain specifications can be translated into constraints on the pole locations in the s -domain. This then permits us to use

the root locus to determine whether a suitable feedback gain exists such that the closed-loop poles are located in a desired *region of satisfaction*. Thus, since the analysis in Sections 5.3 and 5.4 is for first- and second-order systems, in Section 5.5, we discuss how higher-order systems can often be approximated by lower-order systems using the *dominant pole approximation*, and how time-domain specifications can be applied to these approximations.

Finally, a fundamental limitation of proportional feedback control is that since the closed-loop poles are fundamentally constrained by the pole-zero structure of the open loop system — for example, closed-loop poles tend to open-loop zeros — a static feedback gain k typically cannot position the closed-loop poles within the desired *region of satisfaction*. To overcome this limitation, in Section 5.6, we introduce the *PID Controller*, which offers greater flexibility to shape the closed-loop dynamics and achieve the desired performance.

5.2 Types of Specifications

Let's begin by introducing the time-domain specifications that we will consider in this course. They include the *settling time*, *time to peak*, *peak overshoot*, and *rise time*. However, keep in mind that many other specifications exist, and depending on the system and control objectives, different measures may be more appropriate.

Control objectives are typically with respect to the transient behavior of the system, e.g., how fast and in what fashion the time response reaches steady state. Since we are interested in the transient behavior until steady state, we need to define what we mean by steady state. By convention, the steady-state value of the step response, i.e., $y_{ss} = y_{\text{step}}(t \rightarrow \infty)$, is also referred to as the steady-state or DC gain of the system. For stable systems, this value can be obtained using the final value theorem (see Theorem 3.1).

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s G(s) U(s).$$

For the step input of amplitude A , with zero initial condition, the step response is given by $u(t) = Ah(t)$, where $h(t)$ is the Heaviside function, and from Table 3.1 its Laplace transform is $U(s) = A/s$. Substituting this into the final value theorem yields

$$y_{ss} = A \lim_{s \rightarrow 0} G(s).$$

With this in mind we now define the time-domain specifications below; see Figure 5.1 for a visual representation:

- **Settling Time**, T_d : The time it takes for the step response of a system to enter and remain within a specified bound within the steady-state gain. Usually we will specify this bound to be $\pm d\%$ of the steady-state gain.
- **Time to peak**, T_p : The time it takes for the step response to reach its highest value.
- **Peak overshoot**, M_p : The ratio between the peak of the step response, i.e. its highest

value, and the steady-state gain.

- **Rise time, $T_{100\%}$:** The time it takes for the step response to first reach its steady-state gain. Note that you might also see $T_{90\%}$ which is the time it takes for the response to first reach 90% of its steady-state gain.

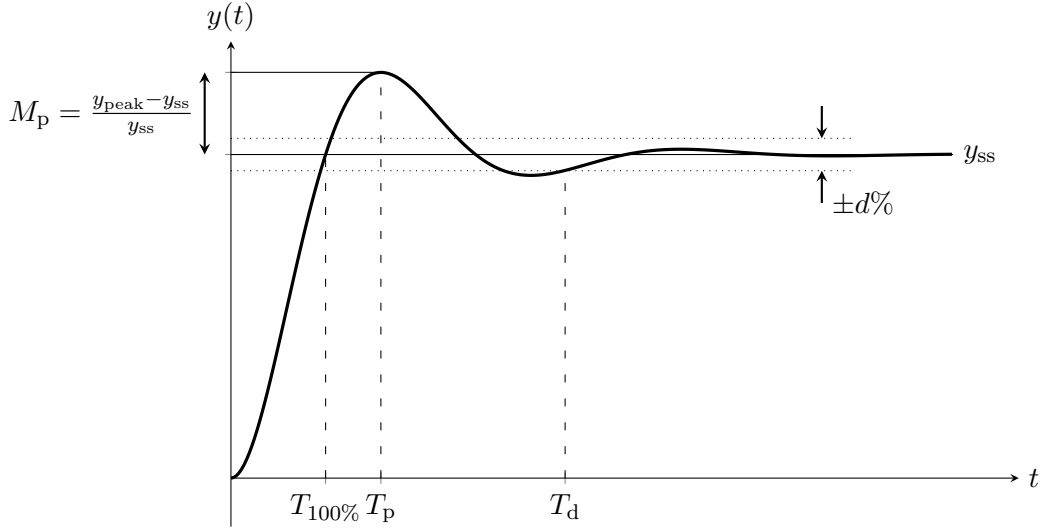


Figure 5.1: Step response of an underdamped second-order system, with the corresponding settling time T_d , time to peak T_p , peak overshoot M_p , and rise time $T_{100\%}$ labeled on the plot.

The introduced time-domain specifications allow us to quantify how a system should behave, transforming qualitative requirements (e.g., the cruise control should not accelerate too quickly) into quantitative ones (e.g., when subjected to a step input, the settling time should be within some desired value). In the next two sections, we will relate these specifications to system properties of first- and second-order systems. This will provide insights into how we should design the controller to meet the specifications.

5.3 First-order Systems

We begin our analysis with a first-order system. Recall from Section 2.2 that such a system has a scalar state vector, and therefore its state-space matrices are also scalars. In this section, consider the general first-order system described by

$$\begin{aligned}\dot{x}(t) &= -\frac{1}{\tau}x(t) + \frac{k_0}{\tau}u(t), \\ y(t) &= x(t),\end{aligned}\tag{5.1}$$

with the state space matrices are given by

$$a = -\frac{1}{\tau}, \quad b = \frac{k_0}{\tau}, \quad c = 1, \quad d = 0,$$

resulting in the transfer function

$$G(s) = \frac{k_0}{\tau s + 1}. \quad (5.2)$$

We adopt this form because, as we will see shortly, it completely characterizes the dynamics of a first-order system. Importantly, recognize that the eigenvalue/pole of the system is at $s = -1/\tau$. The stability of the system therefore depends on the sign of τ ; for asymptotic stability, $\tau > 0$ is required.

We can now investigate how the time-domain specifications, introduced above, apply to this first-order system. Since all specifications were introduced through the step response with zero initial condition, and the amplitude of the step response does not impact the shape of the response, we first derive a general expression for the time response to a unit step input. Starting with the general solution of the LTI state-space form (see Equation (2.5) with the slight modification of ρ being the integration variable)

$$y(t) = Ce^{At}x_0 + C \int_0^t e^{A(t-\rho)} Bu(\rho) d\rho + Du(t),$$

and by plugging in the unit step as input, i.e. $u(t) = h(t)$ and the initial condition $x_0 = 0$,

$$\begin{aligned} y(t)_{\text{step}} &= \int_0^t e^{-\frac{1}{\tau}(t-\rho)} \frac{k_0}{\tau} d\rho, \\ &= k_0 \left(1 - e^{-\frac{t}{\tau}}\right). \end{aligned} \quad (5.3)$$

By analyzing this expression, we observe that for $t \rightarrow \infty$ the steady-state gain y_{ss} approaches k_0 . Furthermore, the rate at which the response approaches the steady-state gain is determined by τ — the *time constant*. Together, k_0 and τ completely characterize the dynamics of a first-order system, since k_0 defines the steady-state value while τ governs the speed of the response. This means that, in this first-order case, the rise time $T_{100\%}$ is not well-defined, since $y(t)_{\text{step}}$ asymptotically approaches the steady-state gain, i.e. $\lim_{t \rightarrow \infty} y(t)_{\text{step}} = k_0$. Furthermore, owing to the lack of sinusoids in Equation (5.3), and assuming asymptotic stability, the step response will monotonically approach the steady-state value without overshooting or oscillations. Consequently, for first-order systems, the time to peak and peak overshoot specifications are also not applicable. We could alternatively deduce this behavior, by observing that the only pole of the system is located at $s = -1/\tau$. Assuming $\tau > 0$ for asymptotic stability, the only pole has a negative real part without any imaginary contribution, and thus has no oscillatory behavior (Figure 5.2). Lastly, the only metric left to consider is the settling time T_d . Recall that the settling time is defined as the time required for the response to enter and remain within $d\%$ of the steady-state value. Mathematically this translates to finding the time T_d where

$$y(T_d)_{\text{step}} = \left(1 - \frac{d}{100}\right) k_0.$$

By plugging this condition into Equation (5.3) we obtain

$$k_0 \left(1 - e^{-\frac{T_d}{\tau}}\right) = \left(1 - \frac{d}{100}\right) k_0,$$

which is rearranged to

$$\begin{aligned}
 e^{-\frac{T_d}{\tau}} &= \frac{d}{100} \\
 -\frac{T_d}{\tau} &= \ln\left(\frac{d}{100}\right) \\
 T_d &= \tau \ln\left(\frac{100}{d}\right). \tag{5.4}
 \end{aligned}$$

By inspecting Equation (5.4) we observe that the settling time is proportional to the time constant τ . If τ is large, T_d will also be large and thus the output will take longer to reach the steady-state. Conversely, if τ is small, the output will approach the steady-state faster. In Figure 5.2 this is graphically shown for a first-order system with some time constant τ and $d = 5$ (Also denoted as $d\% = 0.05$).

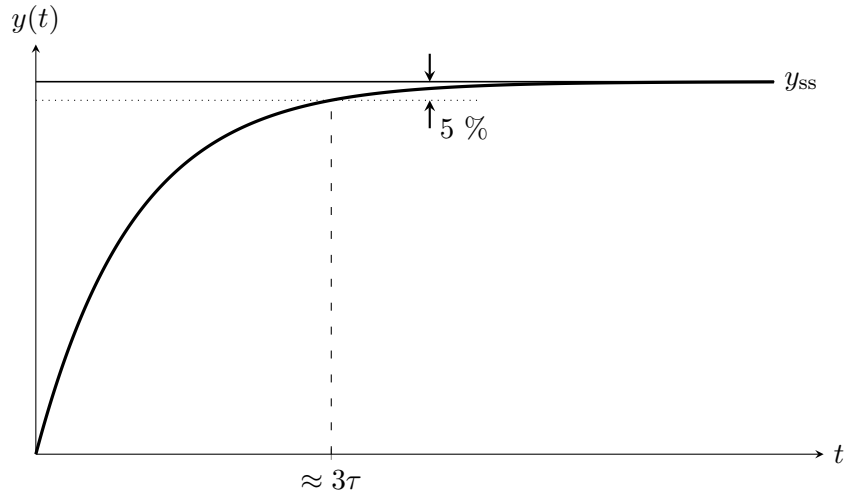


Figure 5.2: Step response of first-order system with time constant τ and the settling time for $d = 5$. The resulting settling time for $d = 5$ is approximately 3τ since $\ln(100/5) \approx 3$.

Besides the direct correlation between τ and T_d the time constant also dictates the location of the pole of the associated transfer function $G(s)$ — from Equation (5.2) we know the pole to be at $s = -1/\tau$. This means that by constraining the settling time we also constrain the location of the pole of $G(s)$ and the other way around. This means that a constraint on the settling time e.g.

$$1 \leq T_d \leq 2$$

can be equivalently rewritten as a function of τ . For example, assuming $T_d = 3\tau$,

$$\begin{aligned}
 1 &\leq 3\tau \leq 2 \\
 -3 &\leq -\frac{1}{\tau} \leq -\frac{3}{2}.
 \end{aligned}$$

This is now a direct constraint on the location of the pole ($p = -1/\tau$) of the transfer function $G(s)$ resulting in a region of satisfaction marking all pole locations that fulfill the settling time requirements. We can also visualize this region of satisfaction in the s -domain as shown

in Figure 5.3. Remember that τ is a model parameter given by the plant and generally cannot

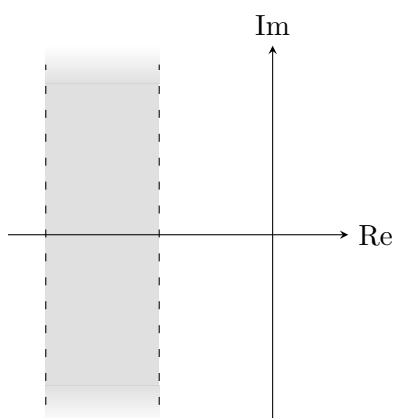


Figure 5.3: Settling time specification expressed as restrictions on the pole location in the s -domain. The gray box indicates the region of satisfaction marking all pole locations that fulfill settling time requirements.

be easily changed. Thus, it is unlikely that an (open-loop) system fulfills all given requirements by itself. Fortunately, by using feedback and a proportional feedback controller, as seen in the previous chapter, we can change the location of the resulting closed loop poles such that specifications are met. Using the root locus we can track how the position of the closed loop poles change as the proportional feedback gain is varied. By overlaying the settling time specification, depicted in Figure 5.3, with the root locus of the first-order system (Equation (5.1)) which has one pole at $s = -1/\tau$, we can now check if there exists some feedback controller gain k for which the specifications are met. This idea is also illustrated in Figure 5.4 where we can see that there exists some feedback gain k that will make the closed loop system fulfill the settling time requirements.

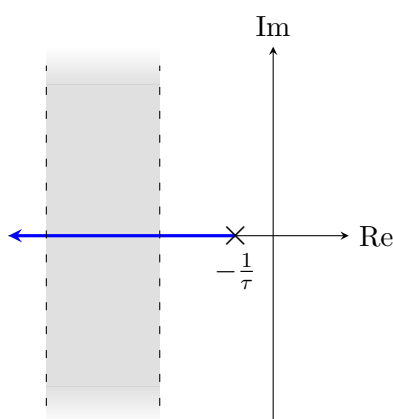


Figure 5.4: Settling time requirements expressed as restrictions on the pole location in the s -domain (shaded region) and the root locus plot of a first-order system with pole at $s = -1/\tau$.

In summary, we conclude that time-domain specifications (in this case the settling time) can directly be translated to restrictions on the pole locations in the s -domain, and with the help of the root locus we can quickly assess whether a proportional feedback controller is able to make the system fulfill the requirements.

5.4 Second-Order Systems

Let us now consider a second-order system, derive analytical expressions for the time-domain specifications, and extend the idea of transforming time-domain specifications to restrictions on the locations of the pole in the s -domain. This will enable us to consider a wider range of systems and also allow us to investigate the other specifications introduced in Section 5.2. To do so we will consider second order systems expressed in the general form

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} u(t) \\ y(t) &= x(t),\end{aligned}\tag{5.5}$$

where ζ is referred to as the damping ratio and ω_n the natural frequency. The damping ratio represents the degree of energy dissipation, e.g. friction in a system, whereas the natural frequency tells us at what frequency the system would oscillate in the absence of any damping and external excitation. From this general form we determine the transfer function for a second-order system to be

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.\tag{5.6}$$

Notably, the poles of this transfer function are given by

$$p_{1,2} = \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}.\tag{5.7}$$

By inspecting the term in the square root of Equation (5.7), we learn that for $|\zeta| < 1$ this system will have complex conjugate poles. As we saw in Section 2.5.1, complex eigenvalues (poles) results in oscillatory behavior in the time response. Conversely, if $|\zeta| \geq 1$, the poles of the system will be real, thus the time-domain behavior is solely characterized by exponential functions. Based on the value of ζ we will thus distinguish between 4 different types of systems:

- *Undamped*, $\zeta = 0$: The poles are located at $p_{1,2} = \pm j\omega_n$. Time response is purely oscillatory with frequency ω_n . This corresponds to a system with no energy dissipation, i.e. no friction or damping.
- *Underdamped*, $0 < \zeta < 1$: The poles have both real and imaginary components. The time response is a mixture of oscillatory and exponential behavior in the time-domain.
- *Critically damped*, $\zeta = 1$: The poles are located at $p_{1,2} = -\zeta\omega_n$. The time response is decomposed into a single exponential. Notably this is the lowest value for ζ for which the system does not oscillate and typically results in the fastest possible convergence to the steady state value y_{ss} .
- *Overdamped*, $\zeta > 1$: The poles are, as in the critically damped case, real valued. The time response is now decomposed into two exponentials with differing exponents according to the location of the pole.

Graphically we can analyze this behavior by considering the step response. Figure 5.5 shows various step responses of stable second order systems for varying values of ζ . For small values

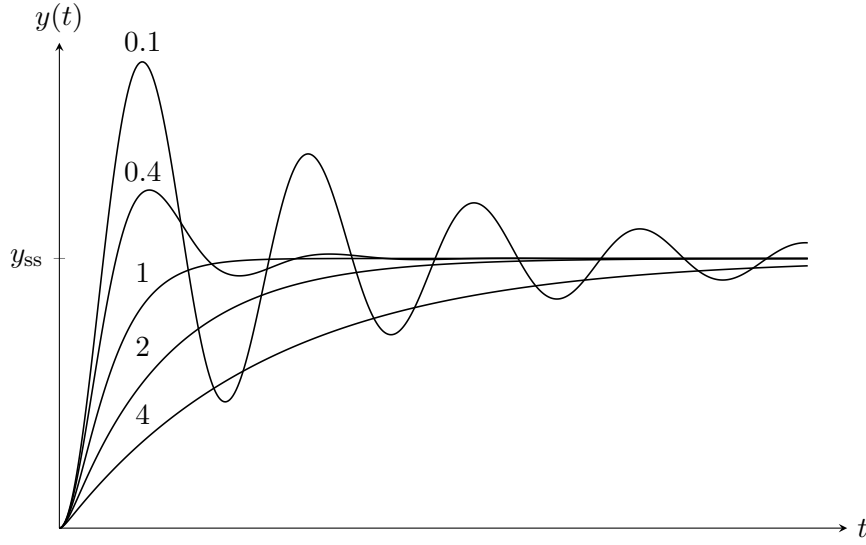


Figure 5.5: Step responses of second order systems with varying damping ratios. The damping ratios ζ are indicated on each curve.

of ζ , e.g. 0.1 and 0.4, we observe oscillations remain within an exponential envelope that decays over time. For $\zeta \geq 1$ the oscillations disappear and the response shows purely exponential behavior. Note that the undamped case $\zeta = 0$ is not depicted since the resulting response is a pure sine wave with no exponential decay.

With this consideration in mind, we can narrow down the types of systems that we want to consider for the time-domain specifications introduced above. Firstly, the undamped case does not converge to a steady state value, as it is not asymptotically stable, and it continually oscillates. Consequently, none of the introduced specifications apply to such a system, as they are all based on a steady-state value. Secondly, similar to the first order case, the critically damped and overdamped systems do not exhibit any overshoot and thus the related time to peak T_p , peak overshoot M_p , and rise time $T_{100\%}$ specifications don't apply. The settling time can be used analogously to the first-order case. Thus, the only system type for us to analyze is when the system is underdamped, i.e. it shows a mix of oscillatory and exponential behavior. In that case, and assuming that the system is stable, we can map out the location of the poles in the complex plane given by its real and imaginary components σ , ω respectively. Figure 5.6 shows the relationship between the model parameters ω_n and ζ to the pole locations $p = \sigma + j\omega$ in the complex plane.

As with the first-order system the next step is to take a closer look at the time-domain solution of the step response. For the underdamped second-order system the unit step response is given by

$$y(t) = 1 - \frac{1}{\cos(\varphi)} e^{\sigma t} \cos(\omega t + \varphi). \quad (5.9)$$

One can verify that this represents the solution to Equation (5.5), with zero initial conditions, i.e. $y(0) = 0$ and $\dot{y}(0) = 0$. From this we now derive relations to find the expressions for the

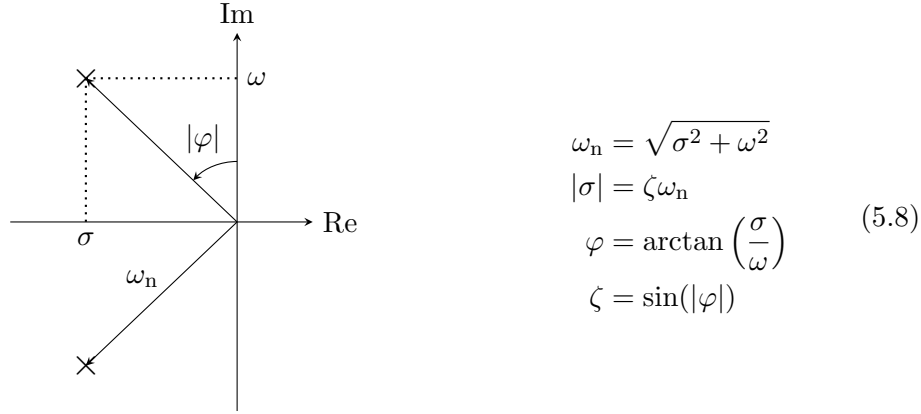


Figure 5.6: Pole location of stable second order system with complex conjugate poles and their relationships to ζ and ω_n .

settling time T_d , time to peak T_p , peak overshoot M_p , and rise time $T_{100\%}$. To achieve this we first need to find the steady-state gain. It can be found by evaluating either $y(t \rightarrow \infty)$ or $G(0)$. In any case, to a unit step input, we get that $y_{ss} = 1$.

First, let's consider the settling time T_d which can be approached similarly to the first-order case. The only difference is that due to the oscillations there might be many points where the system gets within $d\%$ of the steady-state value. Crucially we need to find the last point in time when the system enters this bound and stays within it for all later times. This can be done by considering the exponential envelope $1 - e^{\sigma t}$ that bounds the step response. That way we make sure that even though the system might oscillate, it will remain within its exponential envelope. Thus, we need to check when $1 - e^{\sigma t}$ is within $d\%$ of y_{ss} . This results in the condition

$$1 - e^{\sigma T_d} = \left(1 - \frac{d}{100}\right) y_{ss} ,$$

which can be rearranged to

$$\begin{aligned}
 e^{\sigma T_d} &= \frac{d}{100} \\
 \sigma T_d &= \ln\left(\frac{d}{100}\right) \\
 T_d &= \frac{1}{\sigma} \ln\left(\frac{d}{100}\right) .
 \end{aligned}$$

Since we assume the system to be stable, i.e. $\sigma < 0$, we can further rewrite this as

$$T_d = \frac{1}{|\sigma|} \ln\left(\frac{100}{d}\right) . \tag{5.10}$$

Note the similarity of this expression to that obtained for the first-order system. Similar approaches can be taken to find expressions for the other specifications. In summary, they are

given by

$$T_p = \frac{\pi}{\omega} \quad (5.11)$$

$$M_p = e^{\frac{\sigma\pi}{\omega}} \quad (5.12)$$

$$T_{100\%} = \frac{\frac{\pi}{2} - \varphi}{\omega} \approx \frac{\pi}{2\omega_n}. \quad (5.13)$$

We can use the step response once more to visualize the specifications in the time-domain. Figure 5.7 illustrates this for a stable second order system with some $0 < \zeta < 1$.

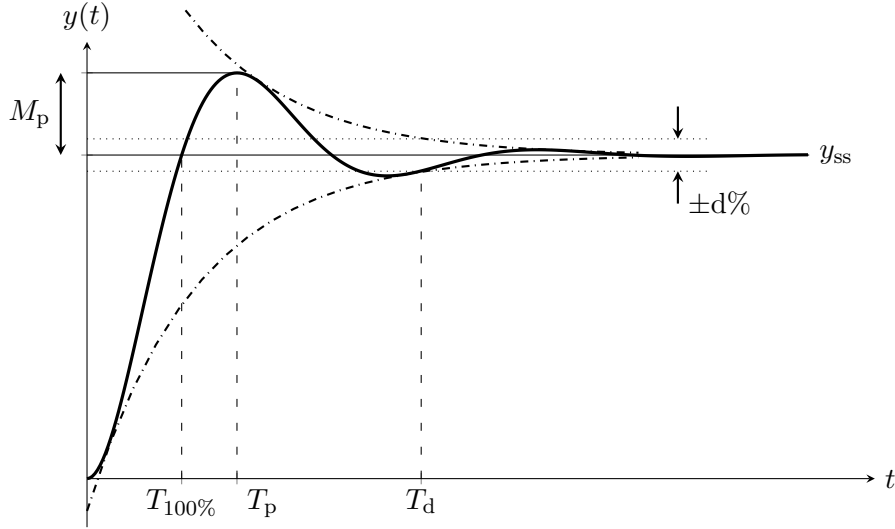


Figure 5.7: Step response of an underdamped second-order system and the corresponding time to peak T_p , peak overshoot M_p , settling time T_d , and rise time $T_{100\%}$.

By inspecting the expressions for the time-domain specifications, given by Equations (5.10) to (5.13) we can recognize that they are all a function of the location of the poles, i.e. σ and ω . Consequently, we can once more translate the time-domain specifications into restrictions on the pole locations in the s -domain, i.e. some region of satisfaction. Let's go through them step by step and see what restrictions they impose.

Firstly, the settling time T_d is inversely proportional to the real part of the poles. So a specification such as

$$T_d < 2,$$

assuming $d = 5$, now translates to

$$3 \frac{1}{|\sigma|} < 2 \implies |\sigma| > \frac{3}{2}.$$

Next, we can analyze the time to peak T_p as described by Equation (5.11). It is inversely proportional to the imaginary part of the pole ω . Thus, the specification

$$T_p > 2,$$

translates to

$$\frac{\pi}{\omega} > 2 \implies \omega < \frac{\pi}{2}.$$

Figure 5.8 visualizes the region of satisfaction corresponding to these two specifications in the s -domain. Note that the restriction on T_p is also plotted for negative ω . This is due to the fact that complex poles always appear in conjugate pairs and thus imposing an upper limit on $+\omega$ automatically also imposes a lower limit $-\omega$.

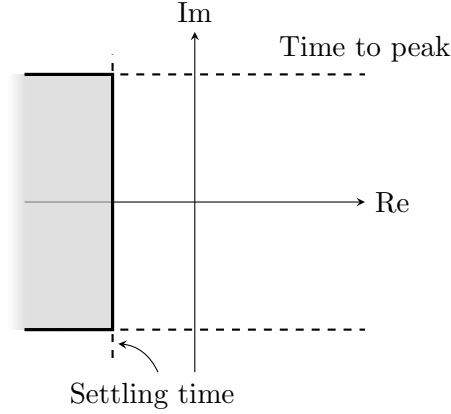


Figure 5.8: Representation of settling time T_d and time to peak T_p in the s -domain. The shaded area represents the region of satisfaction.

Next let us consider the peak overshoot M_p . As we can see in Equation (5.12), this sets a limit on the ratio between the real and imaginary part. By imposing that

$$M_p < 2,$$

the ratio of σ/ω is limited by

$$e^{\frac{\sigma\pi}{\omega}} < 2 \implies \frac{\sigma}{\omega} < \frac{\ln(2)}{\pi}.$$

Graphically this will result in a diagonal constraint in the s -domain as shown in Figure 5.9. Lastly the rise time $T_{100\%}$ is a function of $\omega_n = \sqrt{\sigma^2 + \omega^2}$, the magnitude of the poles. This means that a restriction such as

$$T_{100\%} < 4$$

results in a limitation on the magnitude of the poles, leading to

$$\frac{\pi}{2\omega_n} < 4 \implies \omega_n > \frac{\pi}{8}.$$

In the s -domain this implies a circular constraint as indicated in Figure 5.9. In summary, Figure 5.9 shows the combined region of satisfaction in the s -domain resulting from the time-domain specifications introduced in this section.

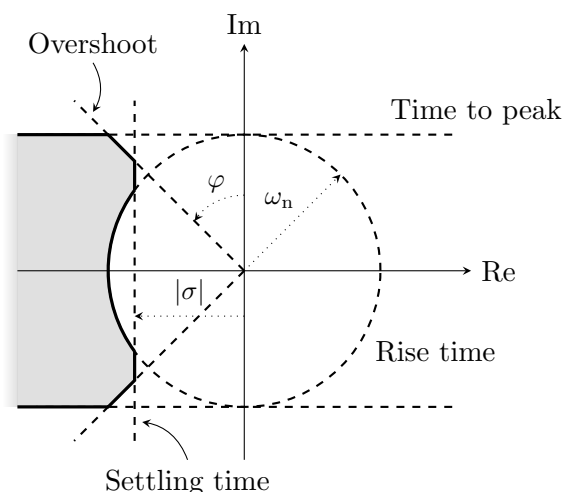


Figure 5.9: Representation of time-domain specifications in the s -domain. The gray area represents the region of satisfaction.

Lastly we can use the root locus to evaluate if a set of time-domain restrictions can be satisfied using some proportional feedback controller k . As before we can overlay the restrictions on the pole locations from Figure 5.9 with the root locus of some system. In Figure 5.10 this is depicted for some second order system with complex conjugate poles at $s = \sigma \pm j\omega$. Here we

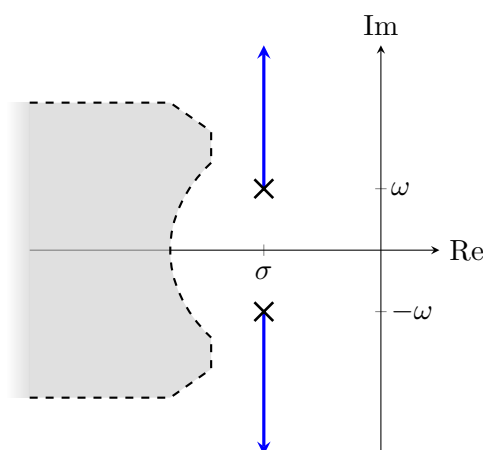


Figure 5.10: Overlay of time-domain specifications (gray area) in the s -domain and a root locus (continuous blue line) with complex conjugate poles at $s = \sigma \pm j\omega$.

can identify that regardless of how big the feedback gain k is chosen, the closed loop system will never be able to satisfy all the time-domain specifications. This is due to the fact that with this control technique we are intrinsically bound by the pole-zero locations of the open-loop. Thus, we conclude that in this case a proportional feedback controller cannot satisfy our time-domain specifications. Note however, that this does not mean that this system will never be able to achieve these specifications. As we will see in Section 5.6, we can employ some more sophisticated controllers to achieve these control objectives.

5.5 Dominant Pole Approximation

In the context of first- and second-order systems, and their respective step responses, we have looked at how time-domain specifications enforce restrictions on the locations of the closed loop poles. Now we turn our attention to higher order systems. In order to apply the same concepts as before, we approximate higher order systems by just looking at their *dominant poles*. To understand what the dominant poles are, we will use the impulse response introduced in Section 2.4.2. Recall that an input described by a unit impulse, i.e. $u(t) = \delta(t)$, corresponds to $U_{\text{imp}}(s) = 1$ in the frequency domain. Hence, the output in the s -domain becomes $Y_{\text{imp}}(s) = G(s) \cdot U_{\text{imp}}(s) = G(s)$. Specifically, we can conclude from this that every pole in a transfer function has a corresponding exponential in the time-domain output, which can be seen by writing the transfer function $G(s)$ in a partial fraction expansion (we will assume no direct feed through with $d = 0$).

$$\begin{aligned} Y_{\text{imp}}(s) = G(s) &= \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_0} \\ &= \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \cdots + \frac{r_n}{s - p_n}. \end{aligned}$$

If we then apply the inverse Laplace transform, we can exactly see the exponential behavior in the time-domain:

$$y_{\text{imp}}(t) = \mathcal{L}^{-1}\{Y_{\text{imp}}(s)\} = r_1 e^{p_1 t} + r_2 e^{p_2 t} + \cdots + r_n e^{p_n t}.$$

The real part of the pole indicates how fast the corresponding exponential grows or decays. We can use this fact to conclude that stable poles — poles in the left half plane — decay faster if they are further away from the imaginary axis. Concretely these poles are going faster to zero since they decay at a higher rate, i.e. they have a “more negative”/smaller exponent. For example e^{-10t} tends to zero faster than e^{-t} . This idea is visualized in Figure 5.11, where a system with a pole at $p = -10$ (dotted line) decays at a much faster rate than the system with the pole at $p = -1$ (dashed line). If we now have a system with two poles, one at $p_1 = -1$ and the other at $p_2 = -10$ (solid line), we can see that the overall behavior is mostly dictated by the “slower” pole at -1 . We can thus conclude that a higher-order system can often be reasonably approximated by considering only the poles that decay the slowest — also referred to as dominant poles. A common design approach is therefore to simplify the system to a first- or second-order model using the dominant pole approximation, which captures the essential dynamics without significant loss of accuracy. The detailed procedure will be illustrated in the following two examples.

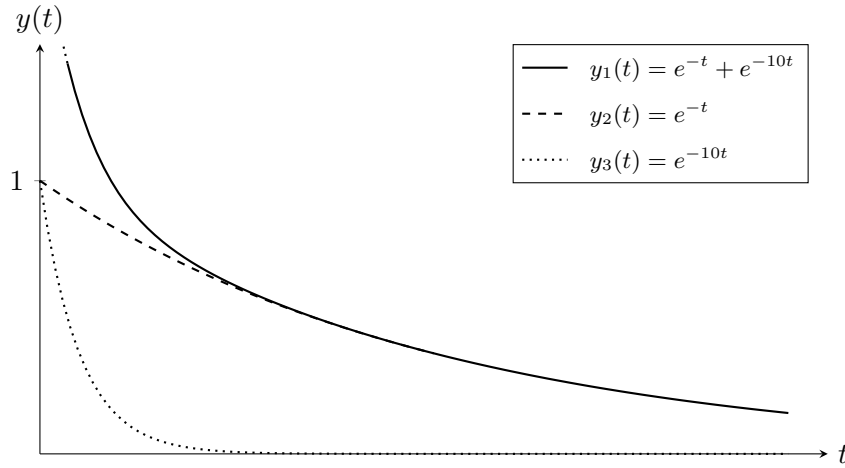
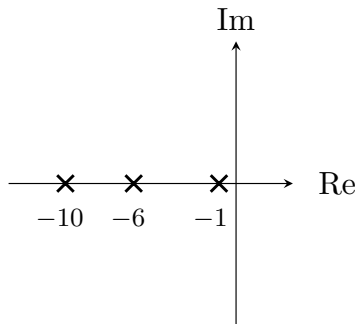


Figure 5.11: Impulse response of three different systems: System 1 has poles $p_1 = -1$, $p_2 = -10$ with $y_1(t) = e^{-t} + e^{-10t}$, System 2 has pole $p = -1$ with $y_2(t) = e^{-t}$, and System 3 has pole $p = -10$ with $y_3(t) = e^{-10t}$. System 1 can be written as a superposition of systems 2 and 3.

Example - Reducing to a first-order system

We will start with the following system with three poles:



$$G(s) = \frac{5}{(s+10)(s+6)(s+1)}$$

Figure 5.12: Poles plotted in the s -domain.

In this example, the pole $p = -1$ has the lowest decay, and consequently, is the dominant pole. Therefore, we neglect the terms with $p = -6$ and $p = -10$, and the approximated transfer function then becomes:

$$G_{\text{dom}}(s) = \frac{\alpha}{s+1}.$$

To determine the coefficient α , we need to ensure that the steady-state response to a unit step input remains the same for the approximated transfer function. We therefore need the steady-state / DC gain of the approximated transfer function to be equal to the steady-state gain of the original transfer function, i.e. $y_{\text{ss}} = y_{\text{ss,dom}}$. Using the final value theorem from Section 3.7 we ensure that

$$G_{\text{dom}}(0) \stackrel{!}{=} G(0).$$

For our example we obtain that

$$\alpha = \frac{1}{12}.$$

So overall the approximation becomes:

$$G_{\text{dom}}(s) = \frac{1/12}{s + 1}.$$

The step response for the full and approximated system are shown in Figure 5.13. We can confirm that the dominant pole approximation is very close to the actual solution, and they tend to converge as $t \rightarrow \infty$.

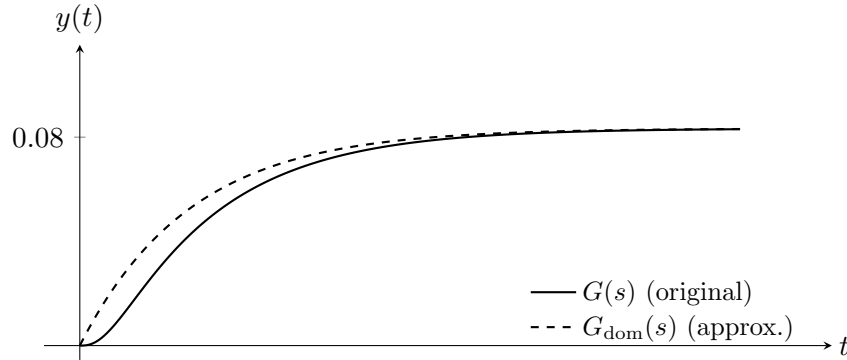
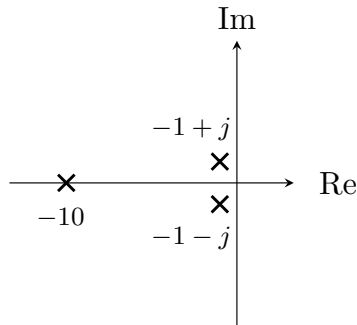


Figure 5.13: Step response of the systems $G(s)$ (solid line) and $G_{\text{dom}}(s)$ (dashed line).

Example - Reducing to a second order system with a complex conjugate pair

For our second example we have the transfer function:



$$G(s) = \frac{1}{(s + 10)(s + 1 + j)(s + 1 - j)}$$

Figure 5.14: Poles plotted in the s -domain.

We can again neglect the pole at -10 since it decays faster than the complex conjugate pair with real part -1 . We can thus approximate the system as the following transfer function

$$G_{\text{dom}}(s) = \frac{1}{10} \frac{1}{(s + 1 + j)(s + 1 - j)}.$$

Note that the factor of $1/10$ ensures that the steady-state gain remains the same for the approximated transfer function, i.e. $G(0) = G_{\text{dom}}(0)$. In Figure 5.15 we can observe the step response of both systems and confer that the dominant pole approximation is close to the real system.

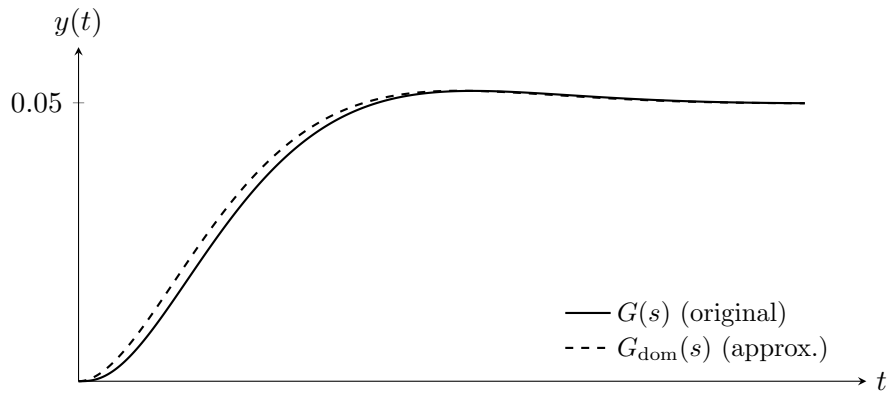


Figure 5.15: Step response of the systems $G(s)$ (solid line) and $G_{\text{dom}}(s)$ (dashed line).

Time-Domain Specifications for Dominant Pole Approximation

To determine whether a higher-order system satisfies the time-domain specifications, we check whether the approximated system meets the specifications. As before, to ease analysis, we can transform the specifications to the s -domain to determine the region of satisfaction. Once we have the region, we can then plot the root locus of the approximated open-loop transfer function to determine whether it is possible to position the closed-loop poles within the region of satisfaction.

This approach is justified for two main reasons. Firstly, as seen in Figures 5.13 and 5.15, the time response of the approximated transfer function closely resembles that of the original higher-order system. Secondly, as illustrated in Figure 5.9, the neglected poles are located further into the region of the s -plane associated with greater satisfaction of the time-domain specifications. Hence, by applying the dominant-pole approximation, we concentrate on the most critical poles, those most likely to violate the specified performance criteria.

However, it is important to keep in mind that solely relying on the approximated system might lead to incorrect conclusions regarding stability of the closed-loop system. This is illustrated in Figure 5.16, which compares the root locus of $G(s)$ with its dominant-pole approximation $G_{\text{dom}}(s)$.

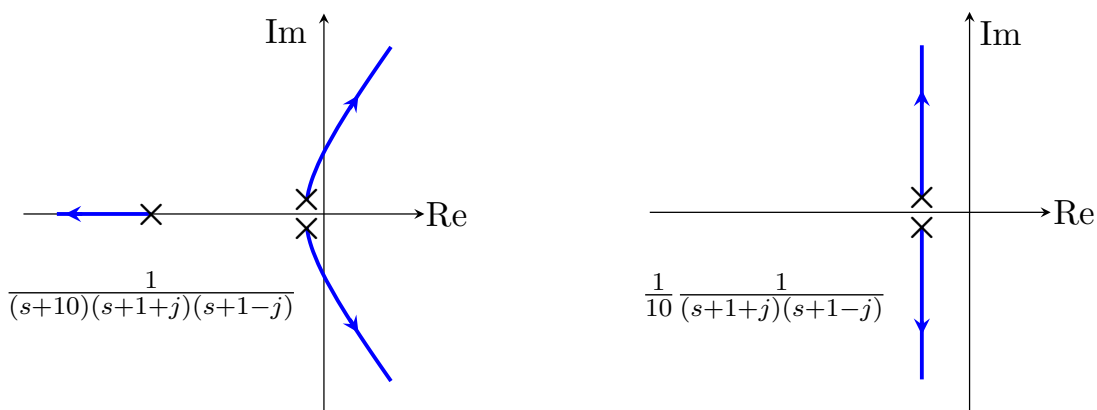


Figure 5.16: Root locus of the full system $G(s)$ on the left and approximated system $G_{\text{dom}}(s)$ on the right.

By neglecting the pole at $s = -10$ the asymptotic behavior of the dominant pole approximation is different from the original system. Specifically, in the original system, there exists a feedback gain that results in an unstable closed-loop system. Thus, relying solely on the dominant pole approximation could lead to an incorrect assessment of stability. Therefore, when using the approximated transfer function $G_{\text{dom}}(s)$ we should keep in mind that the root locus should generally only be used for a back of the envelope calculation and not be applied for a rigorous detailed design.

Finally, one should also keep in mind that the expressions derived for the time-domain specifications are only valid for systems of the respective form. Thus, when using the time-domain specifications for a second-order system on an approximated system, keep in mind that deviations to the exact expressions are possible.

In summary, the dominant pole approximation provides a general overview of the system dynamics, without the need to perform a complete analysis with all the poles. It also lets us readily check whether the system could satisfy the time-domain specifications by analyzing an approximated first- or second-order system.

5.6 PID Control

As we saw in Section 5.4, using solely proportional feedback control is not always sufficient to satisfy the time-domain specifications. Consequently, in this section we introduce the *PID controller* — a controller that in practice is suitable for a wide range of systems and tasks.

To begin with, recall the block diagram of a standard feedback system (Figure 5.17). Since one of the implicit aims of control is to reduce the error $e(t)$, before introducing the PID controller, let us first examine how the error signal evolves according to the open-loop transfer function $L(s) = C(s)P(s)$. This will help us understand the various properties of the PID controller and how they work together.

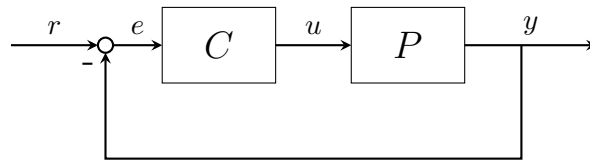


Figure 5.17: Standard feedback architecture with controller C and plant P .

In the standard feedback architecture, the controller C receives the error signal $e(t) = r(t) - y(t)$, representing the difference between the desired reference $r(t)$ and the actual system output $y(t)$. The controller then generates the control input $u(t)$, which acts on the plant with the aim to drive this error towards zero. Intuitively, we would like $e(t) = 0$ for all $t \geq 0$, meaning that the system output perfectly follows the reference signal. Since the time response is composed of a transient and steady-state component, we already know that the error signal will not always be zero. However, understanding the conditions under which the error signal ends up at zero provides valuable insight into what is required from a system (and controller) to achieve this. In

what follows, we analyze the asymptotic behavior of the error signal, referred to as *steady-state error* (SSE), for various types of reference signals. This will allow us to understand what is required from the system as a whole to achieve zero SSE, and consequently, what is required from the controller. Throughout this discussion, we assume that the closed-loop system is asymptotically stable, otherwise no steady-state behavior is present.

To begin with, let the SSE be defined as

$$e_{ss} = \lim_{t \rightarrow \infty} e(t),$$

and consider the canonical case of a unit step reference

$$r(t) = h(t) \xrightarrow{\mathcal{L}} R(s) = \frac{1}{s}.$$

Further, recall that the sensitivity transfer function, $S(s) = \frac{1}{1+L(s)}$ (Equation (4.8)), maps the reference signal to the error signal, i.e. $E(s) = S(s) \cdot R(s)$. With this in mind, and using the final value theorem we can see that the SSE to a unit step input is given by

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s S(s) R(s) = \lim_{s \rightarrow 0} \frac{1}{1 + L(s)}. \quad (5.14)$$

Importantly we recognize that if $L(0) \rightarrow \infty$ then $e_{ss} \rightarrow 0$. Consequently, we want to know under what conditions $L(0) \rightarrow \infty$. To do so, let's look at the Bode form of $L(s)$ and oversee its behavior when $s \rightarrow 0$:

$$L(s) = \frac{k_{\text{Bode}}}{s^q} \frac{\left(\frac{s}{-z_1} + 1\right) \cdots \left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right) \cdots \left(\frac{s}{-p_{n-q}} + 1\right)} \xrightarrow{s \rightarrow 0} L(0) = \frac{k_{\text{Bode}}}{s^q}. \quad (5.15)$$

Importantly we see that if $q \geq 1$ then $L(0) \rightarrow \infty$ and consequently $e_{ss} \rightarrow 0$. So in order to have zero steady-state error to a step, the system needs to have at least one pole at the origin. The poles at the origin are also called *integrators* (The inverse Laplace transform of $1/s$ is an integral).

However, limiting analysis to a step reference is not very practical. To investigate the influence of more complex reference signals, let's consider the case where the reference signal is a ramp signal of order δ , where ramps are defined as

$$r(t) = \frac{1}{\delta!} t^\delta \xrightarrow{\mathcal{L}} R(s) = \frac{1}{s^{\delta+1}}.$$

In Figure 5.18 the first 3 ramps are plotted as a function of time. Note that a first order ramp is equal to a unit step.

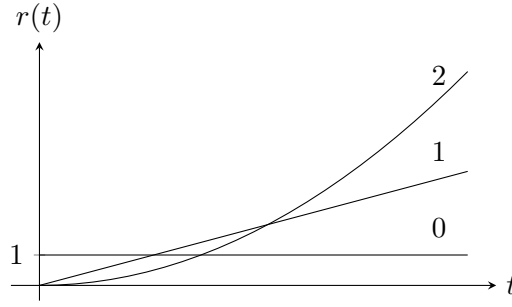


Figure 5.18: Different ramp reference signals and the corresponding order δ as indicated on the curves.

Using this notation we can now extend the expression for the SSE from Equation (5.14) to the general case of ramp references resulting in

$$e_{ss} = \lim_{s \rightarrow 0} s \frac{1}{1 + L(s)} \cdot \frac{1}{s^{\delta+1}}.$$

Then by reusing the expression for the Bode form of $L(s)$ in Equation (5.15) we see that:

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{1 + \frac{k_{\text{Bode}}}{s^q}} \cdot \frac{1}{s^{\delta}}.$$

Thus, if the reference signal is a ramp of order δ , and $q \geq \delta + 1$ then $e_{ss} \rightarrow 0$. So in order to have zero steady-state error to a ramp reference of order δ , the system needs to have at least $\delta + 1$ integrators.

We conclude that the SSE is both a function of the system order q and the order of the reference ramp δ . Their relationship is summarized in Table 5.1.

e_{ss}	$\delta = 0$	$\delta = 1$	$\delta = 2$
$q = 0$	$\frac{1}{1+k_{\text{Bode}}}$	∞	∞
$q = 1$	0	$\frac{1}{k_{\text{Bode}}}$	∞
$q = 2$	0	0	$\frac{1}{k_{\text{Bode}}}$

Table 5.1: Steady-state error depending on the number of integrators q and the order of the ramp δ .

Note that to keep consistency with the previous chapters, we have used the variable q to denote the number of integrators in the open-loop transfer function, and δ to denote the order of the ramp reference signal. In the lecture, q was used to denote the order of the ramp reference signal.

At this point recall that ideally we want the controller to bring the SSE of the closed loop system to zero. Now using Table 5.1 we can already deduce how many integrators are needed to

achieve this goal. Next, we also need to consider the transient behavior of the system, i.e. what happens before we reach the steady-state. To address this, the following sections go through each component of the PID controller and its influence on the closed loop system.

5.6.1 Proportional Controller

The first component of the PID controller is the proportional gain.

$$C_P(s) = k_P.$$

Recall that in proportional control, the closed-loop transfer function and sensitivity transfer function are given by:

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{k_P P(s)}{1 + k_P P(s)}, \quad S(s) = \frac{1}{1 + L(s)} = \frac{1}{1 + k_P P(s)},$$

where k_P is the proportional gain and $P(s)$ the plant transfer function. From this we can see that:

- Since $e_{ss} = \frac{1}{1+k_P P(0)}$, increasing k_P decreases the SSE. However, for any finite k_P , the SSE will never be zero.
- Increasing k_P makes the system approach its steady-state value faster. To build intuition, Let $P(s)$ be a first-order system. Then the pole is given by $-\frac{\tau}{1+k_P}$. Hence, as k_P increases, the pole moves further to the left in the complex plane. Similar deductions can be made for higher-order systems.
- Increasing k_P also increases the sensitivity to noise. (We will derive this in a future chapter when we start to think about the ability of the controller to reject disturbances.)

5.6.2 Integral Controller

In the previous section, we established that the number of integrators present in the open-loop transfer function $L(s)$ determines whether the SSE converges to zero for a given class of reference inputs. More concretely, we saw that for a step reference, a system with one integrator achieves zero SSE. This brings us to the second component of the PID controller, the integrator given by

$$C_I(s) = \frac{k_I}{s}.$$

In practice, the integrator represents the integral of the error signal, i.e., the input to the plant is given by

$$u(t) = k_I \int_0^t e(\tau) d\tau,$$

Intuitively, the integrator accumulates past error over time. As long as a nonzero error persists, the integral term continues to grow, increasing the control input $u(t)$. This accumulating action compels the system output to adjust until the error is eliminated.

Assuming closed-loop stability, an integrator:

- Removes SSE to a step reference.
- Makes the time response more oscillatory.
- Does not influence the system's sensitivity to noise.

5.6.3 Derivative Controller

So far, we have seen that the proportional term reacts instantly to the present error and the integral term eliminates SSE by accumulating past error. However, neither of these terms provides any anticipation of how the error is *changing*. For example, we have no knowledge whether the system is going to overshoot its reference or not.

To counter this effect, recall that in Section 3.6 we saw that zeros can be interpreted as providing “anticipatory behavior” i.e., they react to the *rate of change* of the signal. Thus, we introduce the derivative controller given by

$$C_D(s) = k_D s.$$

In the time domain, this corresponds to

$$u(t) = k_D \frac{de(t)}{dt}.$$

The derivative term produces a control effort that increases when the error changes rapidly, effectively opposing sudden variations in the error. Used with proportional action (PD), it may be possible to reduce overshoot and oscillation without sacrificing speed, and sometimes even with a faster rise. One practical caution is that an ideal differentiator $k_D s$ is not implementable and amplifies high-frequency noise, so a filtered derivative $k_D s / (1/\omega_f s + 1)$ is often used. Nevertheless, in this course we will only consider the ideal differentiator.

Assuming closed-loop stability, increasing k_D in the derivative controller:

- Does *not* influence the SSE to a step, since the derivative of a constant error is zero.
- Makes the response *less oscillatory* by opposing rapid changes in the error, acting as a kind of “viscous damping” in the closed-loop dynamics.
- Increases *sensitivity to measurement noise*, since differentiating a noisy signal amplifies high-frequency fluctuations.

5.6.4 Putting Everything Together

Having discussed the effect of each component individually, we can now combine them to form the proportional–integral–derivative (PID) controller. In block-diagram form, this structure is shown in Figure 5.19. Each term acts on the same error signal in *parallel*: the proportional term reacts to the present error, the integral term removes long-term bias, and the derivative term anticipates future error based on its rate of change.

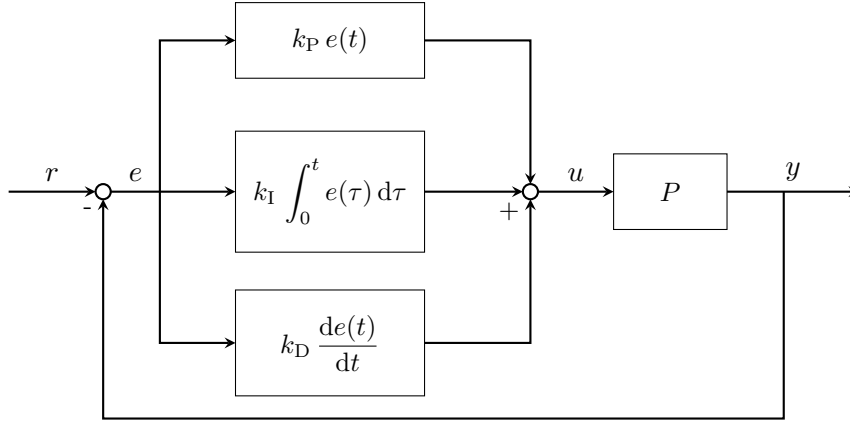


Figure 5.19: The PID controller in a standard feedback architecture.

Mathematically, the input $u(t)$ to the plant in the time domain is given by

$$u(t) = k_P e(t) + k_I \int_0^t e(\tau) d\tau + k_D \frac{de(t)}{dt}.$$

The corresponding transfer function of the controller is therefore

$$C_{\text{PID}}(s) = k_P + \frac{k_I}{s} + k_D s = \frac{k_D s^2 + k_P s + k_I}{s}. \quad (5.16)$$

it should be noted that the PID controller is often expressed in the following filtered form:

$$C_{\text{PID}}(s) = k_P \left(1 + \frac{1}{T_I s} + T_D s \right), \quad k_I = \frac{k_P}{T_I}, \quad k_D = k_P T_D,$$

where T_I is the integral time constant and T_D is the derivative time constant. This representation is convenient when using the root locus, it allows us to interpret the feedback gain directly as the proportional gain of the controller.

It is important to note that the three components of the PID controller are in parallel. Consequently, we can pick whether to have all three components present, or just some of them. The selection of which components to include is governed by what is desired from the controller. For example, if we do not require the controller to remove SSE, we can omit the integral term. Typically, the following configurations are considered:

- **P control:** $k_P > 0$, $k_I = 0$, $k_D = 0$. Suitable if the plant already provides integral action or if small SSE is tolerable.
- **PI control:** $k_P > 0$, $k_I > 0$, $k_D = 0$. Eliminates SSE to step
- **PD control:** $k_P > 0$, $k_I = 0$, $k_D > 0$. Improves damping and reduces overshoot; does not by itself remove step SSE.
- **PID control:** $k_P > 0$, $k_I > 0$, $k_D > 0$. Combines zero SSE with improved transient shaping when properly tuned.

Lastly, it is important to note the tuning of the PID parameters is a non-trivial task. Heuristic tuning rules such as Ziegler-Nichols can serve as useful starting points, though they typically require refinement. Some intuitive visual explanations of PID control can also be found online. A practical approach is:

1. Start with a proportional controller.
2. Draw the root locus of the open-loop transfer function $L(s)$.
3. If the root locus does not go through the “region of satisfaction” region, add a derivative term.
4. Choose a gain that places the dominant poles in the “region of satisfaction”.
5. If the SSE remains too large, add an integral term. Go back to step 2.

It should be noted that for some systems, such as the drone example from Lecture 1, cannot be controlled by a PID controller. We will introduce a couple more advanced controllers later on the course.

To help gain intuition behind the influence of the PID parameters, a nice visualization can be found online at <https://www.youtube.com/watch?v=4Y7zG48uHRo>.

Furthermore, the practical approach introduced above is very visual and requires use of the root locus. Try using the root locus to tune the PID for a selected system.

Chapter 6

Frequency response

6.1 Introduction

In the previous chapter we learned how to verify whether a closed-loop system meets its time-domain control objectives, such as the settling time T_d , and how these translate into restrictions on the location of the closed-loop poles in the s -domain. However, not all control objectives are naturally expressed in the time-domain.

Consider, for instance, the system in Figure 6.1. Here the output is measured by a sensor. However, real sensors are always subject to noise. Thus, the measured output signal becomes $\tilde{y}(t) = y(t) + v(t)$, where $v(t)$ is the measurement noise. In practice, this noise is often modeled as a Gaussian process, which can be characterized by how its energy is distributed across different frequencies. Although developing the spectral representation is beyond the scope of this course, the key idea is that noise contains components at many frequencies, and each of these components is processed by the system in the same way as a sinusoidal input.

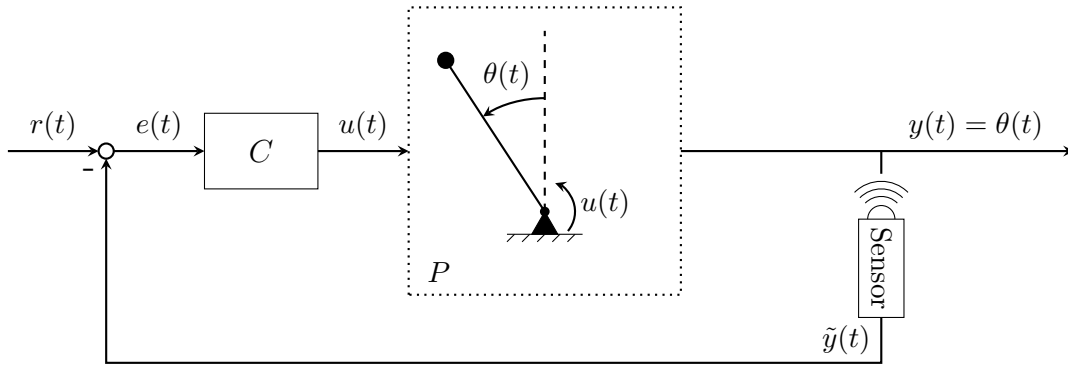


Figure 6.1: Visualization of a standard feedback loop for the inverted pendulum. The sensor is perturbed by noise $v(t) \sim \mathcal{N}(0, \sigma^2)$, resulting in a measured output signal $\tilde{y}(t) = y(t) + v(t)$.

Knowing that the controller will be computing the control signal $u(t)$ based on the error signal $e(t) = r(t) - \tilde{y}(t)$, a requirement for the closed-loop system could be that when the system is subject to a measurement noise of a certain frequency range, the output signal $y(t)$

should remain within some margin of the reference signal $r(t)$. From Chapter 3, we know that the steady-state response of an LTI-system to a sinusoidal reference is a sinusoid at the same frequency, but with modified magnitude and phase. Consequently, we can express this requirement as a restriction on the magnitude and phase of the transfer function at the relevant frequency range. This is where the frequency response comes into play. In this chapter we will introduce tools that permit us to visualize the frequency response of a system. Similar to previous chapters, we will rely on the open-loop system to deduce properties about the closed-loop system. Then, in the following chapter we will introduce frequency-domain specifications and how to synthesize controllers such that the system meets these specifications.

The first step is to consider how a system responds to a broader range of inputs, mainly how it responds to periodic/oscillatory excitations. Such excitations can be found in many use cases beyond noise and must be considered when designing a controller. Take for example a tall building, that is subject to periodic gusts of wind. If some periodic component of the wind matches the eigenfrequency of the building — possibly causing resonance — a catastrophic outcome could be reached. Another example is a house heating system designed to maintain a constant indoor temperature despite fluctuations in the outdoor temperature. Since there is usually a periodic change in temperature between the night and day this system is also affected by a periodic disturbance. In both cases it will be very beneficial to understand how the steady-state of each system behaves when it is affected by some periodic excitation, i.e. when it is excited by an oscillating signal at some frequency.

In this context, the following chapter explores methods to characterize the *frequency response* of a system, i.e. the way a system responds to oscillatory excitations. To this end recall that the steady-state output of a stable system, to some general input e^{st} can be obtained by

$$y_{ss}(t) = G(s)e^{st}. \quad (6.1)$$

Since we are now only interested in the effect of oscillatory inputs we can set the real part σ of the complex variable $s = \sigma + j\omega$ to zero, leaving only the imaginary part, i.e. $s = j\omega$. By substituting $s = j\omega$ into Equation (6.1) we obtain the steady-state output

$$y_{ss}(t) = G(j\omega)e^{j\omega t} = |G(j\omega)|e^{\angle(G(j\omega))} e^{j\omega t}.$$

From this, and as previously seen in Section 3.3, we can conclude that the steady-state output to a purely oscillatory input $e^{j\omega t}$ ¹ is fully characterized by the magnitude $|G(j\omega)|$ and phase $\angle(G(j\omega))$ of the transfer function. Importantly we also observe that the frequency of the steady-state output is equal to the frequency of the input. Figure 6.2 summarizes this behavior by illustrating the effect of a sinusoidal input applied to a system. We can observe that the steady-state output has the same frequency as the input but is shifted in phase by exactly $\angle(G(j\omega))$. Further the magnitude of the steady-state output is given by $|G(j\omega)|$.

From this we can conclude that in order to characterize the frequency response of a system we need to keep track of the magnitude $|G(j\omega)|$ and phase $\angle(G(j\omega))$ as a function of ω . In what

¹Recall that $e^{j\omega t}$ can be thought of as “half” of a sinusoid.

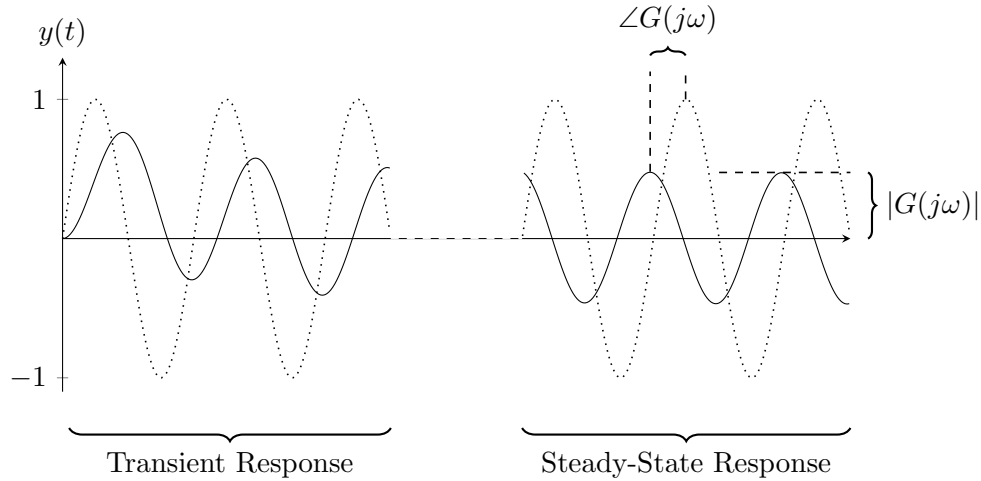


Figure 6.2: Time response (solid line) to sinusoidal input (dotted line) in the transient and steady-state.

follows we will discuss three ways to do so. First, Section 6.2 introduces the Bode plot that explicitly keeps track of both the magnitude and phase of $G(j\omega)$ as a function of the frequency ω . By discussing how a Bode plot can be sketched we also develop some intuition on how individual poles and zeros in the transfer function affect the overall frequency response. Next, the polar plot, introduced in Section 6.3, represents the magnitude and phase in an implicit curve in the complex plane. It serves as a stepping stone to finally obtain the Nyquist plot, covered in Section 6.4. The Nyquist plot will then enable us to assess the closed-loop stability based on the open-loop transfer function. Note that while this chapter introduces both the Bode and Nyquist plot their respective use cases and applications will mainly be discussed in the next chapter.

6.2 Bode plot

In the previous section we reviewed how a transfer function affects a sinusoidal input $u(t) = e^{j\omega t}$, $s = j\omega$: the output is a sinusoid at the same frequency, but with modified magnitude $|G(j\omega)|$ and phase $\angle G(j\omega)$. While evaluating $G(j\omega)$ for a specific frequency tells us how the system responds to one particular sinusoid, in practice we are interested in the behavior of the magnitude and phase over a wide range of frequencies. Low-frequency inputs often represent disturbances, while high-frequency inputs typically correspond to noise from measurements. Thus, understanding how the magnitude and phase of $G(j\omega)$ varies with the frequency ω will help us analyze and synthesize controllers.

A powerful graphical tool for this purpose is the *Bode plot*. Since the steady-state output depends only on the magnitude and phase of the transfer function at that excitation frequency, a Bode plot displays the magnitude and phase of the transfer function as functions of the frequency ω . This allows us to visualize how the system amplifies or attenuates signals across the frequency spectrum, and provides understanding that is difficult to obtain from algebraic expressions alone. Thus, in this section, we will focus on how to construct and interpret Bode

plots. In the next chapter, we will discuss how to use Bode plots to inform the design of controllers.

More concretely, the Bode plot is composed of two individual plots (Figure 6.3). On the x -axis we plot the frequency $\omega > 0$ in [rad/s] using a logarithmic scale. On the y -axis we use the following scales and units for each plot:

- The magnitude is usually given in decibels [dB], which is defined as

$$|G(j\omega)|_{[\text{dB}]} = 20 \cdot \log_{10} |G(j\omega)|.$$

- The phase $\angle G(j\omega)$ is given in degrees.

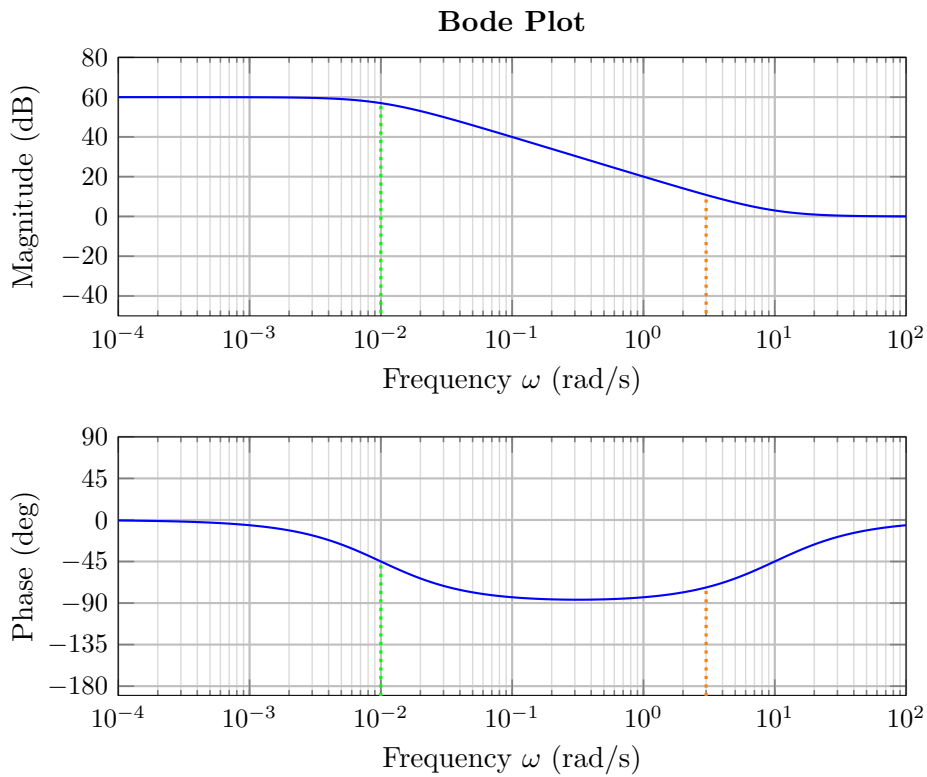


Figure 6.3: The Bode plot of the example $G(s) = 10^3 \cdot \frac{s/10+1}{s/0.01+1}$ in blue. The green line marks the value $\omega = 0.01$ and the orange line marks $\omega = 3$.

One approach towards generating the Bode plot is to evaluate the transfer function for many frequencies.

Example

Consider the transfer function

$$G(s) = 10^3 \cdot \frac{s/10 + 1}{s/0.01 + 1}.$$

Let's pick two values of ω and evaluate the transfer function at them.

Let's start with $\omega = 0.01$ or equivalently $s = 0.01j$. Evaluating $G(j\omega)$ we get that

$$G(0.01j) = 10^3 \cdot \frac{0.001j + 1}{j + 1}.$$

We use the rules from the graphical method, to compute the magnitude of the transfer function at this frequency,

$$|G(0.01j)| = 10^3 \cdot \frac{|0.001j + 1|}{|j + 1|} \approx \frac{10^3}{\sqrt{2}}.$$

Then we convert this to decibels,:

$$|G(0.01j)|_{[\text{dB}]} = 20 \cdot \log_{10} |G(0.01j)| = 20 \cdot \log_{10} \left(\frac{10^3}{\sqrt{2}} \right) \approx 57 \text{ dB}.$$

Meanwhile the phase can be calculated by subtracting the phase of the denominator from the phase of the numerator:

$$\angle G(j \cdot 0.01) = \angle(0.001j + 1) - \angle(j + 1) \approx 0^\circ - 45^\circ = -45^\circ$$

The resulting magnitude and phase are marked in the Bode plot in Figure 6.3.

As another example one could also calculate the magnitude and phase shift at the frequency of $\omega = 3$,

$$G(3j) = 10^3 \cdot \frac{\frac{3}{10}j + 1}{300j + 1}.$$

The magnitude in dB of the transfer function at this frequency becomes:

$$|G(3j)| = 10^3 \cdot \frac{|\frac{3}{10}j + 1|}{|300j + 1|} \approx 10^3 \cdot \frac{\sqrt{(\frac{3}{10})^2 + 1^2}}{300} \approx 3.48$$

$$|G(3j)|_{[\text{dB}]} \approx 20 \cdot \log_{10}(3.48) \approx 10.8 \text{ dB}$$

Doing the calculation for the phase we get:

$$\angle G(3j) = \angle \left(\frac{3}{10}j + 1 \right) - \angle(300j + 1) \approx \frac{180^\circ}{\pi} \cdot \arctan \left(\frac{3}{10} \right) - 90^\circ \approx -73^\circ$$

The complete magnitude and phase curves at any frequency $\omega > 0$ is displayed by the blue line in Figure 6.3.

However, this approach of evaluating the transfer function for many frequencies is not very efficient. Instead, we will decompose the entire transfer function into simpler components that we know how to draw, and then superimpose them to obtain the complete Bode plot. In the following subsection (Section 6.2.1), we will systematically go over some of the most important elements and develop drawing rules and approximations for them.

You might be bothered by the convention of multiplying by 20 instead of 10 in the conversion to [dB]. So from where is a factor of 2 coming, is this just another silly engineering convention? Not really this time however, since there is a physical and historical reason behind this choice. In many cases we are not interested to measure the direct amplitude, and sometimes it's not even possible at all, but instead we try to determine energetic or power quantities, which commonly scale with the square of the amplitude. For example in an electric circuit with only an AC-voltage and a resistance R , the overall electrical power consumed by the resistor scales with the amplitude squared, i.e. $P_{\text{elec}}(t) \propto u(t)^2$. Or in a mechanical harmonic oscillator the elastic energy stored in a spring scales with the displacement squared $E_{\text{el}}(t) \propto x(t)^2$. Hence, for example if the amplitude of the voltage $u(t)$ increases by a decade in magnitude, i.e. 10 dB in the “classical” sense, which we are used to, the consumed power would actually increase by 20 dB. Since by applying a logarithm the exponent becomes a factor:

$$\log(u(t)^2) = 2 \cdot \log(u(t))$$

So we observe that the missing factor of two comes from analyzing energetic quantities combined with a logarithmic magnitude scale, which is the place where the Bode plot was historically originally applied for.

Overall using this logarithmic scale every 20 dB represents a decade in order. An overview with some additional (approximate) values, which are helpful to know is provided in the table 6.1 below.

Decimal scale	[dB] scale
100	40
10	20
3.16	10
2	6.02
1	0
$1/\sqrt{2}$	-3.01
0.1	-20
0.01	-40
0	$-\infty$
$ G(j\omega) $	$20 \cdot \log_{10} G(j\omega) $
$10^{ G(j\omega) _{\text{[dB]}}/20}$	$ G(j\omega) _{\text{[dB]}}$

Table 6.1: Conversion between common values in the decimal and dB scale on the top and the general formulas in the bottom.

6.2.1 Bode plot of simple elements

In this section we will introduce how to draw the Bode plot. Although you will often draw the Bode plots through software tools, such as MATLAB or Python, it is important to understand where the underlying principles of the resulting plot come from. Since most transfer functions can be decomposed into a product of simple elements (poles, zeros, conjugate poles, or conjugate zeros), we will look at drawing the Bode plot for each of these elements. In the next section (Section 6.2.2), we will then discuss how to combine these simple elements to draw the Bode plot of more complex transfer functions.

To help build intuition, feel free to follow along the examples using the provided tool [BodeTool](#).

As a first step, it is important to note that it is useful to represent the transfer function in the Bode form. The reason for this will hopefully become clear in the following examples.

Constant gain

Let's start with a transfer function that represents a scalar multiplication:

$$G(s) = k$$

Notice that the output is always the same regardless of the input frequency $s = j\omega$. Thus, the magnitude and phase also represent constant functions. We can distinguish between the cases where the gain is positive and negative. Figure 6.4 plots the Bode plot for $k = 2$ and $k = -2$. The sign of the gain does not affect the magnitude $|G(s)|$, which in both cases is exactly 2. By table 6.1 this roughly evaluates to 6 dB. Meanwhile, the phase for the positive gain corresponds to a phase of zero, whereas a negative gain results in an angle of 180° in the s -plane.

Integrator

We will now consider the case of

$$G(s) = \frac{1}{s},$$

which corresponds to a pole placed exactly at the origin and can be interpreted in the time-domain as the transfer function of an integrator. If we set $s = j\omega$ to determine the behavior at different frequencies for this transfer function we obtain

$$G(j\omega) = \frac{1}{j\omega} = \frac{-j}{\omega}.$$

The phase is -90° (or equivalently 270°) independent of ω . In comparison, the magnitude $|G(j\omega)| = 1/\omega$ depends on the frequency. To understand how the magnitude behaves, let's plug in different values of ω . For example, if we set $\omega = 10$ we get -20 dB. Doing the same for

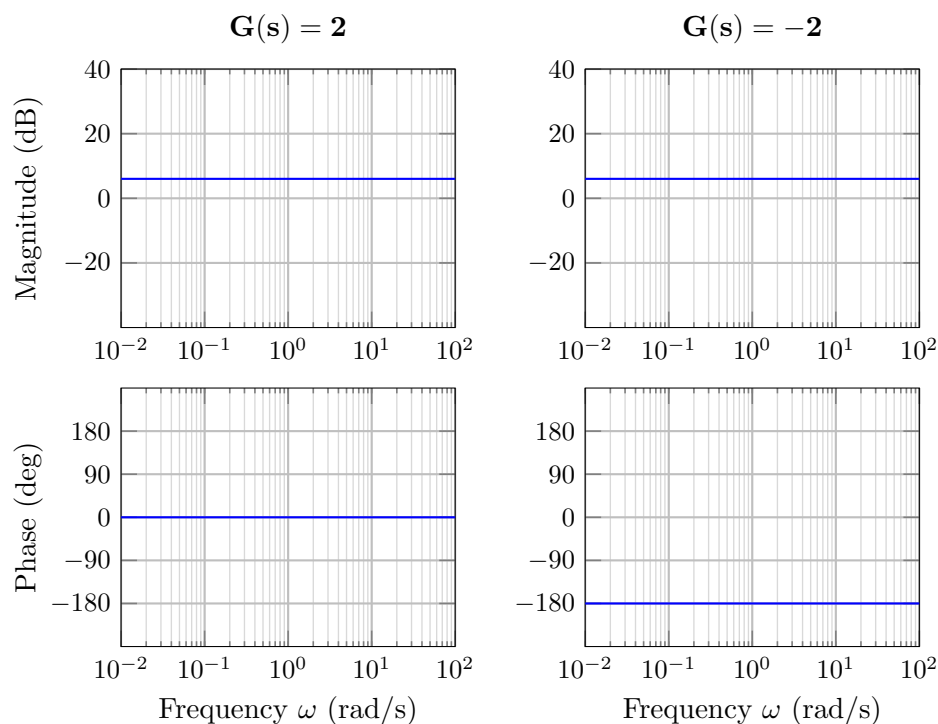


Figure 6.4: Positive gain on the left and negative gain on the right.

$\omega = 100$ we obtain -40 dB. In general, we notice that for an increase in the frequency by a factor of 10 the magnitude drops by 20 dB. This is a very important characteristic behavior of an integrator!

An integrator has a slope of -20 dB/decade.

As it can be seen in Figure 6.5 the overall magnitude curve ends up being a linear line going through 0 dB at $\omega = 1$.

Real pole

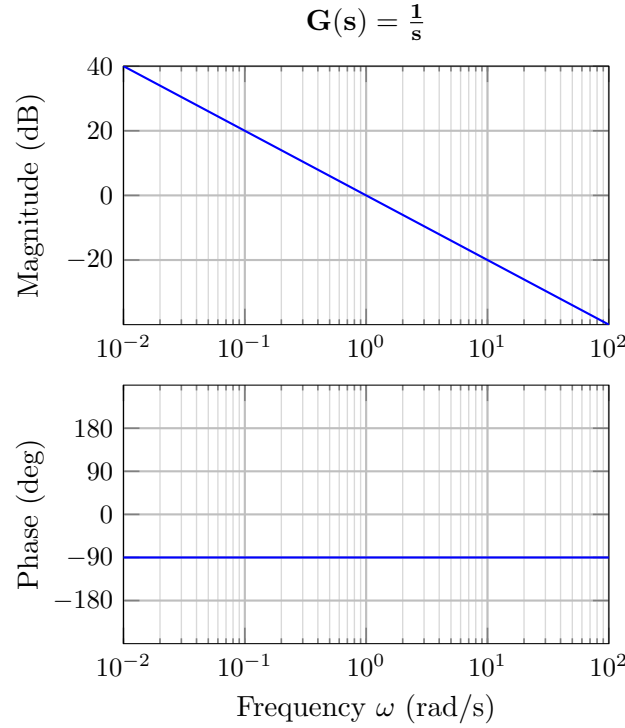
Next, we will consider the case where a pole p is not necessarily placed in the origin, but somewhere on the real axis. This corresponds to the transfer function

$$G(s) = \frac{1}{\frac{s}{-p} + 1}$$

The transfer function for a real pole is often written in the following form:

$$G(s) = \frac{1}{\tau s + 1}$$

with $\tau = -1/p$ referring to the time constant introduced in Section 5.3. For a pole that is not placed at the origin, we generally need to distinguish two cases: whether p is positive or negative. The two different cases are plotted in Figure 6.6 with $p = -10$ and $p = 10$. To provide some

**Figure 6.5:** Bode Plot of an Integrator

understanding as to where these behaviors come from, let's analyze the asymptotic behaviors of the magnitude and phase. Specifically, we will consider how the transfer function behaves as $\omega \rightarrow 0$, $\omega \rightarrow +\infty$, and $\omega = |p|$.

1. In the limit as $\omega \rightarrow 0$:

$$\lim_{\omega \rightarrow 0} |G(j\omega)| = 1 \implies |G(j\omega)|_{[dB]} = 0 \text{ dB},$$

$$\lim_{\omega \rightarrow 0} \angle G(j\omega) = 0^\circ.$$

2. For $\omega \rightarrow \infty$:

$$\lim_{\omega \rightarrow \infty} |G(j\omega)| = 0 \implies |G(j\omega)|_{[dB]} = -\infty \text{ dB},$$

$$\lim_{\omega \rightarrow \infty} \angle G(j\omega) = -90^\circ \cdot \text{sgn}(p).$$

Furthermore, the magnitude at large frequencies corresponds to a line with a slope of -20 dB/decade :

$$|G(j\omega)| \approx \frac{|p|}{\omega} \implies |G(j\omega)|_{[dB]} \approx [20 \log_{10}(|p|) - 20 \log_{10}(\omega)] \text{ dB},$$

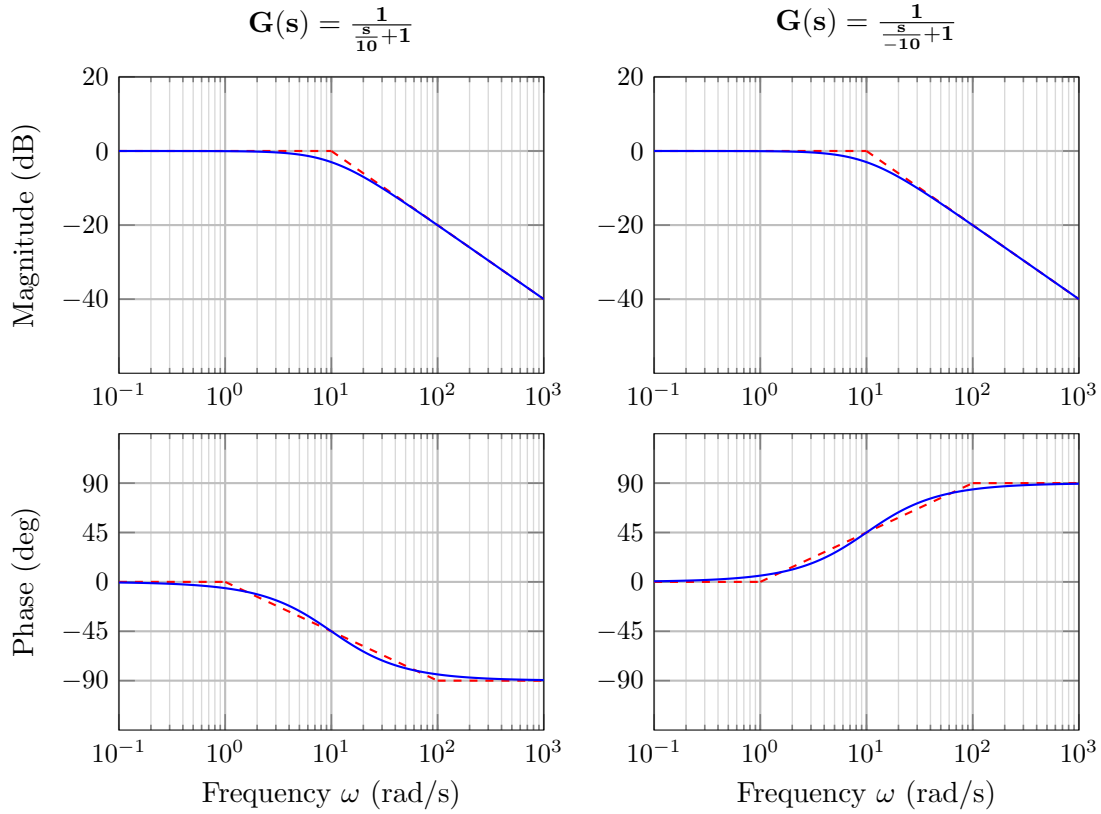


Figure 6.6: Bode plot of a stable pole on the left and an unstable pole on the right (in blue). The straight line approximations are depicted as a dashed red line.

3. If $\omega = |p|$, then

$$\lim_{\omega=|p|} |G(j\omega)| = \frac{1}{\sqrt{2}} \implies |G(j\omega)|_{[dB]} = -3 \text{ dB},$$

$$\lim_{\omega=|p|} \angle G(j\omega) = -45^\circ \cdot \text{sgn}(p).$$

By convention the frequency $\omega = |p|$ is referred to as the *corner/cutoff frequency*.

If drawing the Bode plot by hand, we can use straight line approximations to draw our elements (These are depicted in Figure 6.6 as red dashed lines). The **magnitude** of a real pole can be approximated by two lines:

1. $\omega \leq |p|$: For small frequencies the transfer function behaves like a constant gain.

$$|G(j\omega)|_{[dB]} = 0 \text{ dB}$$

2. $\omega > |p|$: At large frequencies the simple pole behaves like an integrator with the characteristic slope of -20 dB/decade .

Meanwhile, the **phase** can be approximated to be composed of three segments:

1. $\omega \leq 0.1|p|$: A constant line at 0° .

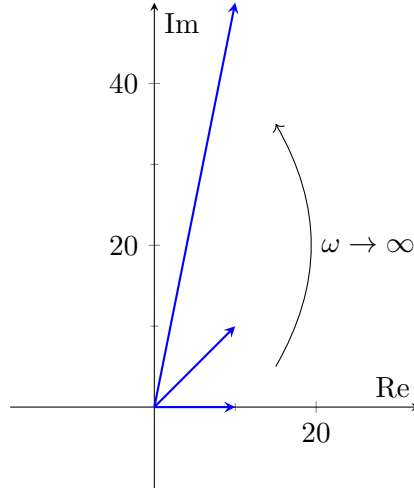


Figure 6.7: Based on the stable pole transfer function $G(s) = \frac{1}{\frac{s}{10} + 1}$, the vectors of the denominator $D(j\omega) = \frac{j\omega}{10} + 1$ are plotted for different values of ω . Notice that as the frequency increases, the angle of the blue arrows representing the denominator approaches $+90^\circ$. However, since we actually divide by the denominator in the transfer function, the angle of the overall transfer function approaches -90° .

2. $0.1|p| < \omega \leq 10|p|$: The phase drops linearly from 0° to -90° over two decades.
3. $10|p| < \omega$: A constant line at -90° .

As it can be seen in the figure the straight line approximations are exact at describing the asymptotic behavior for very small and large frequencies, though are slightly off if ω and p have roughly the same order of magnitude.

Complex conjugate poles

Lastly, the only missing element that could appear in the denominator are complex poles. We know that these always have to appear in complex conjugate pairs. We will restrict ourselves to the case where the real part of the complex conjugate pair is negative, i.e. the case of complex, underdamped, stable poles with $\omega_n > 0$ and $1 > \zeta > 0$:

$$G(s) = \frac{1}{s^2/\omega_n^2 + 2\zeta s/\omega_n + 1} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (6.2)$$

The poles of this transfer function are located at $p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. We represented the Bode plot of this transfer function below in Figure 6.8 for $\omega_n = 1$ and various values of ζ . As a remark this transfer function corresponds to the harmonic oscillator in the time domain

$$\ddot{y}(t) + 2\zeta\omega_n\dot{y}(t) + \omega_n^2 y(t) = u(t),$$

so we expect to see typical characteristics of that specific differential equation like resonance in the Bode plot. We will again try to get a better understanding of this element by investigating the limits of $\omega \rightarrow 0$ and $\omega \rightarrow \infty$, as well as analyzing the behavior if ω approaches the natural frequency ω_n .

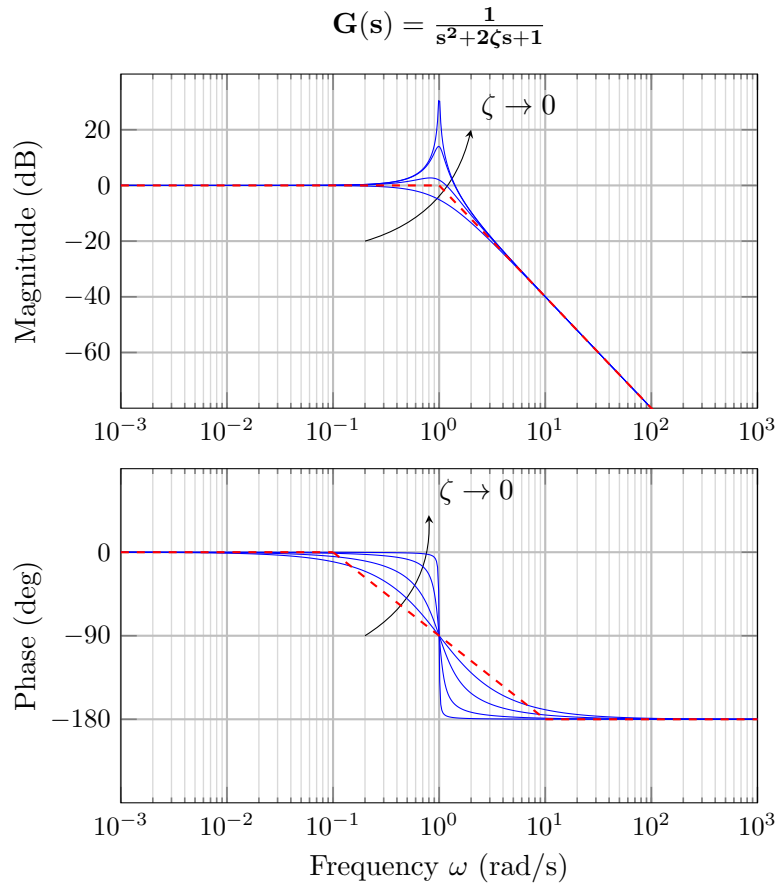


Figure 6.8: The Bode plot for a complex conjugate pair of poles is visualized for various values of the damping ratio ζ . We observe that as $\zeta \rightarrow 0$ the resonance peak increase in magnitude, meanwhile the phase plot approaches a step function-like behavior. In contrast, as the damping ratio approaches 1, the resonance peak disappears and the phase changes over a larger frequency span in a slower manner. A resonance peak only exists if $\zeta < 1/\sqrt{2}$. The straight line approximations are drawn as red dashed lines.

1. If $\omega \rightarrow 0$, then the denominator term $+1$ dominates and the transfer function becomes $G(j\omega) \approx 1$:

$$|G(j\omega)| \approx 1 = 0 \text{ dB}, \quad \angle G(j\omega) \approx 0^\circ$$

2. If $\omega \rightarrow \infty$, then the s^2/ω_n^2 term start to dominate the behavior, i.e. the $\left(\frac{j\omega}{\omega_n}\right)^2$ term. The transfer function becomes $G(j\omega) \approx -\frac{\omega_n^2}{\omega^2}$:

$$|G(j\omega)| \approx \frac{\omega_n^2}{\omega^2} = [40 \ln(\omega_n) - 40 \ln(\omega)] \text{ dB}, \quad \angle G(j\omega) \approx -180^\circ$$

We see that for large frequencies the magnitude behaves like a linear line with a slope of -40 dB/decade .

3. If our input frequency matches the natural frequency, i.e. $\omega = \omega_n$, we obtain the transfer function $G(j\omega) = \frac{1}{2\zeta j} = -\frac{j}{2\zeta}$, the magnitude and phase evaluate to:

$$|G(j\omega)| = \frac{1}{2\zeta}, \quad \angle G(j\omega) = -90^\circ$$

In particular we observe here that as the damping ratio approaches zero, our magnitude start to explode. This is the expected physical result for an undamped system excited about the resonance frequency.

For completeness, we will state without any derivations that the resonance peak for the under-damped case is slightly of the natural frequency and located at

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$$

with a peak value for the amplitude of

$$|G(j\omega)|_{max} = \frac{1}{2\zeta\sqrt{1 - \zeta^2}}.$$

We can again approximate the Bode plot via linear segments. We can dissect the magnitude plot into two linear lines. One for $\omega < \omega_n$, representing the behavior of a constant gain of 0dB. And secondly a line of slope -40 dB/decade for $\omega > \omega_n$ which represents a double integrator. In comparison, for the phase we split it up into three segments again. A first part with zero phase for $\omega \leq 0.1 \cdot \omega_n$, a second part for $0.1 \cdot \omega_n < \omega < 10 \cdot \omega_n$ where the phase drops linearly from 0° to -180° , and thirdly a constant phase of -180° for $\omega \geq 10 \cdot \omega_n$.

We notice that in this case that the asymptotic behavior is captured completely accurately again, while in the case of the complex poles the magnitude and phase around the order of the pole might be quite significantly off. One particular point is that we completely neglected the resonance peaks in the straight line approximations. Even though there exist better line approximations for this second order system, the one we introduce here is easy to use and remember and for a detailed analysis it's better to use computer tools like MATLAB.

Inverting the transfer function

Before analyzing the elements in the numerator of the transfer function, we will exploit an important behavior of complex numbers and an important property of logarithms. Specifically the transfer function in polar coordinates $G(s) = r \cdot e^{i\theta}$ always represents a complex number, so if we invert it we get $1/G(s) = 1/r \cdot e^{-i\theta}$, so the angle gets negated and the radius inverted. But taking the logarithm of the inverted radius results in the negative magnitude of the original transfer function, i.e.

$$\left| \frac{1}{G(s)} \right|_{[\text{dB}]} = 20 \cdot \log_{10} \left| \frac{1}{G(s)} \right| = -20 \cdot \log_{10} |G(s)| = -|G(s)|_{[\text{dB}]}.$$

Graphically we can interpret this as:

We can get the transfer function of $1/G(s)$ by mirroring the Bode plot of $G(s)$ around 0 dB for the magnitude- and 0° for the phase plot.

An example is illustrated in Figure 6.9. So overall we don't need to re-derive any sketching

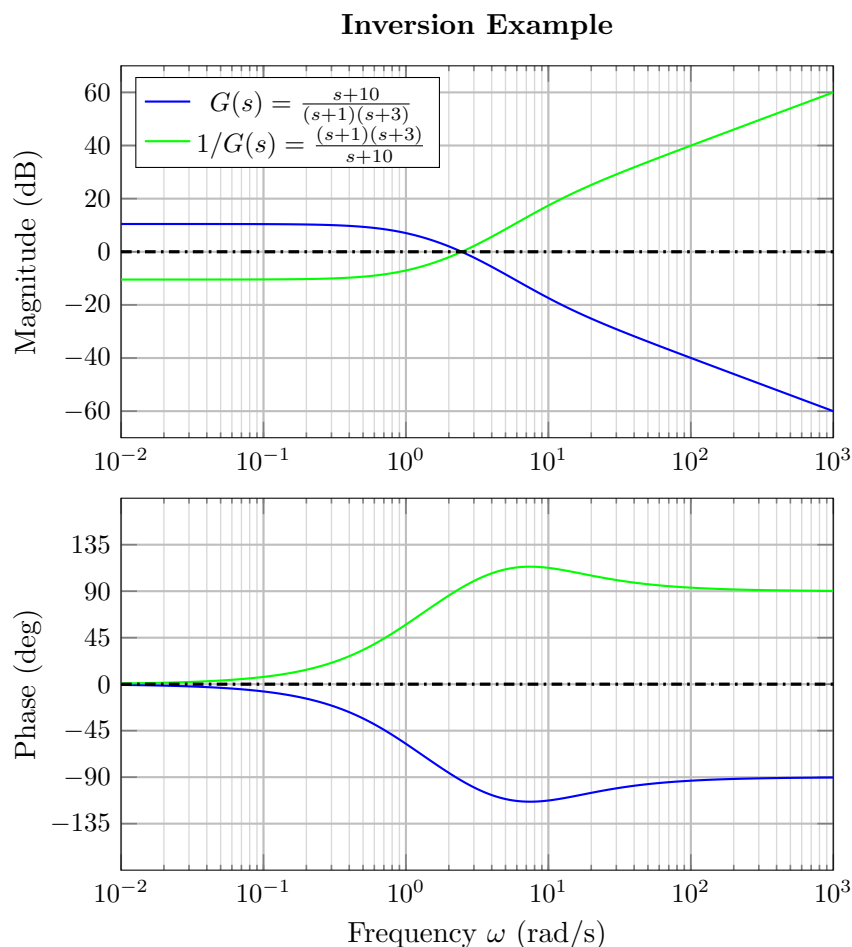


Figure 6.9: We notice that the inversion is equivalent to mirroring the Bode plot around the horizontal axis (dashed-dotted line representing axis of symmetry) for both plots.

rules or characteristic behaviors for elements appearing in the numerator, instead we can just mirror the plots from the one's of the denominator which we already know. Thus, we will just state the characteristics and give the plots without further details on the exact derivation steps to avoid unnecessary repetition.

Differentiator

The differentiator in the time domain corresponds to a zero at the origin, that is

$$G(s) = s.$$

Evaluated at $s = j\omega$ we see that $G(j\omega) = j\omega$ always has a constant phase of 90° and a linearly growing magnitude with frequency. The typical characteristic behavior of a differentiator is its consistent positive slope and quantitatively:

A differentiator has a slope of $+20$ dB/decade.

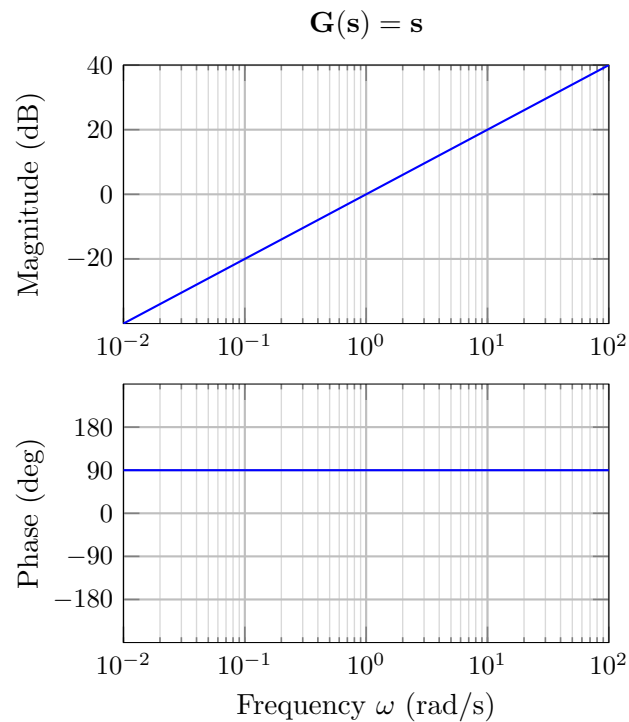


Figure 6.10: Bode plot of a differentiator

Real zero

A real zero describes the following class of elements:

$$G(s) = \frac{s}{-z} + 1$$

Similarly to the case of the real pole, we need to distinguish between the case of the minimum phase zero and non-minimum phase zero. The Bode plots are shown below in Figure 6.11 for the values of $z = -10$ and $z = 10$.

Overall for the magnitude plot we can approximate a real zero to be a constant gain for small frequencies and having the behavior of a differentiator at large frequencies. Whereas for the phase plot we can approximately trisect the curve into three parts in a similar fashion like the real pole. Interestingly one also observe that the non-minimum phase zero has a negative phase shift, which means that it lags behind the input and behaves similarly as if it reacted delayed to the applied input. This result is consistent with our interpretation of zeros as derivative action, which we saw in Section 3.6.2.

Complex conjugate zeros

Also in this case we will only consider complex zeros which are minimum phase, i.e. they have negative real part:

$$G(s) = s^2/\omega_n^2 + 2\zeta s/\omega_n + 1$$

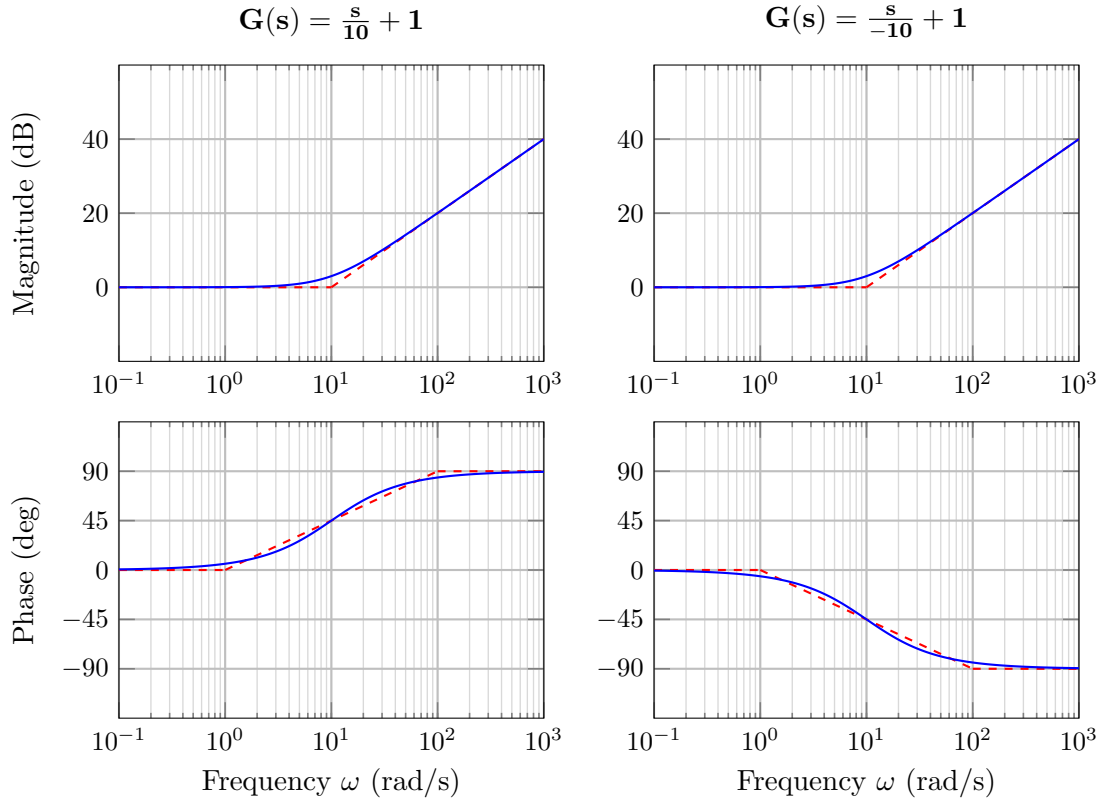


Figure 6.11: Bode plot of a minimum phase zero on the left and non-minimum phase zero on the right (in blue). The straight line approximations are depicted as a dashed red line.

In comparison to the underdamped complex poles, the resonance peak is actually a valley, since it's flipped and represents a minimum instead of a maximum. We also notice that this valley can again only exist if $\zeta < 1/\sqrt{2}$. A characteristic of the complex pair of zeros is their behavior as constant gain for small frequencies and as a double differentiator for larger frequencies with a slope of +40 dB/decade. We plotted a few different possibilities of these kinds of transfer functions with $\omega_n = 1$ and varying ζ in Figure 6.12.

6.2.2 Combining simple elements

We still haven't discussed how to draw any kind of arbitrary complicated transfer function. You might notice that the example in Figure 6.3 is not of any kind of "simple element" form previously discussed. To recreate the Bode plot shown in there, we will need to exploit the superposition principle for constructing the Bode plot of more complicated functions based on known ones. Specifically for any value of s , where we evaluate the transfer function it ends up representing a complex number, and we know that for a multiplication of complex number the magnitudes multiply and the phases add up. But if we apply a logarithm to the magnitude they also add up! Concretely mathematically this means if $G(s) = G_1(s) \cdot G_2(s)$, then for the magnitude

$$|G(s)| = |G_1(s)| \cdot |G_2(s)| \quad \Rightarrow \quad |G(s)|_{[\text{dB}]} = |G_1(s)|_{[\text{dB}]} + |G_2(s)|_{[\text{dB}]}$$

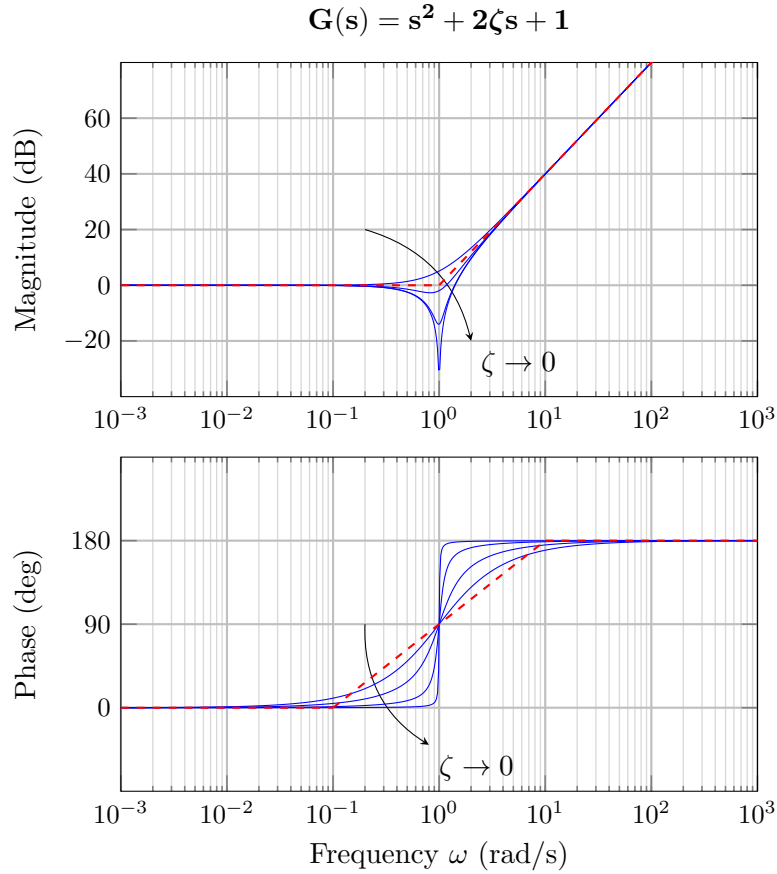


Figure 6.12: In the figure above the Bode plot for a complex conjugate pair of zeros is visualized for various values of the damping ratio ζ . We observe that as $\zeta \rightarrow 0$ the “resonance valley” gets lower and eventually goes to a magnitude of $-\infty$ dB, meanwhile the phase plot becomes closer to a step function. In contrast, as the damping ratio approaches 1, the dip disappears and the phase changes over a larger frequency span in a slower manner. The straight line approximations are drawn as red dashed lines.

holds, which is the case since

$$20 \cdot \log_{10} |G(s)| = 20 \cdot \log_{10} (|G_1(s)| \cdot |G_2(s)|) = 20 \cdot \log_{10} |G_1(s)| + 20 \cdot \log_{10} |G_2(s)|$$

As a consequence the Bode plot exhibits the extremely powerful property of superposition! We can always split up the entire transfer function into simpler factors and can then add them up later using linearity. Since we know how any of these simple elements behave, we can draw the Bode plot of any kind of system regardless of their complexity. We will illustrate this method on a few examples in the following.

Example: Double Integrator

Consider the case of

$$G(s) = \frac{5}{s^2}$$

we can rewrite this as $G(s) = 5 \cdot \frac{1}{s} \cdot \frac{1}{s}$, so the overall transfer function is composed of a

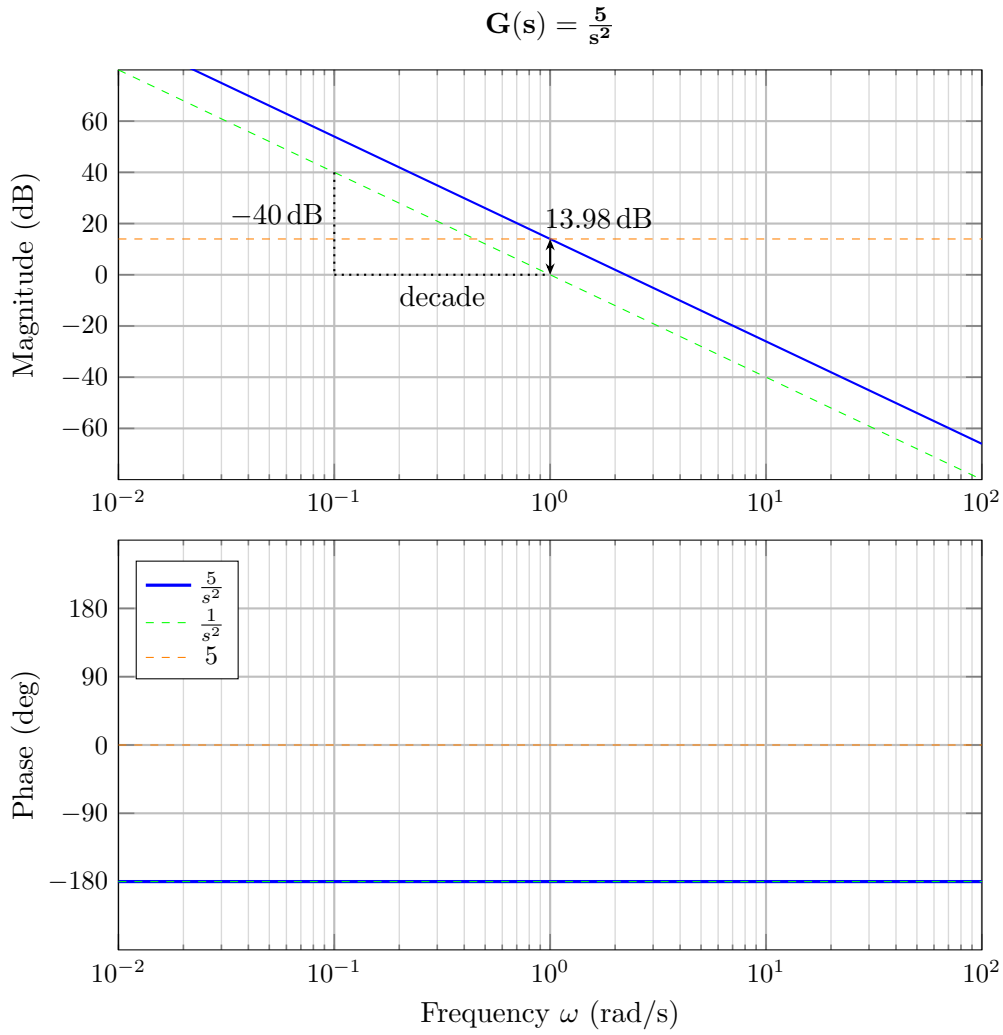


Figure 6.13: Bode plot of $G(s)$ in blue, a gain combined with a double integrator

gain and a double integrator. The gain has a phase of zero and a magnitude of $20 \log_{10} 5 \approx 13.98$ dB. Meanwhile, the double integrator is just twice the value of a normal integrator, so it has a phase of -180° and its magnitude represents a linear line going through 0 dB at $\omega = 1$ with a slope of -40 dB/decade.

The overall phase can be calculated by adding up the two contributions, which ends up being $\angle G(j\omega) = \angle 5 - \angle (j\omega)^2 = -180^\circ$. Similarly, we know that the constant gain just adds a constant shift to the y -axis of the magnitude plot. Shifting the magnitude plot of $1/s^2$ with 13.98 dB gives us the total magnitude of $G(s)$.

Example: Second order high pass filter

We will analyze the transfer function

$$G(s) = \frac{s^2}{s^2 + 10s + 100}$$

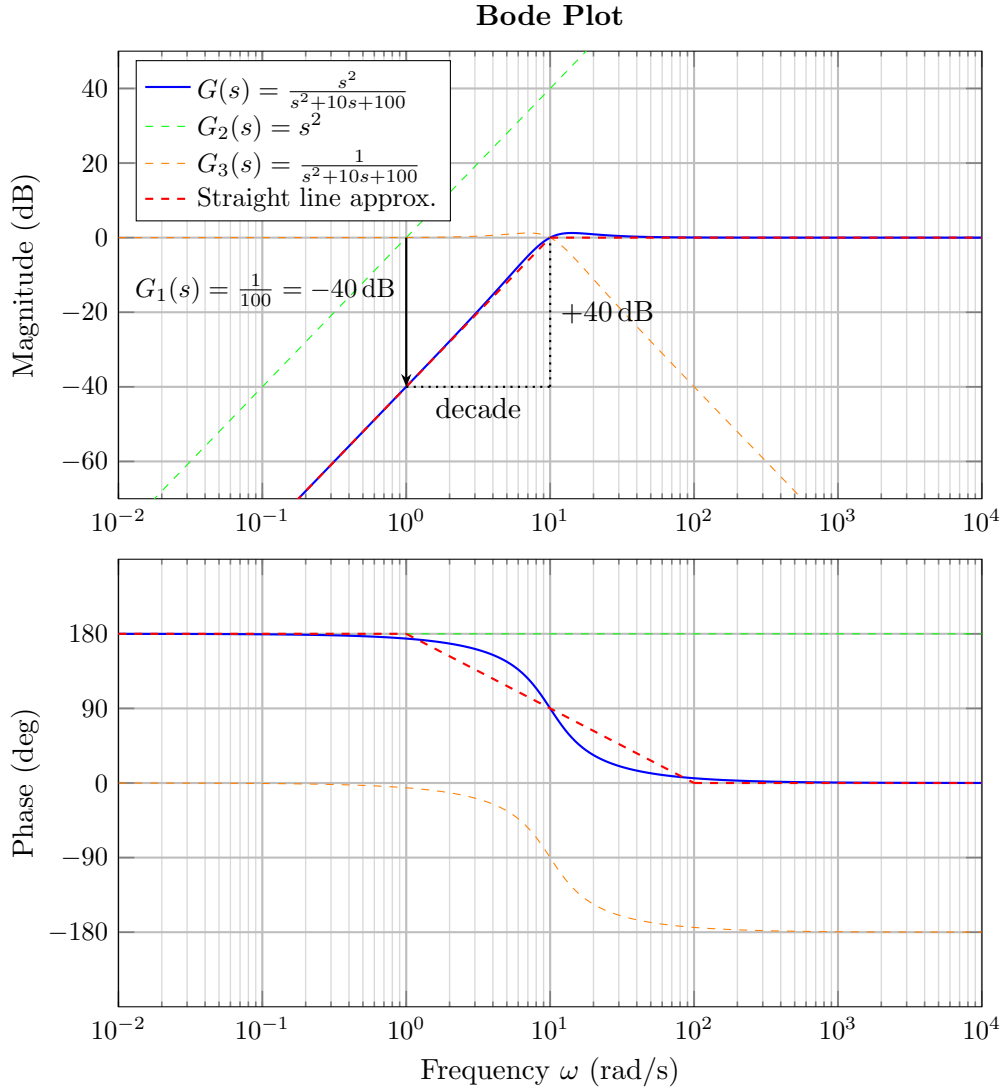


Figure 6.14: Second order high pass filter in blue and straight line approximation in red.

in this example. Here, we can again proceed by factoring the entire function into simpler bits, where we already know the solution to the Bode plot.

$$G(s) = \underbrace{\frac{1}{100}}_{G_1(s)} \cdot \underbrace{s^2}_{G_2(s)} \cdot \underbrace{\frac{100}{s^2 + 10s + 100}}_{G_3(s)}$$

One can start out by interpreting the first term, which represents a constant. Its only influence is a shift by -40 dB in the magnitude plot. Also observe that the second factor corresponds to a double differentiator, i.e. it has a phase of 180° and its magnitude represents a linear line crossing 0 dB at $\omega = 1$ with a constant slope of $+40 \text{ dB/decade}$. So the first two elements could be summarized as a single line portraying the double differentiator, but shifted by -40 dB as a consequence of the gain. This is pictured by the green dashed line and black arrow in Figure 6.14.

Meanwhile, the third factor depicts a stable complex conjugate pair of poles. By com-

paring with the general form in Equation (6.2), we can conclude that $\omega_n = 10$ and $\zeta = 0.5$. With the given value of the damping ratio we will observe a resonance peak since $\zeta < 1/\sqrt{2}$, even though in this case the peak is very shallow. Additionally, the complex poles will approximately induce a decay of -40 dB/decade in the magnitude once ω reaches the natural frequency ω_n . This will essentially cancel out the increase in magnitude produced by the double differentiator. For the phase we observe that the phase shift of -180° induced by the complex poles will also cancel out the $+180^\circ$ of the double differentiator approximately one decade after the resonance frequency ω_n . The overall curve is presented in blue and the straight line approximation one would use to draw the Bode plot by hand

We notice that high frequencies are transmitted almost without any changes, so they just pass through the transfer function, as if it didn't exist. This can be observed by the fact that the magnitude is one and the phase shift is zero at high frequencies, so for an input of the form $u(t) = \sin(\omega t)$ the output will also be close to identical, i.e. $y(t) = |G(j\omega)| \sin(\omega t + \angle G(j\omega)) \approx \sin(\omega t)$. Meanwhile, low frequencies tend to get blocked by the transfer function since a sine wave with a low frequency gets multiplied with a very small amplitude making the signal go close to zero. This behavior is thus referred to as a “high pass filter”, allowing high frequencies to pass through undisturbed, while filtering out low frequencies.

The term “second order” refers to the feature of having a slope of $+40$ dB/decade until the natural frequency. In this case the natural frequency indicates up to until which point we want to block frequencies and from which point on we want to allow frequencies to pass. In particular how sharp this switch happens, i.e. the slope of the drop defines the order of the filter. In our case the fall of the magnitude for decreasing frequencies behaves like a second order zero in regions below the “switch” frequency.

6.2.3 Drawing tips and general procedure

We generally recommend the following procedure when drawing the Bode plot:

1. Bring transfer function into Bode form.
2. Determine the contributions of the individual factors using simple elements.
3. Add the different factors together.

Lastly for adding up the different elements it helps to think about, when certain contributions kick in. By doing this we can often skip over finding the exact Bode form and drawing all the individual contributions and just think about their effect and from which point on they should be added. Restating what we found out so far in the Bode plot is:

In an approximate manner poles and zeros start to play a role from the following time points on.

- For the magnitude the effect of any pole or zero on the Bode plot starts approximately when the frequency ω matches exactly the magnitude of the pole/zero. Where any single pole creates a slope of -20 dB/decade, while any single zero creates a slope of $+20$ dB/decade.
- For the phase plot any pole or zero creates changes from one decade before the pole/zero up to one decade after the pole/zero. The change over this period for any single individual pole/zero is $\pm 90^\circ$.

An exact overview can be found in tables 6.2 and 6.3.

	real pole	real zero	complex poles	complex zeros
slope [dB/decade]	-20	+20	-40	+40

Table 6.2: Slopes of different elements once the frequency reaches the magnitude of the pole/zero.

	stable pole, non-min. phase zero	unstable pole, min. phase zero	stable com- plex poles	min. phase complex zeros
phase change	-90°	$+90^\circ$	-180°	$+180^\circ$

Table 6.3: Total phase change over two decades for different elements.

As a sanity check even for complicated systems it's also advisable to consider the edge cases of $\omega \rightarrow 0$ and $\omega \rightarrow \infty$, which as an example helps us to check that no factor was forgotten. Starting from the general Bode form

$$G(s) = \frac{k_{\text{Bode}}}{s^q} \frac{\left(\frac{s}{-z_1} + 1\right) \left(\frac{s}{-z_2} + 1\right) \cdots \left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right) \left(\frac{s}{-p_2} + 1\right) \cdots \left(\frac{s}{-p_{n-q}} + 1\right)}$$

typical characteristics for any complicated system at these extreme values can be summarized as:

- If $\omega \rightarrow 0$:

$$G(j\omega) \approx \frac{k_{\text{Bode}}}{(j\omega)^q} \quad (6.3)$$

In particular if there are no integrators or poles the magnitude has to start out as a constant gain. Concretely in this case this means (assuming no integrators/differentiators)

$$G(0) = k_{\text{Bode}}. \quad (6.4)$$

- If $\omega \rightarrow \infty$ the slope of the magnitude can be described by the difference in poles n and zeros m :

$$\text{slope} = -20(n - m) \text{ dB/decade} \quad (6.5)$$

These statements about the edge cases just formalize our previous discussion into a mathematical formula, which we can use to validate our hand-drawn Bode plot. Using these insights we can systematically go over another example:

Example: Efficient method for drawing

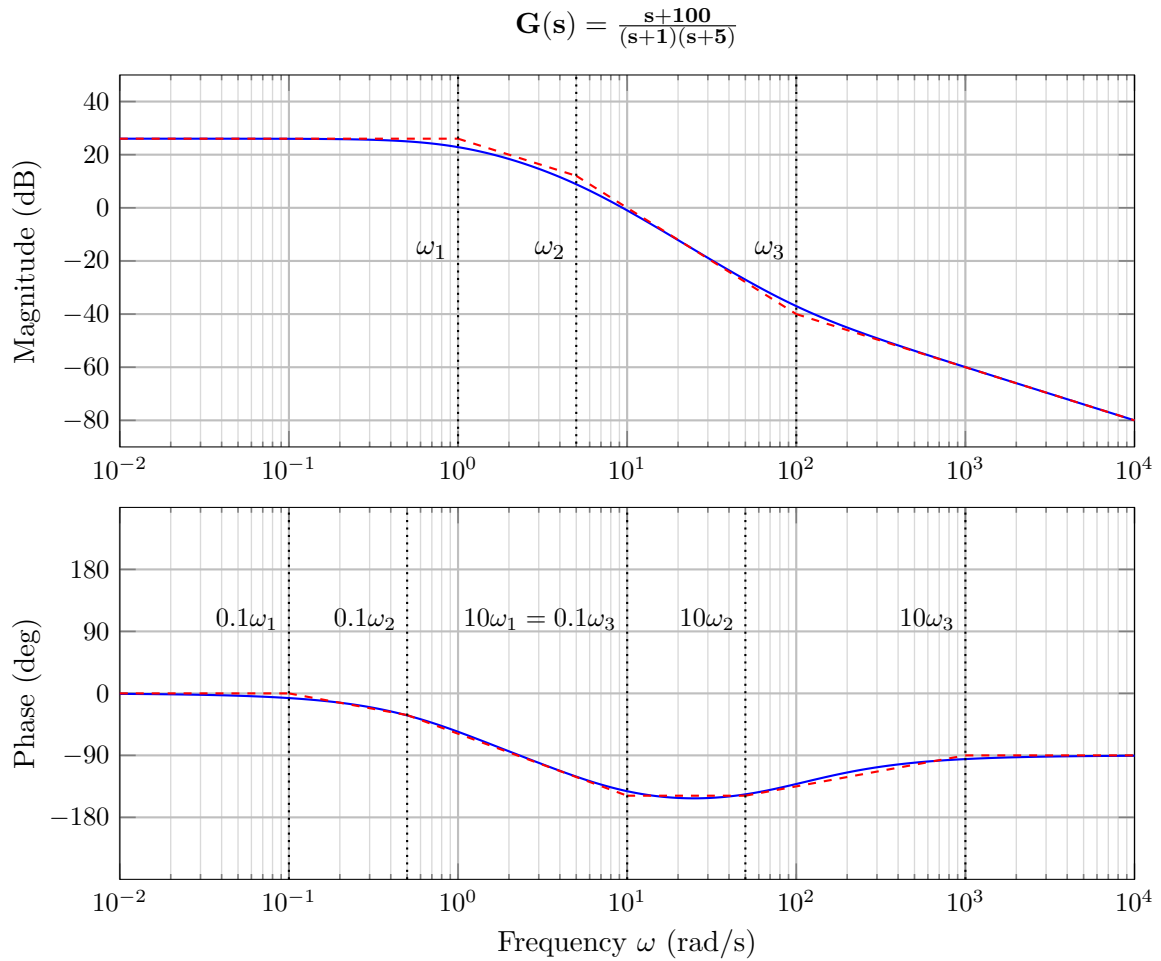


Figure 6.15: Exact Bode plot of the example in blue. Straight line approximations used for drawing in red. The analyzed regions are indicated with dotted lines.

We will look at a few tricks and procedures on how one could draw the Bode plot of the transfer function

$$G(s) = \frac{s + 100}{(s + 1)(s + 5)}$$

efficiently. In the above transfer function the poles lie at $p_1 = -1$, $p_2 = -5$ and a zero at $z = -100$. Proceeding from here, this time around instead of drawing out the Bode plot of every single element, we will try to draw the Bode plot in terms of influence regions of the poles and zeros. For this we will define three helper frequencies for drawing namely

$\omega_1 = 1$, $\omega_2 = 5$, $\omega_3 = 100$. By dividing the plots into different frequency regions we can construct an approximate Bode plot. We can now draw the magnitude and phase plot separately.

Magnitude plot:

1. $0 < \omega \leq \omega_1$: Since no integrators or differentiators are present the magnitude plot has to start out at $G(0) = k_{\text{Bode}} = 20$. We can thus draw a constant line of $20 \ln_{10} 20 \approx 26$ dB until ω_1 .
2. $\omega_1 < \omega \leq \omega_2$: At ω_1 the first pole starts to effect the Bode plot leading to a slope of -20 dB/decade.
3. $\omega_2 < \omega \leq \omega_3$: From ω_2 on the second pole also start to influence the magnitude, combined with the first pole we will have an overall slope of -40 dB/decade in this region.
4. $\omega_3 < \omega$: Lastly once we reach the zero it will add 20 dB to the slope, hence combined with the two poles we will have an overall gradient of -20 dB/decade for any ω greater than ω_3 . This result is consistent with Equation (6.5).

Phase plot:

1. $0 < \omega \leq 0.1 = 0.1\omega_1$: We know that for small frequencies the transfer function can be approximated by the constant Bode gain k_{Bode} , which is positive constant. Hence, the phase starts out at 0° .
2. $0.1\omega_1 = 0.1 < \omega \leq 0.5 = 0.1\omega_2$: The first pole starts to effect the phase. A single stable pole introduces a drop in phase of 90° over two decades, i.e. a slope of $-45^\circ/\text{decade}$.
3. $0.1\omega_2 = 0.5 < \omega \leq 10 = 10\omega_1$: In this region two stable poles produce a reduction in phase. Which doubles the slope of the previous segment, i.e a gradient of $-90^\circ/\text{decade}$.
4. $10\omega_1 = 0.1\omega_3 = 10 < \omega \leq 50 = 10\omega_2$: The complete phase shift of the first pole has occurred by this point and also the minimum phase zero starts to have an effect from this point on. The line will be flat in this segment, since the slope of the second pole and zero cancel out.
5. $10\omega_2 = 50 < \omega \leq 1000 = 10\omega_3$: The effect of the two stable poles has settled in and only the zero still induces changes in the phase with a slope of $+45^\circ/\text{decade}$.
6. $10\omega_3 = 1000 < \omega$: Finally the phase shift for all elements has settled in, and we reach a constant phase of -90° in this region.

Lastly we would also like to relate and apply the Bode plot for a physical system. We will

continue with the pendulum, which we used so far over the entire notes.

Example – Pendulum

Assume we would like to analyze the pendulum's response to oscillatory excitation. We have already established the transfer function of the pendulum linearized at $(0, 0)$ in Equation (3.13) as

$$G(s) = \frac{0.2}{s^2 + s + 2}. \quad (6.6)$$

To graphically find the magnitude and phase shifts as a function of excitation frequency, a Bode plot is to be constructed.

First, we can extract useful information by writing the system in a standard form. Since we have a second order system present, we will rearrange Equation (6.6) to take the form

$$G(s) = k_{\text{Bode}} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \quad (6.7)$$

Taking the given system parameters into account, we write $G(s)$ as

$$G(s) = 0.1 \frac{2}{s^2 + s + 2},$$

with $k_{\text{Bode}} = 0.1$, $\omega_n = \sqrt{2}$ and $\zeta = \frac{1}{2\sqrt{2}}$.

With this information about the system, we can start constructing the Bode plot. We will do so by looking at different points on the plot and the contributions of both the Bode gain k_{Bode} as well as the poles. Finally, we find the resonance spike through the damping coefficient.

1. **Bode gain:** The Bode gain is found by evaluating $G(0)$, which results in

$$G(0) = k_{\text{Bode}} = 0.1.$$

From this we find

$$|G(0)| = 0.1 = -20\text{dB}, \quad \angle G(0) = 0^\circ.$$

Therefore, a straight line is drawn at -20dB on the magnitude plot since the Bode gain remains unchanged for all frequencies. The phase experiences no shift for this frequency, from which we conclude that the phase plot starts at 0° .

2. **Pole contribution:** For systems of 2nd order form as shown in Equation (6.7), we have that the amplitude starts decreasing by $-40 \frac{\text{dB}}{\text{dec}}$ ($-20 \frac{\text{dB}}{\text{dec}}$ per pole) at the natural frequency ω_n which corresponds to the absolute value of the stable complex conjugate poles. Additionally, the phase drops by -180° , around ω_n . The frequency range over which the phase drops is approximated by the range $[0.1\omega_n, 10\omega_n]$.

3. Resonance Spike: The resonance spike occurs at the resonance frequency ω_r , which is calculated as

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} \approx 1.225.$$

At this frequency, the plot rises by a maximum value of

$$-20 \log_{10}(2\zeta \sqrt{1 - \zeta^2}) \approx -3.59 \text{dB}.$$

With this information, we can now construct an approximation of the Bode plot as shown in Figure 6.16. In that specific figure, the dashed lines represent the individual elements from which an approximation can be constructed. The true Bode Plot of the system is shown in blue.

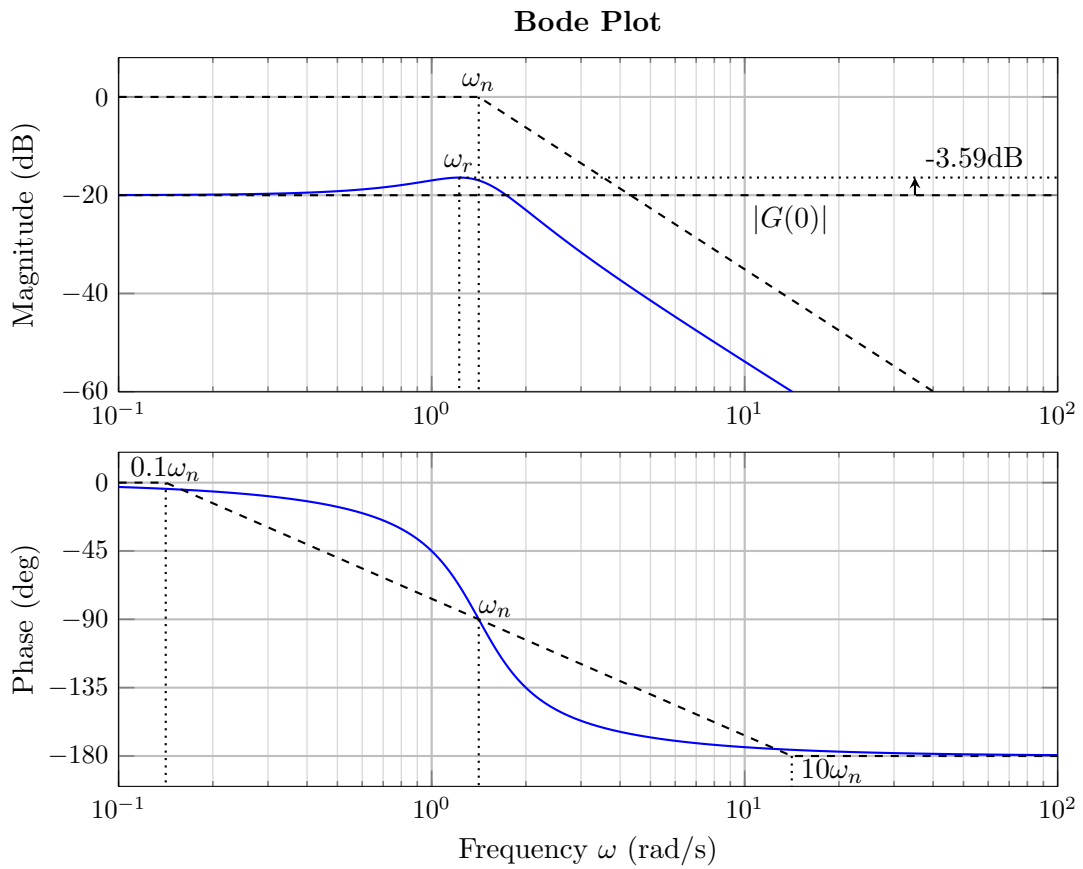


Figure 6.16: Bode plot of the Pendulum, with sketching aid lines.

First, we will construct the magnitude plot: We begin by drawing a straight line representing the Bode gain k_{Bode} . Then, a second line at 0dB up to the natural frequency ω_n , from where on we continue with the evaluated decrease of $40 \frac{dB}{dec}$. This line represents the effect of the complex pole. Due to the logarithmic nature of the plot, we can add up the dashed lines to arrive at a approximate trend line for the magnitude plot. To take into account resonance effects, we raise the magnitude plot at the resonance frequency ω_n by the evaluated amount. Secondly, the phase plot starts out at 0° , as neither integrators nor

differentiators are present. A pair of stable complex conjugate poles results in a phase shift of -180° . To sketch the frequency range over which the phase shift occurs, we approximate the beginning of the shift at $0.1\omega_n$ and return to a constant phase at around $10\omega_n$. Both plots can now be approximated by drawing a smooth line through the evaluated segments.

As stated before, you will not have to draw the exact Bode plots completely accurately by hand, as there exists many fast computational tools. Especially details like the precise increase at the resonance spike and the steepness of the phase shift are not important to be able to calculate and draw by hand.

The Bode plot now allows us to analyze the frequency response of the pendulum around the linearization point. We can conclude that up prior to ω_r , the system attenuates inputs by -20 dB and no significant phase shift occurs. At the resonance frequency ω_r , the least attenuation occurs. The height of the resonance spike is relatively small, from which we can conclude that strong damping is present. Here, this corresponds to a comparatively large friction coefficient. Further increasing excitation frequency leads to stronger attenuation of the oscillation. The pendulum is ‘too slow’ to react to the fast oscillations. Additionally, the phase starts to lag behind the input and eventually reaches a full shift of π . This general behavior is common for physical systems and referred to as having ‘2nd order low-pass filter’-characteristics.

6.3 Polar plot

The Bode plot showed the phase and magnitude of $G(j\omega)$ as a function of ω on two separate plots. Since the frequency is explicit in this plot, it is very useful to analyze the effects that specific frequencies have on the system. This makes Bode plots an accessible tool for the first steps of system analysis, eventually leading to controller design. On the other hand, the Bode plot has the disadvantage that it can result in wrong conclusions, e.g., when considering stability of open-loop unstable systems or systems with non-minimum phase zeros. Thus, a more reliable way to portray steady-state behavior is desired. To achieve this, we introduce the polar plot which depicts the phase and magnitude as a parametric curve in the complex plane where ω is implicit. Even though this is just another way of representing $|G(j\omega)|$ and $\angle(G(s))$, the polar plot will be fundamental in understanding a fundamental criterion used to assess stability of a closed-loop system as discussed in Section 6.4.3.

6.3.1 Polar Plot Construction

Instead of showing the magnitude $|G(j\omega)|$ and the phase $\angle G(j\omega)$ on two separate plots, the polar plot takes the magnitude and phase for a given ω and represents it as a point in the complex plane. Conveniently we can then attribute the magnitude of the resulting complex number to be $|G(j\omega)|$ and the phase to be $\angle G(j\omega)$. For ease of notation we write $\rho = |G(j\omega)|$ as the radial coordinate and $\phi = \angle G(j\omega)$ as the angular coordinate, and represent the complex

values as a vector with length ρ and angle ϕ . These vectors are also referred to as *phasors*. To obtain the polar plot we now plug in all possible frequencies $\omega \in [0, +\infty)$, and plot the resulting trajectory. Due to the frequency being implicit, it cannot be inferred from the plot. Therefore, the polar plot only shows us the general trajectory of the system along an increasing frequency in the complex plane.

To show the construction of the Polar plot for a general system, let us look at an example. To that end consider the plant

$$P(s) = \frac{1}{s+1}. \quad (6.8)$$

By substituting different values of ω into Equation (6.8), we can then obtain the corresponding values of ρ and ϕ . Figure 6.17 illustrates this for three different values of ω plotting the respective phasors in the complex plane. We can see the phasors of the plant taking different directions and lengths depending on the frequency the system is excited with. For higher frequencies, the phasors become shorter and turn clockwise. We can interpret this as the output having a smaller amplitude and larger phase shift for higher frequencies.

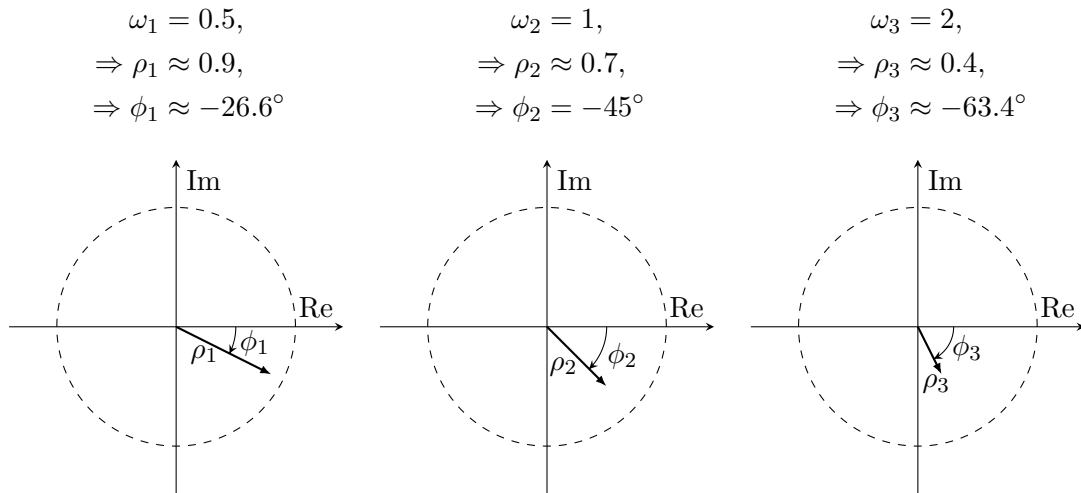


Figure 6.17: Phasors of $P(s)$ as described in Equation (6.8) for three different frequencies $\omega \in \{0.5, 1, 2\}$. The unit circle is represented with the dashed line.

By plotting all magnitudes and phases for all frequencies $\omega \in [0, +\infty)$ we will end up with a continuous curve. Figure 6.18 shows the resulting plot for the example from above. This plot lets us extract useful information about the plant. For example, since the Polar plot arc is fully contained within the unit circle, we can conclude that the plant reduces the input signal amplitude for all frequencies. Similarly, we find that for increasing frequency, the amplitude of the response is continuously dampened. For $\omega \rightarrow \infty$, the polar plot approaches the origin, from which we conclude that output becomes zero. Regarding phase, we find that the plant increasingly shifts from 0° for $\omega \rightarrow 0$ towards -90° for $\omega \rightarrow \infty$. Therefore, the maximum phase shift that the plants output can have in comparison to the input, is $-\pi/2$.

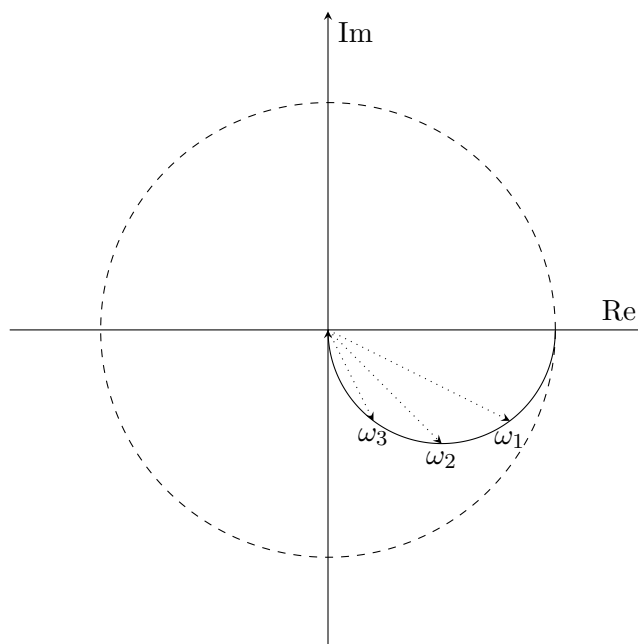


Figure 6.18: Full polar plot of $P(s)$ from Equation (6.8). To clarify construction, the samples from Figure 6.17 are included as dotted arrows. Note that the directionality of the plot is indicated to show in which direction of the plot ω approaches $+\infty$.

6.3.2 From the Bode Plot to the Polar Plot

Recall that the Bode plot shows the magnitude and phase as an explicit function of ω , allowing us to read off the values for ρ and ϕ all frequencies. Most often, it will prove simpler to first construct a Bode plot of a system and then derive a polar plot from it. In this subsection, the procedure is demonstrated for cases where the Bode plot is already given.

Consider the example in Figure 6.19, which relates the Bode plot of Equation (6.8) to its polar plot. The same sample frequencies from before ($\omega_{1,2,3} \in \{0.5, 1, 2\}$) are shown as dashed lines in both the magnitude and phase plots. We find the magnitudes at the different frequencies as the phasor lengths ρ_i and the phases as the phasor angles ϕ_i .

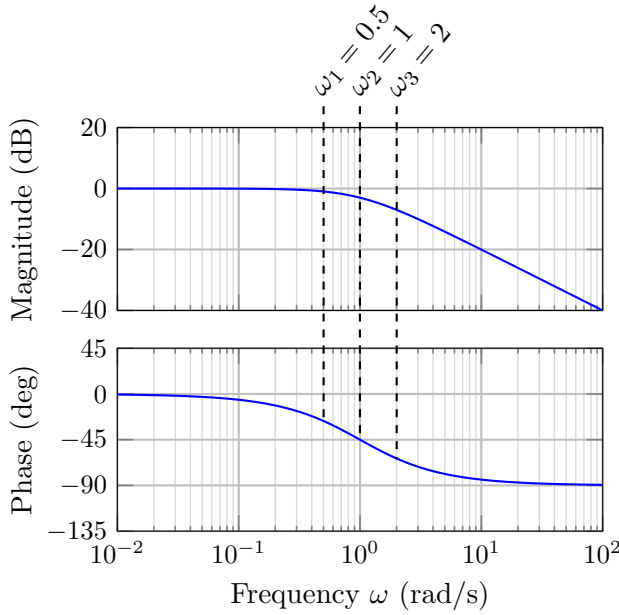


Figure 6.19: Bode plot of the Pendulum with samples at $\omega \in \{0.5, 1, 2\}$.

Reading from the magnitude and phase plots we find approximately:

$$\omega_1 = 0.5 : \quad \rho_1 \approx -0.97\text{dB} \approx 0.9 \\ \phi_1 \approx -26.6^\circ$$

$$\omega_2 = 1 : \quad \rho_2 \approx -3.0\text{dB} \approx 0.7 \\ \phi_2 \approx -45^\circ$$

$$\omega_3 = 2 : \quad \rho_3 \approx -7.0\text{dB} \approx 0.4 \\ \phi_3 \approx -63.4^\circ$$

Additionally, we will consider the points on the Bode plot for very small and very large frequencies. Here we find the following results:

- $\omega_0 = \omega \rightarrow 0$: The magnitude plot approaches 0 dB, which corresponds to a ρ of 1. The phase plot shows 0° . Therefore, the polar plot must start at 1 on the real axis.
- $\omega_\infty = \omega \rightarrow \infty$: The magnitude plot approaches $-\infty$ dB, which corresponds to $\rho = 0$. The phase plot approaches -90° . Thus, the polar plot approaches the origin with an angle of -90° .

As shown in Figure 6.20a we can mark these points on the complex plane and sketch the polar plot by fitting a curve through all points. In this case the five points suffice to draw the Polar plot as shown in Figure 6.20b.

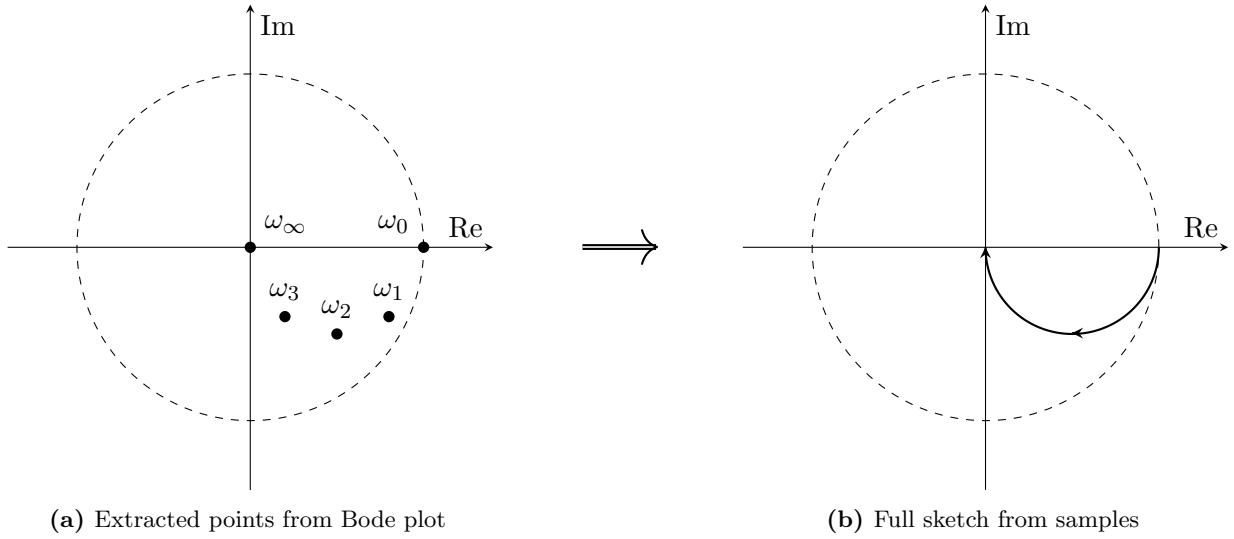
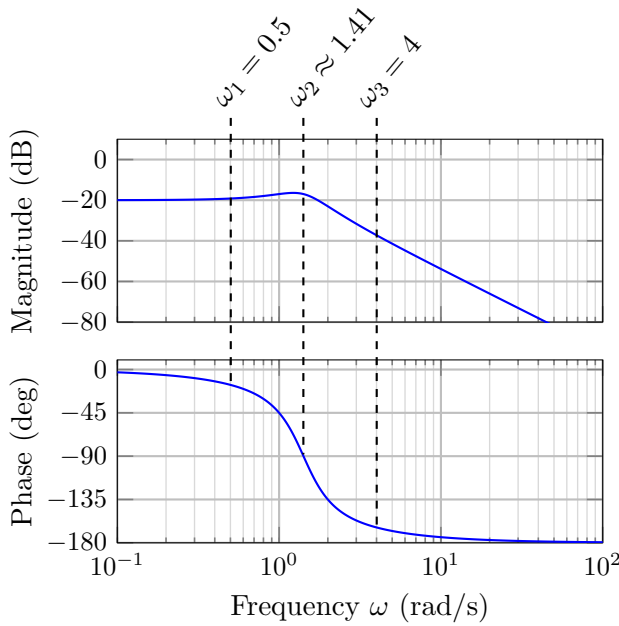


Figure 6.20: The polar plot of the system is constructed by connecting the points extracted from the Bode plot.

Example - Polar Plot of Pendulum

Previously, we constructed the Bode plot of the pendulum. From this, we can now derive the polar plot by strategically sampling characteristic points and plotting them on a complex plane. For reference, the Bode plot is shown again in Figure 6.21. To use the Bode plot to find the approximate structure of the Polar plot, we sample points and plot them on the complex plane. Three points are chosen such that the plots general behavior is captured.



Reading three samples from magnitude and phase plots we find approximately:

$\omega_1 = 0.5 :$	$\rho_1 \approx -19.18\text{dB} \approx 0.11$
	$\phi_1 \approx -15.95^\circ$
$\omega_2 = \omega_n \approx 1.41 :$	$\rho_2 \approx -16.99\text{dB} \approx 0.14$
	$\phi_2 \approx -90^\circ$
$\omega_3 = 4 :$	$\rho_3 \approx -37.24\text{dB} \approx 0.01$
	$\phi_3 \approx -164.06^\circ$

Figure 6.21: Bode plot of pendulum with samples at $\omega \in \{0.5, \sqrt{2}, 4\}$.

To find the behavior for very small and very large frequencies, we find the magnitude and phase for the extreme cases:

$$\begin{aligned}\omega_0 : \quad \rho_0 &= -20 \text{ dB} = 0.1, & \phi_0 &= 0^\circ, \\ \omega_\infty : \quad \rho_\infty &= -\infty \text{ dB} = 0, & \phi_\infty &= -180^\circ.\end{aligned}$$

With the gathered information, we can proceed to construct the polar plot of the pendulum as shown in Figure 6.22. We find that the plot starts out on the real axis, indicating that initially there is no phase shift. Increasing omega leads to a negative angle as well as an initial increase of the distance to the origin, which corresponds to the magnitude spike at the resonance frequency. As soon as the negative imaginary axis is crossed magnitude starts to strongly decrease, up to the point where it approaches the origin from the negative real axis.

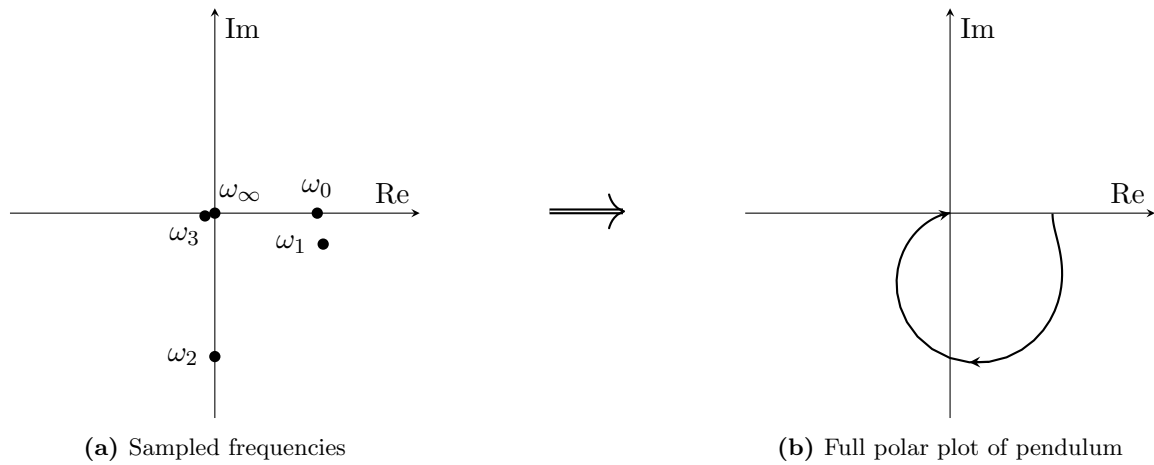


Figure 6.22: Sampled points plotted on the complex plane and connected to result in the polar plot of the pendulum.

6.4 Nyquist plot

6.4.1 Introduction to the Nyquist Plot

Previously introduced tools such as root locus and Bode plots are excellent for many tasks, yet each comes with scope limits for certain plants and controllers. For example, in plants with pure time delays or other irrational dynamics, where root locus isn't defined, and Bode margins can mislead, Nyquist still applies. In fact, Nyquist applies universally to linear time-invariant SISO systems: it lets us reason about closed-loop behavior using only the open-loop description, providing a direct link between frequency response and closed-loop behavior for a very large class of systems.

From frequency response to a parametric curve in the complex plane For a linear time invariant system with transfer function $P(s)$, the frequency response is obtained by evalu-

ating $P(j\omega)$ on the imaginary axis, where $\omega \in \mathbb{R}$. This maps each sinusoidal input at frequency ω to a complex gain with magnitude $|P(j\omega)|$ and phase $\angle P(j\omega)$. The Nyquist plot takes these complex values and places them as a parametric curve in the complex plane.

Relation to other frequency-domain plots We already saw two graphical representations of frequency response:

- **Bode plot:** Magnitude and phase are shown as separate functions of frequency (usually on a logarithmic scale), from which we can observe trends, low/high-frequency limits and asymptotes. However, pairing magnitude and phase at the exact same ω requires cross-referencing the two charts.
- **Polar plot:** The complex path of $P(j\omega)$ for $\omega \geq 0$ by common convention. It shows the magnitude and phase relationship directly, but unless explicit frequency labels are added it is harder to associate a point with a specific ω . *Omitting the negative-frequency branch and a traversal direction introduces ambiguity:* The conjugate symmetry of real LTI systems is hidden, and the sequence in ω in which the curve crosses reference lines (e.g., real axis, -180° ray) cannot be determined, so the evolution of the response with frequency is unclear.

Like the polar plot, the Nyquist diagram shows the complex path of $P(j\omega)$ and thus ties magnitude and phase together at each frequency. What distinguishes the Nyquist diagram is that it traces the *full* sweep $\omega : -\infty \rightarrow \infty$ with a prescribed orientation (and detours if needed), yielding a closed, oriented curve that is conventionally annotated with direction and key frequencies and that highlights important points, which we will introduce in Subsection 6.4.3. This makes it straightforward to associate points with specific ω and to relate geometric features of the curve to closed-loop properties, while remaining an open-loop construction.

Why do we include negative frequencies? As already mentioned, the Nyquist plot is defined over the full sweep $\omega \in (-\infty, \infty)$. There are two complementary reasons:

1. **Completeness of the mapping** The Nyquist construction conceptually maps a closed contour in the complex s -plane into the complex $P(s)$ -plane. Traversing both $\omega \geq 0$ and $\omega < 0$ corresponds to tracing both directions along the imaginary axis, which closes the contour and preserves the correct orientation at high frequency.
2. **Conjugate symmetry for real systems** For real values of P , $P(-j\omega) = \overline{P(j\omega)}$. Including negative ω therefore makes the Nyquist plot symmetric about the real axis. This symmetry is not merely aesthetic; it ensures the plotted curve reflects the full frequency response of the real system.

Open loop in service of closed loop Control design often infers closed-loop properties from open-loop information to avoid solving for closed-loop poles directly. The Nyquist plot is built

from the open-loop transfer function $L(s) = C(s)P(s)$, yet its geometry encodes precisely the features needed to reason about closed-loop stability and robustness at a glance. More details on this will follow in subsection 6.4.3.

6.4.2 Drawing the Nyquist Plot

The *Polar plot* traces the Bode plot of $G(j\omega)$ for $\omega \in [0, \infty)$ on the complex plane, showing magnitude and phase together while leaving the frequency implicit. The *Nyquist plot* is nothing more than a completion of this idea into a *closed, oriented* curve suitable for stability tests:

1. Start with the Polar plot: trace $G(j\omega)$ for $\omega : 0 \rightarrow \infty$ (positive branch).
2. Add the negative-frequency branch. For real-coefficient systems, $G(-j\omega) = \overline{G(j\omega)}$, so the Nyquist curve mirrors the positive branch about the real axis and is traversed in the reverse direction.
3. Handle any poles on the $j\omega$ axis by inserting a small right-half-plane detour in the s -plane; this maps to a corresponding arc in the G -plane.
4. Include the limiting behavior as $\omega \rightarrow 0$ and $\omega \rightarrow \infty$ to close the contour and annotate the curve orientation.

End behavior ($\omega \rightarrow \infty$)

Let $G(s) = \frac{N(s)}{D(s)}$ with $\deg N = n$ and $\deg D = m$. As $\omega \rightarrow \infty$,

$$G(j\omega) \sim \frac{c_n(j\omega)^n}{a_m(j\omega)^m} = \frac{c_n}{a_m} (j\omega)^{n-m}$$

so

$$|G(j\omega)| \sim \left| \frac{c_n}{a_m} \right| \omega^{n-m} \quad \angle G(j\omega) \sim (n-m) \cdot 90^\circ + \angle \left(\frac{c_n}{a_m} \right)$$

This tells you whether the Nyquist curve ends at the origin, at a nonzero finite point, or at infinity, and *from which angle* it approaches.

Mini-example: Nyquist for pendulum

Starting from the measured/simulated polar samples of the pendulum $G(j\omega)$ for $\omega \geq 0$, form the Nyquist *positive-frequency branch* by mirroring across the real axis, i.e., $(\text{Re}, \text{Im}) \mapsto (\text{Re}, -\text{Im})$.

For the pendulum model the relative degree is $r = 2$ (two more poles than zeros) and the leading coefficients are real with $\frac{c_n}{a_m} > 0$. Therefore, the high-frequency limit is

$$G(j\omega) \sim \frac{c_n}{a_m} (j\omega)^{-2}, \quad \implies \quad |G(j\omega)| \xrightarrow{\omega \rightarrow \infty} 0, \quad \angle G(j\omega) \xrightarrow{\omega \rightarrow \infty} -180^\circ,$$

i.e., the curve approaches the origin along (just above) the negative real axis at an angle very close to 180° .

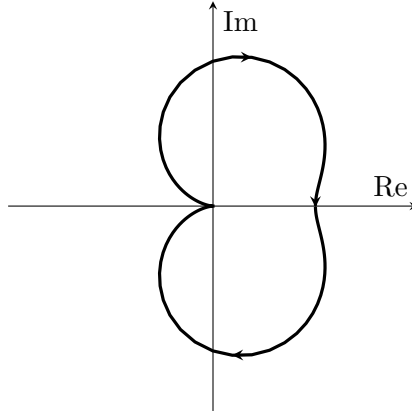


Figure 6.23: Nyquist plot of the pendulum. The positive branch is below the real axis, while the negative branch is mirrored to be above the real axis with arrows indicating increasing ω . Notice that the head of the positive branch approaches the origin along the negative real axis, consistent with $r = 2 \Rightarrow \angle G(j\infty) = -180^\circ$.

Special cases and quick rules for the endpoints

- *Poles on the imaginary axis (including $s = \pm j\omega_0$):* remove a small interval around ω_0 and connect the endpoints with a detour arc corresponding to a small RHP semicircle in the s -plane. The orientation of this arc follows the contour direction (upwards on $j\omega$ for $\omega > 0$).
- *Integrators (k poles at the origin):* near $\omega \rightarrow 0$, $|G(j\omega)| \sim \omega^{-k} \rightarrow \infty$ and $\angle G(j\omega) \rightarrow -k \cdot 90^\circ$ (for $k > 0$). You must detour around $s = 0$ in the RHP; in the G -plane this appears as a large arc connecting the two branches.

Example. Let

$$G(s) = \frac{(s+1)^2}{s+1}.$$

Here $\deg N = 2$, $\deg D = 1$, so the relative degree is $r = m - n = -1 < 0$ and $|G(j\omega)| \xrightarrow{\omega \rightarrow \infty} \infty$. A short calculation shows the frequency response collapses to a line:

$$G(j\omega) = \frac{(1 - \omega^2) + j2\omega}{1 + j\omega} = \frac{(1 - \omega^2) + j2\omega}{1 + j\omega} \cdot \frac{1 - j\omega}{1 - j\omega} = \frac{(1 + \omega^2) + j\omega(1 + \omega^2)}{1 + \omega^2} = 1 + j\omega$$

Thus the polar/Nyquist branch for $\omega \geq 0$ is the vertical ray at $\text{Re} = 1$, going *up* from $(1, 0)$ to $1 + j\infty$. The negative-frequency branch is its mirror, going to $1 - j\infty$. Because the two ends are at $\pm j\infty$, the Nyquist curve is closed by an *arc at infinity* that connects $1 + j\infty$ to $1 - j\infty$. (Geometrically, on the standard Nyquist contour the big right-half-plane semicircle $s = Re^{j\theta}$, $\theta : \frac{\pi}{2} \rightarrow -\frac{\pi}{2}$)

In short: Nyquist = Polar plot (for $\omega > 0$) + its conjugate mirror (for $\omega < 0$) + prescribed detours and orientation.

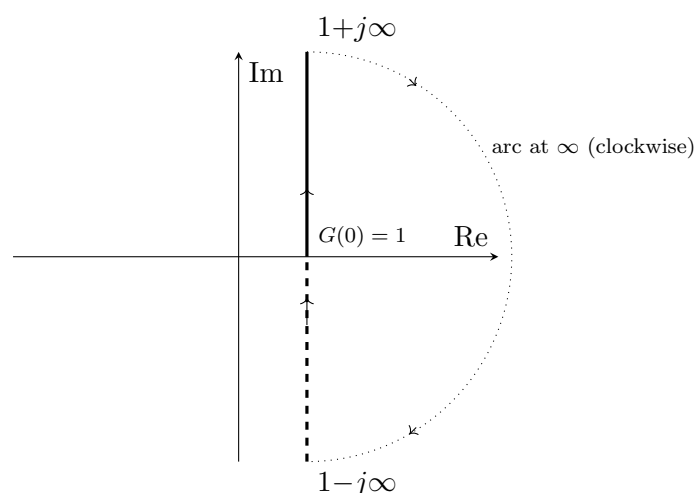


Figure 6.24: Case $G(s) = \frac{s^2+1}{s+1}$: the Nyquist branches run to $\pm j\infty$ at $\text{Re} = 1$, so the contour is closed by an *arc at infinity*. Solid = $\omega : 0 \rightarrow \infty$; dashed = mirror branch.

6.4.3 Nyquist Stability Criterion

In order to assess the stability of the closed loop system using the Nyquist plot we need to take a small detour into complex analysis. More precisely we will need to understand the *principle of variation of the argument*, sometimes also referred to as *Cauchy's argument principle*. We won't prove it but instead try to develop an intuition for it and see how it helps us understand stability in the Nyquist plot.

Principle of Variation of the Argument

To gain some intuition for this theorem consider some transfer function $G(s) = s + 2$, with one zero at $s = -2$. As we previously learned, this transfer function can be depicted in the s -plane by indicating its poles and zeros with crosses and circles respectively. Now let's observe what happens when we plug in specific values of s into $G(s)$, e.g., $s_0 = -1 + j$ resulting in

$$G(-1 + j) = 1 + j,$$

which is another complex number. Next we will plot this resulting complex number in a new plane and call it the $G(s)$ -plane since it contains the points obtained by evaluating $G(s)$ at some specific value $s = s_0$. We can now refer to the transfer function $G(s)$ as a *map* between the s - and $G(s)$ -plane since it maps each point s_0 in the s -plane to its counterpart $G(s_0)$ in the $G(s)$ -plane. We can further extend this mapping and map continuous curves — which are essentially made up of infinitely many points. Particularly interesting are closed curves — curves where the initial and final point coincide — that describe a bounded, simply connected region² D in the complex plane. We will then define a *contour*, denoted Γ , as the curve obtained by travelling along the boundary of D in the clockwise direction. In short, a contour Γ is a closed oriented curve in the complex plane. You can think about the mapping of such a contour

²Simply connected in this context means that the region has no holes in its interior.

as mapping all the individual points that make it up. Figure 6.25 illustrates such a mapping of a contour Γ from the s -plane to the transformed contour $\tilde{\Gamma}$ in the $G(s)$ -plane. In particular, we see that the contour Γ goes through the point $s_0 = -1 + j$ which gets mapped to the point $G(s_0) = 1 + j$ — confirming the calculation from above. An important observation is that a closed continuous curve (contour) in the s -plane corresponds to a closed continuous curve (contour) in the $G(s)$ -plane, and this holds true for all contours in the s -plane.

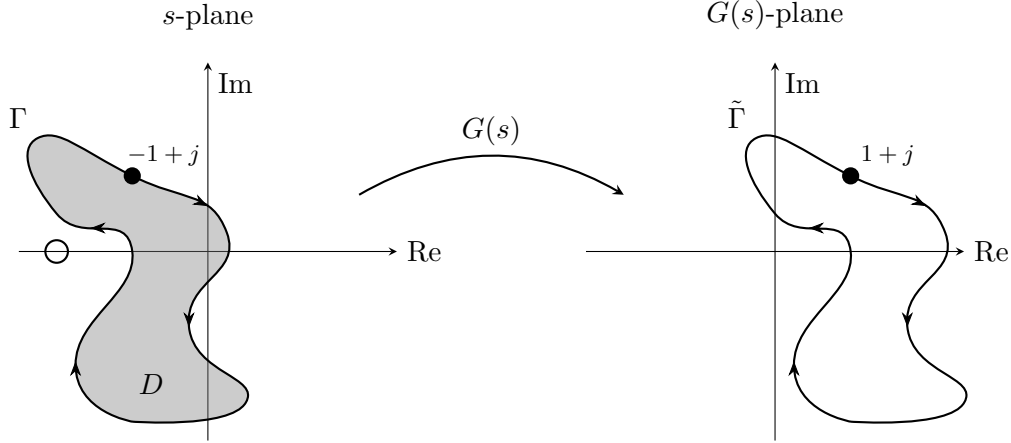


Figure 6.25: Mapping of contour Γ between the s -plane and $G(s)$ -plane given by the transfer function $G(s) = s + 2$.

Next let's explore why the mapping of contours might be of interest to us. Remember that the mapping is described by some transfer function $G(s)$. Consequently, if we are given some contour Γ in the s -plane and its counterpart $\tilde{\Gamma}$ in the $G(s)$ -plane, we should be able to extract some information about the transfer function from them. To see what information we can gain from the contours, let's first consider once more the case of mapping a single point and then, since contours are made up of infinitely many points, generalize the learnings to contours.

We begin by examining the phase of a mapped point $G(s_0)$. As a concrete example we will use the transfer function

$$G(s) = \frac{s + 2}{s + 1} = \frac{s - z_1}{s - p_1}.$$

We can evaluate the phase of any point $G(s_0)$ that is mapped by this transfer function, by evaluating the phase $\angle(G(s_0))$. In our case this case can be expressed as

$$\angle(G(s_0)) = \angle\left(\frac{s_0 - z_1}{s_0 - p_1}\right) = \angle(s_0 - z_1) - \angle(s_0 - p_1). \quad (6.9)$$

From this expression we see that the phase of the mapped point can be expressed as a function of the locations of the poles and zeros of the transfer function $G(s)$. More specifically we recognize that zeros add to the phase and poles subtract from the phase. Figure 6.26 shows a graphical interpretation of this result where the point $s_0 = -1 + j$ gets mapped to $G(s_0) = 1 - j$. Here we can also see that by adding the angle of the line between the zero z_1 and s_0 , and subtracting the angle of the line between the pole p_1 and s_0 , we get the angle of the phasor of $G(s_0)$, i.e. $\angle(G(s_0)) = 45^\circ - 90^\circ = -45^\circ$. By generalizing the expression from Equation (6.9) we can in

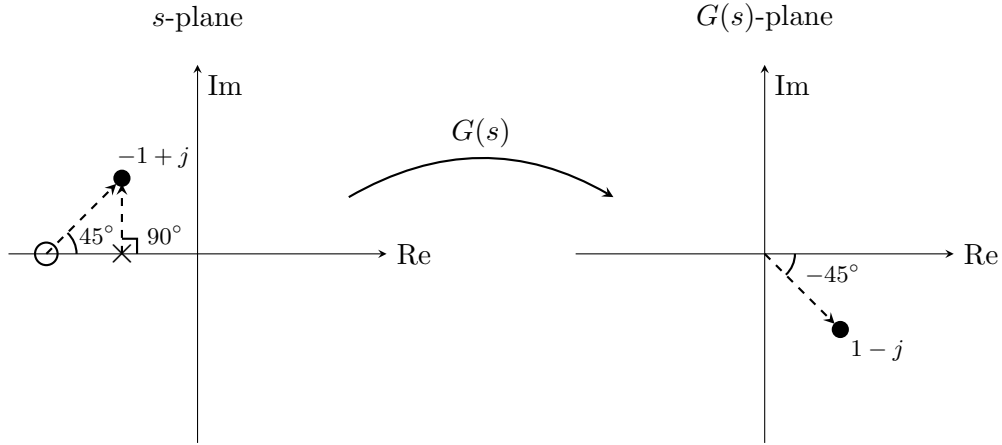


Figure 6.26: Mapping of point $s_0 = -1 + j$ from the s -plane to the $G(s)$ -plane described by the transfer function $G(s) = \frac{s+2}{s+1}$.

fact evaluate the phase of any point mapped by some arbitrary transfer function with m zeros and n using

$$\angle(G(s)) = \sum_m \angle(s - z_i) - \sum_n \angle(s - p_i). \quad (6.10)$$

With this in mind we can once more turn our attention to contours where it will be especially interesting to understand what happens if the contour contains the poles or zeros of the transfer function within itself. Take the example depicted in Figure 6.27 where some contour Γ is mapped by $G(s) = 1/(s + 1)$ where notably Γ does not contain the pole of $G(s)$ in its interior. In this case we can see that the angles of the lines between all possible points s_i on the contour Γ , and the pole p of the transfer function ($\angle(s_i - p)$), will never exceed $\pm 90^\circ$. In return, the phase $\angle(G(s_i)) = -\angle(s_i - p)$ (Equation (6.10)) can never exceed $\pm 90^\circ$. In other words, the mapped contour $\tilde{\Gamma}$ can not contain the origin in its interior. Recall that we defined the contour Γ as

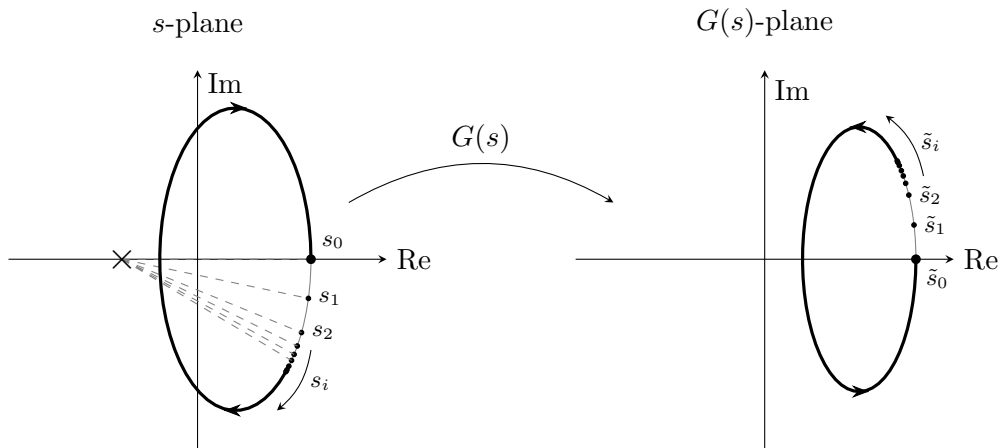


Figure 6.27: Mapping of contour Γ from the s -plane to the $G(s)$ -plane described by the transfer function $G(s) = \frac{1}{s+1}$.

going in the clockwise (CW) direction around some region D . To emphasize this directionality of the contour we usually say that a contour *encircles* a point instead of containing it.

Now we can take the same transfer function $G(s) = 1/(s + 1)$ and move the contour Γ such that it encircles the pole (see Figure 6.28). By applying similar logic as above, we can see that the angles of the lines between all possible points s_i on the contour Γ , and the pole p of the transfer function ($\angle(s_i - p)$), will vary between $0^\circ - 360^\circ$. In return, the phase $\angle(G(s_i)) = -\angle(s_i - p)$ (Equation (6.10)) will also vary between $0^\circ - 360^\circ$. In other words, the mapped contour $\tilde{\Gamma}$ must encircle the origin once. In this example we can even determine the direction in which

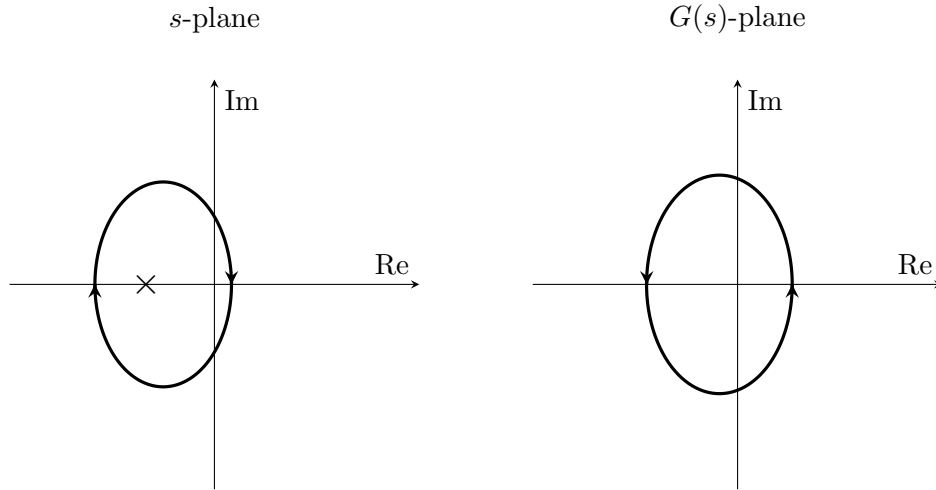


Figure 6.28: Mapping described by the transfer function $G(s) = \frac{1}{s+1}$ between the s -plane and $G(s)$ -plane.

the mapped contour $\tilde{\Gamma}$ encircles the origin. Recall that $\angle(G(s_i)) = -\angle(s_i - p)$, which means that if the contour Γ encircles the pole CW, then $\tilde{\Gamma}$ must encircle the origin counter-clock wise (CCW). We can also add another pole to the transfer function and observe in Figure 6.29 that by encircling two poles CW in the s -plane we will lead to two CCW encirclements of the origin in the $G(s)$ -plane.

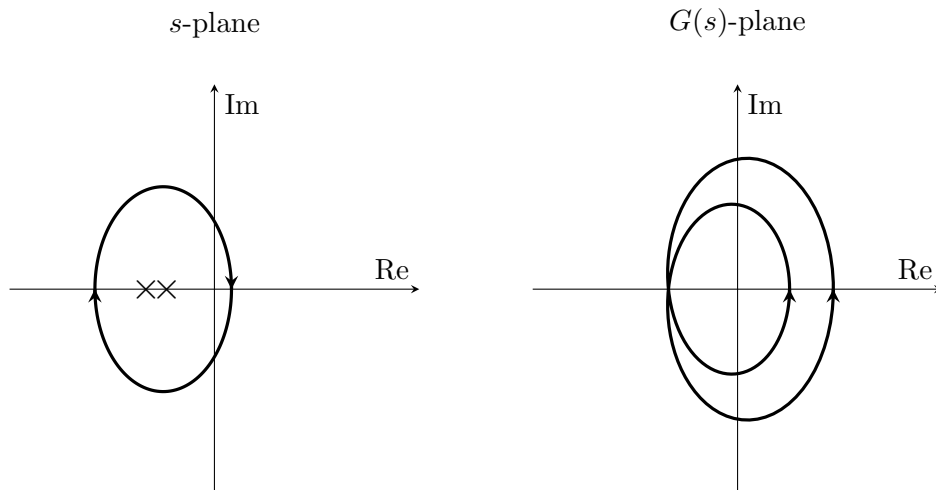


Figure 6.29: Mapping described by the transfer function $G(s) = \frac{1}{(s+1)(s+0.8)}$ between the s -plane and $G(s)$ -plane.

The same idea can also be applied to transfer functions containing zeros. With the difference that now for each CW encirclement of a zero in the s -plane, we will have one CW encirclement

of the origin in the $G(s)$ -plane.

In summary, each zero in the contour Γ adds a CW encirclement of the origin in the $G(s)$ -plane and each pole in Γ adds a CCW encirclement of the origin in the $G(s)$ -plane.

Consequently, if the contour contains as many poles as zeros we end up with no encirclements, i.e. they cancel out. If there are $n - m$ more poles than zeros in Γ then we will have $n - m$ CCW encirclements of the origin. Conversely, if there are $m - n$ more zeros than poles we end up with $m - n$ CW encirclements of the origin. Further examples are shown in Figure 6.30. With

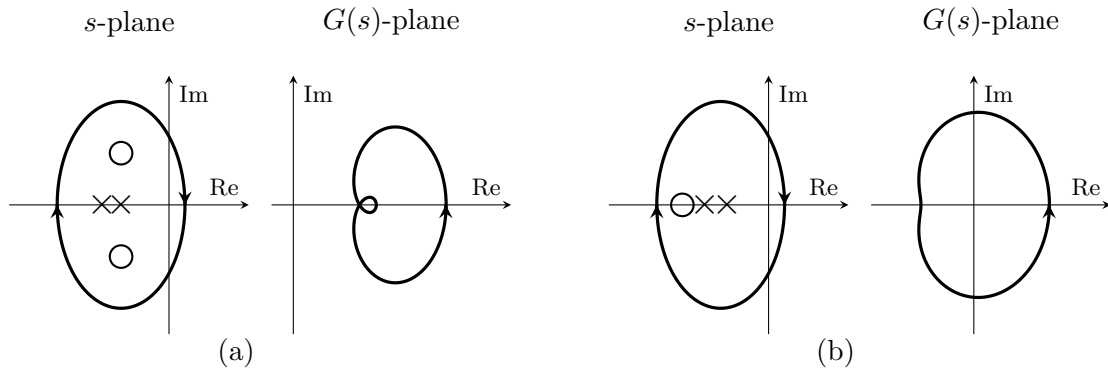


Figure 6.30: (a) Mapping of contour Γ described by transfer function $G(s) = \frac{s^2 + 1.5s + 0.8125}{(s+1)(s+0.8)}$.
 (b) Mapping of contour Γ described by transfer function $G(s) = \frac{(s+1.2)}{(s+1)(s+0.8)}$.

this we now have a way to tell the relative difference of poles and zeros inside a given contour Γ by counting how many times the mapped contour encircles the origin and in which direction. Formally this is given by the following theorem.

Theorem 6.1 (Variation of the Argument) *The number N of times that $G(s)$ encircles the origin of the complex plane as s moves along the contour Γ of a bounded simply-connected region D of the plane satisfies*

$$N = Z - P \quad (6.11)$$

where Z and P are the numbers of zeros and poles of $G(s)$ in D , respectively. Note that the encirclements are counted positive if in the same direction as s moves along Γ , and negative otherwise.

Nyquist Stability Criterion

Let us now use this principle of the variation of the argument to get some insights to the stability of a given closed-loop system. Recall that for an open-loop system we can check stability by looking at the poles of the open-loop transfer function $L(s)$. In particular, if there are any poles in the right half-plane (RHP) we can conclude that the open-loop system is unstable. If we now introduce some feedback, i.e. we close the loop, we have to check the poles

of the closed-loop transfer function

$$T(s) = \frac{L(s)}{1 + L(s)}$$

instead. This means finding all values of s where the denominator of $T(s)$ is zero, i.e. we need to evaluate the expression $1 + L(s) = 0$. We can now see that the stability of the closed-loop can be verified by checking if $1 + L(s)$ has any *zeros* in the RHP³.

With this in mind, we can now leverage the knowledge from the principle of the variation of the argument and choose the contour Γ to encircle the entire RHP of the s -plane. Such a contour is also referred to as the Nyquist- or D-contour. In fact if we choose our contour to be the Nyquist-contour and map all its points using some transfer function $G(s)$, the resulting plot will be equal to the Nyquist-plot of $G(s)$. Figure 6.31 illustrates the Nyquist-contour in the s -plane. By choosing the contour like this and using the principle of variation of the argument

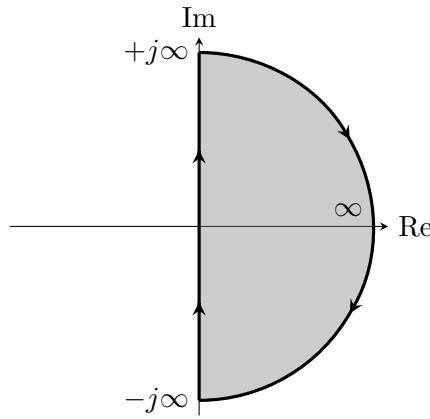


Figure 6.31: Nyquist-contour shown in the s -plane

we can now tell the relative difference of poles and zeros of $G(s)$ in the RHP by counting the number of encirclements (enc.) of the Nyquist-plot of $G(s)$ around the origin.

In particular, we can choose $G(s) = 1 + L(s)$ and use the resulting Nyquist-plot to give us some information about the zeros of $1 + L(s)$ in the RHP. More specifically we can use the principle of the variation of the argument to conclude that the number of enc. of the origin of the Nyquist plot of $1 + L(s)$ is equal to the number of zeros of $1 + L(s)$ in the RHP minus the number of poles of $1 + L(s)$ in the RHP. Per convention, we define a CW encirclement to be positive number and CCW encirclement to be negative number. In a shorter form this condition can be expressed as

$$\begin{aligned} & \# \text{ Enc. of origin of Nyquist plot of } 1 + L(s) \\ & = \\ & \# \text{ RHP zeros of } 1 + L(s) - \# \text{ RHP poles of } 1 + L(s) \end{aligned} \tag{6.12}$$

In what follows, we will analyze each component of this relationship and try to transform

³Since the zeros of $1 + L(s)$ are equal to the poles of $T(s)$

it into quantities related to the open-loop transfer function enabling once more the analysis of the closed-loop system by characteristics of the open-loop.

The first thing to recognize is that given the Nyquist-plot of the function $1 + L(s)$ we can find the Nyquist-plot of the function $L(s)$ by shifting the whole plot by -1 in the complex plane. This means that counting the encirclements of the Nyquist plot of $1 + L(s)$ around the origin is equal to counting the encirclements of the Nyquist plot of $L(s)$ around the point -1 .

$$\# \text{ Enc. of origin of Nyquist plot of } 1 + L(s) = \# \text{ Enc. of } -1 \text{ of Nyquist plot of } L(s). \quad (6.13)$$

This is very useful since we already know how to draw the Nyquist plot of the open-loop function $L(s)$.

Next let's consider the number of poles of $1 + L(s)$ in the RHP. Recall that the open-loop transfer function is a rational function that can be expressed as the ratio

$$L(s) = \frac{N(s)}{D(s)}.$$

Importantly the poles of $L(s)$ are now defined by the denominator $D(s)$. With this definition we can also write $1 + L(s)$ as

$$1 + L(s) = 1 + \frac{N(s)}{D(s)} = \frac{D(s) + N(s)}{D(s)},$$

which implies that $L(s)$ and $1 + L(s)$ have the same poles, i.e. all points where $D(s) = 0$. Notably we can conclude that the number of poles of $1 + L(s)$ in the RHP is equal to the number of poles of $L(s)$ in the RHP, i.e. the unstable open-loop poles. Thus, we come to the following conclusion.

$$\# \text{ RHP poles of } 1 + L(s) = \# \text{ Unstable open-loop poles} \quad (6.14)$$

For the last step in simplifying the initial expression from Equation (6.12), we need to think back to the closed-loop transfer function

$$T(s) = \frac{L(s)}{1 + L(s)},$$

and recall that the poles of the closed-loop system are given by evaluating $1 + L(s) = 0$, i.e. finding the zeros of $1 + L(s)$. This means that the zeros of $1 + L(s)$ are equal to the poles of the closed-loop system. In particular this means that the number of zeros of $1 + L(s)$ in the RHP is equal to the number of unstable closed-loop poles.

$$\# \text{ RHP Zeros of } 1 + L(s) = \# \text{ Unstable closed loop poles} \quad (6.15)$$

Finally, we can combine Equations (6.13) to (6.15) to simplify the initial expression (Equation (6.12)) and rearrange to finally arrive at the Nyquist stability criterion. It allows us to assess the stability of the closed-loop system by considering the open-loop poles and the Nyquist plot.

Nyquist Stability Criterion

$$Z = N + P \quad (6.16)$$

where Z is the number of unstable closed loop poles, N the number of enc. of -1 of the Nyquist plot of the open-loop transfer function, and P the number of unstable open-loop poles.

As we have seen in previous sections, we can also include some proportional feedback gain k in the system. Consequently, the closed-loop transfer function becomes

$$T(s) = \frac{kL(s)}{1 + kL(s)} = \frac{L(s)}{\frac{1}{k} + L(s)}.$$

Since we still want to express everything as a function of the open-loop transfer function $L(s)$, we need to slightly adapt the condition from Equation (6.13). Instead of counting the enc. of the Nyquist plot of $L(s)$ around -1 , we have to count the enc. around the point $-1/k$ to account for the feedback gain k ⁴. Note that the other two conditions are not affected by introducing the feedback gain k .

Importantly, this provides us with a convenient way to find a stabilizing feedback gain k given the Nyquist plot of $L(s)$. By using the Nyquist stability criterion, we can choose k such that the point $-1/k$ gets encircled $-P$ times forcing Z — the number of unstable closed-loop poles — to be zero.

Example - Nyquist Plot of Pendulum

For a more insightful analysis, let us consider the pendulum linearized around the unstable equilibrium. Furthermore, the polar plot of this system is given, which we can extend to a Nyquist plot by mirroring about the real axis as seen in Figure 6.32.

When linearized around the unstable equilibrium point, the transfer function contains two real-valued poles, one of which is unstable, i.e. $P = 1$. We have already seen in the Feedback chapter, that a stabilizing feedback gain exists. We can now reassure this discovery by using Nyquist' stability criterion. For stability, we need that the number of unstable closed-loop poles is $Z = 0$. Therefore, by Equation (6.16), we require that

$$N = Z - P = -1,$$

i.e. the Nyquist plot of $L(s)$ should have one CCW encirclement of the point $-\frac{1}{k}$. As the gain increases, it shifts this point towards the origin, eventually crossing the Nyquist

⁴By setting $k = 1$ we can recover the nominal case.

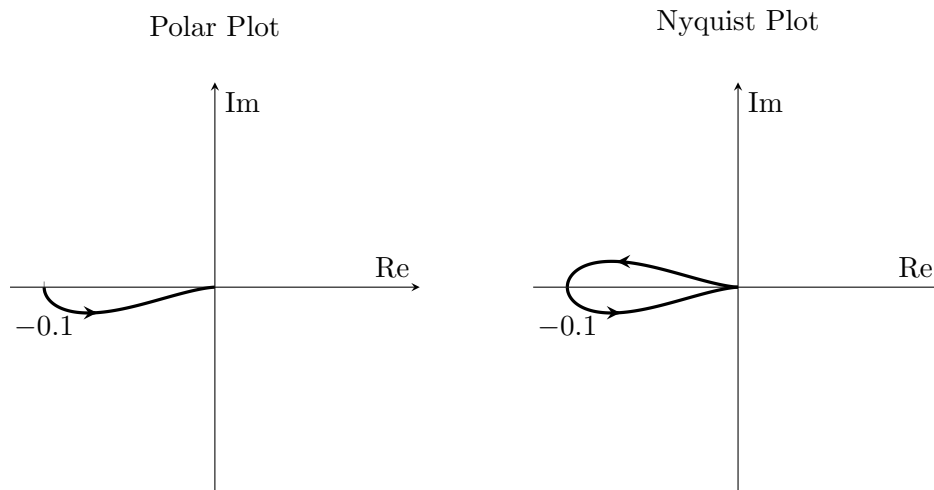


Figure 6.32: Polar and Nyquist plots of $L(s) = \frac{1}{s^2 + s - 10}$.

plot. In this configuration this results in one counter-clockwise encirclement of $-\frac{1}{k}$, thus stabilizing the closed-loop. Figure 6.33 shows the locations of the $-\frac{1}{k}$ point for various values of k .

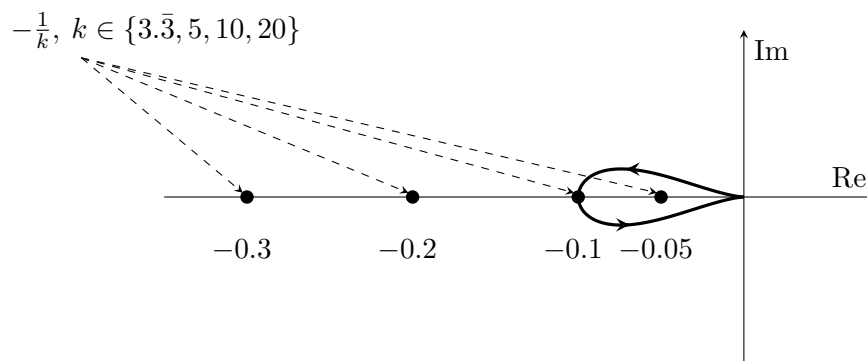


Figure 6.33: Nyquist plots of $L(s) = \frac{1}{s^2 + s - 10}$ with indicated locations of the feedback gain for various values.

As the Nyquist plot crosses the negative real axis at -0.1 , we can analytically find the point where the plant is stabilized by solving

$$\begin{aligned} -\frac{1}{k} &= -0.1, \\ \implies k &= 10. \end{aligned}$$

For a stable closed-loop, we therefore require that $k > 10$. This result corresponds to the stabilizing feedback gain found by analyzing the root locus plot in Section 4.4.4.

Chapter 7

Frequency Domain Specifications

7.1 Introduction

In the previous chapter, we saw how to visually interpret a system’s frequency response. This chapter will build on those visualizations and show how they can be useful for controller design.

Previously we saw that the output of an LTI system to some sinusoidal input, is also a sinusoid. Notably the output sinusoid will have the same frequency as the input while being scaled in magnitude and having a shift in phase. This scaling and shifting is completely defined through the transfer function’s magnitude and phase at the frequency of the input. This behavior is illustrated as a block diagram in Figure 7.1 below. Thus, we can fully characterize the frequency response by keeping track of the magnitude and phase of the transfer function as a function of the frequency.

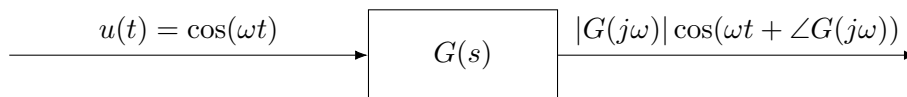


Figure 7.1: LTI system response to a sinusoidal input.

Considering how systems behave as a function of frequency can prove to be very useful when considering system specifications, since certain specifications are naturally described in the frequency-domain. Similar to what we saw for time-domain specifications (e.g. rise time, peak overshoot, etc.), we can use the tools we’ve learned in frequency response analysis to design a controller that will meet frequency-domain specifications.

As an example, sensors used to measure the output of our system corrupt the measurement with noise – which is usually quantified as high-frequency (“white”) noise. Since this noise is not desirable, we would like our controller to *attenuate* (reduce) the influence of noise on the closed-loop system. Specifically, we could require that all high frequencies, e.g. $\omega > 100$ rad/s, should be reduced in magnitude by a factor of 10. This would reduce the added high-frequency noise and minimize its propagation throughout the system. Using the fact that $T(s)$, the closed-

loop transfer function of $L(s)$, maps the noise to the output, we obtain the following frequency domain specification:

$$|T(j\omega)| < 0.1 = 20 \log_{10}(0.1) = -20 \text{ dB}, \quad \forall \omega > 100 \text{ rad/s}.$$

Note that while this requirement would be difficult to plot in time-domain, it nicely applies constraints on the frequency response of our system. As an example Figure 7.2 shows aforementioned condition on the Bode plot.

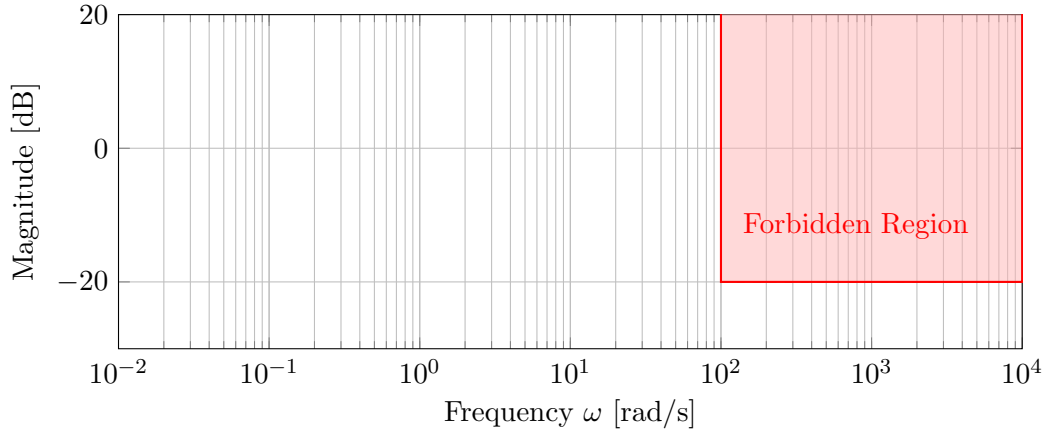


Figure 7.2: Bode plot of the requirement on the closed-loop transfer function $T(s)$ due to the frequency domain specification.

However, plotting the Bode plot of the closed-loop transfer function $T(s)$ is quite tedious, and thus, following the course's line of motivation, we'll look to express these requirements in terms of the open-loop transfer function $L(s)$. Thus, the previous specification can be restated as

$$\begin{aligned} T(s) &= \frac{L(s)}{1 + L(s)} \quad \Rightarrow \quad T(j\omega) = \frac{L(j\omega)}{1 + L(j\omega)}, \\ |T(j\omega)| < 0.1 &\iff \left| \frac{L(j\omega)}{1 + L(j\omega)} \right| < 0.1, \\ &\iff |L(j\omega)| < 0.1 |1 + L(j\omega)|. \end{aligned}$$

Which, using the triangle inequality $|a + b| \leq |a| + |b|$, can be transformed into an explicit condition on $L(s)$

$$\begin{aligned} |L(j\omega)| &< 0.1 (1 + |L(j\omega)|), \\ 0.9 |L(j\omega)| &< 0.1, \\ |L(j\omega)| &< \frac{1}{9}. \end{aligned}$$

Although this is an over-approximation, by expressing the condition in terms of $L(s)$, we can now look at the Bode plot of the open-loop transfer function to enforce frequency domain specifications of the closed-loop. This allows us to design a controller $C(s)$ for the closed-loop, by checking if the open-loop transfer function $L(s) = C(s)P(s)$ fulfils the specifications.

Thus, the rest of this chapter is laid out as follows. In Section 7.2, we begin by introducing stability margins and how they reflect system robustness. Then, we look at what it means for a system to have good command tracking, disturbance rejection, and noise rejection and how we can think about these in the frequency-domain. We then use the main takeaways from this subsection to establish a Bode obstacle course, which reflects our frequency domain specifications. In Section 7.3, we look at loop shaping and how to choose the right controller to meet frequency domain specifications.

7.2 Frequency Domain Specifications

There are various ways to specify the desired performance and robustness of a control system in the frequency domain. These specifications typically address four key aspects: robustness margins, which quantify how far the system is from closed-loop instability; disturbance rejection and command tracking, which measure how well the system follows reference signals and rejects disturbances; noise rejection, which determines how effectively the system attenuates high-frequency sensor noise; and bandwidth, which characterizes how fast the system can respond to changes in the reference signal. In the following subsections, we will introduce each of these specifications formally and show how they can be visualized on the Bode plot to guide controller design.

7.2.1 Stability Margins

As we learned in the previous chapter, the Nyquist plot can be used to assess closed-loop stability. Now, considering that we usually expect some uncertainty in our system, e.g., due to noise or an imprecise model, we also have to be able to evaluate how *close* we are to being unstable. This is important because if our system is very *close* to instability, even a small amount of noise or a slight modeling error could make the system unstable. In this sense we would like to have a system that is far away from instability such that it is *robust* against uncertainty, noise, disturbances, etc.

To quantify this notion of robustness consider a stable open-loop transfer function $L(s)$ and the corresponding Nyquist plot as shown in Figure 7.3. As usual, we are interested determining

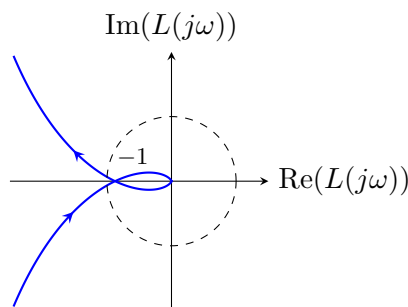


Figure 7.3: Nyquist plot of stable open-loop transfer function $L(s)$.

whether the closed-loop system is stable. By applying the Nyquist criterion we can check the stability of the resulting closed-loop system. In particular, Figure 7.3 shows that although the closed-loop system is stable, it is very close to being unstable since the Nyquist plot passes near the critical point at -1 . Thus, a small variation in $L(j\omega)$ may cause this closed-loop system to become unstable. To identify such behavior we will consider 2 critical measures to quantify the *robustness* of the closed-loop stability, i.e. how far away the system is from being unstable.

First we check the magnitude of the Nyquist plot as it crosses the negative real axis, i.e. when $\angle L(j\omega_{pc}) = -180^\circ$, where ω_{pc} is called the *phase crossover frequency*. For closed-loop stability the magnitude at ω_{pc} has to be smaller than 1, else the Nyquist plot would encircle the point -1 causing the closed-loop to be unstable. In other words, we need to ensure that $|L(j\omega_{pc})| < 1$. The closer $|L(j\omega_{pc})|$ gets to 1, the closer the closed-loop system is to instability.

A way to quantify this distance is through the gain margin, defined as

$$g_m = \frac{1}{|L(j\omega_{pc})|}. \quad (7.1)$$

From this definition we can see that if $|L(j\omega_{pc})| = 1$, then the gain margin $g_m = 1 = 0$ dB. Furthermore, when $|L(j\omega_{pc})|$ gets smaller, i.e. we move away from -1 on the real axis, the gain margin gets bigger. Thus, we can conclude that a big gain margin g_m leads to robustness of the closed-loop system. We can think about the gain margin as the amount of feedback gain we can add to the system before it becomes unstable.

Secondly, we consider the point where the Nyquist plot has a magnitude of 1, i.e. when $|L(j\omega_c)| = 1 = 0$ dB, where ω_c is called the *crossover frequency*¹. For closed-loop stability we require the phase at ω_c to be more negative than -180° , i.e., $\angle L(j\omega_c) < -180^\circ$. We can observe that the closer $\angle L(j\omega_c)$ gets to -180° , the closer we get to the -1 point and therefore instability. Similar to before we quantify this distance through a quantity called the phase margin defined as

$$\varphi_m = 180^\circ + \angle L(j\omega_c). \quad (7.2)$$

We observe that if $\angle L(j\omega_c) = -180^\circ$, we have zero phase margin and an unstable closed-loop system since we are on the critical point -1 . As the phase at the crossover frequency increases (becomes less negative) the phase margin will increase. Consequently, we conclude that a big phase margin will indicate robustness of the closed-loop system. In summary, Figure 7.4 shows both the phase and gain margins for an open-loop stable system.

Note that the definitions above cover open-loop stable systems which, according to the Nyquist stability criterion, require no encirclement of -1 to be closed-loop stable. Conversely, it can also be that the open-loop has n unstable poles, requiring n CCW encirclements of -1 for closed-loop stability. In that case the stability margins need to quantify how close the

¹Sometimes this is also referred to as the gain crossover frequency, ω_{gc} , to better distinguish it from the phase crossover frequency.

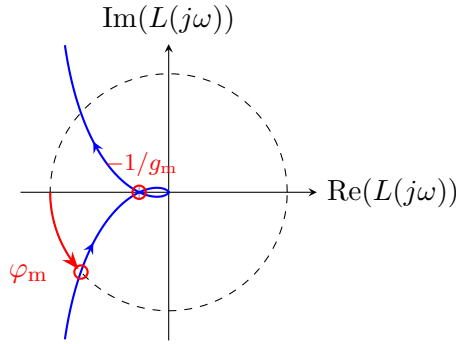


Figure 7.4: Nyquist plot of an open-loop stable system, and its corresponding gain margin g_m , and phase margin φ_m .

Nyquist plot is to encircling at least -1 n -times in the CCW direction. Thus, the way to define the stability margins depends on how many encirclements of -1 are required for closed-loop stability. Nevertheless, in this course we will focus on the case of a stable open-loop system.

As we have seen in Equations (7.1) and (7.2), the gain- and phase-margins can be analytically calculated using the open-loop transfer function. This will always give an exact result, but it can be tricky to compute depending on the transfer function. Alternatively, as shown in Figure 7.4, the gain- and phase margins can also be read-off from the Nyquist plot directly. This will work for any open-loop system, but obtaining exact results from the plot can be difficult.

In the special case of stable, minimum-phase, open-loop systems we can also use the bode plot to determine the gain- and phase margins. The gain margin g_m can be obtained by marking the point where $\angle(L(j\omega_{pc})) = -180^\circ$, and reading of the magnitude at that frequency, i.e. $|L(j\omega_{pc})|$. The gain margin (in dB) is then the difference between $|L(j\omega_{pc})|_{\text{dB}}$ and 0 dB. Similarly, we can get the phase margin φ_m by marking the point where $|L(j\omega_c)| = 0$ dB, and reading of the phase at that frequency, i.e. $\angle(L(j\omega_c))$. The phase margin is then given by the difference in phase between -180° and $\angle(L(j\omega_c))$. In Figure 7.5 this process is also graphically visualized for a stable, minimum-phase, open-loop system.

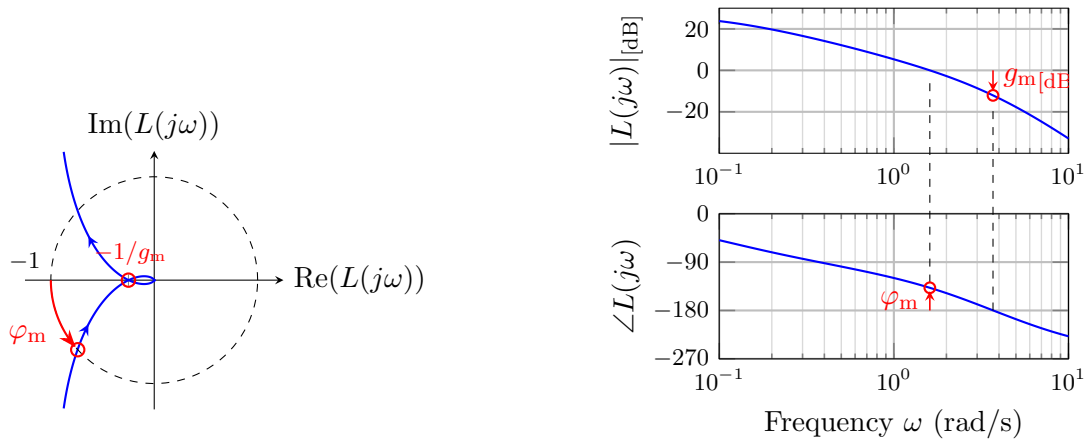


Figure 7.5: Visualization of gain margin g_m and phase margin φ_m for a minimumphase open-loop stable system on the Nyquist plot (left) and the Bode plot (right).

7.2.2 Disturbance Rejection / Command Tracking

Consider a cruise-control system for a car whose objective is to maintain a specified reference speed. When the reference speed is set, the system should follow it closely, rejecting external disturbances, such as wind resistance and variations in the slope of the road. These two desired behaviors (track reference and reject disturbances) motivate the *command tracking* and *disturbance rejection* objectives. Command tracking refers to how accurately the systems output follows a desired reference signal, while disturbance rejection measures how effectively the system maintains its performance despite exogenous disturbances. In practice, a well-designed feedback system should simultaneously follow reference commands accurately and not be significantly affected by disturbances.

We can formalize the description of these specifications by considering the standard feedback architecture with noise and disturbances (see Figure 7.6). Since we deal with SISO systems, we look to derive the relevant transfer function upon which to apply these specifications.

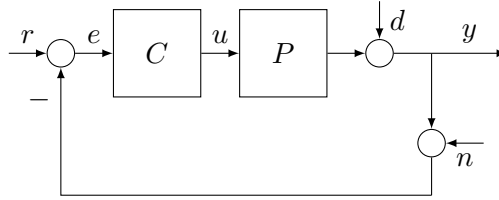


Figure 7.6: Standard feedback architecture with noise and disturbances.

Command Tracking

For command tracking we want the error signal to be small, and hence are interested in the mapping from the reference to the error, i.e., $e(t) = r(t) - y(t)$. As we saw in Section 4.2, this is given by the sensitivity function $S(s)$, i.e., $E(s) = S(s) \cdot R(s)$. As the goal is to minimize the tracking error, we want the magnitude of $S(s)$ to be *small*. We can structure this requirement into the following form:

$$|S(j\omega)| \ll \text{small value.}$$

Importantly, the sensitivity function $S(s)$ can be expressed as a function of the open-loop transfer function $L(s)$. This means that we can translate the closed-loop command tracking requirement into a requirement on the open-loop transfer function $L(s)$.

$$|S(j\omega)| = \frac{1}{|1 + L(j\omega)|} \ll \text{small value} \quad \implies \quad |L(j\omega)| \gg \text{large value.} \quad (7.3)$$

Example:

Suppose that a closed-loop system of the architecture shown in Figure 7.6 needs to track commands with max. 10% error. According to Equation (7.3) this corresponds to the

requirement

$$|S(j\omega)| \leq 0.1 \implies |1 + L(j\omega)| \geq 10.$$

Often $|L(j\omega)| \gg 1$, and so we approximate the requirement as

$$|L(j\omega)| \geq 10.$$

Thus, to ensure good command tracking, we need to design the open-loop transfer function $L(s)$ with large magnitude. However, as we will see, this is only true for a specific frequency range.

Disturbance Rejection

For disturbance rejection we want the output to be robust to disturbances. In this SISO setup, this means that the effect of disturbances $d(t)$ on the output $y(t)$ is small. Using Figure 7.6, we can derive $Y(s) = S(s) \cdot D(s)$ which implies that if we want $d(t)$ to have a small influence on the output $y(t)$, we require the magnitude of $S(s)$ to be small. Since disturbances typically act at low frequencies, usually under 10Hz, we can structure this requirement as

$$|S(j\omega)| \ll \text{small value}, \quad \forall \omega \in [0, \text{low frequency threshold}]. \quad (7.4)$$

This varies from the aforementioned command tracking requirement in that we are now interested in the magnitude of $S(s)$ being small for a specific frequency range. However, since large open-loop gain at high frequencies typically reduces disturbance attenuation and reduces the error to reference, command tracking and disturbance rejection are often expressed in the same fashion.

Example:

Suppose that a closed-loop system of the architecture shown in Figure 7.6 needs to track commands with max. 10% error up to a frequency of 10 Hz (≈ 60 rad/s). This corresponds to the requirement

$$|S(j\omega)| \leq 0.1 \quad \forall \omega \in [0, 60 \text{ rad/s}] \implies |1 + L(j\omega)| \geq 10 \quad \forall \omega \in [0, 60 \text{ rad/s}]. \quad (7.5)$$

Often $|L(j\omega)| \gg 1$ for $\omega < 60$ rad/s, and so we approximate the requirement as

$$|L(j\omega)| \geq 10 \quad \forall \omega \in [0, 60 \text{ rad/s}]. \quad (7.6)$$

Thus, to ensure good disturbance rejection, we need to design the open-loop transfer function $L(s)$ such that it is large at low frequencies. Since the command tracking / disturbance rejection requirement can be expressed solely in terms of the open-loop transfer function $L(s)$, we can visualize these requirements directly on the Bode plot as shown in Figure 7.7.

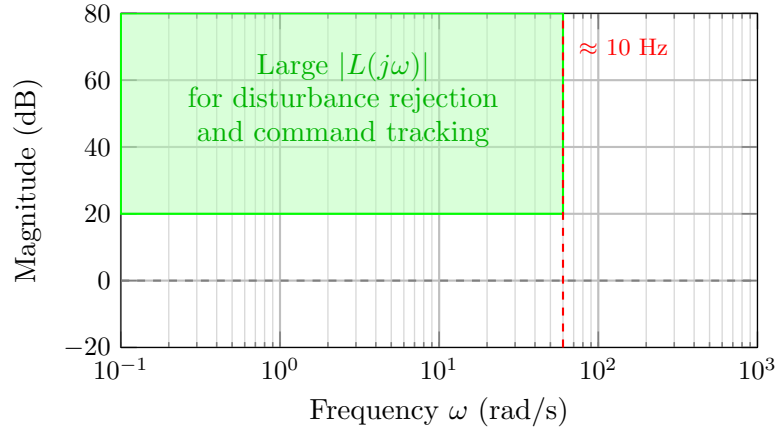


Figure 7.7: Bode magnitude plot illustrating the disturbance rejection requirement. The green region shows where $|L(j\omega)| \geq 20$ dB for frequencies below 10 Hz, ensuring disturbances are attenuated by at least a factor of 10.

7.2.3 Noise Rejection

Returning to the previous cruise-control example, the speed sensor used to measure the car’s velocity is never perfect, and is subject to measurement noise. This noise, caused by factors such as sensor imperfections and vibrations, appear as rapid fluctuations in the measured speed signal (see Figure 7.8). Intuitively, if the controller were to react to these high-frequency fluctuations, it would cause rapid and unnecessary adjustments to the throttle, causing a jerky driving experience.

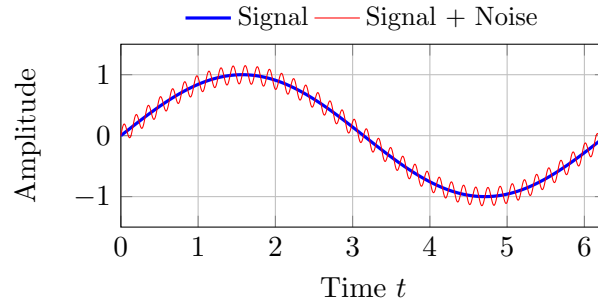


Figure 7.8: Example of a low-frequency signal (blue) and the same signal corrupted by high-frequency noise (red).

From the desire to avoid such a reaction, we can derive the *noise rejection* requirement. Unlike disturbances, which act directly on the plant and are typically low-frequency phenomena, *measurement noise* is a high-frequency phenomenon that corrupts the sensor readings. Ensuring that the noise is not amplified by the system is crucial so that the true signal remains distinguishable from the noise.

To evaluate our system’s noise rejection performance, let’s revisit Figure 7.6. We are now interested in the mapping from the noise signal $n(t)$ to the output $y(t)$. This is given by

$$Y(s) = T(s) \cdot N(s).$$

From this we can see that to ensure good noise rejection, we want the magnitude of $T(s)$ to attenuate noise as much as possible. Since noise is typically a high-frequency phenomenon, we can structure this requirement as:

$$|T(j\omega)| \ll 1, \quad \forall \omega \in [\text{high frequency threshold}, \infty).$$

Similar to the sensitivity function $S(s)$, the complementary sensitivity function $T(s)$ can be expressed as a function of the open-loop transfer function $L(s)$. This means we can once again translate requirements on $T(s)$ into specifications on the open-loop transfer function $L(s)$. To ensure noise rejection, we want $T(s)$ to be *small*, meaning that we also want to minimize the magnitude of $L(s)$ at high frequencies.

$$|T(s)| = \frac{|L(s)|}{|1 + L(s)|} \ll 1 \quad \implies \quad |L(s)| \ll 1 \quad \text{as} \quad |T(s)| \approx |L(s)|. \quad (7.7)$$

Example:

Suppose that a closed-loop system of the architecture shown in Figure 7.6 needs to attenuate noise by a factor of 10 at frequencies above 100 Hz ($\approx 628 \text{ rad/s}$). This corresponds with the requirement

$$|T(j\omega)| \leq 0.1 \quad \forall \omega \in [628 \text{ rad/s}, \infty).$$

Since $|T(j\omega)| \approx |L(j\omega)|$ for high frequencies, we can approximate this requirement as

$$|L(j\omega)| \leq 0.1.$$

Thus, to ensure good noise rejection, we need to design the open-loop transfer function $L(s)$ such that it is small at high frequencies. Similarly to the command tracking / disturbance rejection requirement, the noise rejection requirement can be expressed solely in terms of the open-loop transfer function $L(s)$ and can be visualized directly on the Bode plot as shown in Figure 7.9.

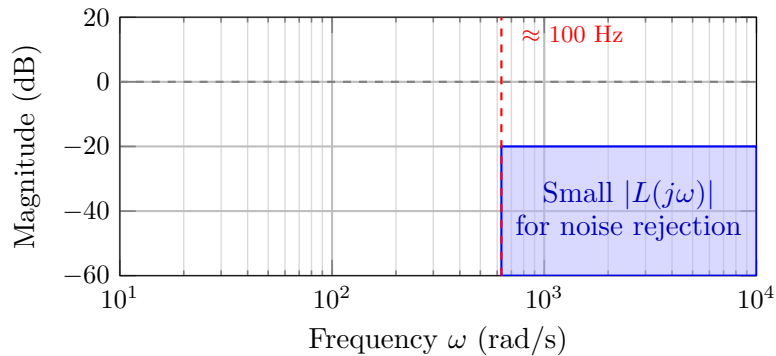


Figure 7.9: Bode magnitude plot illustrating the noise rejection requirement. The blue region shows where $|L(j\omega)| \leq -20 \text{ dB}$ for frequencies above 100 Hz, ensuring noise is attenuated by at least a factor of 10.

7.2.4 Bode Obstacle Course

Thus far, we have posed our requirements on the sensitivity function $S(s)$ and the complementary sensitivity function $T(s)$ in terms of the open-loop transfer function $L(s)$ and used the Bode plot to visualize these requirements (see Figures 7.7 and 7.9). In order to minimize $S(s)$ we aimed for large $|L(j\omega)|$, while to minimize $T(s)$ we aimed for small $|L(j\omega)|$. This allows us to visualize these requirements on the Bode plot. Before we do so however, we need to introduce a fundamental limitation associated with command tracking and noise rejection.

As described in the previous subsections, ideally we want both the sensitivity function $S(s)$ and the complementary sensitivity function $T(s)$ to be small. However, these two cases cannot be achieved simultaneously. To see why this is the case, consider the definitions of $S(s)$ and $T(s)$:

$$S(s) = \frac{1}{1 + L(s)}, \quad T(s) = \frac{L(s)}{1 + L(s)}.$$

Adding the two yields

$$S(s) + T(s) = \frac{1}{1 + L(s)} + \frac{L(s)}{1 + L(s)} = \frac{1 + L(s)}{1 + L(s)} = 1. \quad (7.8)$$

Since $S(s) + T(s) = 1$, this means that we cannot simultaneously achieve small $S(s)$ and $T(s)$ at the same frequency, i.e. making one of them small forces the other to be close to one. As control designers, we must choose appropriate trade-offs between the two in the frequency domain. To do so, we can visualize both requirements simultaneously on the Bode plot (see Figure 7.10). This gives us a *Bode obstacle course* that the open-loop transfer function $L(s)$ must navigate.

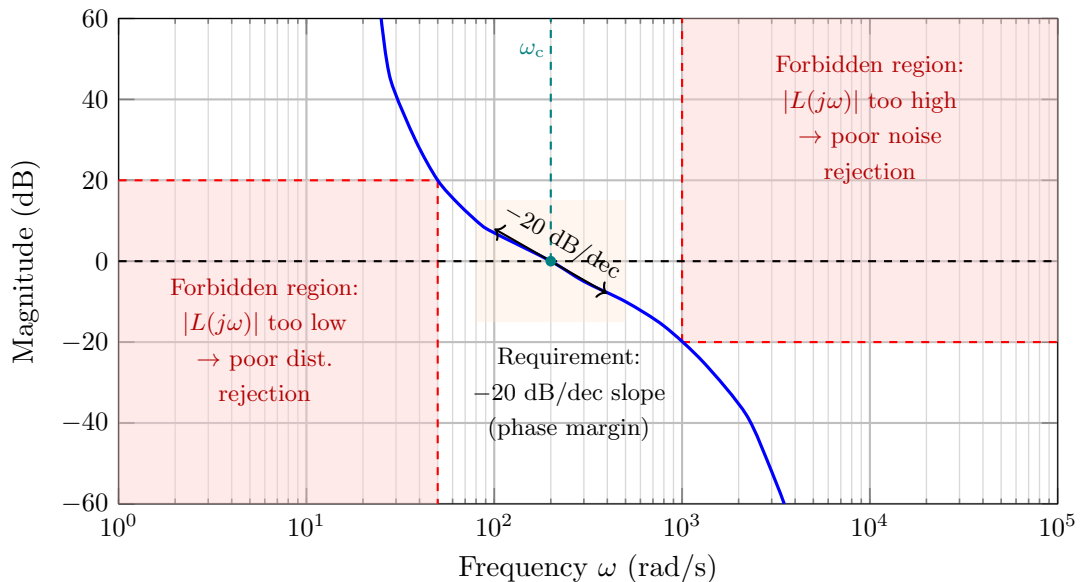


Figure 7.10: Bode obstacle course showing the three main design requirements: (1) high loop gain at low frequencies for disturbance rejection and command tracking, (2) low loop gain at high frequencies for noise rejection, and (3) approximately -20 dB/decade slope around the gain crossover frequency ω_c for good phase margin (More in the next subsection).

It should be mentioned that this limitation is formalized as the *waterbed effect*. Pushing down the sensitivity $S(s)$ at low frequencies inevitably causes it to increase in another frequency range. This is derived from *Bode's integral theorem*, which we will not cover further in this script. For readers who wish to learn more, see Åström and Murray, *Feedback Systems*, Sec. 14.2 for more details.

It is important to keep in mind that when working on controller design, we must always make an appropriate trade-off. Improving performance in one aspect inevitably limits what we can achieve in another.

7.2.5 Bandwidth

Previously, the requirements for desirable performance of a closed-loop system were posed. Due to the known structure of the feedback architecture, we are able to translate closed-loop requirements to the open-loop system. One of the points of interest on the open-loop Bode plot, was the crossover frequency ω_c , which describes the frequency at which signals transition from being amplified to getting attenuated (weakened), i.e. when we cross the 0 dB line. This frequency and the open-loop Bode plots characteristics in its neighborhood are closely related to the *bandwidth*.

The bandwidth is the maximum ω , such that $|T(j\omega)| > \frac{1}{\sqrt{2}}$ (≈ -3 dB).

In other words, the bandwidth denotes the range of frequencies for which a closed-loop system can track references with a maximal attenuation of $\frac{1}{\sqrt{2}} \approx 0.7$. In a physical sense, this can be interpreted as how fast the closed-loop can react to changing references. A large bandwidth means that even high frequencies are passed with ease, whereas a low bandwidth indicates that even slow oscillations already experience attenuation of the reference amplitudes. As the bandwidth is a closed-loop system specification that we can't directly choose, we must investigate how it is affected by open-loop design choices. By looking at the complementary sensitivity function, the connection to the crossover frequency ω_c arises:

$$|T(j\omega)| = \frac{|L(j\omega)|}{|1 + L(j\omega)|} = \frac{1}{\sqrt{2}},$$

which is solved for $L(j\omega) = \pm j$, i.e. frequencies that have a gain of $|L(j\omega)| = 1$ with a phase of $\angle L(j\omega) = \pm 90^\circ$. It is common to require a phase margin $\varphi_m \approx 90^\circ$ for robust performance of the closed-loop, as it is at this point that the slope of the magnitude plot is approximately -20 dB/decade. If an open-loop system satisfies the phase margin conditions approximately, then the frequency that meets the amplitude and phase conditions of the bandwidth frequency is in fact close to the crossover frequency:

$$|L(j\omega_c)| = 1, \quad \angle L(j\omega_c) \approx 90^\circ.$$

And therefore

$$\omega_c \approx \omega_b.$$

The better the phase margin conditions are met, the more the ω_b and ω_c match.

Example:

To visualize the relationship of the crossover frequency and the bandwidth of a system, consider a second order open-loop system given as $L(s) = \frac{2}{s^2+s+1}$. We find the crossover frequency of this system by solving

$$\left| \frac{2}{-\omega_c^2 + j\omega_c + 1} \right| = 1, \quad \text{which gives } \omega_c \approx 1.87 \text{ rad/s.}$$

To find the bandwidth, we first establish the closed-loop as

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{3s^2 + 3s + 3}{s^4 + 2s^3 + 6s^2 + 5s + 4}.$$

Now we solve

$$\left| \frac{-3\omega_b^2 + 3j\omega_b + 3}{\omega_b^4 - 2j\omega_b^3 - 6\omega_b^2 + 5j\omega_b + 4} \right| = \frac{1}{\sqrt{2}}, \quad \text{which gives } \omega_b \approx 3 \text{ rad/s,}$$

which is reasonably close to the crossover frequency.

To assess this in a visual manner, Figure 7.11 shows the Magnitude plot of the open-loop and closed-loop with the corresponding crossover frequency and bandwidth.

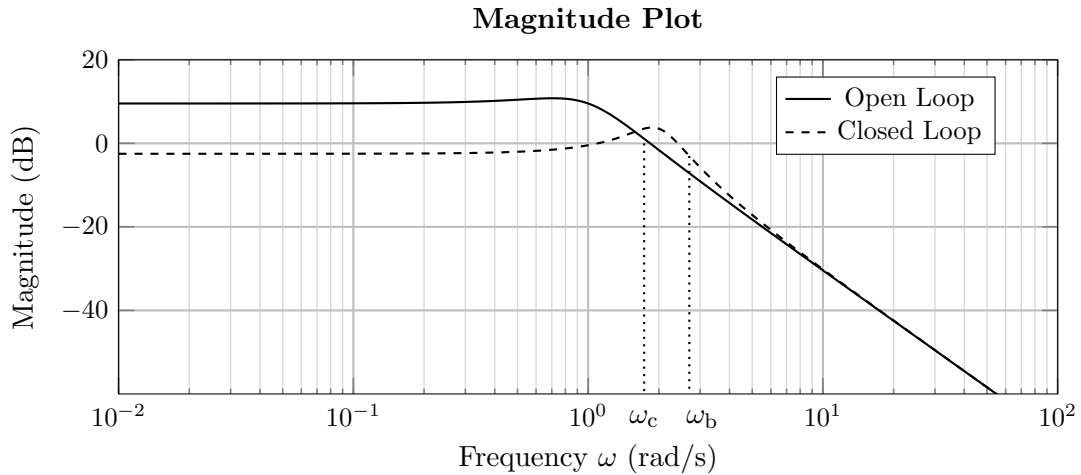


Figure 7.11: Open-loop and closed-loop of $L(s) = \frac{2}{s^2+s+1}$, with crossover frequency ω_c and the bandwidth ω_b .

7.3 Control Synthesis via Loop Shaping

In most applications, designing a controller should not only stabilize a system but also meet a handful of other criteria. While the desired closed-loop behavior is often characterized with

similar traits, the controller achieving these traits is heavily dependent on the plant. We have already seen in Chapter 5, how desirable behavior of a step response is described in the form of pole and zero locations. The root locus then proved useful to analyze how the choice of feedback gain could be leveraged to achieve desired closed-loop behavior. Unfortunately, the root locus can only be used to tune a single gain at a time, thus proving to be limited when dealing with dynamic compensators such as a PID controller. A workaround is to tune one gain, while keeping the remaining fixed and then iterating through all gains until a satisfactory response is achieved. While possible in theory, this process is predominantly a ‘Trial and Error’-Approach and is therefore neither systematic nor does it guarantee any robustness.

In the following, alternative controller synthesis methods using the frequency response are proposed. In Section 7.2 we established criteria that have to be met to achieve desirable closed-loop behavior. To choose a controller that meets these criteria, we have two methods. The *Brute Force Approach*, which reverse engineers a controller in such a way that the closed-loop transfer function matches a predefined structure and the *Loop Shaping Approach*, that builds a controller systematically using a set of standard control elements.

7.3.1 Brute Force Approach

The brute force approach assumes a given plant $P(s)$, and chooses a controller $C(s)$ for it such that the open-loop takes a predefined desired form $L_{\text{des}}(s)$. In a first step, one has to find an open-loop transfer function that adheres to the specifications presented in Section 7.2. Secondly, since for a general setup we have that

$$L(s) = C(s)P(s),$$

we find the controller that makes the open-loop behave in the desired way as

$$C(s) = \frac{L_{\text{des}}(s)}{P(s)}. \quad (7.9)$$

While this procedure is computationally intense, it only works under the following circumstances:

- Open-loop must be stable and minimum-phase. If this is not given, the found controller will cancel the instabilities leading to false conclusions regarding stability of the closed-loop.
- The compensator may be unnecessarily complicated as it essentially contains an inverse of the plant.
- If the desired open-loop transfer function is not chosen well, the compensator may not be causal.

Example - Brute Force Controller Synthesis of the Pendulum:

Recall the previously derived transfer function of the pendulum linearized at the origin:

$$P(s) = \frac{0.2}{s^2 + s + 2}.$$

Since the pendulum is linearized at a stable point, a brute force approach is chosen to make the open-loop take a desired shape. The following specifications are given for the open-loop such that the closed-loop inherits desirable behavior:

- Disturbance rejection: $|L(s)| > 40$ dB for $\omega < 0.1$ rad/s,
- Noise rejection: $|L(s)| < -26$ dB for $\omega > 100$ rad/s,
- Tracking speed: $\omega_c = 10$ rad/s,
- Phase margin: -20 dB/dec at ω_{gc} .

A proposed transfer function that satisfies these requirements is found as

$$L_{des}(s) = \frac{10}{s^2/50 + s}.$$

The Plant, requirements and the desired open-loop transfer function are shown in Figure 7.12.

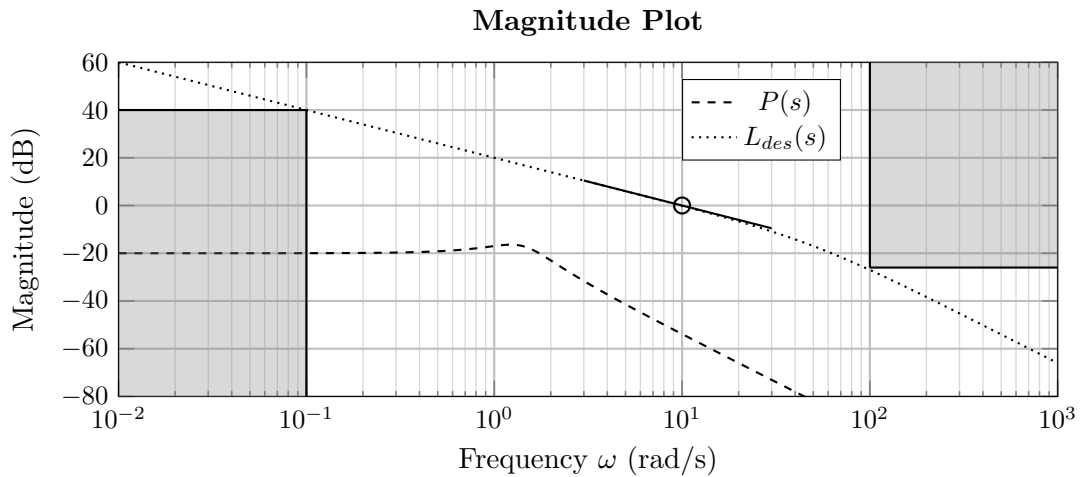


Figure 7.12: Magnitude plot of the Plant $P(s)$, the desired open-loop transfer function $L_{des}(s)$ and the restrictions in the frequency response for the desired closed-loop behavior.

Applying Equation (7.9) to the given plant, we find $C(s)$ as

$$C(s) = \frac{10s^2 + 10s + 20}{0.004s^2 + 0.2s}.$$

The open-loop using this controller i.e. $L(s) = C(s)P(s)$ now perfectly matches the desired open-loop $L_{des}(s) = L(s)$. In the present case, this technique is applicable as the plant is both stable and minimum-phase, but remember that it should generally be used with care. The controller contains an inverse of the plant and could therefore cancel unstable behavior or just take an unnecessarily complex form.

7.3.2 Loop Shaping Approach

Where the brute-force approach generated a controller that cancelled out the plant to introduce the desired dynamics, the loop shaping approach will configure a series of standard control elements with known characteristics in such a way that the requirements are met. While more involved than the brute-force approach, it produces predictable behavior, its design is systematic, and can be generally applied.

The dynamic compensator is built step-by-step, where we individually tune each element. Each element is then added in a serial configuration such that our final controller will have a structure of the form

$$C(s) = C_1(s) \cdot C_2(s) \cdot \dots \cdot C_n(s).$$

Since the standard elements affect the loop in a known way, we can *shape* the frequency response to arrive at a desirable controller. Many control problems can be solved with a relatively small selection of standard elements. The most fundamental ones that will be covered in this chapter are

- Proportional gain: k ,
- Integrator: $\frac{1}{s}$,
- Lead element: $\frac{s/a+1}{s/b+1}$, with $0 < a < b$,
- Lag element: $\frac{s/a+1}{s/b+1}$, with $0 < b < a$.

The proportional and integrator elements have already been introduced in Section 6.2. In terms of their frequency response, a proportional controller shifts the magnitude over all frequencies with no influence on the phase. An integrator adds a slope of -20 dB/dec over all frequencies and shifts the phase plot down by -90° .

Example - Loop Shaping the Pendulum (Gain/Integrator):

Consider again the pendulum and the desired specifications as given in the brute force pendulum example. When attempting to meet the specifications for the open-loop, neither a proportional control setup $L(s) = kP(s)$ nor an open-loop setup with an integrator $L(s) = \frac{1}{s}P(s)$ will suffice. In Figure 7.13 the effects on the magnitude plot for these compensators are shown. With a proportional controller, the open-loop magnitude plot is shifted upward by the gain factor. Here, the requirements for disturbance and noise rejection are met with a gain of $k = 1000$, but the crossover frequency requirements can not be satisfied. When using an integrator, a slope of -20 dB/dec is added to the magnitude plot over all frequencies, but it still fails to meet any of the stated requirements. We conclude that neither a proportional controller nor an integrator (and in fact any combination of them) will lead to the desired behavior. We must thus resort to other additional standard elements for compensator construction.

In addition to the already established proportional controller and the integrator, lead and lag elements will prove to be a powerful way to construct complex desired behavior in a structured

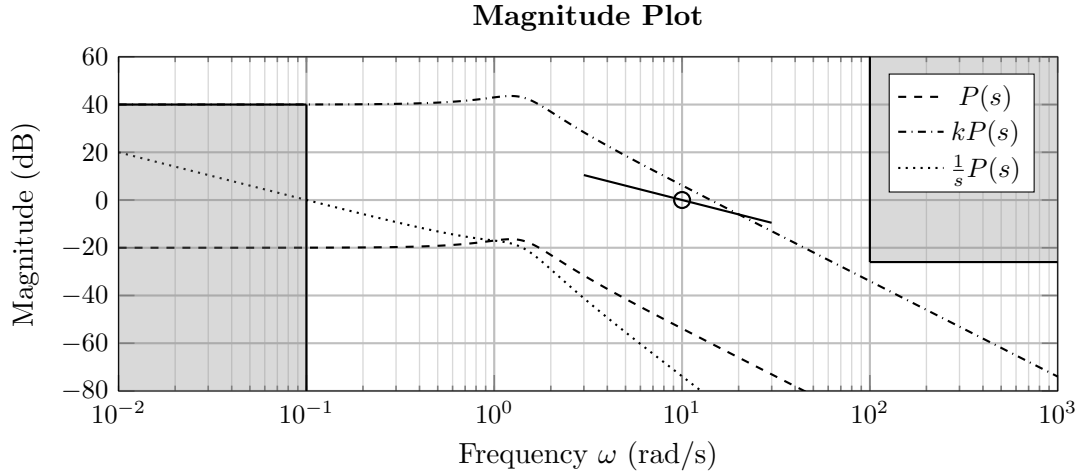


Figure 7.13: Magnitude plot of the Plant $P(s)$ and the open-loops for a proportional controller and a pure integrator. The line centered about a circle at $\omega = 10$ rad/s denotes a slope of 20dB/dec.

way. By respectively introducing a pole and a zero at strategically chosen locations, these compensators will be able to manipulate both the magnitude and the phase response of the system systematically.

Lead Compensator

The lead compensator is a combination of a stable pole and a minimum-phase zero, where the pole is placed closer to the origin than the zero, i.e:

$$C_{\text{lead}}(s) = \frac{s/a + 1}{s/b + 1}, \quad 0 < a < b.$$

The parameters a and b allow for the tuning of the way the lead compensator affects the magnitude and phase plot of the frequency response. Its effects are characterized in the following ways:

- **Magnitude response:** The magnitude at high frequencies is increased by b/a . Low frequencies are not affected. The slope of the plot is increased by +20dB/dec between a and b .
- **Phase response:** The phase is increased around $\sqrt{ab}$ by up to 90° . The maximum phase increase is found by

$$\varphi_{\text{max}} = 2 \arctan \left(\sqrt{\frac{b}{a}} \right) - 90^\circ.$$

In Figure 7.14 the Bode plot of a lead element with $a = 1$ rad/s and $b = 10$ rad/s is shown. For this choice of parameters, we see a shift upward in magnitude for high frequencies of

$$\frac{b}{a} = 10 \implies 20\text{dB},$$

and a maximum phase increase of

$$2 \arctan(\sqrt{10}) - 90^\circ \approx 54.9^\circ,$$

around $\omega = \sqrt{10} \approx 3.16 \text{ rad/s}$.

Bode Plot of a Lead Element

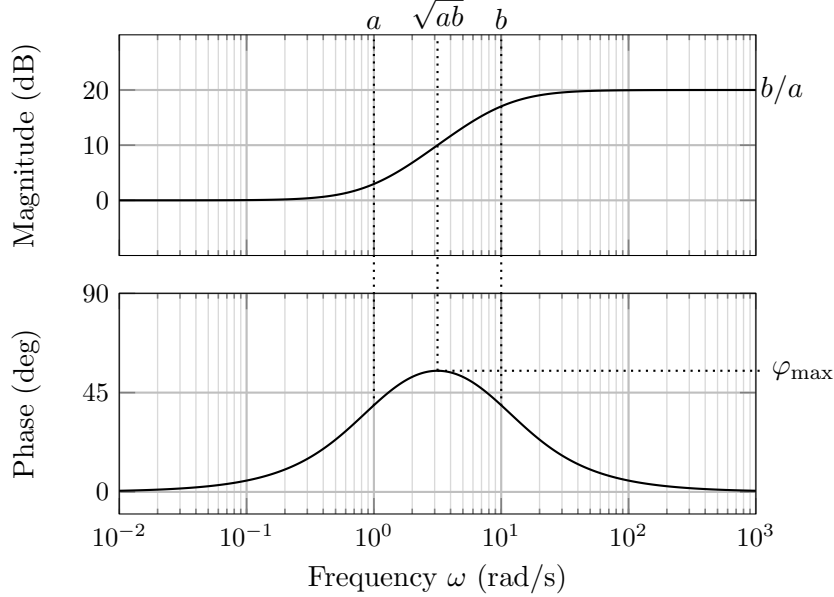


Figure 7.14: Bode Plot of a lead element with $a = 1$ and $b = 10$

Typically, lead compensators are used in combination with a proportional controller k to increase the phase margin at specifiable crossover frequency. To achieve this, the following process is followed:

1. Pick $\sqrt{ab}$ at the desired crossover frequency ω_c ,
2. Pick b/a such that the desired phase increase is achieved,
3. Adjust k , such to set the crossover frequency to the desired frequency.

Since this also increases the magnitude at high frequency, noise rejection may be compromised by a lead compensator.

Lag Compensator

A lag compensator is in many ways an inverse of the lead compensator. It has the same transfer function structure with a stable pole and a minimum-phase zero, with the difference, that here the zero is closer to the origin,

$$C_{\text{lag}}(s) = \frac{s/a + 1}{s/b + 1}, \quad 0 < b < a.$$

It affects the magnitude and phase response in the following ways:

- **Magnitude response:** The magnitude at high frequencies is decreased by b/a . Low frequencies are not affected. The slope of the plot is decreased by 20 dB/dec between b and a .
- **Phase response:** The phase is decreased around $\sqrt{ab}$ by up to 90° . The maximum decrease is found as

$$\varphi_{\min} = 2 \arctan\left(\sqrt{\frac{b}{a}}\right) - 90^\circ.$$

The frequency response of a lag compensator with $b = 1$, $a = 10$ is shown in Figure 7.15. For this choice of parameters, we see a drop in magnitude for high frequencies of

$$\frac{b}{a} = 0.1 \implies -20\text{dB},$$

and a maximum phase decrease of

$$2 \arctan\left(\sqrt{0.1}\right) - 90^\circ \approx -54.9^\circ,$$

around $\omega = \sqrt{10} \approx 3.16 \text{ rad/s}$.

Bode Plot of a Lag Element

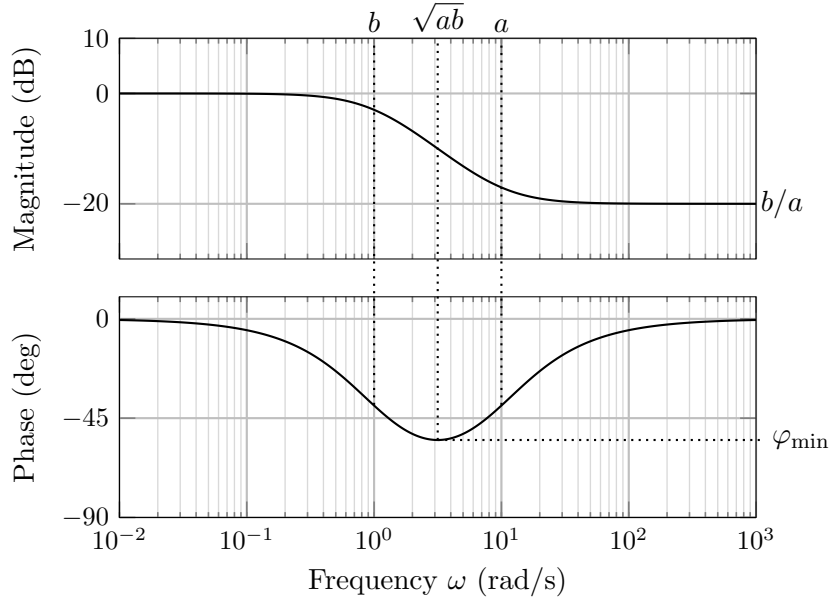


Figure 7.15: Bode Plot of a lag element with $b = 1$ and $a = 10$

Lag compensators are typically used for disturbance rejection and command tracking. In Section 7.2.2 it was found that the open-loop system should have a high magnitude at low frequencies. Using a lag compensator in addition to a proportional gain, we can achieve this by following the design procedure:

1. Pick a/b as the desired increase in magnitude at low frequencies,
2. Multiply the proportional gain k by a/b . This ensures that the response at high frequencies is unaffected and increases the magnitude at low frequencies.
3. Pick a sufficiently smaller than the crossover frequency such that phase margins are not affected too much.

This may also introduce phase lag at low frequencies and potentially reduce phase margins.

General Procedure

With the established controller elements, we will be able to systematically design a dynamic compensator of the following general structure:

$$C(s) = k \frac{1}{s^q} \prod_{i,j} C_{\text{lead},i}(s) C_{\text{lag},j}(s),$$

where q denotes the necessary amount of integrators, and i, j denote the possibly multiple lead- and lag elements respectively. To make the closed-loop behave in the desired way, we will now try and meet the requirements that were translated to the open-loop with our constructed compensator. As Bode plots are additive, we can choose each element and step-by-step achieve the desired form. In general, the compensator will be designed from ‘left-to-right’, i.e. the response is assessed from low to high frequencies in the following manner:

1. Choose the number of integrators p such that the reference can be followed with zero steady-state error. Recall that the necessary number corresponds to the reference ramp order.
2. Fix the proportional gain such that the low-frequency Bode obstacle is cleared.
3. Add lead and lag elements such that the magnitude plot crosses the 0 dB line with a slope of -20 dB/dec (this would result in a phase margin of 90°). Instead of tweaking a and b parameters of the lead/lag-elements, it may be simpler to add terms of the form $(\tau s + 1)$, which are normalized on the zero-order term such that frequencies to the left of the pole/zero are not affected.
4. Past the crossover frequency, add poles as needed to clear the high-frequency Bode obstacle (noise rejection/uncertainty).

Example - Full Loop Shaping for the Pendulum

Continuing with the pendulum example, we aim to find a compensator that can follow a step-reference satisfying the same requirements as in the previous example. To do so, we will follow the general procedure outlined above. Lastly, to make the step-by-step construction easier to follow, each intermediate step of the controller derivation is denoted

as $C_i(s)$ and the corresponding open-loop as $L_i(s) = C_i(s)P(s)$. Feel free to follow along using the [BodeTool](#). Recall that the plant is given by,

$$P(s) = \frac{0.2}{s^2 + 0.2s + 0.2}.$$

1. The pendulum should be able to follow step-response references without a steady-state error. Hence, we add a single integrator to the dynamic compensator:

$$C_1(s) = \frac{1}{s}$$

2. Using the [BodeTool](#) and plotting $C_1(s)P(s)$, we choose $k = 185$, to avoid the low-frequency Bode obstacle:

$$C_2(s) = 185 \frac{1}{s}$$

3. To achieve the desired crossover frequency, lead-elements must be introduced. The current open-loop transfer function decays at ω_c with -60 dB/dec, hence two lead elements are needed to reduce the slope to the desired -20 dB/dec. To increase the phase margin enough to stabilize the (currently unstable) closed-loop, a satisfactory trade off is found by choosing the parameters for the two lead elements as $a = 2$ rad/s, and $b = 120$ rad/s. Thus, the controller reads as

$$C_3(s) = 185 \frac{1}{s} \left(\frac{s/2 + 1}{s/120 + 1} \right)^2.$$

4. Finally, to ensure that we avoid the high-frequency obstacle, a lag element is added with the parameters $a = 80$ rad/s, and $b = 30$ rad/s. The final dynamic compensator is now written as:

$$C_4(s) = 185 \frac{1}{s} \frac{s/80 + 1}{s/30 + 1} \left(\frac{s/2 + 1}{s/120 + 1} \right)^2$$

Figure 7.16 shows the Bode plots of all the intermediate steps of the controller design. Note that the first two iterations are unstable, as in these cases the phase margin is $\varphi < -180^\circ$. Adding the lead elements solves this issue, but brings the open-loop plot dangerously close to the high-frequency obstacle. The lag element then successfully addresses this issue.

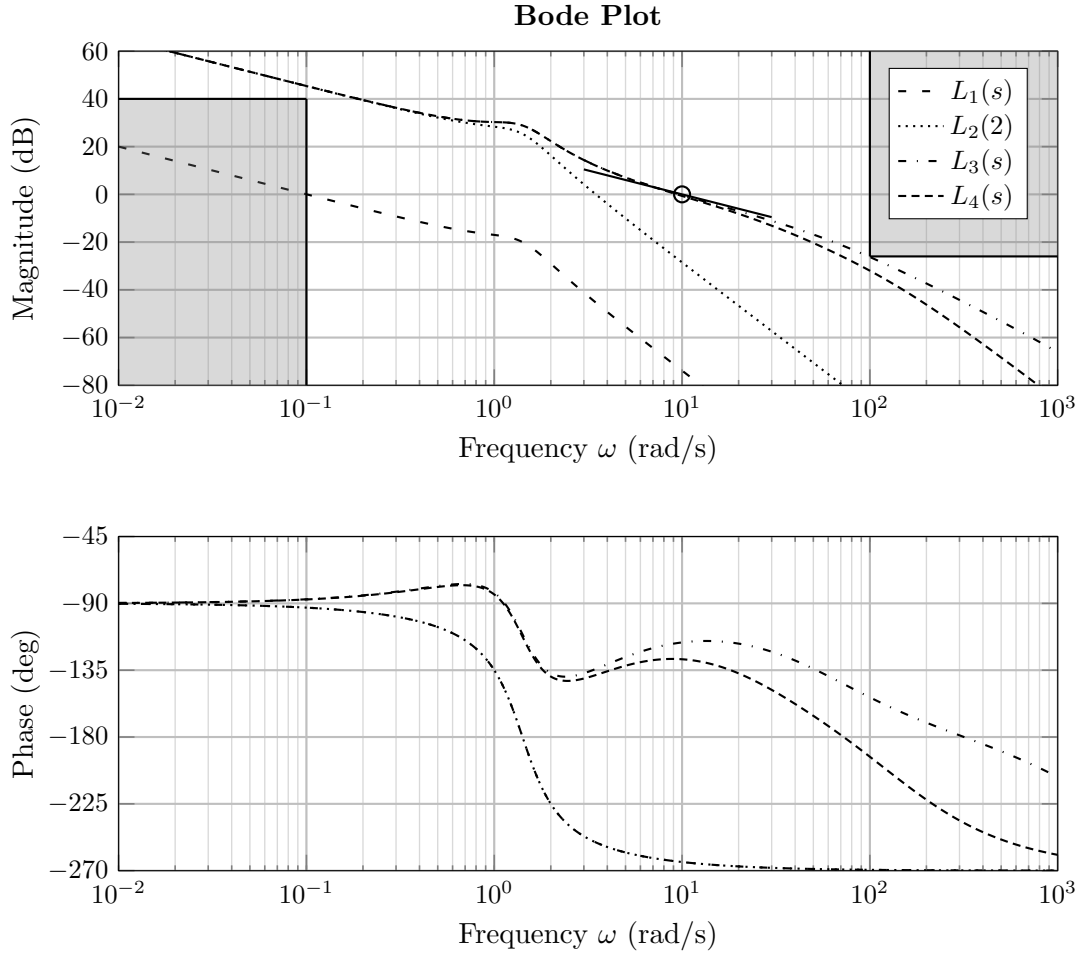


Figure 7.16: Bode plots of all design iterations of the dynamic compensator using the loop shaping approach.

Why does PID work so well?

In a previous chapter we said that PID control works well in practice. But why is that? In order to understand this recall the general form of a PID controller:

$$C_{\text{PID}}(s) = K_P + \frac{K_I}{s} + K_D s,$$

where K_P is the proportional gain, K_I the integral gain, and K_D the derivative gain. For a closed-loop system in the time-domain, this is equivalent to manipulating the error in different ways and feeding it back into the system. Specifically

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt},$$

where $u(t)$ is the controller output applied to the plant based on the proportional, integral, and derivative actions on the error $e(t)$. We can now remember the Bode plots of the proportional, integral, and derivative controllers as shown in Figure 7.17.

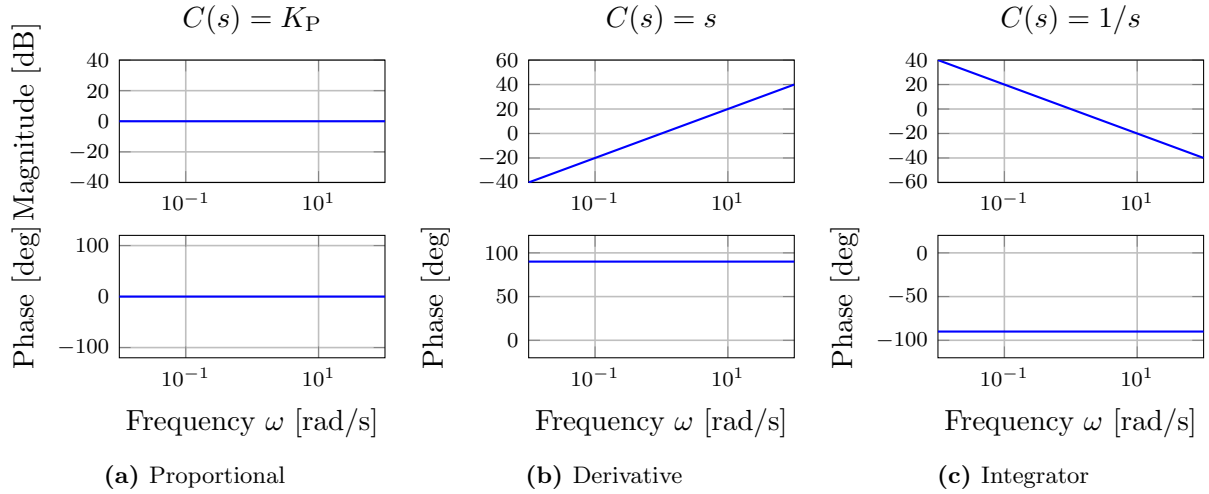


Figure 7.17: Bode plots of the proportional, derivative, and integral building blocks.

Therefore, looking through the lens of the frequency-domain, we can interpret the tuning of a PID controller as adding or removing a certain amount of each Bode plot. For example, if we want to add phase and are fine with losing some noise rejection, we could increase our derivative gain K_D . Inversely, if we were fine with losing some phase, but wanted to increase our command tracking and noise rejection, we could increase our integral gain K_I . Since the PID controller contains derivative action, that is not a causal operator (although it can be approximated), we often consider the following implementable version of a PID controller:

$$C_{\text{PID}}(s) = K \cdot \frac{\left(\frac{s}{z_1} + 1\right) \cdot \left(\frac{s}{z_2} + 1\right)}{s \left(\frac{s}{p} + 1\right)},$$

where K is a scalar gain, z_1 and z_2 denote controller zeros, and p is a high-frequency pole used to regularize the differentiating action. This factorization can be recognized as

$$C_{\text{PID}}(s) = K \underbrace{\frac{\frac{s}{z_1} + 1}{s}}_{\text{lag with pole at } s=0} \underbrace{\frac{\frac{s}{z_2} + 1}{\frac{s}{p} + 1}}_{\text{lead with pole at } s=-p},$$

so the integrator furnishes the lag behavior, while the zero-pole pair acts as a lead compensator whose pole is typically chosen at $-p$ with $p \gg 1$ to approximate an ideal differentiator over the bandwidth of interest.

Example - PID Controller Design for a First-Order Plant

Consider the following plant and its Bode plot (Figure 7.18):

$$P(s) = \frac{1}{s + 2}.$$

This first-order model captures a stable plant with a single real pole at -2 . Its Bode

magnitude rolls off with a slope of -20 dB/dec past 2 rad/s while the phase approaches -90° .

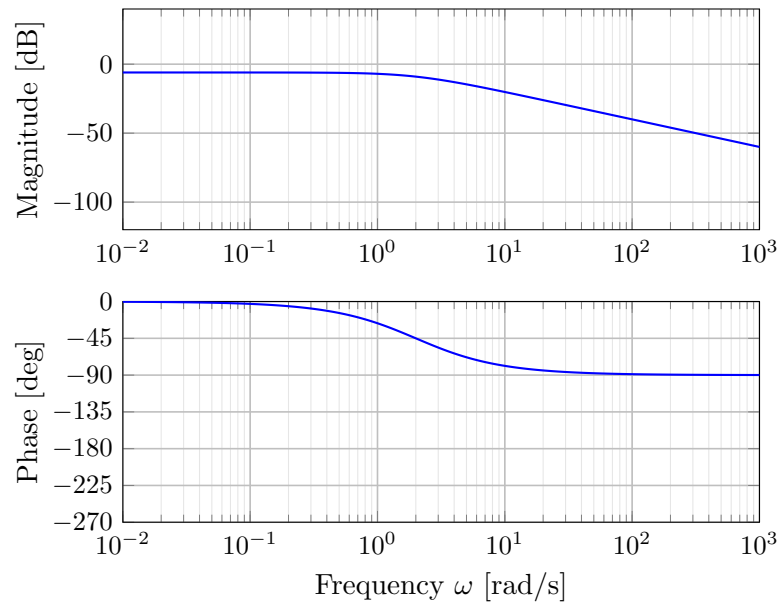


Figure 7.18: Bode magnitude and phase of the sample plant $P(s) = 1/(s + 2)$.

A loop-shaping motivated PI controller that adds low-frequency gain for tracking could be:

$$C(s) = \frac{6s + 4}{s}.$$

This could be expressed as:

$$C(s) = 6 + \frac{4}{s},$$

which is a PI controller with proportional gain $K_P = 6$ and an integral gain $K_I = 4$.

The corresponding Bode plot of $C(s)$ is shown in Figure 7.19. We can see that the -20 dB/dec slope from the integral action at low frequencies and a mid-band plateau set by the proportional gain. Because there is no derivative term, the phase never exceeds $+90^\circ$, but the controller zero at $-2/3$ still provides a mild phase boost near crossover.

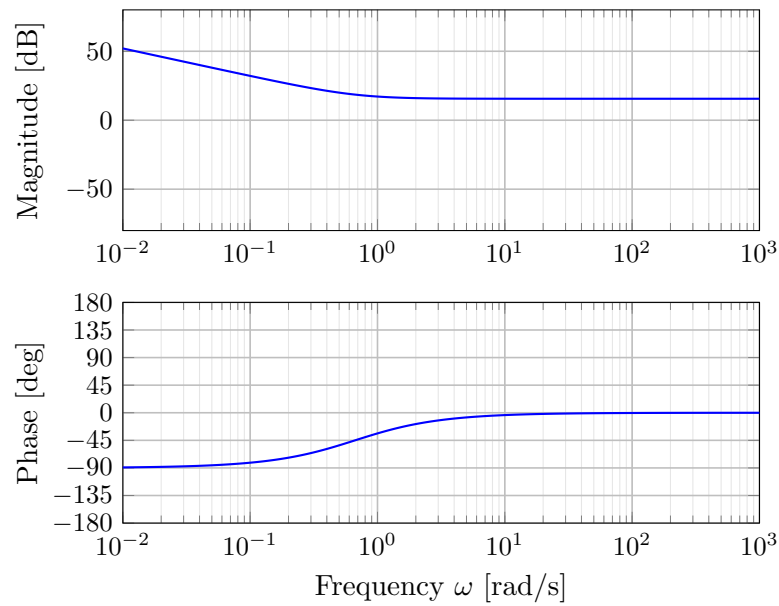


Figure 7.19: Bode magnitude and phase of the tuned PI controller $C(s)$.

Multiplying the two transfer functions yields the open-loop response

$$L(s) = C(s)P(s) = \frac{6s + 4}{s(s + 2)},$$

whose Bode plot is shown in Figure 7.20. The design achieves a crossover close to 4 rad/s with roughly 60° of phase margin while providing an additional 40 dB of low-frequency gain for disturbance rejection.

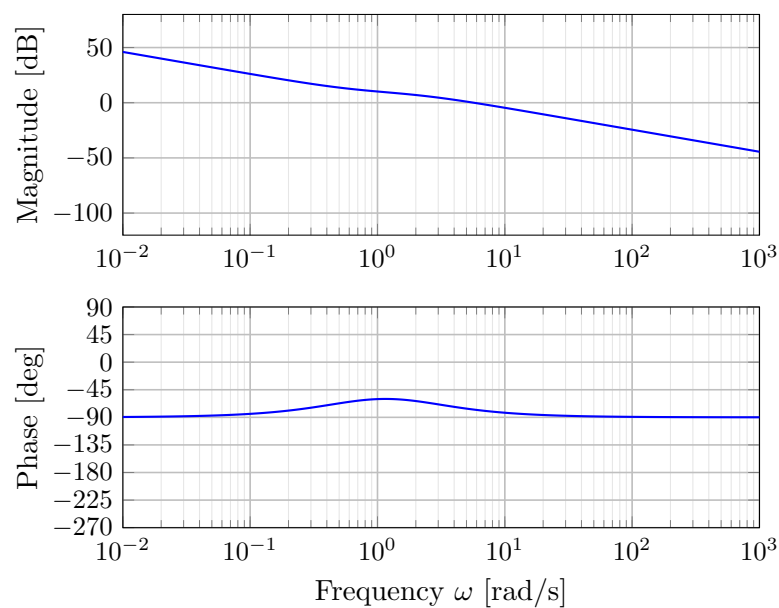


Figure 7.20: Bode magnitude and phase of the open-loop transfer function $L(s) = C(s)P(s)$.

From this example we observe three useful takeaways:

- The integrator in $C(s)$ guarantees zero steady-state error by raising the low-frequency magnitude of $L(s)$;
- The proportional term in $C(s)$ sets the desired crossover while keeping high-frequency gain moderate;
- Even without derivative action the controller zero shapes the phase enough to maintain a healthy phase margin.

PI tuning therefore provides an inexpensive way to trade off tracking authority, robustness, and noise amplification by looking solely at the Bode plot of the open-loop system.

Non-minimum-phase and Unstable Systems

Previously we stated that the Bode plot can not be used to arrive at conclusions about the stability of a system if the system is non-minimum-phase or unstable. So how can we use loop shaping for these systems? The idea here is to refactor the plant's transfer function into a minimum-phase and “all-pass” part. This looks as follows:

$$P(s) = P_{\text{mp}}(s) D(s),$$

$P_{\text{mp}}(s)$ is obtained by replacing all poles and zeros in the right half plane with their mirror image with respect to the imaginary axis, while $D(s)$ contains all the original unstable and non-minimum phase terms and their inverses introduced in the refactoring. Note that this architecture means that by definition:

$$|D(j\omega)| = 1, \quad \forall \omega \in \mathbb{R}$$

This constant magnitude in frequency response is why we call $D(s)$ an “all-pass filter”. Note that the all-pass filter does not affect the magnitude plot, but does influence the phase plot.

For example, consider the unstable and non-minimum phase transfer function:

$$P(s) = \frac{s - z}{s - p}, \quad \text{with } z, p > 0.$$

We refactor it as follows:

$$\begin{aligned} P(s) &= \frac{s - z}{s - p} \cdot \frac{s + z}{s + z} \cdot \frac{s + p}{s + p} \\ &= \underbrace{\frac{s + z}{s + p}}_{P_{\text{mp}}(s)} \cdot \underbrace{\frac{s - z}{s + z} \cdot \frac{s + p}{s - p}}_{D(s)}. \end{aligned}$$

Therefore,

$$P_{\text{mp}}(s) = \frac{s + z}{s + p},$$

$$D(s) = \frac{s - z}{s + z} \cdot \frac{s + p}{s - p},$$

where $P_{\text{mp}}(s)$ is minimum-phase/stable and $D(s)$ mirrors the problematic zero/pole while preserving unit magnitude. We can then use the loop shaping techniques we have learned to shape the minimum-phase part $P_{\text{mp}}(s)$.

When we shape the minimum-phase part of the plant, $P_{\text{mp}}(s)$, we are shaping the portion we can influence, but this is never sufficient on its own because the non-minimum-phase factor $D(s)$ still injects unavoidable phase lag, delays, and right-half-plane dynamics into the final loop. After designing $C(s)P_{\text{mp}}(s)$ we must always reincorporate $D(s)$ and re-examine the actual loop $L(s) = C(s)P_{\text{mp}}(s)D(s)$. The phase plot becomes the critical checkpoint: the fixed phase lag contributed by $D(s)$, from RHP zeros, delays, or lightly damped poles can easily erode phase margin and reduce robustness, no matter how cleanly the minimum-phase part was shaped. Since we can compute this lag explicitly, we must keep it in mind when targeting crossover frequency or desired margins. Finally, we need to verify that the design is stable. This is done by checking the Nyquist plot. The Nyquist plot accounts for unstable poles, non-minimum-phase behavior, and all the distortions introduced by $D(s)$, and only the final Nyquist contour can confirm internal stability.

In short: shape the minimum-phase part, but always check the true loop's phase and close the loop with Nyquist to ensure the design survives the non-minimum-phase realities of the plant.

Chapter 8

Time Delays

8.1 Introduction

In the previous chapter we saw how to synthesize a controller for a given dynamical system, and hence, given a system and some control objectives, we can design a controller to achieve those objectives. Nonetheless, so far we have considered control systems in which inputs, disturbances, and references are applied instantaneously. However, in practice, real systems frequently exhibit delays originating from sensing, computation, communication, or intrinsic physical propagation. For example, when driving a car, there is a slight time delay between applying the throttle and the car accelerating (a physical propagation delay).

As discussed in Chapter 1, the idea behind feedback control is that one does not require a high-fidelity model of the system to achieve the control objectives, but rather a model that contains all the *key* elements. Time delays are one such element and, depending on the system, they may need to be taken into account when designing a controller for a dynamical system. Consequently, the purpose of this chapter is to understand how to model time delays, and how to analyze the influence of time delays on the system behavior.

In Section 8.2 we introduce the time-delay operator, derive its transfer function, and examine its frequency-domain characteristics through Polar, Nyquist, and Bode representations. Building on this, in Section 8.2.3 we investigate how delays alter the loop transfer function and reduce the phase margin, highlighting their impact on closed-loop stability. In Section 8.2.4 we then outline a practical controller-design procedure that accounts for this degradation. Finally, in Section 8.3 we discuss how the root-locus method can be applied despite the non-rational nature of delay operators: in Section 8.3.1 we analyze a naive Taylor-series approximation and its limitations, while Section 8.3.2 presents the Padé approximation as a more accurate rational representation that enables classical analysis tools.

8.2 Introduction to Time Delays

To move from the intuitive examples above to a quantitative treatment, we now model time delays as operators and analyze their influence on closed-loop behavior. Specifically we represent a time delay as an operator that shifts an input signal $u(t)$ by $T \geq 0$ units of time, resulting in the output

$$y(t) = u(t - T).$$

The time-delay operator is linear, i.e., the delayed output of a linear combination of inputs is the same linear combination of the delayed outputs. It is also time-invariant, meaning that shifting the input in time results in an equivalent shift in the output. Thus, time delays fit within the framework of linear time-invariant (LTI) systems, and we can continue to rely on the tools developed in the previous chapters to analyze their influence on closed-loop behavior.

To find the transfer function of such a delay, consider the general input $u(t) = e^{st}$, which when shifted in time, results in the output

$$y(t) = e^{s(t-T)} = e^{st}e^{-sT} = e^{-sT}u(t).$$

We can deduce that the transfer function of a time delay of T seconds is thus

$$H(s) = e^{-sT}. \quad (8.1)$$

Although linear and time-invariant, the transfer function from Equation (8.1) is not rational. This complicates the use of classical tools such as the root-locus method, which are tailored to rational transfer functions. For this reason, frequency-domain techniques are often more suitable for determining closed-loop stability and performance in the presence of delays. Thus, in Section 8.2.1 and Section 8.2.2 we analyze the frequency response of a time delay and its influence on closed-loop stability.

8.2.1 Polar and Nyquist Plot of a Time Delay

To understand how a time delay behaves in the frequency domain, we evaluate its transfer function on purely sinusoidal inputs, i.e., at $s = j\omega$. Substituting $s = j\omega$ into Equation (8.1) gives

$$H(j\omega) = e^{-j\omega T}.$$

We now compute its magnitude and phase. Using Euler's formula $e^{j\theta} = \cos \theta + j \sin \theta$, we see that $e^{-j\omega T}$ has unit magnitude and an angle of $-\omega T$, i.e.,

$$|e^{-j\omega T}| = 1, \quad \angle(e^{-j\omega T}) = -\omega T. \quad (8.2)$$

This has two important consequences:

- The magnitude is constant and equal to 1 for all frequencies. In a polar plot (Figure 8.1),

this means the locus lies on the unit circle centred at the origin.

- The phase decreases linearly with ω . Thus, higher frequencies experience a larger phase lag. In other words, the delay does not attenuate or amplify sinusoidal signals; it only shifts their phase, and the amount of phase shift grows with frequency.

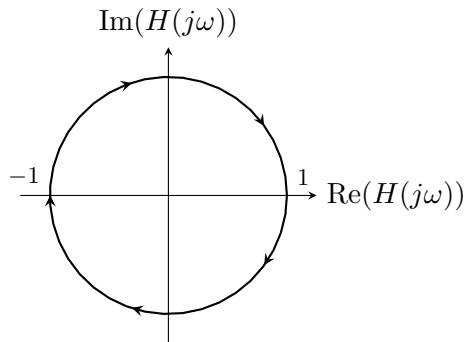


Figure 8.1: Polar plot of a time delay.

This behavior is consistent with what we observe directly in the time domain. If the input is a sinusoid $u(t) = \sin(\omega t)$, then the time-delayed output is

$$y(t) = \sin(\omega(t - T)) = \sin(\omega t - \omega T),$$

which is again a sinusoid of the *same* frequency and amplitude, but shifted in phase by $-\omega T$.

8.2.2 Bode Plot of a Time Delay

As shown in section 8.2.1, the magnitude and phase of a time delay are 1 and $-\omega T$, respectively. This leads to the following Bode plot characteristics:

- The magnitude plot is a straight horizontal line at 0 dB (i.e., magnitude of 1).
- The phase plot is linear in ω , namely $\varphi(\omega) = -\omega T$. To avoid unbounded phase values, it is common to wrap the phase plot to the interval $(-\pi, \pi]$, as shown in Figure 8.2.

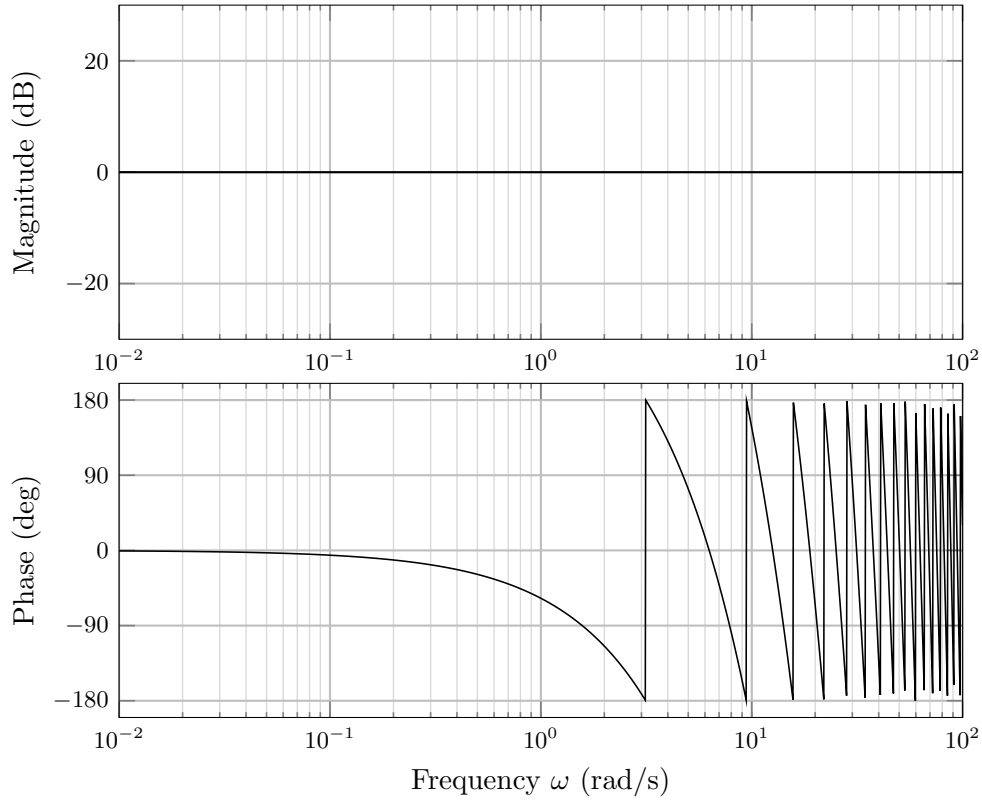


Figure 8.2: Bode plot of a time delay wrapped within the interval $(-\pi, \pi]$.

8.2.3 Effect of a Time Delay on the Frequency Response

We now examine how a time delay affects the frequency response of a feedback system, and in particular its closed-loop stability.

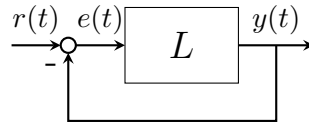


Figure 8.3: Standard feedback architecture

Consider a standard feedback system (Figure 8.3) with loop transfer function $L(s)$. If we introduce a time delay of duration T seconds in the open-loop (Figure 8.3), the open-loop transfer function becomes

$$L'(s) = e^{-sT} L(s).$$

To understand the effect on the frequency response, we evaluate $L'(s)$ to purely sinusoidal inputs, $s = j\omega$:

$$L'(j\omega) = e^{-j\omega T} L(j\omega).$$

From Section 8.2.1, we know that

$$|e^{-j\omega T}| = 1, \quad \angle e^{-j\omega T} = -\omega T.$$

Therefore, the magnitude and phase of the modified loop transfer function are

$$\begin{aligned} |L'(j\omega)| &= |e^{-j\omega T}| |L(j\omega)| = |L(j\omega)|, \\ \angle L'(j\omega) &= \angle e^{-j\omega T} + \angle L(j\omega) = \angle L(j\omega) - \omega T. \end{aligned}$$

These two properties are crucial:

- The *magnitude* of the loop transfer function is unchanged for all frequencies. In particular, the crossover frequency ω_c (defined by $|L(j\omega_c)| = 1$) is the same with or without the delay.
- The *phase* is reduced by an additional term $-\omega T$ that grows linearly with frequency.

Since the magnitude is unchanged, we are only interested in how the phase margin is affected by the time delay. Recall that the phase margin $\varphi_{m,0}$ of the original loop (without delay) is defined as

$$\varphi_{m,0} = 180^\circ + \angle L(j\omega_c),$$

where ω_c is the crossover frequency. With the time delay, the phase at ω_c becomes

$$\angle L'(j\omega_c) = \angle L(j\omega_c) - \omega_c T,$$

so the new phase margin is

$$\varphi_{m,T} = 180^\circ + \angle L'(j\omega_c) = 180^\circ + \angle L(j\omega_c) - \omega_c T = \varphi_{m,0} - \omega_c T.$$

Thus, we can conclude that:

- For a fixed delay T , increasing the crossover frequency ω_c (i.e., making the loop faster) *reduces* the phase margin.
- For a fixed crossover frequency, increasing the delay T reduces the phase margin.

Thus, the phase margin decreases linearly with both the crossover frequency and the time delay. If the delay is too large, the phase margin may become zero or negative, causing the closed-loop system to become unstable.

8.2.4 Design Procedure for Systems with Time Delays

The considerations made in the previous sections lead to a procedure for designing feedback controllers for systems with time delays. The procedure can be summarized as follows:

1. Design the feedback controller assuming zero time delay.
2. Evaluate the effective phase margin using $\varphi_{m,T} = \varphi_{m,0} - \omega_c T$. If the resulting phase margin is too small or negative (implying instability), modify the controller accordingly by either:

- Increasing the phase at crossover (e.g. via a phase lead controller);
 - Reducing the crossover frequency (e.g. by lowering the gain or using a phase lag controller).
3. Iterate until satisfactory closed-loop performance is achieved.

This procedure follows directly from the effect a time delay has on the phase response and phase margin of the system, and is commonly used in practice.

8.3 Time Delays and the Root Locus method

In previous chapters, we introduced the root-locus method as a powerful tool for assessing the feasibility of a controller design. However, as the transfer function of a time-delay is not rational, the root-locus method cannot be applied directly (since the root locus requires a rational transfer function). Instead, we need to approximate the time delay as a rational transfer function. Below we introduce two such approximations: the Taylor series approximation and the Padé approximation.

8.3.1 Taylor Series Approximation

A first idea for approximating a time delay is to expand the exponential term in a Taylor series around $sT = 0$. Recall that a pure delay of duration T has the transfer function

$$e^{-sT}.$$

One may attempt a direct approximation by using the Taylor expansion of the exponential:

$$e^{-sT} = 1 - sT + \frac{1}{2}(sT)^2 - \frac{1}{6}(sT)^3 + \dots$$

If we truncate this series, for instance keeping only terms up to second order, we obtain

$$e^{-sT} \approx 1 - sT + \frac{1}{2}(sT)^2.$$

However, this truncated expression introduces several issues. Firstly, it's not a proper transfer function, secondly it has two non-minimum-phase zeros, and thirdly, unlike the true delay whose magnitude satisfies

$$|e^{-j\omega T}| = 1 \quad \text{for all } \omega,$$

the magnitude of the Taylor approximation diverges as $\omega \rightarrow \infty$.

For these reasons, the naive Taylor approximation results in a poor representation of the delay, especially at higher frequencies, and hence is generally unsuitable for stability analysis and controller design.

8.3.2 Padé Approximation

A more accurate way to approximate a time delay is to use a Padé approximant. Instead of truncating the Taylor series of e^{-sT} directly, we seek a (proper) rational function whose Taylor expansion matches that of the exponential up to a chosen order.

In other words, given a function $f(x)$, the *Padé approximant* of order $[m/n]$ is the rational function,

$$R(x) = \frac{\sum_{k=0}^m a_k x^k}{1 + \sum_{p=1}^n b_p x^p} = \frac{a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m}{b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n},$$

for which $f^{(d)}(0) = R^{(d)}(0) \forall d = \{0, 1, \dots, n+m\}$.

The Padé approximant is desirable because it matches the Taylor series of the function up to the chosen order, whilst ensuring that the rational function is proper. Although in practice this is usually computed through your chosen software package, it is useful to understand the process behind it and why it is suitable for the time delay approximation. Thus, below we derive the first-order Padé approximant for the time delay.

For a first-order Padé approximation, we assume an expression of the form

$$e^{-sT} \approx k \frac{s+p}{s+q},$$

where the constants k , p , and q are to be determined. Since we are looking for a first-order approximation (i.e., $m = 1$ and $n = 1$), we need to match the expression to the first two terms of the Taylor series, i.e.,

$$k \frac{s+p}{s+q} = 1 - sT + \frac{1}{2}(sT)^2.$$

We can then simplify the expression to obtain

$$ks + kp = s + q - s^2T - qsT + \frac{1}{2}s^3T^2 + \frac{1}{2}q(sT)^2.$$

By comparing the coefficients on both sides, we can solve for k , p , and q ,

$$k = 1, \quad p = -\frac{2}{T}, \quad q = \frac{2}{T},$$

so that the first-order Padé approximation becomes

$$e^{-sT} \approx \frac{\frac{2}{T} - s}{\frac{2}{T} + s} = \frac{2/T - s}{2/T + s}.$$

This rational approximation preserves an important property of the true delay, which is that its magnitude remains equal to one for all frequencies,

$$\left| \frac{\frac{2}{T} - j\omega}{\frac{2}{T} + j\omega} \right| = 1.$$

At the same time, the approximation introduces one pole at $s = -\frac{2}{T}$ and one zero at $s = +\frac{2}{T}$, the latter being a non-minimum-phase zero, which we can represent in the root locus analysis.

Thus, by replacing the non-rational delay term e^{-sT} with this rational expression, classical tools such as the root-locus method can once again be applied. Nevertheless, the accuracy of the approximation remains limited, particularly at higher frequencies. Higher-order Padé approximations improve fidelity by matching more terms of the Taylor series, although this comes at the cost of introducing additional poles and zeros. As a result, while Padé approximations provide useful intuition and enable the use of the root-locus for a time delayed system, the conclusions drawn from them should be interpreted with care. Hence, it is useful for back of the envelope calculations, with Bode or Nyquist plots being used to confirm the results.

Chapter 9

Describing Functions

9.1 Introduction

So far, this course has predominantly covered linear systems, where the plant, controller, and the open- and closed-loop systems are all linear. However, most real-world systems are *nonlinear*. Until now, we have handled nonlinearities in the plant by *linearizing* around equilibrium points and then designing (linear) controllers to achieve some control objectives¹. Because the behavior of the nonlinear system near an equilibrium is often well approximated by its linearization, this approach allows us to use linear tools to understand key aspects of the nonlinear system, such as stability, robustness margins, and time- and frequency-domain performance with respect to specifications.

However, linear models cannot capture all phenomena that nonlinear models may exhibit. One important example is the presence of self-sustained oscillations, known as *limit cycles*. Unlike oscillations in marginally stable LTI systems (which neither grow nor decay, and whose amplitude is set by the initial conditions), limit cycles arise from the nonlinear dynamics themselves: the system converges to a periodic motion of a particular amplitude that persists even in the absence of external input. Understanding such phenomena requires tools that go beyond the linear analysis we have covered so far.

Thus, this chapter offers a brief introduction to analyzing nonlinear systems. Since the behavior of nonlinear systems can be very complex, and because there is still no single general method to analyze arbitrary nonlinear dynamics, we will focus on a special case: systems with a single static nonlinearity in the feedback path. In Section 9.2, we introduce static nonlinear gains and explain how to incorporate them into our control system. In Section 9.3, we develop tools to assess the stability of such systems. Then, in Section 9.4, we introduce *describing functions* as a tool that approximates the effect of the nonlinear element by an amplitude-dependent transfer function. Finally, in Section 9.5, we use describing functions to identify *limit cycles* in our system and determine whether these are stable or unstable.

¹Recall that transfer functions represent LTI systems, so specifying a transfer function for a controller results in a linear controller.

9.2 Static Nonlinearities

In a real control loop, many elements can be nonlinear (the plant, sensors, actuators, even the controller implementation). In this chapter, however, we restrict attention to a restricted but very relevant class of systems: those with a *single static nonlinear feedback gain* NL in an otherwise linear feedback loop, as shown in Figure 9.1.

Before introducing tools for analyzing nonlinear systems, it is helpful to clarify the specific type of nonlinearity we will study and where it arises in practice. Rather than defining this structure abstractly right away, we first illustrate it through a concrete example and will then return to formalize what we mean by a static nonlinear feedback gain.

Consider again the cruise-control problem for a car. The objective is to regulate the vehicle's speed, and the controller produces a command representing the desired traction force (or torque) delivered by the engine. In practice, however, the engine cannot generate arbitrarily large forces: the engine output saturates at a physical maximum. This creates a static nonlinearity in the feedback loop — the actual force is approximately proportional to the control input for small inputs, but becomes “clipped” once the input exceeds the engine's capability (Figure 9.2a is an example of this clipping behavior).

Diagrammatically, we represent this nonlinearity as a gain NL in the feedback loop, as shown in Figure 9.1: the controller receives an error signal e and produces a control input u , which passes through the nonlinearity NL to produce an output y . The nonlinear element NL clips the input u to ensure that it remains within a maximum and minimum value. This output y is then passed through the plant P to produce the feedback signal y_{fb} .

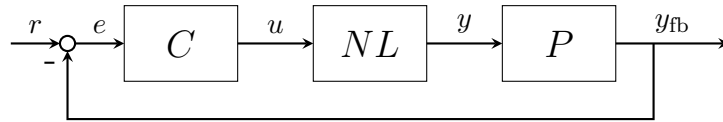


Figure 9.1: Feedback architecture for the cruise control system. The nonlinear element NL represents the engine saturation, which saturates the minimum and maximum actuation levels, while C and P represent the linear controller and plant respectively.

The saturation nonlinearity is an example of a *static nonlinear feedback gain*. In this context, “static” means that it is a memoryless input-output map

$$NL : u \mapsto y(u).$$

At each time instant, the output of the nonlinearity NL depends only on the current value of its input, not on past values. Such nonlinear elements often arise in practical systems and can significantly affect system behavior. A few examples of static nonlinearities are listed below with the corresponding input-output relationship visualized in Figure 9.2.

Saturation

Output is limited to a maximum and/or minimum value. Example: an electric motor can only deliver torque up to a certain limit.

Relay / Switch

The output switches between two distinct levels. Example: A thermostat turns the heater on when the temperature drops below a set point, and off when the temperature rises above that same set point.

Dead zone

Small inputs around zero produce no output change. Example: a joystick may have a small range around the neutral position where motion has no effect.

Quantization

A continuous input is mapped to discrete output levels. Example: a digital thermometer rounds temperature to the first decimal place.

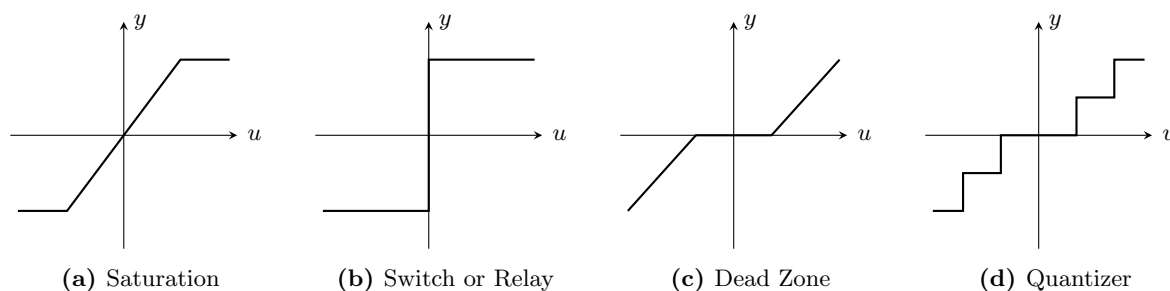


Figure 9.2: Input-Output behavior of various static nonlinear elements.

On the other hand, a *dynamic* nonlinearity is one that depends on the direction (derivative) of the input. A few examples of dynamic nonlinearities are listed below with the corresponding input-output relationship visualized in Figure 9.3.

Hysteresis

The output depends on the history (direction) of the input as well as its current value. Example: a thermostat turns heating on below one temperature and off only after exceeding a higher temperature.

Schmitt Trigger

The output switches between two distinct levels. Example: a thermostat turns the heater on when the temperature drops below a set point, and off when the temperature rises above a *different* set point.

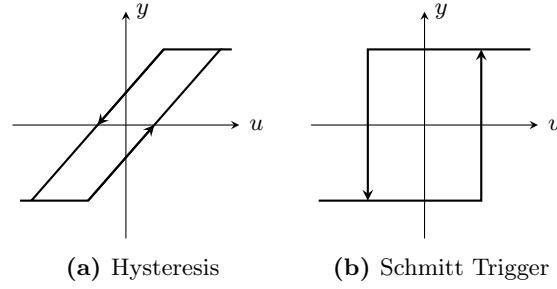


Figure 9.3: Input-Output behavior of various dynamic nonlinear elements.

9.3 Absolute Stability with Static Nonlinearities

Similar to the linear case, we would like to assess the stability of feedback systems that contain a nonlinear element. Recall that for linear systems, stability is a property of the closed-loop dynamics and does not depend on the particular reference signal. The same idea carries over to the nonlinear setting: our goal is to determine whether the closed-loop system is stable, independent of the reference signal. This motivates studying stability of the closed-loop system through the Nyquist plot of the (nonlinear) open-loop transfer function.

Recall that in the purely linear setting, the open-loop transfer function appears in the form $kL(s)$. In contrast, the cruise-control architecture in Figure 9.1 is expressed as $C(s) \cdot NL \cdot P(s)$. Because we are working with linear SISO systems, all transfer functions are scalar and hence commute. This permits us to rewrite the open-loop interconnection as $NL \cdot L(s)$ where $L(s) = C(s)P(s)$, and leads us directly to the equivalent architecture shown in Figure 9.4. This rearrangement isolates the nonlinearity in a standard form and enables the application of (an extended) Nyquist criterion.

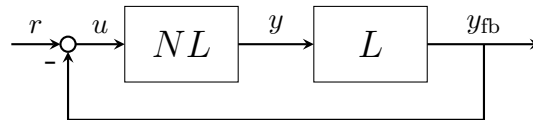


Figure 9.4: Equivalent feedback architecture for the cruise control system. The non-linear element NL represents the throttle, which is limited to a maximum and minimum value. The linear open-loop transfer function L represents the plant and controller.

To apply Nyquist-type criteria to a nonlinear feedback system, we require a description of how the nonlinearity NL may influence the loop gain. A standard assumption in absolute stability theory is that the static nonlinearity lies in a *sector* $[k_1, k_2]$, meaning that it is bounded by two constant gains. Formally, we assume

$$NL(0) = 0 \quad \text{and} \quad k_1 \leq \frac{NL(u)}{u} \leq k_2 \quad \forall u \neq 0. \quad (9.1)$$

Graphically, the input-output curve of NL must lie in the shaded regions between the lines $k_1 u$ and $k_2 u$ (see Figure 9.5).

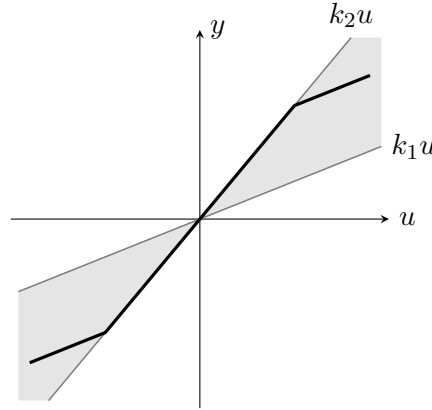


Figure 9.5: Static nonlinearity (thick black line) bounded by gains k_1 and k_2 . The nonlinearity will be contained in the shaded region for all u .

Definition 9.1 (Absolute Stability) Consider the feedback interconnection shown in Figure 9.4, consisting of a linear time-invariant (LTI) system $L(s)$ with $r(t) = 0$ and a static nonlinearity NL that satisfies (9.1).

The closed-loop system is said to be absolutely stable in the sector $[k_1, k_2]$ if $u = 0$ is a globally asymptotically stable equilibrium of the closed-loop system.

In other words, the interconnection of L and NL is absolutely stable in the sector $[k_1, k_2]$ if, for every admissible nonlinearity NL and every initial condition of the linear subsystem L with no reference, the corresponding closed-loop trajectories satisfy

$$\lim_{t \rightarrow \infty} u(t) = 0, \quad \text{and} \quad \lim_{t \rightarrow \infty} y(t) = 0.$$

Having formalized both the structure of the static nonlinearity and the notion of absolute stability, in the following sections we provide conditions that allow us to verify this property in practice. The key idea is that the sector bounds $[k_1, k_2]$ effectively restrict the possible “gain” of the nonlinearity NL , allowing us to adapt the Nyquist criterion to nonlinear feedback systems. In Section 9.3.1 we introduce a necessary condition for absolute stability, and in Section 9.3.2 we build upon this to introduce a sufficient condition that checks whether the Nyquist plot of the linear subsystem L avoids a disk defined by the sector bounds $[k_1, k_2]$.

9.3.1 Necessary Condition for Absolute Stability

A necessary condition for absolute stability follows directly from the sector bound on the nonlinearity. Since NL satisfies the condition

$$k_1 \leq \frac{NL(u)}{u} \leq k_2,$$

the nonlinearity behaves, at every operating point, like an effective gain k_{NL} lying somewhere in the interval $[k_1, k_2]$. Since absolute stability must hold for every such admissible gain, we

need to verify that the Nyquist plot of $L(s)$ avoids all points in the interval $[-1/k_1, -1/k_2]$.

Geometrically, this means that instead of avoiding a single point in the complex plane, the Nyquist plot must avoid an entire *line segment* between $-1/k_1$ and $-1/k_2$, as illustrated in Figure 9.6. This condition is necessary for absolute stability.

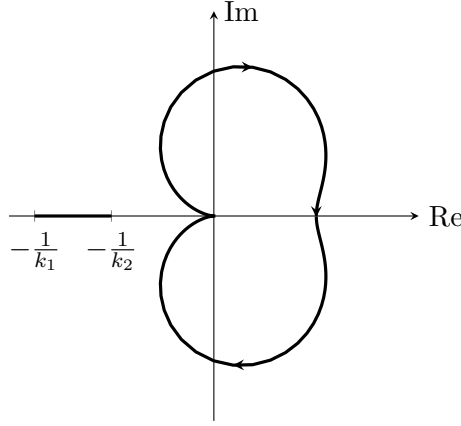


Figure 9.6: Nyquist plot illustrating the necessary condition for absolute stability. The line segment between $-1/k_1$ and $-1/k_2$ indicates the range of points that must be considered for closed-loop stability.

9.3.2 Sufficient Condition for Absolute Stability

The necessary condition showed that the Nyquist plot of $L(s)$ must avoid the entire line segment between the points $-1/k_1$ and $-1/k_2$. However, avoiding this segment alone guarantees stability only for every *constant* gain in the sector $[k_1, k_2]$. To ensure stability for any varying gain within the sector, a stronger geometric constraint is required.

This is because a general sector-bounded nonlinearity may behave like a time-varying gain. Hence, its gain may take any value within the sector, and its interaction with the phase of $L(j\omega)$ can be more destabilizing than any fixed gain.

The Circle Criterion captures this effect by enlarging the exclusion region from a line segment to the entire *closed disk* with the segment $[-1/k_1, -1/k_2]$ as its diameter. In other words, for absolute stability it is sufficient that the Nyquist plot of $L(s)$ does not enter this disk and encircles the disk counterclockwise a number of times equal to the number of unstable poles of $L(s)$. This geometric condition is illustrated in Figure 9.7. It is important to emphasize that this condition is *sufficient* but not *necessary*. If the Nyquist plot avoids the line segment but intersects the surrounding disk, the system may still be absolutely stable. Determining stability in such cases typically requires more advanced tools. In the following section we will introduce describing functions as a tool to determine stability in such cases.

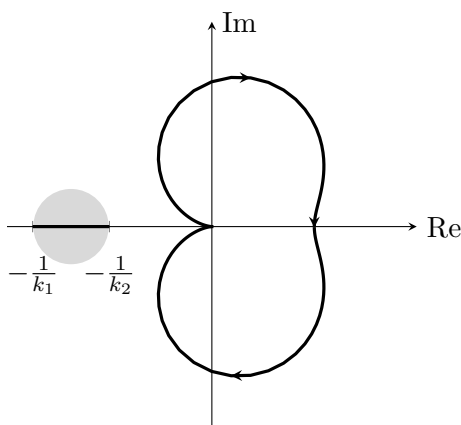


Figure 9.7: Nyquist plot illustrating the sufficient condition for absolute stability. The circle between $-1/k_1$ and $-1/k_2$ indicates the range of points that must be considered for closed-loop stability.

9.4 Describing Functions

In the previous section we introduced necessary and sufficient conditions for absolute stability for systems with a static nonlinearity. However, in the case that the sufficient condition is not met, but the necessary condition is, we saw that we could not conclude anything about the stability of the system. This section introduces describing functions as an *approximate* method for analyzing stability in such cases.

A *describing function* is an amplitude-dependent complex gain that approximates the input-output behavior of a nonlinearity under sinusoidal inputs.

By obtaining such an approximation for the nonlinearity, it permits us to analyze the stability of the system using tools we have already seen (such as the Nyquist plot)

The rest of this section is dedicated towards introducing how to derive the describing function for a given nonlinearity. In particular, we first focus on the case of static nonlinearities to sinusoidal inputs, and then later extend the idea to general (potentially dynamic) nonlinearities. However, before we do so, we introduce the key idea behind how describing functions work below.

Suppose that a system with a nonlinearity NL is subject to a sinusoidal input

$$u(t) = A \sin(\omega t).$$

The output of the system is then

$$y(t) = f(A \sin(\omega t)). \quad (9.2)$$

In other words, the output is a nonlinear function that depends on the amplitude A and frequency ω of the input. The exact relationship depends on the type of nonlinearity present in the system, such as those presented in Section 9.2. Since the input is periodic, the output is

also periodic, and we may therefore express it using a Fourier series expansion to decompose the nonlinear response into harmonic components. We will then approximate the nonlinearity by retaining only its first harmonic, neglecting higher-order harmonics. This then provides us with a way to approximate the nonlinearity as an amplitude dependent gain.

9.4.1 Describing Functions for Static Nonlinearities

Recall that a periodic function can be expressed through a Fourier Series as

$$y(t) = \frac{a_0}{2} + \sum_{i=1}^{+\infty} [a_i \cos(i\omega t) + b_i \sin(i\omega t)],$$

where

$$a_i = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \cos(i\omega t) d(\omega t),$$

$$b_i = \frac{1}{\pi} \int_{-\pi}^{\pi} y(t) \sin(i\omega t) d(\omega t),$$

and i denotes the index of the Fourier coefficient — not to be confused with the imaginary unit j . Note that depending on function symmetry we can make the following simplifications:

$$\begin{cases} y(t) \text{ odd} & \implies a_i = 0 \quad \forall i, \\ y(t) \text{ even} & \implies b_i = 0 \quad \forall i. \end{cases}$$

Physical systems typically behave as low-pass filters because inertia, energy storage, and finite actuation speed prevent them from responding to rapid (high-frequency) variations in input. Therefore, higher frequency components in the output are typically attenuated, and we can approximate the nonlinearity by only considering its fundamental component, i.e. $i = 1$. By normalizing the resulting amplitude dependent gain, with the input amplitude A , we then arrive at the definition for a describing function for static nonlinearities.

The describing function for a static, symmetric nonlinearity NL is defined as:

$$N(A) = \frac{b_1(A) + ja_1(A)}{A}$$

where $b_1(A)$ and $a_1(A)$ are the coefficients of the first harmonic of the output.

When the nonlinearity is odd or even, then $a_1 = 0$ and $b_1 = 0$ respectively, and we can simplify the describing function to:

$$\begin{cases} y(t) \text{ odd} : & N(A) = \frac{b_1(A)}{A} = \frac{1}{\pi A} \int_{-\pi}^{\pi} y(t) \sin(\omega t) d(\omega t), \\ y(t) \text{ even} : & N(A) = \frac{ja_1(A)}{A} = \frac{j}{\pi A} \int_{-\pi}^{\pi} y(t) \cos(\omega t) d(\omega t). \end{cases}$$

Example – Saturation Nonlinearity

Let us consider a saturation nonlinearity as shown in Figure 9.8. If we apply an input $u = A \sin(\omega t)$ to the nonlinearity NL , then the output $y(u)$ will be unaffected if the amplitude is smaller than 1, i.e. $A < 1$. If the amplitude of the input signal is larger than the saturation threshold, i.e. $A > 1$, then the output will be truncated at values that are larger than the threshold.

$$y(u) = \begin{cases} 1 & \text{if } u \geq 1 \\ u & \text{if } -1 < u < 1 \\ -1 & \text{if } u \leq -1 \end{cases}$$

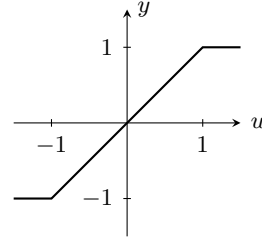


Figure 9.8: Mathematical description of the saturation nonlinearity and the corresponding graph.

The effect of the nonlinearity on inputs of different amplitudes is portrayed in Figure 9.9. Low amplitude frequencies are passed with no effect, whereas signals with higher amplitudes get cut off.

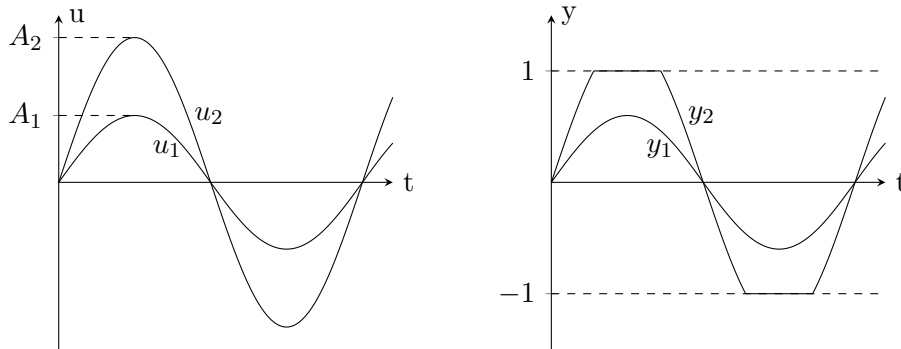


Figure 9.9: Comparison of inputs and outputs of a saturation nonlinearity.

We recognize two areas of distinct effect on an input: the signal is unaffected or truncated. In the truncated case, finding the first harmonic gives us an approximation for what the true truncated signal will look like. This is visualized in Figure 9.10. The larger the amplitude of the input, the more truncation will occur and therefore the farther away from the original signal the first harmonic approximation will be.

To find the Fourier coefficients for the different ranges of A , we first note that the saturation nonlinearity is an odd function, and therefore we have $a_0 = 0$. In the first case, finding the Fourier coefficient proves simple, as the first harmonic fully captures the signal already. In other words, we find that

$$\text{if } A \leq 1 : \quad b_1(A) = A.$$

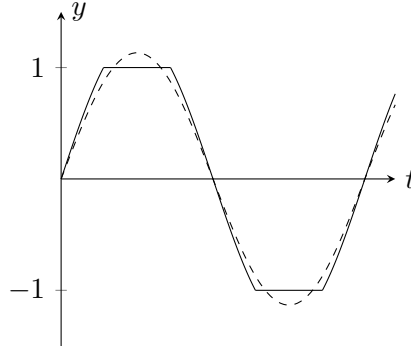


Figure 9.10: Truncated output y (solid line) and its first harmonic approximation (dashed line).

In the second case, where $A > 1$, the calculation of the Fourier coefficient proves more difficult. First, we get from the symmetry features that b_i simplifies to

$$b_i = \frac{4}{\pi} \int_0^{\pi/2} y(t) \sin(i\omega t) d(\omega t).$$

Next we identify the segments of the output that are truncated as a function of the amplitude. We define ψ as the value at which the signal starts to get truncated, i.e.

$$\begin{aligned} A \sin(\psi) &= 1, \\ \implies \psi &= \arcsin\left(\frac{1}{A}\right). \end{aligned}$$

Now, the individual segments can be found as:

$$b_i = \frac{4}{\pi\omega} \int_0^\psi A \sin(\psi) \sin(i\psi) d\psi + \frac{4}{\pi} \int_\psi^{\pi/2} \sin(i\psi) d\psi.$$

But since we assumed that only the first harmonic is needed, we instead evaluate the following integral:

$$\begin{aligned} \text{if } A > 1: \quad b_1(A) &= \frac{4A}{\pi} \int_0^\psi \sin(\psi)^2 d\psi + \frac{4}{\pi} \int_\psi^{\pi/2} \sin(\psi) d\psi \\ &= \frac{4A}{\pi} \int_0^\psi \frac{1 - \cos(2\psi)}{2} d\psi + \frac{4}{\pi} \cos(\psi) \\ &= \frac{2A}{\pi} \arcsin\left(\frac{1}{A}\right) + \frac{2}{\pi} \sqrt{1 - \left(\frac{1}{A}\right)^2}. \end{aligned}$$

Normalizing by the amplitude we arrive at the describing function for the truncated case:

$$N(A) = \frac{b_1(A)}{A} = \frac{2}{\pi} \left(\arcsin\left(\frac{1}{A}\right) + \frac{1}{A} \sqrt{1 - \left(\frac{1}{A}\right)^2} \right).$$

The full describing function for the saturation nonlinearity is now given as:

$$N(A) = \begin{cases} 1, & A \leq 1 \\ \frac{2}{\pi} \left(\arcsin\left(\frac{1}{A}\right) + \frac{1}{A} \sqrt{1 - \left(\frac{1}{A}\right)^2} \right), & A > 1 \end{cases}$$

Plotting this function, visualizes the gain effect for different input amplitudes. In essence, this visualizes how strongly the saturation nonlinearity affects our input signal. The plot for this describing function is shown in Figure 9.11.

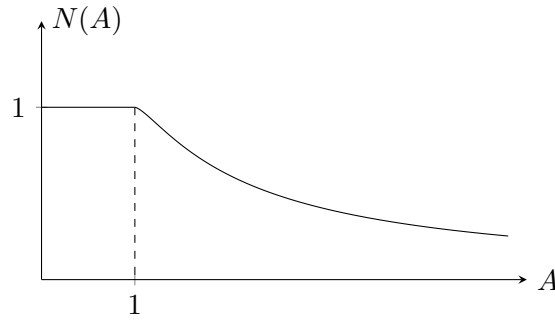


Figure 9.11: Describing function of the saturation nonlinearity with a saturation point at 1.

9.4.2 General Describing Functions

In the more general non-static case of nonlinearities, the describing function can also be a function of frequency. In order to account for this, the describing function of a general nonlinearity NL is defined as the amplitude and frequency dependent normalized approximation:

$$N(A, \omega) = \frac{C_1(A, \omega)}{A} e^{j\phi(A, \omega)},$$

where

$$c_1(A, \omega) = \sqrt{a_1^2 + b_1^2}, \quad \phi(A, \omega) = \arctan\left(\frac{a_1}{b_1}\right).$$

9.5 Limit Cycles

One of the applications of describing functions is in analyzing the existence and characteristics of limit cycles. Limit cycles can occur in any physical nonlinear feedback system.

A limit cycle is a self-sustaining oscillation that arises when the feedback loop stabilizes at a specific amplitude and frequency.

Intuitively, a (stable) limit cycle is a periodic motion such that the system state repeats after a fixed period. Figure 9.12 shows a phase portrait of a system exhibiting a limit cycle. A

phase portrait is a vector field of the system trajectories in the state space, each light gray arrow in Figure 9.12 represents the direction of the state vector at that point in time. Compared with a marginally stable LTI system, a stable limit cycle does not depend on the initial conditions and also attracts nearby trajectories. In the phase portrait of a system exhibiting a limit cycle, as illustrated in Figure 9.12, trajectories (shown in solid black lines) starting from a wide range of initial conditions converge toward the closed orbit (shown in blue). Once the trajectory reaches a stable limit cycle, it remains on that periodic trajectory for all future time.

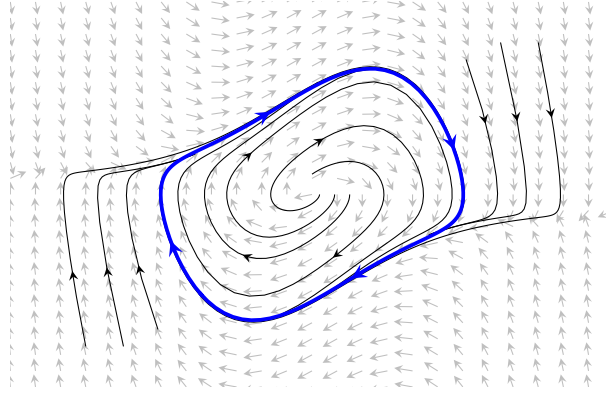


Figure 9.12: Phase portrait of a stable limit cycle (Van der Pol oscillator)

Limit cycles may also be unstable; in that case, trajectories are repelled from the closed orbit. In the context of describing functions, we are primarily interested in stable limit cycles, since they correspond to sustained oscillations that the system settles into.

It should be noted that the presence of a limit cycle is not in itself undesired. Consider that it may be preferable to have a system oscillate at certain frequencies to minimize friction. Nevertheless, it could also be undesired, causing failure of mechanical parts or operator discomfort.

To identify whether a limit cycle is present, we can use the describing function as an approximate transfer function of the nonlinear element. Now let us observe what happens when the input to the NL is given as $Ae^{j\omega t}$ as shown in Figure 9.13. From this we can observe that if

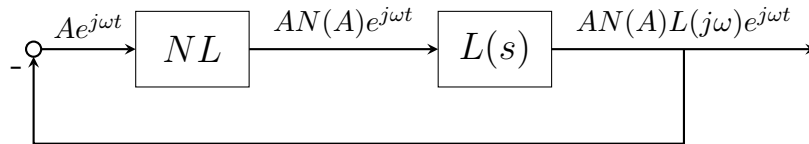


Figure 9.13: Signal propagation through the feedback loop. The nonlinear element NL is approximated by its describing function $N(A)$, and the linear element $L(s)$ modifies the signal according to its frequency response.

$$A = -AN(A)L(j\omega), \quad (9.3)$$

then the feedback loop is self-sustaining. The input $Ae^{j\omega t}$ at NL produces an output at $L(s)$ of $-Ae^{j\omega t}$, which is then negatively fed-back to the input of NL . This causes the system to oscillate indefinitely, causing a limit cycle. A condition for the existence of such a limit cycle is

then given by Equation (9.3) which can be rearranged to obtain

$$-\frac{1}{N(A)} = L(j\omega). \quad (9.4)$$

This condition also provides a graphical way to detect limit cycles. By plotting the Nyquist plot of $L(j\omega)$ and $-1/N(A)$ in the same plane (Figure 9.14) we can verify the condition from Equation (9.4) by looking at the intersections of the two plots which indicate whether limit cycles are present.

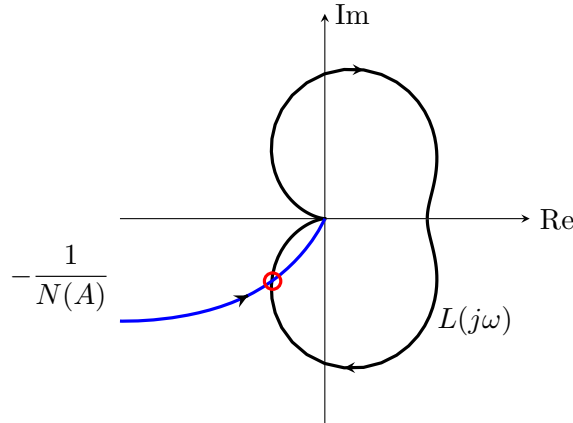


Figure 9.14: Graphical method for detecting limit cycles. The intersection (marked in red) between the Nyquist plot of $L(j\omega)$ and the negative inverse describing function $-1/N(A)$ indicates the presence of a limit cycle at the corresponding amplitude A and frequency ω .

9.5.1 Stability of Limit Cycles

By checking if $L(j\omega)$ and $-1/N(A)$ intersect each other we can determine if there is a limit cycle present in the closed-loop system. By this graphical analysis we can further determine the amplitude and frequency at which this limit cycle occurs. However, recall that the describing function is only an approximation, and consequentially, this graphical approach gives approximate values of amplitude and frequency at which the limit cycle might occur.

Besides the existence of limit cycles one might further be interested in their stability properties. Specifically we will refer to stable limit cycles as those that, after a small perturbation in amplitude, return to their previous value — in a sense they attract nearby trajectories. Conversely, unstable limit cycles are those that, after a small perturbation in amplitude, continue to increase/decrease until another (potential) equilibrium state is reached — such limit cycles can be thought of as repelling nearby trajectories.

To determine the stability of a limit cycle we can use the combination of the Nyquist plot and the plot of $-1/N(A)$ as shown above. As an example consider the case depicted in Figure 9.15. Here the black (solid and dotted) lines indicate the Nyquist plot of an open loop stable system with open-loop transfer function $L(s)$. Stable regions of the closed-loop, i.e. those where there is no net encirclement, are indicated with S while unstable regions are marked with U . The function $-1/N(A)$ is plotted in blue and its arrow indicates the direction in which the function

value moves as A is increased. As before we can now identify limit cycles which occur at the intersections marked with red circles.

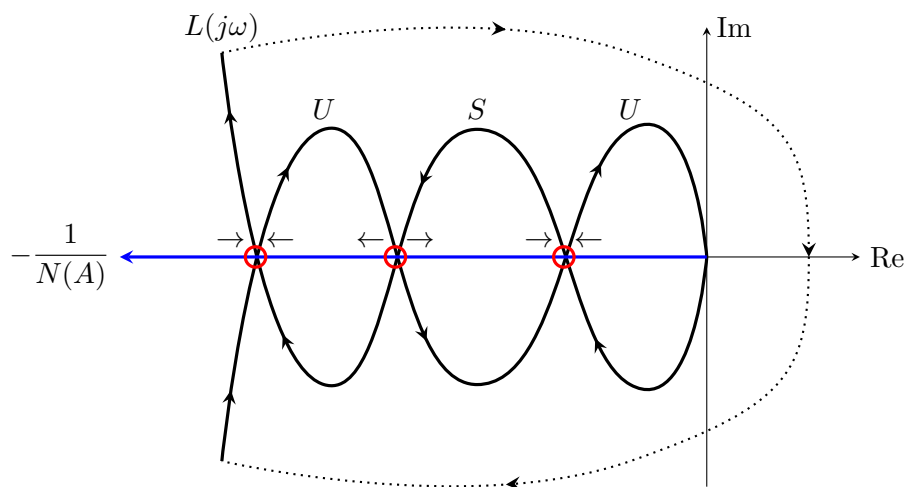


Figure 9.15: Nyquist plot illustrating the stability of limit cycles. The black lines (solid and dotted) represent the Nyquist plot of the open-loop system, while the blue line indicates the trajectory of $-1/N(A)$. Regions marked S and U denote stable and unstable areas, respectively, with red circles indicating points of intersection that correspond to limit cycles.

Let us now consider what will happen at the limit cycles when we perturb the amplitude by a small amount. Starting with the one closest to the imaginary axis we can see that if the amplitude gets smaller we will enter an unstable region of the Nyquist plot which will then increase the amplitude again. Similarly, if we increase the amplitude we will enter a stable region and thus the amplitude will decrease again. As a result whenever the amplitude is changed by some small amount we will be pushed back to the limit cycle as indicated by the arrows in the figure. The same is also true for the leftmost limit cycle. For changes in amplitude we will be pushed back to the limit cycle. In these two cases we can say that the limit cycles are stable since any nearby trajectories will be attracted by it.

Analogously, we can also analyze the behavior around the middle limit cycle. Here we can see that if we reduce the amplitude by a small amount we will be pushed into a stable region which further reduces the amplitude, driving us away from the limit cycle. By increasing the amplitude we will enter an unstable region which further increases the amplitude also driving us further away from the limit cycle. Here we can see that any small perturbation in amplitude will drive us away from the point of the limit cycle, and thus it is unstable.